

Contemporary Dental Pharmacology

Evidence-Based Considerations

Arthur H. Jeske
Editor

Second Edition

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 Springer

Editor
Arthur H. Jeske
UTHealth School of Dentistry
Houston, TX, USA

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Preface

Up-to-date clinical-scientific information remains an absolute requirement for the healthcare practitioner, and dental education is now undergoing a transformation that reflects this. In dental schools around the world, the profession is moving academically from a discipline based on tradition and experience to one that still requires the practitioner's psychomotor skills and judgment and the needs and preferences of the patient, and which also must integrate new research findings and scientific validation of procedures, materials, and patient management.

And perhaps no other component of dental education requires more frequent scientific revision of information than pharmacology. The basic array of medications used in dentistry has changed little in the past 100 years—local anesthetics, analgesics, and antibiotics comprise the majority of drugs administered to and prescribed for dental patients. However, practitioners who graduated from dental school just 10 years ago must now revise their prescribing patterns to reflect an ever-growing list of new classes of medical drugs and new information on optimal effectiveness.

There are four major objectives of this book:

1. Update the advanced dental practitioner on current, high-level scientific information regarding traditionally prescribed dental drugs.
2. Provide the advanced dental practitioner with important information on selected classes of medically prescribed drugs for cardiovascular disease (e.g., novel oral anticoagulants) and neurologic conditions, with special emphasis on incorporating this information into safe and effective patient management.
3. Identify the most important sources of information on the dental and medical drugs covered, in order to enable the advanced dental practitioner to periodically assess new scientific information.
4. Summarize the state of current scientific evidence for the use of basic dental drugs, including the level(s) of evidence for their applications and the strength of recommendation taxonomies (SORT), when those are available.

This second addition focuses on the medications most frequently prescribed in dentistry, as well as important classes of agents which often dictate changes in both regular dental treatments and dental pharmacotherapy, such as anticoagulant/antiplatelet agents and drugs for neurologic disorders, and

we are pleased to present a new chapter on biologic drugs. We are pleased to add two new chapters to the book, with up-to-date information on pediatric considerations and biologic agents. Rather than serving as a comprehensive pharmacology textbook, this book is designed to provide the practitioner with scientific evidence and assess the current evidence-based indications, contraindications, etc. for the drugs included. Finally, it is hoped that the reader will utilize the internet-based resources found in Chap. 11 to build upon the information presented in the book and continue to consult the scientific literature in the future management of patients.

Houston, TX, USA

Arthur H. Jeske

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About the Author

Arthur H. Jeske is a professor (tenured) in the Department of General Practice and Dental Public Health at the University of Texas School of Dentistry at Houston, where he is also Associate Dean for Strategic Planning and Continuing Dental Education. He is Fellow of the International College of Dentists, the American College of Dentists, the Academy of Dentistry International, and the Pierre Fauchard Academy. Dr. Jeske is a Consultant to the American Dental Association's Council on Scientific Affairs and has served as an Evidence Reviewer for the American Dental Association Center for Evidence-Based Dentistry. He is the Pharmacology Section Editor for *The Journal of Craniomandibular & Sleep Practice* and has served as Editor-in-Chief for the 8th through 14th editions of *Mosby's Dental Drug Reference*. Dr. Jeske is also a general dentist with over 40 years of clinical experience.

Introduction to Contemporary Dental Pharmacology and Special Topics

Arthur H. Jeske

The US Food and Drug Administration Drug Approval Process

The dental profession is able to incorporate evidence-based information into the use of drugs owing to the considerable scientific data generated by the US government's Food and Drug Administration Drug Approval Process (www.fda.gov/Drugs/DevelopmentResources/Labeling/ucm425415) as illustrated in Fig. 1.

Monographs produced as labelling information for drugs approved for the US pharmaceutical market contain detailed information about all FDA-approved agents and .pdf versions of the entire document can be easily accessed using the following internet search term:

fda prescribing information [drug name]

These documents are organized in the following format:

Description. The official chemical name of the drug and its structural formula are found here, along with detailed descriptions of all of the ingredients found in all of various dose forms of the drug. Additionally, the dose forms and their imprinted information are described. If the dose

form requires additional preparation (e.g., reconstitution into a suspension), that information is also found in this section.

Clinical Pharmacology. All of the pharmacokinetic information about the drug is found here (e.g., peak blood levels, maximum serum concentrations), as well as routes of metabolism and excretion. The specific actions of the drug are also found in this section (e.g., bacterial susceptibility data for antibiotics).

Indications and Usage. This section lists the various conditions for which the drug has been approved for therapeutic use, including indications for its use in combination with other approved drugs. Uses not included in this list would be considered as "off label," meaning that the drug was not specifically approved by the FDA for such a use.

Contraindications. The conditions under which the drug should not be used are described in this section, such as allergy, and known diseases which may be worsened by administration of the drug.

Warnings. Serious outcomes (e.g., potentially fatal), which can occur as a result of administration of the drug, are listed in this section, along with a description of the emergency measures which must be undertaken to manage the adverse outcome(s). Signs and symptoms of the development of these serious outcomes may also be included in this section.

A. H. Jeske (✉)
UTHealth School of Dentistry, Houston, TX, USA
e-mail: Arthur.H.Jeske@uth.tmc.edu

THE BIOPHARMACEUTICAL RESEARCH AND DEVELOPMENT PROCESS

From drug discovery through FDA approval, developing a new medicine takes at least 10 years on average and costs an average of \$2.6 billion.* Less than 12% of the candidate medicines that make it into Phase I clinical trials will be approved by the FDA.

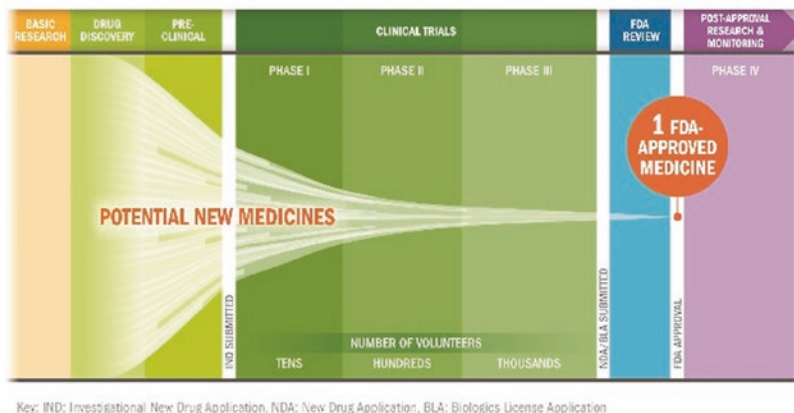


Fig. 1 Biopharmaceutical Research and Development Process. (Reproduced with permission, Pharmaceutical Research and Manufacturers of America (PhRMA). The Complex Biopharmaceutical R&D Process. July 2018.

http://phrma-docs.phrma.org/industryprofile/2018/pdfs/2018_IndustryProfile_DynamicResearchandDevEcosystem.pdf

Precautions. This section is comprised of a detailed list of preexisting conditions that may limit or preclude the use of the drug, recommendations for laboratory tests that may be indicated to monitor the development of adverse outcomes, drug interactions, drug-laboratory test interactions, possible carcinogenic and mutagenic effects of the drug, and effects of the drug on fertility. This is also where the “Pregnancy Category” of the drug is found. Possible adverse outcomes result from the use of drug during labor and delivery, and nursing is also described here, and precautions required for special populations (pediatric, geriatric) are described. Finally, “Information for Patients” is provided here, including dose intervals, duration of therapy, and the need for compliance with dosing instructions.

Adverse reactions. All of the possible untoward reactions to the drug are listed here, along with their incidence derived from clinical trials. These adverse reactions are organized by organ

system, and special notes about adverse reactions observed when the drug is used in combination with other agents are included.

Overdosage. Instructions for managing overdose are generally provided in the first part of this section, and specific outcomes from various levels of intoxication are described. Also included is information on the ability of the drug to be removed by appropriate measures, for example, hemodialysis and the use of antidotal (reversal agents).

Dosage and administration. Doses for adults and special populations for the drug’s indication(s) are listed here, both in gross dosage units (e.g., adults 500 mg) or weight-adjusted dosage (e.g., mg/kg/day for children >3 months). Recommendations for variations in dosage are described in this section, including recommendation for patients with impaired renal and/or hepatic function, hemodialysis patients, and, if appropriate, instructions for mixing products that require reconstitution.

How supplied. This section describes all of various dose forms and strengths of the drug, with their respective NDC code numbers, and recommendations for storage conditions.

Clinical studies. The outcomes from significant clinical studies used to develop specific dosing recommendations, combination uses, etc. are included in this section. These data are very detailed, statistical format to inform practitioners about the evidence for use of the drug for these conditions.

References. List of references for laboratory standards, special uses, etc. is found here.

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Evidence-Based Dentistry

The current definition of evidence-based dentistry, as promulgated by the American Dental Association, is:

an approach to oral healthcare that requires the judicious integration of systematic assessments of clinically relevant scientific evidence, relating to the patient's oral and medical condition and history, with the dentist's clinical expertise and the patient's treatment needs and preferences.

Various types of evidence are generally presented as a pyramid, with the lowest (weakest) levels of evidence at or near the base of the pyramid and the highest (strongest) levels at the apex of the pyramid, as shown in Fig. 2.

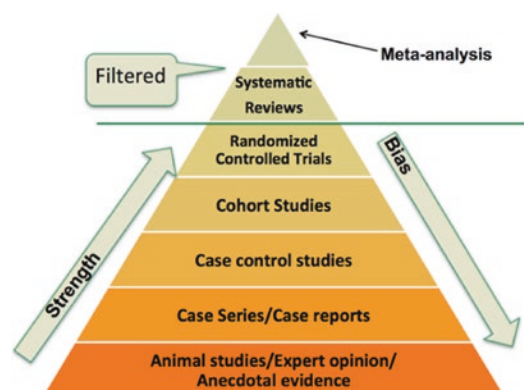


Fig. 2 Evidence pyramid showing hierarchical relationships between various levels of evidence (strongest evidence with lowest bias at apex). (From (Higgins 2011))

When interpreting various types of scientific evidence, it is important to keep in mind the definitions of various types of studies and evidence, as follows.

Meta-Analysis. A statistical analysis that combines or integrates the results of multiple independent clinical trials considered by the analyst(s) to “combinable” to the level of re-analyzing the original data in a single data pool, an approach also referred to as quantitative synthesis.

Systematic Review. A review of a body of data that utilizes explicit methods to locate the relevant, primary studies applicable to a specific research question and explicit criteria to assess their quality (including risk of bias, etc.).

Randomized Controlled Trials (RCTs). Studies in which individual participants (subjects) are allocated to a control group and an experimental group who receive a specific intervention (treatment). The two groups are otherwise identical for any significant variables. The groups are followed (assessed) for specific therapeutic endpoints used to measure the efficacy of a given treatment when compared to the controls.

Cohort Studies. In these studies, groups of people are selected on the basis of their exposure to a particular agent or condition and followed for specific outcomes at various intervals.

Case Control Studies. Individual patients (cases) with a specific condition are matched with controls (without the condition), and a retrospective analysis is used to evaluate difference between the two groups of cases.

Case Study. A report based on a single patient (case) with a specific condition and having had a specific intervention or having been followed over a specific period of time. Multiple cases may be reported in a single publication as a short series of cases.

Expert Opinion, Anecdotal Evidence. Evidence generated as an opinion from a thought leader or expert in a given field of study, typically based on the expert's clinical experiences in a variety of individual patient cases which were not standardized or controlled. Stronger expert opinion is based upon groups of experts achieving consensus through rigorous discussion of avail-

able case and treatment information, frequently under the auspices of a respected professional organization.

As noted above, this textbook was developed to present current or very recent, high-level scientific evidence about the use of drugs in dentistry, and about specific types of medical drugs which may impact the use of dentally-useful therapies. As such, it represents a synthesis of high-level scientific evidence by experts and not opinions of experts.

Special Considerations in the Administration and Prescription of Medications in Dental Patients

There are several critical factors which the dentist must take into consideration prior to the administration or prescription of a drug. These include (Jeske 2017):

1. Use of a medical drug(s) by the patient. The use of a drug prescribed by a physician indicates the presence of a systemic medical condition that may predispose the patient to serious adverse drug-drug interactions and may limit the patient's ability to tolerate dental appointments, particularly stress. The prescribing physician(s) should be involved in the determination of the patient's ability to tolerate specific dental procedures, particularly those categorized as American Society of Anesthesiologists' Physical Status Classification III or IV.
2. Any changes to a patient's medical drug therapy must be done by the prescribing physician, especially as this relates to drugs for serious neurologic and cardiovascular diseases, such as elevated risk of thromboembolism if antiplatelet or anticoagulant therapy is modified.
3. Vital signs and other appropriate physical assessments should be made at any dental visit in which a drug will be administered.
4. The adverse effects of medically prescribed drugs must be monitored and managed appro-

priately. Hyposalivation is a common example of this, and the patient and/or the patient's primary caretakers must be alerted to this consideration and should play a role in minimizing the impact of these conditions, primarily through effective oral hygiene.

5. References should be consulted to obtain detailed information about the management of patients with special needs (Wasserman 2009).

Medication-Related Osteonecrosis of the Jaw

In 2014, the American Association of Oral and Maxillofacial Surgeons (AAOMS) updated its *Position Paper on Medication-Related Osteonecrosis of the Jaw* (MRONJ), formerly termed "bisphosphonate-related osteonecrosis of the jaw (BRONJ)" (American Association of Oral and Maxillofacial Surgeons 2022). This update expanded the list of drugs known to be associated with an increased risk for MRONJ, including antiangiogenic drugs (e.g., denosumab, Prolia®) and corticosteroids. The updated document provides estimates of risk for MRONJ, comparisons of the risks and benefits of medications related to osteonecrosis, and guidance for clinicians on the differential diagnosis of MRONJ and prevention measures, as well as management strategies for patients with disease-stage MRONJ.

According to the AAOMS document, risk for MRONJ is increased in cancer patients who have been treated with zoledronate (Reclast®, Zometa®) and anti-angiogenic monoclonal antibodies (e.g., denosumab) and tyrosine kinase inhibitors (e.g., sunitinib, Sutent®), although the incidence is lower in patients treated with the same drugs for osteoporosis.

Local risk factors for MRONJ include operative treatment (e.g., tooth extraction), anatomic features (e.g., mandibular bone supporting a complete denture), and concomitant oral disease (e.g., inflammatory dental diseases, periodontitis).

The AAOMS position paper provides additional information on genetic, demographic, and

systemic factors in MRONJ and a summary of dental management strategies for patients at-risk of MRONJ, including the following:

- Extraction of non-restorable teeth and those with a poor prognosis prior to initiation of antiresorptive/antiangiogenic therapy
- Elimination of mucosal trauma caused by removable prostheses
- Consultation with the patient's physician(s) in order to follow patient-specific MRONJ-prevention protocols
- Maintenance of good oral hygiene and dental care
- Avoidance of dental implant placement in oncology patients receiving intravenous anti-resorptive or antiangiogenic medications

For patients taking oral bisphosphonates (e.g., alendronate, Fosamax®), specific guidance for cases based on length of medication use includes the following:

- For individuals who have taken an oral bisphosphonate for less than 4 years and have no clinical risk factors, no alteration or delay in planned oral surgery is necessary (this includes any and all procedures common to oral and maxillofacial procedures, periodontists, and other dental providers).
- For those patients who have taken an oral bisphosphonate for less than 4 years and have also taken corticosteroids or antiangiogenic medications concomitantly, the prescribing physician should be contacted to consider discontinuation of the oral bisphosphonate ("drug holiday") for at least 2 months prior to oral surgery, if systemic conditions permit.
- For those patients who have taken an oral bisphosphonate for more than 4 years with or without any concomitant medical therapy, the prescribing physician should be contacted to consider discontinuation of the antiresorptive medication for 2 months prior to oral surgery, if systemic conditions permit.

The complete position paper should be consulted for detailed information, including infor-

mation on the management of patients with established MRONJ.

Biologic Therapies

More detailed coverage of biologic therapies, such as monoclonal antibodies, can be found in a separate chapter of this book. Monoclonal antibodies, anti-TNF agents, and other preparations are now in widespread use and account for a relatively high proportion of drug sales in the United States. While limitations on their use frequently include the need for injection, they have had a significant impact on the management of several important disorders, particularly rheumatoid arthritis and Crohn's disease. They are generally large proteins that can be manufactured via recombinant DNA methodologies. As the number of these agents increases, their impact on dental care and dental drug therapy will become clearer. At this time, the reader is provided with a current list of examples of these in Table 1 agents to call attention to the very serious diseases for which biologic therapies are indicated. The types of agents may be recognized generally by the suffixes of their official ("generic") names, for examples, "-mab" indicates "monoclonal antibody," "-ib" indicates "inhibitor," etc.

Biologic agents can be classified as follows, based on their specific targets:

1. T-cell modulators (e.g., abatacept, Orencia®)
2. B-cell cytotoxic agents (e.g., rituximab, Rituxan®)
3. IL-1 (interleukin) blockers (e.g., anakinra, Kineret®)
4. Anti-IL-6 receptor antibody (e.g., tocilizumab, Actemra®)
5. JAK (Janus kinase) inhibitors (e.g., tofacitinib, Xeljanz®)
6. TNF (tumor necrosis factor)-alpha blockers (e.g., adalimumab, Humira®)

For dental patients taking biologic therapies, the following guidelines should be followed:

- The prescribing physician(s) should be consulted to assess the status of the patient's dis-

Table 1 Examples of monoclonal antibodies approved for use in the United States (Jeske 2017)

Official name	Tradename	Indication(s)
Abciximab	ReoPro®	Adjunct for prevention of thromboembolism
Adalimumab	Humira®	Rheumatoid arthritis
Alemtuzumab	Campath®	Chronic lymphocytic leukemia
Basiliximab	Simulect®	Anti-rejection (renal transplant)
Bevacizumab	Avastin®	Metastatic colorectal and other cancers
Canakinumab	Ilaris®	Cryopyrin-Associated Periodic Syndrome
Certolizumab	Cimzia®	Rheumatoid arthritis
Cetuximab	Erbitux®	Squamous cell carcinoma
Daclizumab	Zenapax®	Anti-rejection (renal transplant)
Denosumab	Prolia®	Osteoporosis (high fracture risk)
Eculizumab	Soliris®	Nocturnal hemoglobinuria
Guselkumab	Tremfya®	Plaque psoriasis
Golimumab	Simponi®	Rheumatoid arthritis
Ibritumomab	Zevalin®	Non-Hodgkin's lymphoma
Infliximab	Remicade®	Rheumatoid arthritis
Ipilimumab	Yervoy®	Unresectable metastatic melanoma
Muromonab	Orthoclone®	Anti-rejection (renal transplant)
Natalizumab	Tysabri®	Multiple sclerosis Crohn's disease
Ofatumumab	Arzerra®	Chronic lymphocytic leukemia
Omalizumab	Xolair®	Allergic asthma
Palivizumab	Synagis®	Respiratory syncytial virus
Panitumumab	Vectibix®	Metastatic colorectal cancer
Ranibizumab	Lucentis®	Macular degeneration
Rituximab	Rituxan®	Non-Hodgkin's lymphoma
Tocilizumab	Actemra®	Rheumatoid arthritis
Trastuzumab	Herceptin®	Breast and gastroesophageal cancers
Ustekinumab	Stelara®	Plaque psoriasis
Vedolizumab	Entyvio®	Ulcerative colitis, Crohn's disease

ease and the ability of the patient to tolerate dental procedures.

- Immunosuppression is associated with biologic therapies and may predispose the patient to a higher incidence and severity of oral and systemic infections (e.g., tuberculosis), including fungal infections.
- Because the biologic agent must be injected, injection site discomfort and acute symptoms may accompany administration (e.g., nausea, diarrhea).

FDA Pregnancy and Lactation Drug Labeling

In 2015, the US Food and Drug Administration (FDA) required major changes in the labeling and information for the use of prescription drugs and

biologic agents in pregnancy, including risks of exposure to pregnant and lactating females, review of the data supporting these risks, and other information to assist practitioners and patients in making informed decisions about medication use during pregnancy and lactation (U.S. Food and Drug Administration 2015; Watkins and Archambault 2016; Katzung and Trevor 2021). These new labeling requirements mandated that manufacturers replace the old pregnancy risk categories (A, B, C, D, and X) with the updated information within 3 years [FDA]. Various sections of the new labeling information are illustrated in Table 2.

Additionally, the FDA now requires manufacturers to list pregnancy registry information (a pregnancy registry being an ongoing, systematic, epidemiologic study that collects and assesses data on a mother's, fetus' or infant's adverse

Table 2 FDA labeling information for pregnant and lactating patients

Pregnancy
• Pregnancy exposure summary
• Risk summary
• Clinical considerations
– Disease-associated maternal and/or embryo/fetal risk
– Dose adjustments during pregnancy and postpartum
– Maternal adverse reactions
– Fetal/neonatal adverse reactions
– Labor or delivery
• Data
– Human
– Animal
Lactation
• Risk summary
• Clinical considerations
• Data
Females and males of reproductive potential
• Pregnancy testing
• Contraception
• Infertility

reactions to medications, biologic agents and vaccines) (www.fda.gov/).

Conclusion

Dental drug therapy typically follows and evolves from the development of new drug entities (or improvements in or new indications for existing

drugs) in medicine. The FDA approval process generates a tremendous amount of information that is foundational to the evidence-based use of drugs in dental practice. When new drugs are approved, it is incumbent upon the prudent dentist to determine not only what adverse drug interactions may occur between the new medical drug and existing dental drugs, but to ascertain the implications that the new drug has for the comprehensive management of the patient’s dental treatment.

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Local Anesthetics

Arthur H. Jeske

Mechanism of Action and Clinical Implications

Unlike most other pharmaceuticals, local anesthetics begin their therapeutic action immediately upon application to a target tissue, usually via injection into an anatomic space adjacent to a nerve trunk (e.g., nerve block) or nerve endings (e.g., infiltration). This requirement results from the need to abolish pain impulses in small areas of the body, and the poor oral bioavailability obviates any utility of these drugs for systemic effects.

Beginning with the pioneering work of Ritchie and Greengard, which established the active form of conventional local anesthetic drugs and the influence of pH on their clinical behavior, work over the last three decades of the twentieth century has led to an understanding of the principal site of action as the voltage gated sodium channel (Ritchie and Greengard 1966). A folding diagram of the channel is illustrated in Fig. 1.

Voltage-gated sodium channels are ubiquitous in nature and are designated by the symbol Na_v, followed by a number 1.1 to 1.9, representing isoforms found in various tissues (Alexander et al. 2017). They underlie the functioning of all electrically excitable tissues, including periph-

eral and central neurons and the pacemaker and conducting tissues of the heart, accounting for the typical signs and symptoms of systemic toxicity observed at elevated blood levels (see section “Comparative Efficacy of Injectable Local Anesthetics”).

Under pathologic conditions (e.g., nerve trauma), neurons can become phenotypically altered, and there can be expression of channels that are pharmacologically characterized as “tetrodotoxin resistant,” indicating extreme resistance to block, including local anesthetic block (Alexander et al. 2017; Waxman and Zamponi 2014). This is one possible explanation for the difficulty encountered by clinicians in attempting to anesthetize teeth, which have been chronically infected and inflamed due to caries, trauma, etc.

Nerve block caused by local anesthetics begins to occur immediately following injection of the drug near the nerve. Following equilibration of the local anesthetic solution with the extracellular fluid, the uncharged species of the anesthetic penetrates the nerve membrane and enters the intraneuronal space. There, the anesthetic once again equilibrates into positively charged and uncharged forms, according to the Henderson-Hasselbalch equation:

$$\text{pKa} = \text{pH} - \log \left[\frac{\text{base}}{\text{conjugate acid}} \right]$$

When the nerve membrane is depolarized, voltage sensors alter the conformation of the channel, the sodium channels open, and local

A. H. Jeske (✉)
UTHealth School of Dentistry, Houston, TX, USA
e-mail: Arthur.H.Jeske@uth.tmc.edu

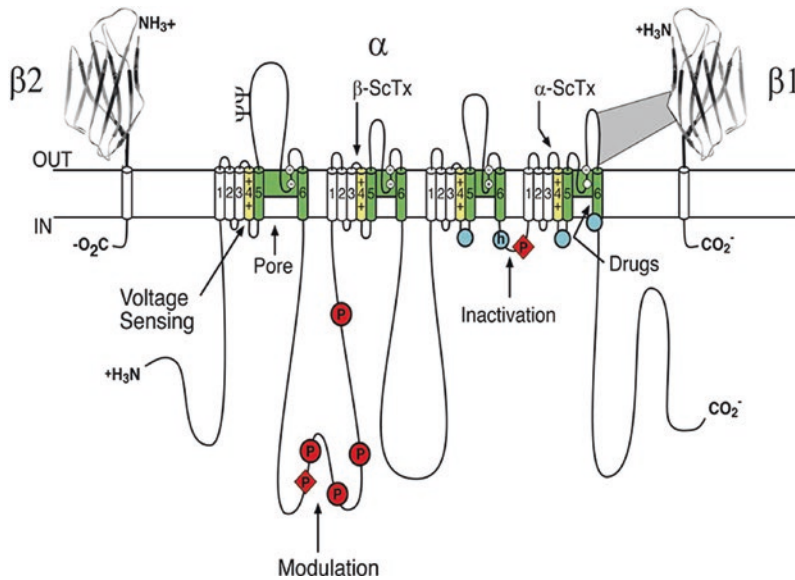


Fig. 1 Folding diagram of the mammalian voltage gated sodium channel. The alpha subunit is comprised of four domains of six alpha-helically coiled amino acid sequences. These groups span the nerve membrane and surround and form the ion “pore.” A voltage-sensing sequence of positively charged amino acids in each domain responds to changes in polarity of the adjacent nerve membrane, resulting in a transient opening, followed by inactivation (modulated by the inactivation gate) and closing of the channel during a depolarization. Local anesthetic binding occurs at the sixth transmembrane helix, at a site near the internal opening of the channel (indicated by “drugs”). Reprinted from *Neuron* Volume 26, No. 1, Catterall WA, From Ionic Currents to Molecular Mechanisms: The Structure and Function of Voltage-Gated Sodium Channels, pp. 13–25, 2000, with permission from Elsevier

vation (modulated by the inactivation gate) and closing of the channel during a depolarization. Local anesthetic binding occurs at the sixth transmembrane helix, at a site near the internal opening of the channel (indicated by “drugs”). Reprinted from *Neuron* Volume 26, No. 1, Catterall WA, From Ionic Currents to Molecular Mechanisms: The Structure and Function of Voltage-Gated Sodium Channels, pp. 13–25, 2000, with permission from Elsevier

anesthetic molecules can access and bind to their active site, which lies within the channel, near the inner opening, at the sixth transmembrane helices of domains 1, 3, and 4 of the channel’s alpha subunit (Catterall 2012). Local anesthetic binding stabilizes the channel in the inactivated form, which cannot conduct Na^+ ions. Accumulation of a sufficient number of inactivated channels prevents subsequent conduction of nerve impulses, effecting a state of nerve block, or local anesthesia.

Two additional mechanisms of local anesthetic action have been proposed, including the following:

1. Direct access to the anesthetic binding site from within the lipid core of the neuronal membrane. This mechanism, also known as the “hydrophobic pathway,” explains the nerve-blocking actions of benzocaine, an

anesthetic that only exists in an uncharged form (Catterall 2012; Lirk et al. 2014).

2. Entry of charged anesthetic molecules into the neuron via large-pore vanilloid-1 (TRPV-1) channels. This pathway, also known as the “hydrophilic pathway,” explains the ability of investigational fixed-charge anesthetics to induce nerve blockade when these channels are activated by acute applications of capsaicin.

Local anesthetics may also act via secondary targets, including potassium and calcium channels, and G protein-coupled and N-methyl-D-aspartate (NMDA) receptors (Reader et al. 2017), although it is unlikely that these targets play a major role in the blockade of peripheral nerve fibers (Salinas et al. 2004).

Rapidly firing neurons are blocked earlier than those with lower firing rates, and smaller neurons

are generally blocked before larger ones. In the case of rapidly firing neurons, the rapid opening of the voltage-gated sodium channels increases the likelihood that local anesthetic molecules can access their binding sites within the channel, in a phenomenon known as “frequency dependence” (Drasner 2015).

Neuronal diameter and other intrinsic traits may alter the susceptibility of nerves to local anesthetic block. For example, for nerves with similar cross-sectional area, those with myelin would tend to be blocked sooner than those that lack myelin, as the myelin may serve as a “reservoir” for uncharged local anesthetic. Theoretically, larger diameter fibers are blocked later than smaller ones because the distance over which the larger fibers can passively propagate an electrical impulse is greater. Generally, the order of blockade of sensory fibers is described as (Drasner 2015):

Temperature > pain > light touch > pressure > proprioception.

Motor fibers are not blocked by dental local anesthetics unless the anesthetic is placed into close proximity to a motor nerve, for example, cranial nerve VII, as occurs when the parotid gland is accidentally entered during mandibular block techniques, resulting in temporary facial nerve paralysis. Motor nerve block is a much more important consideration when local anesthetics are used in the setting of epidural and spinal anesthesia (Drasner 2015; Neal et al. 2018).

Physicochemical Characteristics

Injectable dental local anesthetics are relatively small molecular structures (molecular weights ranging from 234 to 321 Da). These values and their other physicochemical properties are presented in Table 1.

Relatively high lipid solubility and protein binding for bupivacaine and tetracaine are the basis for their relatively long duration of action, that is, they possess high affinity for the proteo-lipid complex structure of the neuronal cell membrane. Lipid affinity is also the basis for the use of lipid emulsion for treating systemic toxicity of local anesthetics (see section “Comparative Efficacy of Injectable Local Anesthetics”).

Chemically, conventional local anesthetics are classified as amides or esters, based upon the linkage of the aromatic (lipophilic) moiety at one end of the molecule to the intermediate, aliphatic chain. Amides are metabolized primarily via hepatic microsomal oxidation, while esters, with a relatively weak intermediate bond, are rapidly hydrolyzed by plasma cholinesterases (Reader et al. 2017). Their relatively small molecular size limits their allergenicity, although metabolic products of ester-type agents can be allergenic by token of their molecular similarity to para-amino benzoic acid (PABA). Recently, sensitization to amide local anesthetics has been noted to occur in connection with the application of large amounts of topical preparations to skin to reduce pain associated with cosmetic enhancements of the body (hair removal, etc.) (To et al. 2014).

Table 1 Physicochemical characteristics of common local anesthetics

Drug	Lipid partition coeff.	Protein binding (%)	Mol. wt. (Da)	Concentration in dental cartridges (%)
Articaine	17	70	321	4
Bupivacaine	346	95	288	0.5
Lidocaine	2.4	64	234	2
Mepivacaine	21	77	246	2–3
Prilocaine	25	55	220	4

Effect of pH and pK/Buffering

Dental local anesthetic drugs are weak bases and possess a tertiary amine terminus, which can be protonated (except for benzocaine). This highly polar group confers water solubility upon the drugs, although the positively charged form (protonated) of the anesthetic apparently is also essential for the binding of the anesthetic to its active site in the voltage-gated sodium channel. Similarly, the aromatic ring at the opposite end of the local anesthetic molecule confers lipid solubility upon the drug, allowing it to traverse the cell membrane. Diffusion into the nerve does not occur via ion pores but through passive diffusion through the neuronal membrane by the uncharged form of the anesthetic. Thus, the higher the pKa of the anesthetic, the greater the ionization of the drug in the extracellular fluid and the lower the fraction of uncharged form (Table 2). For the long-acting agents bupivacaine and tetracaine, with relatively high pKa, their potency derives from their high lipid solubility and protein-binding affinity (Table 1), a critical factor that explains not only their higher potency, that is, their nerve-blocking concentration, but also their long duration (Drasner 2015).

When considering the values presented in Table 1, the reader should consider the fact that these numbers apply to plain solutions. For preparations with epinephrine, the pH of the solution is adjusted downward, and such solutions may have pH values as low as 3.3, with correspondingly greater proportions of charged versus uncharged molecules. However, once injected into body tissues, the pH quickly equilibrates to normal tissue fluid pH. This re-equilibration is not immediate, with the result that some patients

may experience a burning sensation during the injection. This consideration, along with a theoretical advantage that increasing the relative proportion of anesthetic molecules in the uncharged form can accelerate the onset of the local anesthetic effect, has led to the development of buffering systems for dental local anesthetic cartridges.

The potential benefits of buffering local anesthetic solutions (to raise their pH) theoretically include faster onset and greater efficacy (owing to a larger fraction of uncharged anesthetic available to diffuse into the nerve), as well as a reduction in injection pain associated with acidic solutions. Reduced pH is also believed to play a role in the failure of local anesthetic to block nerves when their tissues of innervation are inflamed and the pH of the extracellular fluid may drop by as much as 0.5 pH units (Lirk et al. 2014).

Recent studies on the effect of bicarbonate buffering of the local anesthetic solution prior to injection have produced equivocal results. For example, Saatchi et al. (2015) studied the effect of buffering 2% lidocaine with 1:80,000 epinephrine for the success rate of the inferior alveolar nerve block (IANB) in patients with symptomatic irreversible pulpitis. Comparing the administration of two cartridges of the non-buffered anesthetics with the same dose buffered with 0.18 ml of 8.4% sodium bicarbonate solution, the investigators achieved success rates of 62.5% for the buffered solution versus 47.5% for the non-buffered one, without achieving statistical significance. Similar outcomes were obtained in a study of the effect of 2.8 ml buffered 4% lidocaine for the IANB (versus the same protocol without buffering) and showed no significant difference for success in patients with symptomatic irreversible pulpitis (32% and 40%, respectively), nor were the ratings of pain during the injection different (Schellenberg et al. 2015).

Interestingly, one study has demonstrated that injection of sodium bicarbonate in a 2% lidocaine solution via buccal infiltration of irreversibly inflamed lower first molars followed by a conventional IANB block with 2% lidocaine with epinephrine produced significantly better success

Table 2 Relationship between local anesthetic pKa and fraction of unionized drug at pH 7.4

Drug	pKa	% Drug uncharged
Articaine	7.8	28
Bupivacaine	8.1	15
Lidocaine	7.9	25
Mepivacaine	7.6	37
Prilocaine	7.9	24
Tetracaine	8.6	7

rates versus infiltration of lidocaine with distilled water (78% vs 44%) (Saatchi et al. 2016). The mechanism by which this difference occurred may have been due simply to the additional buffered lidocaine at the tooth site, with improved diffusion from neutralization of inflamed extracellular fluid, and it is tempting to speculate that alteration of the function of acid-sensing ion channels may also have played a role. In oral surgery, the effect of buffering 2% lidocaine with epinephrine on various clinical parameters has been investigated, including the impact on blood levels of the local anesthetic.

It has been shown that the buffered preparation resulted in significantly lower blood levels after IANB, and subjects reported lower injection-pain scores and shorter time to lip numbness with the buffered solution (Phero et al. 2017). Theoretically, the buffered solutions would contain smaller fractions of charged local anesthetic molecules, which are hydrophilic and would be expected to be absorbed by blood more extensively than the uncharged, less polar forms. This could become a significant factor when large amounts of local anesthetic are required for surgical procedures or for postoperative pain control, for example, intercostal block.

Recently, a meta-analysis of the efficacy of sodium bicarbonate buffered versus non-buffered lidocaine with epinephrine for the IANB was conducted, and the outcomes of the analysis were based on 11 included studies (Guo et al. 2018). While this report found that buffering significantly decreased time of onset of anesthesia (48 s shorter) and injection pain (five-point reduction on the 0–100 VAS), there was no significant difference for anesthetic success rate or the incidence of painless injections. The investigators cautioned that the quality of evidence was low to moderate due to statistical heterogeneity and low sample sizes.

Shurtz et al. evaluated buffered 4% articaine for infiltration of the mandibular first molar and observed no differences in pain of injection or onset of anesthesia when compared with unbuffered 4% articaine (Shurtz et al. 2015). Most recently, a systematic review of buffered versus unbuffered local anesthetic solutions in children

concluded that buffering *may* reduce pain and accelerate onset of anesthesia associated with the inferior alveolar nerve block in children but that sample sizes and number of randomized controlled trials are needed (Tirupathi and Rajasekhar 2020).

At this time, practitioners who adopt buffering as part of their IANB local anesthetic protocols for routine dental procedures can expect some reductions in onset time and injection pain but little advantage for overall pulpal anesthesia success rates. The impact of buffering on the success rates for other types of injections will require additional scientific studies. While the buffering of local anesthetics is considered to be a standard of care in medicine to minimize the discomfort of intradermal local anesthetic injections, the literature in dentistry suggests that it is not a standard of care for intraoral injections at this time.

Structure-Activity Relationships

Structural modifications to the basic local anesthetic molecule that increase lipid solubility have a marked impact on potency, protein binding affinity, and duration of action. An example is that of bupivacaine, in which substitution of the methyl group of mepivacaine with a butyl group results in a significant increase in potency and duration of action, the latter effect owing to the increased binding affinity of the drug for the neuronal membrane. In the case of bupivacaine, two additional considerations arise from the increase in proteolipid affinity—first, bupivacaine possesses an inherently higher cardiotoxicity than other local anesthetics, and second, bupivacaine can be produced in a liposomal delivery formulation, in which slow release from liposomes introduced into a surgical site produce prolonged postoperative pain relief.

Articaine is distinguished from other amide local anesthetics by two unique structure features—first, it possesses a sulfur-containing thiophene ring, which apparently contributes to its ability to diffuse through both soft and hard tissues more readily than other conventional local

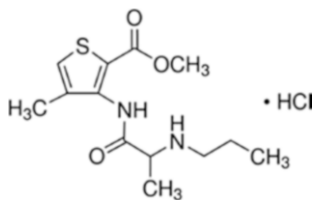


Fig. 2 Structural formula of articaine

anesthetics (Fig. 2). It also possesses an ester side chain, which, because it is quickly hydrolyzed by non-hepatic esterases, results in more rapid inactivation after it leaves its therapeutic site of action, contributing to a relatively short (23 min) plasma half-life. Since the resulting hydrolysis product articainic acid is not pharmacologically active, systemic toxicity is reduced relatively rapidly, as opposed to the prototype amide lidocaine, whose initial oxidative metabolite (monoethyl glycine xylidide) is an active intermediate than may prolong systemic toxicity.

Comparative Efficacy of Injectable Local Anesthetics

Beginning in 2009, there have been several systematic reviews focusing on the comparative efficacy of articaine, lidocaine, and some other local anesthetic preparations. Among the most informative, of course, are those which performed and reported the outcomes of meta-analyses (Corbella et al. 2017; Katyal 2010; Kung et al. 2015; Paxton and Thomas 2010; George et al. 2018; Zhang et al. 2018). Particular interest in the comparative efficacy of articaine versus lidocaine arose from the increasingly widespread use of articaine and its availability, beginning in 1999, in the US dental marketplace.

The primary outcome measure used to compare efficacy among injectable local anesthetics is complete pulpal anesthesia, as assessed by a lack of response of test teeth to electric pulp testing or ability to perform endodontic treatment on teeth presenting with symptomatic irreversible pulpitis with no or only mild procedural discomfort. A variety of additional outcome measures have been reported, most frequently including

speed of onset and duration of local anesthesia and local and systemic adverse effects. A common secondary measure is the global assessment of subjects' satisfaction with the pain control during a procedure.

For success of anesthesia, it now appears that 4% articaine with 1:100,000 epinephrine is superior to 2% lidocaine with 1:100,000 epinephrine for posterior teeth, which present with symptomatic irreversible pulpitis (31% vs 49% incidence of successful anesthesia, RR* 1.60, 95% CI* 1.10 to 2.32) (George et al. 2018).

For studies of soft- and hard-tissue (e.g., tooth extraction) procedures, clear evidence favoring one particular local anesthetic is lacking (George et al. 2018), and variations in volume of injected anesthetic (one versus two cartridges for the inferior alveolar nerve block) did not reveal any significant differences for success between lidocaine and articaine, although in the most recent systematic review specifically of the inferior alveolar nerve block, an additional benefit of supplementary buccal infiltration was evident (Corbella et al. 2017). There is moderate quality evidence that for surgical and periodontal procedures, 2% lidocaine with 1:100,000 epinephrine was superior to 4% prilocaine plain, although comparisons for 4% prilocaine with 1:200,000 epinephrine are not available, and comparative results for surgical outcomes using 0.5% bupivacaine with 1:200,000 epinephrine versus other anesthetic agents are reported as "uncertain" (George et al. 2018).

Earlier systematic reviews (Katyal 2010; Kung et al. 2015) clearly demonstrated a significant contribution of greater anesthetic success for 4% articaine with 1:100,000 epinephrine for buccal infiltration compared with 2% lidocaine with 1:100,000 epinephrine to the overall estimated success rate for articaine. For example, Brandt et al. determined that the odds ratio (OR) for the overall success rate for 4% articaine versus 2% lidocaine was 2.44 (95% CI 1.59–3.76, $p < 0.0001$), while when only buccal infiltration was considered, the advantage for articaine increased to OR 3.81 (95% CI 2.71–5.3, $p < 0.00001$) (Brandt et al. 2011).

More recently, the anesthetic effectiveness of articaine and lidocaine were compared in a systematic review and meta-analysis of trials involving children <14 years old, and while articaine proved to superior to lidocaine, the differences were small based on outcomes for Facial Pain Scores (Taneja et al. 2020). In studies involving the extraction of mandibular third molars, a systematic review and meta-analysis comparing articaine versus lidocaine concluded that for articaine, the success rate, subjective onset time, and duration of anesthesia were somewhat superior, although intraoperative pain scores and objectively measured onset times were not significantly different (Zhang et al. 2018). In 2020, an umbrella review and a systematic review and meta-analysis of the comparative efficacy of articaine versus lidocaine in cases of symptomatic irreversible pulpitis both found significantly higher pulpal anesthesia success rates for articaine (Larocca de Geus et al. 2020; Nagendrababu et al. 2020).

For virtually all studies of local anesthetics, adverse events are rare, and differences in pain on injection do not appear to be significant when comparing 4% articaine to 2% lidocaine preparations (George et al. 2018). The practitioner has several options for local anesthetic preparations, and based on current scientific evidence, one practical approach is to stock both 2% lidocaine with 1:100,000 epinephrine (for blocks and maxillary buccal infiltrations), 4% articaine with epinephrine (for mandibular buccal infiltration anesthesia), and a plain solution, to be used in cases in which epinephrine is absolutely contraindicated (e.g., 3% mepivacaine plain, 4% prilocaine plain).

A 0.5% bupivacaine does not produce reliably long-acting anesthesia in the maxilla, and it is not recommended for use in children. Supplementary anesthetic techniques (e.g., intraosseous injections) can also be employed successfully to help overcome failures of block and or infiltration anesthesia and to limit the total dose of anesthetic, if necessary. A 2% lidocaine with 1:100,000 epinephrine or 3% mepivacaine plain is recommended for use with these techniques (Reader et al. 2017).

Local Anesthetic-Related Nerve Injury

Interest in local anesthetic-related nerve injury peaked in the 1990s, at a time when the use of 4% articaine solutions had become widespread, both in Europe and Canada, following retrospective estimation of the incidence of paresthesias in the dental patient population versus the estimated frequency of use of various types of dental anesthetics (Haas and Lennon 1995). These concerns have persisted into the twenty-first century (Garisto et al. 2010), although limited scientific evidence now suggests that the initial assessments of neuronal toxicity may have been overestimated.

In fact, a second formulation of 4% articaine (with 1:200,000 epinephrine) was subsequently approved by the US Food and Drug Administration.

It is generally accepted that local anesthetics can be neurotoxic when relatively concentrated solutions, for example, 2–4%, are placed in close proximity to small nerves or in tissues with limited or depleted buffering capability (e.g., cerebrospinal fluid). Cauda equina syndrome is well known in medicine as a complication of spinal anesthesia (Drasner 2015). In dentistry, local anesthetic administration is among five procedures associated with the majority of cases of nerve injury, including implant placement, endodontic treatment, bone grafting, and dentoalveolar surgery (particularly lower third molar removal) (Pogrel 2017). Mechanisms that underlie these injuries include induction of apoptosis at lower concentrations and late apoptotic or necrotic cell death at higher ones (Verlinde et al. 2016; Werdehausen et al. 2009). In a neuroblastoma cell line, all conventional local anesthetics were shown to induce apoptosis and neurotoxicity in direct relationship to the anesthetic concentration and appears to be related to the fat solubility and potency of the drug, with no significant differences between amide- and ester-type anesthetics (Werdehausen et al. 2009).

A recent literature review of local anesthetic-induced neurotoxicity identified multiple risk factors for nerve damage, including the block

technique used, patient risk factors (e.g., preexisting neuropathy or neurological diseases), and surgical factors (e.g., inadvertent nerve compression). Cellular mechanisms that may be associated with neurotoxicity include effects of local anesthetics on the intrinsic caspase pathway, PI3k-pathway, and MAPK-pathways. For peripheral nerve blocks, this review estimated the incidence of local anesthetic-related neurological complications as less than 3% and concluded that most are “transient sensory deficits” (Verlinde et al. 2016).

In the United States, the Department of Oral and Maxillofacial Surgery at the University of California San Francisco has perhaps the longest-running case series of patients with iatrogenic injuries to the maxillofacial area (Pogrel and Thamby 2000). This case series now includes 324 patients whose nerve injuries could only be the result of local anesthetic injections, and the following are reported as consistent features in these cases (Pogrel 2017):

- The lingual nerve is the most frequently affected nerve (twice the rate as the inferior alveolar nerve).
- 33% of the patients experience dysesthesia as a result of the injury.
- Recovery, if it occurs, does so in three months, and later recoveries are rare.
- Virtually, all cases involved the inferior alveolar nerve block, and among permanent injuries, the vast majority involved this block injection.

There is considerable variation in the estimates of the actual incidence of these injuries, now ranging from 1 in 6000 for temporary neural deficits with the IANB to 1 in 30,000 for permanent injuries (Pogrel and Thamby 2000).

Contrary to some expert opinion, this type of nerve damage is not associated with a specific anesthetic drug, although it does appear to predominantly affect the lingual nerve. There appears to be no established relationship to the volume of anesthetic injected, and there is no beneficial treatment known, despite unfounded suggestions that corticosteroids may improve the long-term prognosis, nor is there evidence to sup-

port a beneficial effect of vitamin and other dietary supplements. Attempts at surgical correction of dysesthesias have not been successful, and the surgery has typically not revealed macroscopic damage to the nerves.

At this time, there appears to be no known method of preventing local anesthetic-related nerve injury (Pogrel et al. 2011; Pogrel and Thamby 1999; Pogrel 2007, 2002).

The medicolegal considerations involving iatrogenic lingual nerve damage have recently been addressed (Pippi et al. 2018) and highlight the difficulties encountered when attempting to establish the etiology of the injury, that is, differentiation of anesthetic, mechanical, chemical, and even thermal mechanisms.

It is also noteworthy that clinical tests aimed at assessing lingual nerve sensory function have low sensitivity and only moderate specificity. Experts have concluded that the patient must be warned of the possibility of nerve injury during the informed consent process, regardless of the relatively low frequency of local anesthetic-related injury. Unfortunately, there are no good alternatives to local anesthetic injection for pain control for routine, outpatient dental procedures.

Local Anesthetic Systemic Toxicity (LAST) and Management

Local anesthetic systemic toxicity (LAST) is a serious and potentially life-threatening reaction typically associated with high doses and/or multiple injections of local anesthetics in a relatively short period of time. Occasionally, inadvertent intravenous or intraarterial injection can occur, in which case even relatively small amounts of local anesthetics may precipitate systemic toxicity. There are several mechanisms that play a role in the development of this scenario, which can result in the clinical presentation of the reaction being atypical (El-Boghdadly and Jinn Chinn 2016). The current literature suggests that LAST is relatively rare (2.0–2.8 cases per 10,000 peripheral nerve blocks) and is even less likely for infiltration anesthesia utilized in routine dental procedures (Barrington and Kluger 2013).

Factors related to the occurrence of LAST include the following (El-Boghdadly and Jinn Chinn 2016):

- Patient factors. In elderly patients with reduced hepatic and renal clearance of drugs, even though plasma protein binding and peak plasma concentration of anesthetics may not differ from those seen in younger adults. Other patient conditions that may contribute to LAST include cardiac disease (which reduces blood flow to vital organs and, therefore, reduces drug clearance), liver disease (a consideration when doses are repeated or local anesthetic is continuously infused), and pregnancy (e.g., increased cardiac output beginning in the second trimester of pregnancy may increase local anesthetic absorption from injection sites and increase plasma concentrations). The well-established relationship between bupivacaine-induced LAST and pregnancy dictates extra precautions and played a role in the development of lipid emulsion therapy of LAST, described later in this section.
- Drug doses. Appropriate dosing of local anesthetics, based on patient body weight and systemic risk factors, is the single most important factor in preventing LAST. Maximum doses for common dental local anesthetics are found in Table 3 (based on manufacturer's data in FDA-approved labeling).
- Pharmacokinetics. There are only a few serious pharmacokinetic drug interactions involv-

ing amide local anesthetics (see section "Local Anesthetic-Related Nerve Injury").

- In one case, some classes of histamine H-2 receptor antagonists, for example, cimetidine, inhibit the metabolic disposition of lidocaine by the hepatic cytochrome P450 CYP 3A4. Propranolol (Inderal®) and halogenated hydrocarbon anesthetics may reduce cardiac output and hepatic perfusion sufficiently to reduce the clearance of amide local anesthetics. In both of these cases, a careful medical history and avoidance of potentially interactive agents is warranted (Moore 1999).
- Anesthetic technique. Certain intraoral block injections, particularly the posterior superior alveolar and inferior alveolar nerve block, carry a higher probability of penetration of a blood vessel (based on likelihood of positive aspiration). Careful aspiration, slow injection, and increasing use of buccal infiltration anesthesia for routine, single-tooth dental procedures are important contributions to a reduction in LAST in the dental outpatient population.

The signs and symptoms of LAST may present in temporal variations (Lirk et al. 2014). "Instant" LAST involves rapid-onset seizures and cardiovascular depression and is usually associated with inadvertent intravascular administration of relatively high volumes of local anesthetic solution, or smaller amounts injected intraarterially. "Slow" LAST typically follows overdose, rapid absorption of anesthetic by the circulation, reduced metabolism/hepatic clearance, or reduced plasma protein binding, taking up to 30 min or longer to manifest.

The classical signs and symptoms of impending LAST include sensory disturbances, following by early signs of CNS stimulation attributable to depression of central inhibitory neurons (muscle twitching and fasciculation). As plasma levels increase further, tonic-clonic seizures ensue, culminating in post-seizure depression with possible respiratory arrest. Cardiovascular parameters during LAST vary with the stage of toxicity, with increases in blood pressure during the phase of CNS excitation, and profound loss of blood pres-

Table 3 Maximum recommended doses (MRDs) for conventional dental local anesthetics (manufacturer's data)

Drug	Dose mg/kg	Absolute max dose (mg) ^a
Articaine	7	NA ^b
Bupivacaine	2	90
Lidocaine ^c	7	500
Mepivacaine	6.6	400
Prilocaine	8	600

^a Dose that should not be exceeded in a single appointment

^b The manufacturer of articaine does not specify an absolute maximum dose

^c Applies to lidocaine preparations with epinephrine

sure with severe bradycardia in the terminal stage. Exceptions to this cascade may occur, for example, if the patient is being sedated with a benzodiazepine (which possesses anticonvulsant activity), CNS stimulation and seizures may be obtunded or not occur at all. LAST is potential fatal, but the employment of preventive measures reduces the occurrence of cardiac arrest (El-Boghdadly and Jinn Chinn 2016).

The management of LAST involves rapid and, depending on the severity of the toxicity, aggressive rescue measures. Lipid emulsion infusion is becoming widely accepted as an interventional measure in LAST, although it does not substitute for airway management and ventilation with 100% oxygen. Intravenous infusion of lipid emulsion (e.g., 20–30% long-chain triglycerides produced from soybean oil) creates a “lipid sink” in the blood, which allows highly lipophilic local anesthetics such as bupivacaine to partition into the plasma and out of cardiac tissue, thus reducing the cardiac depression that is characteristics of this type of anesthetic. Additional mechanisms that contribute to the beneficial effects of lipid emulsion therapy of LAST-related cardiac depression include promotion of mitochondrial fatty acid metabolism and ATP synthesis and direct increases in cardiac contractility related to an increased intracellular content of calcium in myocardial cells (Fettiplace et al. 2014). Additionally, lipid emulsion can inhibit nitric oxide release and reverse cardiac sodium channel blockade (Lieblich and Danesi 2017).

A recent systematic review based on case reports has established the efficacy of intravenous lipid emulsion therapy for the management of local anesthetic-induced cardiotoxicity (Cao et al. 2015). CNS toxicity is also ameliorated by lipid emulsion therapy and has been demonstrated to be effective not only for LAST due to bupivacaine but also for lidocaine and mepivacaine. At this time, there is no evidence to confirm or refute a beneficial effect of adding epinephrine to lipid emulsion.

An example of a protocol for lipid emulsion infusion for a patient weighing 70 kg is a bolus infusion of 100 ml of Intralipid® 20% over 1 min,

followed by a continuous infusion of 1000 ml/h (Ok et al. 2018).

To summarize, the management of LAST in the dental outpatient setting would include the following (El-Boghdadly and Jinn Chinn 2016):

- **Prevention:** Appropriate local anesthetic dosing, use of less cardiotoxic anesthetics, aspiration before injection.
- **Detection:** Observe for CNS signs (agitation, confusion, seizures, mental depression, sensory disturbances) and CVS signs (hypertension, tachycardia, hypotension, bradycardia, asystole).
- **Initial Management:** Stop injection and dental procedure, summon help and emergency kit, open and maintain airway, ventilated with 100% oxygen, obtain vascular access (i.v. or i.o), terminate seizures (administer parenteral benzodiazepine), consider intravenous infusion of lipid emulsion if symptoms persist or recur.

The emergency use of benzodiazepines for seizure termination is described in chapter “Pediatric Considerations in Clinical Pharmacology” of this book.

Beneficial and Adverse Drug Interactions Involving Local Anesthetics, Reversal of Soft-Tissue Anesthesia

Two beneficial drug-drug interactions involving dental local anesthetics include the pharmacodynamics interaction of epinephrine (a vasoconstrictor in the submucosal tissues) and most local anesthetics, which are vasodilators. By reducing local blood flow in the vicinity of the site of local anesthetic injection, the vasoconstrictor prolongs the duration of action of the anesthetic by reducing vascular uptake and “washout” (responsible for the offset and termination of the local anesthetic effect) and tends to reduce peak plasma levels of local anesthetic as the latter is more slowly absorbed.

A second beneficial interaction is the direct competitive receptor blockade of alpha adrenoceptors produced by phentolamine mesylate (e.g., Oraverse®) to reduce the time to recovery of soft-tissue sensation. The therapeutic basis for the use of phentolamine is competitive blockade of epinephrine at alpha-1 adrenoceptors (mediating vasoconstriction) and was established in a multicenter, double-blind, randomized Phase II clinical trial with 122 adult participants (Laviola et al. 2008). In the study, the participants had received one or two cartridges of dental local anesthetics with epinephrine for a dental procedure. Immediately following the procedures, a cartridge (1.8 ml) with 0.4 mg phentolamine mesylate or placebo was injected directly into the site previously used for the local anesthetic. Subjects who had received two cartridges of local anesthetics also received a second cartridge of phentolamine. Adverse events were uncommon, and the injection of the alpha-1 adrenoceptor blocker reduced the time to soft-tissue recovery by approximately 50%. Despite this significant reduction in recovery from soft-tissue anesthesia, Oraverse® has not been widely adopted in dental practice in the United States, probably because of the additional expense and the need to perform one or two additional injections.

Adverse drug-drug interactions with local anesthetics are rare, although interactions with the vasoconstrictor component of dental local anesthetics must be considered as a second category of interaction. Interactions involving local anesthetic include summation of adverse CNS and CVS effects by the co-administration of another local anesthetic or a medically prescribed drug with local anesthetic actions (e.g., a Class I anti-dysrhythmic drug). These are among the most potentially dangerous interactions and are categorized as Significance Rating 1/Potentially Life-Threatening (Moore 1999).

Another life-threatening interaction is that which occurs when additive or supra-additive CNS depression as a result of co-administration of local anesthetics with sedatives and/or opioids, particularly in children with relatively low maximal allowable doses of local anesthetic (Moore 1999). An additional factor to consider is the pre-

vention of this dangerous action, in which is the CNS-depressant effect of the sedative may mask the signs of CNS excitation seen in the early stages of LAST, a factor that delays recognition of the toxicity and effective intervention.

Interactions involving potentiation of the CVS-stimulant effects of epinephrine involve the following (Yagiela 1999):

- Cocaine
- Amphetamines
- Methylphenidate
- Tricyclic antidepressants

Monoamine oxidase inhibitors do *not* contraindicate the use of epinephrine in local anesthetic. Halogenated hydrocarbon general anesthetics and chloral hydrate sensitize the myocardium to the arrhythmogenic actions of epinephrine. Administration of epinephrine-containing local anesthetics to patients taking non-cardioselective beta blockers (e.g., propranolol, Inderal®) can result in clinically significant bradycardia and even cardiac arrest.

The mechanism appears to be the elimination of beta-2 receptor-mediated vasodilation ordinarily produced by epinephrine, combined with alpha-adrenoceptor-mediated vasoconstriction. Increases in blood pressure resulting from the latter activate the unaffected baroreceptor reflex, causing unopposed, vagally mediated cardiac depression, since the beta-1 receptors at the heart have also been blocked by the nonselective beta blocker (Yagiela 1999).

Less significant drug interactions are also possible, and the clinician should consult the complete prescribing information for all drugs prescribed. For elderly and medically compromised patients with systemic disease that could impact drug metabolism and/or excretion, consultation with the patient's physician is recommended.

Effect of Vasoconstrictors

In the United States, epinephrine is the most widely employed vasoconstrictor used in local anesthetic preparations in dentistry. Its actions

are well known and predictable, and the dose limitations recommended in standard textbooks on local anesthetic are generally accepted as safe for adults, based on the American Society of Anesthesiologists' Physical Status Classification, as follows:

- ASA I: 0.2 mg total dose (equivalent to the amount of epinephrine contained in approximately 11 1.8-ml cartridges with a 1:100,000 concentration)
- ASA II: 0.1 mg total dose (equivalent to the amount of epinephrine contained in approximately 5.5 1.8-ml cartridges with a 1:100,000 concentration)
- ASA III: 0.04 mg total dose (equivalent to the amount of epinephrine contained in approximately two 1.8-ml cartridges with a 1:100,000 concentration)

Concentrations of epinephrine greater than 1:100,000 do not appear to confer any advantage for dental local anesthesia, either in terms of prolongation of duration of action or reduction in peak plasma levels of anesthetic. The initial physiologic response to intravascular injection of epinephrine is tachycardia, and it has been reported that an intravascular dose of 15 ug (equivalent to 3 ml of a 1:200,000 concentration) will produce a transient increase in heart rate of 10 beats or more per minute (BPM), with an approximate transient increase in systolic pressure of 15 mmHg or more (El-Boghdadly and Jinn Chinn 2016). Felypressin, not currently available in dental local anesthetic solutions in the United States, is essentially devoid of the sympathomimetic effects seen with epinephrine, as it locally reduces blood flow via a peptidergic mechanism on venous smooth muscle, and not by activation of alpha or beta adrenoceptors.

Nasally Administered and Topically Applied Anesthetics

In the past decade, attempts to eliminate the need for injections and improve the effectiveness of topical agents have resulted in two major devel-

opments—first, an intranasal preparation to achieve dental anesthetic for maxillary teeth and supporting tissues was recently introduced (Kovanaze®). This product is a combination of a lipophilic, long-acting injectable ester local anesthetic (tetracaine 3%) with oxymetazoline 0.05%, an alpha-1 adrenoceptor agonist commonly available as a vasoconstrictor in nasal decongestants (e.g., Afrin®). Based on the Phase 2 clinical trial in 45 adult subjects in need of a single maxillary tooth restorative procedures, 83% of subjects receiving the nasal spray anesthetic did not require rescue local anesthetic injection (Ciancio et al. 2013). A subsequent parallel design study comparing the tetracaine/oxymetazoline combination with a tetracaine-only and saline (placebo) spray, based on need for rescue anesthetic, determined a success rate of 84% for the combination preparation, versus 27.3% success rates for both the tetracaine-only preparation and placebo (Ciancio et al. 2016). In both studies, no serious adverse events were reported.

Initially, these studies were to have included electric pulp testing as a secondary outcome measure, but those data were not reported, and the product's FDA label indications are limited to routine periodontal and restorative and do not include endodontic procedures in maxillary premolar and anterior teeth.

Intranasal local anesthesia has several advantages, the most significant being avoidance of needle insertion, ease of administration, and ability of dental auxiliaries to administer maxillary anesthesia in licensing jurisdictions in which topical anesthetic administration is a permitted duty. There are, however, several disadvantages of the product:

- Limited types of teeth that can be effectively anesthetized (maxillary premolars, canines, and incisors)
- Limited types of procedures that can be accomplished under intranasal local anesthesia
- Significantly higher cost than injectable local anesthetics
- Significantly longer time for onset of anesthesia (up to 14 min total for the two sprays required)

- Need for refrigeration of the product
- Lack of approved indication for younger children

Intranasal anesthesia can be considered as a viable approach for routing restorative procedures and periodontal procedures when needle phobic patients are being treated or when excessive soft-tissue anesthesia is to be avoided.

The second major development in this area is the widespread availability of combination topical anesthetics, typically produced by compounding pharmacies as extemporaneous preparations made only by prescription from a practitioner. These “stronger” topical agents have also been diverted to non-healthcare setting by the newer cultural emphasis on cosmetic enhancements of the human body, that is, tattoos, piercings, and hair removal. For years, 5% lidocaine ointment and 20% benzocaine have served as the mainstays of topical dental anesthesia, but reports of efficacy of combination topicals for insertion of orthodontic intraosseous anchorage devices (resembling mini implants) have also driven the increased use of combination preparations (Reznik et al. 2009). Typically, these preparations combine two topically effective local anesthetics with a vasoconstrictor (e.g., TAC Alternate Gel®, lidocaine + tetracaine + phenylephrine). While effective for obtaining soft-tissue anesthesia for routine procedures, such as SRP and pre-injection anesthesia, these combination products do not facilitate invasive dental procedures. Additionally, the combination of high percentages of local anesthetics results in doses as high as 240 mg of anesthetic per ml. One such preparation contains lidocaine 12.5%, tetracaine 12.5%, prilocaine 3%, and phenylephrine 3%, for a total of 280 mg of local anesthetic per ml. While brief applications of small amounts of these preparations can be tolerated by adults, this quantity of local anesthetic, if swallowed or if allowed to contact large areas of skin for prolonged periods, could be disastrous in some settings, particularly small children or infants who could swallow the anesthetic.

Exparel®, a liposomal form of bupivacaine designed to provide relief of surgical pain in the

immediate postoperative period, was introduced in 1972, but its potential applications in dentistry have only recently been investigated. When used by buccal infiltration in endodontically involved teeth, a randomized, double-blind trial of liposomal bupivacaine versus infiltration of aqueous bupivacaine HCl failed to demonstrate significantly better pain control by the liposomal preparation. In third molar impaction surgery, liposomal bupivacaine provided better postoperative pain control than placebo, with no greater incidence of adverse events, although this study could not be considered conclusive due to extensive protocol violations in the phase 3 placebo-controlled study arm (Lieblich and Danesi 2017). Finally, a recent systematic review with 63 included studies concluded that the majority of randomized controlled trials of liposomal bupivacaine for postoperative pain relief versus placebo or active drugs do not support claims of superiority of liposomal bupivacaine (Yisi et al. 2021). At this time, dental practitioners should carefully consider this evidence, as well as the cost of liposomal bupivacaine before incorporating it in to their practices on a regular basis. Dentists should continue to consider emerging evidence for the use of intranasal and combination topical anesthetics for potential applications in general dentistry.

Conclusion

Conventional dental local anesthetics provide excellent pain control for routine dental procedures, and there is little financial incentive for pharmaceutical manufacturers to develop new local anesthetic drug entities for dental indications only. The safe use of these agents must be predicated on the utilization of all appropriate measures to prevent LAST, as the management of severe systemic toxicity is challenging and may require advanced techniques and agents not ordinarily available in dental offices. Dental practitioners should continue to monitor outcomes from randomized, controlled trials and systematic reviews in order to refine their selection and use of currently available local anesthetics.

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Non-opioid Analgesics in Dental Practice

Arthur H. Jeske

Principles of Analgesic Therapy

There are several important principles that must be considered prior to prescribing an analgesic for a dental patient:

1. Analgesics are adjuncts to caries removal and surgical interventions (tooth extraction, pulp-ectomy, incision-and-drainage) and should not be used in place of these procedures in the management of acute dental pain.
2. The selection of an analgesic must be based upon the patient's medical history and current disorders and take into account the possibility of adverse events and adverse drug/drug interactions.
3. An analgesic regimen should be based upon the expected level and duration of pain, taking into consideration systemic conditions such as cardiovascular, gastrointestinal, and allergic conditions and defined clinical endpoints (reduction of pain, swelling).
4. While no longer recommended as drugs of first choice for dental pain, the opioids, particularly those marketed in combination with acetaminophen, remain important alternatives when NSAIDs are inappropriate due to allergy and other medical conditions.

A. H. Jeske (✉)
UTHealth School of Dentistry, Houston, TX, USA
e-mail: Arthur.H.Jeske@uth.tmc.edu

Pharmacologic Characteristics of NSAID Analgesics for Acute Postoperative Dental Pain

There are several chemical classes of NSAIDs, many of which are used in various medical conditions, especially osteoarthritis. The most important class for the control of pain in dentistry is the propionic acid group, which includes ibuprofen (Advil®), naproxen (Aleve®), and ketoprofen (Orudis®). These agents possess analgesic, anti-inflammatory, antipyretic, and antiplatelet therapeutic actions and are ideally suited for applications in inflammatory dental conditions (acute apical periodontitis, symptomatic irreversible pulpitis). They act by inhibition of cyclooxygenase (COX-1 and COX-2), which reduces the synthesis of prostaglandins (PGs), and their analgesic action is attributable primarily to a reduction of PGE2 and F2 α , which are synthesized rapidly after tissue damage and sensitize nociceptive nerve endings to a wide variety of noxious stimuli. Importantly, inhibition of PG synthesis occurs relatively early in painful inflammatory conditions, so NSAID therapy should commence as early as feasible in pain management.

A recent overview of systematic reviews provides practitioners with a wealth of high-level evidence for the comparative efficacy of various oral analgesics, based primarily on the third molar impaction surgery model of assessing acute postoperative pain (Moore et al. 2015a).

When using this reference, it is helpful to consider the number needed to treat (NNT) values for various analgesics, with values <2 being generally considered among the best for single-dose outcomes. In regard to these values, NNT refers to the number of subjects who need to receive an intervention in order to see a defined beneficial effect, which in these pain trials is a reduction of pain level by at least 50%. Mathematically, NNT is defined simply as:

$$\text{NNT} = 1 / \text{ARR}$$

where ARR is the absolute risk reduction (difference between active treatment and placebo).

For example, where an active (investigational) drug produces a 50% or greater reduction in pain level in 70% of the subjects and the placebo produces a 50% or greater reduction in pain in 15% of the subjects, the NNT is:

$$\text{NNT} = 1 / 0.7 - 0.15, \text{ or } 1 / 0.55, \text{ or } 1.8.$$

In this example, NNT would predict that the active drug would need to be given to only 1.8 individual patients (less than 2) to see the defined benefit. NNT has both advantages and disadvantages, and an advantage is that it addresses both clinical and statistical significance in an easily understood manner. However, its practical significance may vary with the significance of the disease/patient conditions, and it is only useful to compare NNT values among various interventions when the treatments are for the same condition, severity, and outcome measure (Schechtman 2002). It is also important to note that in order for a drug or drug combination to have been included in this study, outcomes must be from at least two randomized, controlled studies with a total of at least 200 subjects. The outcomes from (Aminoshariae et al. 2016a) are summarized in Table 1. When considering these outcomes, it is important to note that results from studies of single-dose oral analgesics used in the third molar-impaction model may differ from the findings of studies done in endodontic patients. Typically, studies of relief of postoperative endodontic pain have focused on higher (e.g., 600 mg) doses of ibuprofen than those from studies

Table 1 Selected NNT values for common single-dose non-opioid oral analgesics used in dentistry, as reported in (Jose et al. 2022)

Drug	NNT	Confidence interval 95%
Diclofenac 100 mg	1.9	1.7–2.3
Diflunisal 1000 mg	2.1	1.8–2.6
Ibuprofen 200 mg	2.9	2.7–3.2
Ibuprofen 200 mg + 100 mg caffeine	2.1	1.9–3.1
Ibuprofen 400 mg	2.5	2.4–2.6
Ibuprofen 200 mg + APAP ^a 500 mg	1.6	1.5–1.8
Ibuprofen 400 mg + APAP 1000 mg	1.5	1.4–1.7
Ketoprofen 100 mg	2.1	1.7–2.6
Naproxen 500–550 mg	2.7	2.3–3.3
APAP 500 mg	3.5	2.7–4.8
APAP 1000 mg	3.6	3.2–4.1

^a APAP denotes acetaminophen/acetyl para amino phenol

highlighted in Table 1. This may be based upon the possibility that higher doses of NSAIDs provide a greater anti-inflammatory effect, in addition to their analgesic effect, or the perceived need for higher dosages may be based upon the fact that pain of endodontic origin is typically associated with a much longer duration of preoperative pain, as opposed to the pain of third-molar impaction surgery, which is usually exclusively postoperative and which does not involve the prolonged period of bacterial-induced inflammation or invasive dental procedures associated with pulpal disease.

When considering the NNT values presented in Table 1, the reader should consider the fact that these numbers apply to *single* oral doses of these analgesics, with pain responses assessed over a 4- to 6-h postoperative time interval, and that the subjects in the studies used in the calculations of these NNT values were *fasting*.

Acetaminophen

High-level scientific evidence from systematic reviews of acute postoperative pain in adults suggests that, based on number-needed-to-treat (NNT), acetaminophen administered alone is

Table 2 Comparative pharmacologic characteristics of nonsteroidal anti-inflammatory drugs (NSAIDs),^a acetaminophen and opioid analgesics available in the United States (Jeske 2017)

Characteristic	NSAIDs	Opioids	Acetaminophen	Comment
Available OTC	Yes	No	Yes	
Tolerance, dependence	No	Yes	No	Opioid tolerance and dependence are unlikely to occur with short-term use (<5 days)
CNS depression	No	Yes	No	A sedative effect of opioids may be desirable in some circumstances
Anti-inflammatory	Yes	No	No	Principal advantage of NSAIDs
Anti-pyretic	Yes	No	Yes	Beneficial in the presence of infection
Analgesic	Yes	Yes	Yes	NSAIDs act primarily peripherally at the site of tissue injury, while opioids act centrally (CNS). Combinations are generally superior to single-agent regimens
Antiplatelet agents (including aspirin)	Yes	No	No	Increased bleeding risk is associated with NSAIDs, primarily if taken preoperatively; the effect is reversible (unlike aspirin, which is irreversible)
GI dysfunction	Yes	Yes	No	In addition to nausea and vomiting, opioids are associated with constipation. NSAID-related GI irritation is typically seen during prolonged administration (>5 days)

^a Propionic acid class

not a particularly good analgesic (Moore et al. 2015a; Moore et al. 2016). This has been confirmed in randomized controlled trials of post-endodontic pain as well (Ciancio 2014). However, when used in combinations, acetaminophen appears to act synergistically with both NSAIDs and opioid analgesics and the combination of 200–400 mg ibuprofen plus 500–1000 mg acetaminophen (taken at the same time) results in the best NNT values in the oral surgical pain model (Moore et al. 2015a). Additionally, this combination does not result in adverse effects greater than those observed in placebo groups when used on a short-term basis (Moore et al. 2015b). However, it should also be noted that the maximum daily adult dose of acetaminophen from all sources (Rx and OTC) should not exceed 4000 mg (Jeske 2017; Moore et al. 2015a). Hepatotoxicity may occur from excessive acetaminophen intake or from interactions with chronic alcohol use. While unusual, allergy to acetaminophen can occur and would absolutely contraindicate use of this agent.

A summary of the comparative pharmacologic characteristics of propionic acid NSAIDs, acet-

aminophen, and opioid analgesics is presented in Table 2.

Prescribing Considerations

1. For the management of acute postsurgical dental pain (including pain of endodontic origin), in the absence of any significant contraindications, therapy should begin with a standard dose of a combination, orally administered first-choice agents (ibuprofen with acetaminophen) (Aminoshariae et al. 2016a, b; Elzaki et al. 2016; Smith et al. 2017).
2. For optimal pain relief, the combination of 200–400 mg ibuprofen with 500–1000 mg acetaminophen has been shown to provide pain relief that is superior to virtually all acetaminophen/opioid combinations and COX-2 selective NSAIDs (Moore et al. 2015a).
3. Because peak pain associated with dental extractions appears to occur within the first 4–8 h postoperatively and then decline over the next 2–3 days, short-term administration of the ibuprofen/acetaminophen combination can be employed.

4. A need for a sedative effect, especially in the first 24 h postoperatively, may warrant the addition of an opioid analgesic in combination with the NSAID, when the patient's activities would not be affected by possible CNS depression.
5. Warnings with analgesic therapy should be issued verbally and in writing on the prescription. They should include the possible development of allergic reactions, as well as GI disturbances, increased bleeding risk, and the risk of adverse interactions between acetaminophen and alcohol and acetaminophen overdose.
6. Whenever possible, analgesics should be taken on an empty stomach with a glass of water in order to hasten the dissolution of the dose form and delivery of the drug from the stomach to the small intestine.
7. To extend duration of action to >8 h, a long-acting NSAID at a higher dose (diflunisal 1000 mg) or an NSAID-acetaminophen combination at a higher dose (ibuprofen 400 mg + acetaminophen 1000 mg) can be employed.
8. Caution is advised in patients who are regularly taking NSAIDs for systemic disorders, for example, osteoarthritis, as adding a second NSAID to their medication regimens put them at risk for serious gastrointestinal irritation and possible ulceration, as well as nephritis and renal failure, and/or severe bleeding. In these patients, alternative analgesics (e.g., acetaminophen, opioids) should be considered and can be added to the patient's NSAID regimen if no contraindications exist.

Adverse Effects

NSAID analgesics, as prescribed in dentistry, are generally well tolerated. With the exception of allergy, most adverse effects from short-term use of NSAIDs are related to their effects on the gastrointestinal tract and platelets. NSAIDs inhibit the formation of gastroprotective PGs, and this irritant effect, combined with their antiplatelet effect, can result in ulcerations and GI

bleeding. Short-term use of NSAIDs has been shown to be relatively safe when administered for dental pain (Aminoshariae et al. 2016a). Furthermore, robust systematic reviews consistently report significantly fewer adverse effects of NSAIDs when compared to opioids (Chou et al. 2020).

Ingestion of high doses of NSAID analgesics is associated with nephropathy, and the risk of this complication increases in elderly patients, as well as patients who are dehydrated or have pre-existing renal insufficiency, heart failure, or diabetes (Kharasch 2004). It should be noted that dehydration could be present in individuals with symptomatic irreversible pulpitis who have experienced diarrhea and/or nausea and vomiting (possibly induced by self-prescribed antibiotics and/or analgesics) and who are not well nourished/hydrated due to dental pain. This is particularly problematic when patients have ingested over-the-counter NSAIDs or acetaminophen before receiving dental treatment.

Since renal blood flow and urine formation are partly regulated by physiologic PGs, blood pressure may be elevated by the ingestion of NSAIDs, and this should be considered when designing an analgesic regimen for patients with hypertension and other cardiovascular disorders.

Bleeding is associated with all NSAIDs, and increased intraoperative and postoperative bleeding must be anticipated and dealt with effectively, including the use of careful surgical technique, suturing, and other hemostatic measures (oxidized cellulose packs).

The risk of allergic and adverse respiratory reactions to NSAIDs should be evaluated through a careful medical history, especially in patients with a prior history of aspirin allergy, asthma, and reactive airway disease.

Pregnancy constitutes a contraindication to the use of NSAIDs, particularly in the first and third trimesters. Among the commonly used NSAIDs, observed differences in GI irritation only become manifest after prolonged therapy (>30 days). Typically, another drug to reduce GI

irritation (e.g., misoprostol) is only prescribed during longer-term administration.

In summary, when used short term for the management of acute postoperative pain in dentistry, most NSAIDs and NSAID-acetaminophen combinations produce no greater incidence or severity of adverse effects than placebo. The evidence for this is found in the overview of systematic reviews published by Moore et al. It should be noted, however, that this same review determined that some commonly utilized analgesics produce significantly more adverse events than placebo, and these agents include aspirin 1000 mg, diflunisal 1000 mg, all opioids, and fixed-dose combination products containing opioids (Moore et al. 2015b).

Adverse Drug Interactions

NSAID analgesics are capable of adversely interacting with other dental and medical drugs, both through pharmacodynamic and pharmacokinetic mechanisms. The most significant adverse interactions for commonly prescribed NSAID analgesics are listed in Table 3.

Corticosteroids

In some cases, anti-inflammatory corticosteroids may be beneficial in providing short-term pain relief, particularly in cases of inflammation associated with apical periodontitis, acute apical abscess, and third-molar extraction. In a prospective, triple-blinded, split-mouth randomized trials, Al-Dajani (Al-Dajani 2017) evaluated single i.m. doses of dexamethasone versus placebo administered preoperatively in patients prior to third

Table 3 Clinically significant drug interactions involving NSAID analgesics used in dentistry (modified from (Ciancio 2014))^a

Primary drug	Action	Interaction (and effect)
Alcohol	Enhanced by NSAIDs	Increased GI irritation, nausea, GI pain, bleeding
Diuretics, antihypertensive drugs	Antagonized by NSAIDs	Increased salt and water retention with increased blood pressure
Coumarins (including warfarin)	Enhanced by NSAIDs	Increased risk of bleeding
Antiplatelet agents (aspirin, clopidogrel)	Enhanced by NSAIDs	Increased risk of bleeding; increased risk of thromboembolism (ibuprofen blocks the antiplatelet effect of aspirin when the drugs are taken concurrently)
Direct oral anticoagulants (rivaroxaban, dabigatran)	Enhanced by NSAIDs	Increased risk of bleeding
Potassium-sparing diuretics	Enhanced by NSAIDs	NSAIDs may increase serum potassium levels
Potassium supplements	Enhanced by NSAIDs	NSAIDs may increase serum potassium levels
Cancer chemotherapeutic agents	Enhanced by NSAIDs	Increased risk of GI ulceration
Selective serotonin Reuptake inhibitors (SSRIs)	Enhanced by NSAIDs	Increased risk of GI ulceration and bleeding
Corticosteroids	Enhanced by NSAIDs	Increase salt and water retention; increased risk of GI ulceration

^a Less significant drug interactions are also possible—the clinician should consult the complete prescribing information for all drugs prescribed. For elderly and medically compromised patients with systemic disease that could impact drug metabolism and/or excretion, consultation with the patient’s physician is recommended

molar surgery and concluded that this method can be safely used to decrease postoperative pain and improve daily functioning. By implication, corticosteroids may also spare opioid use, thus avoiding opioid-related issues such as constipation and CNS depression. Based on recent scientific evidence, both methylprednisolone and dexamethasone can be utilized in such situations, although an injection is required (Bane et al. 2016; Chen et al. 2017). A recent systematic review determined that preoperative administration of corticosteroids (4 or 8 mg oral dexamethasone, 40 mg parenteral dexamethasone) was effective in reducing in post-endodontic pain (Aminoshariae et al. 2016b). Orally administered corticosteroids can be used to suppress inflammation, both pre- and postoperatively, and this use is commonly exploited in third-molar surgery.

In periodontics, for dental implant surgery, postoperative glucocorticoids used as primary analgesics can produce pain reduction equivalent to NSAIDs used alone (Khouly et al. 2021). It is important to note in regard to the preemptive use of corticosteroids that best results are obtained with parenteral (rather than oral) administration. At this time, the optimal regimens for these applications have not been established (Wagner et al. 2022).

The characteristics and uses of corticosteroids are described in greater detail in chapter “Endocrine Drugs of Significance in Dentistry”.

Pre-emptive Analgesia

NSAIDs have been evaluated for use pre-emptively (preoperative administration to reduce postoperative pain). The reader is referred to an excellent update on this topic by Lieblisch (Lieblisch 2017). Typically, single-dose ibuprofen 400–600 mg, administered approximately 1 h before a procedure, is recommended, based on limited evidence, although there is limited evidence that single NSAID administration, including ibuprofen and ketorolac, is not effective for this use (Aminoshariae et al. 2016b). When used in this manner, intraoperative bleeding is increased, and the practitioner should be pre-

pared to take additional hemostatic measures during the procedure in patients who are undergoing surgical procedures and who have received preoperative NSAID medications, and this effect has also been demonstrated in orthodontic patients following placement of bands or separators. It should be noted that evidence is lacking for a similar pre-emptive analgesic effect of acetaminophen when administered preoperatively. In endodontics, a recent systematic review demonstrated that oral preoperative administration of a corticosteroid provides superior postoperative pain relief when compared to NSAIDs as oral premedication, with the additional benefit of improving success rates of local anesthetic nerve blocks (Jose et al. 2022).

Other strategies for pre-emptive analgesia include administration of long-acting local anesthetics (e.g., bupivacaine), a positive effect occurring even when used with general anesthesia for extracted of impacted third molar teeth.

Conclusion

Dentists should continue to consider emerging evidence for the use of non-opioid analgesics, especially in view of the ever-increasing problem of opioid abuse and diversion.

Dentists can now confidently prescribe an NSAID or recommend OTC (e.g., ibuprofen + acetaminophen) combinations for excellent relief of acute dental pain, based on high-level scientific evidence. This evidence is readily accessible by searching the PubMed database of the US National Library of Medicine, and entering the name of a specific drug within the following search terms:

Single dose oral _____ for acute postoperative pain in adults.

Current evidence supports the use of an NSAID (ibuprofen), in combination with acetaminophen if possible as first-choice therapy for the management of acute postoperative dental pain in adults. In children, a single NSAID is currently supported by the American Academy of Pediatric Dentistry’s Guideline on Pain Management as the

first-line agent in the treatment of acute mild to moderate postoperative pain (American Academy of Pediatric Dentistry 2017).

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Opioid Analgesics, Benzodiazepines, and Other Controlled Substances

Arthur H. Jeske

Principles of Opioid Prescribing

There are several important principles that must be considered prior to prescribing an opioid analgesic for a dental patient (American Association of Oral and Maxillofacial Surgeons 2017):

1. A nonsteroidal anti-inflammatory drug (NSAID) may be used preoperatively to reduce the severity of postoperative pain.
2. Perioperative corticosteroids can be used to reduce swelling and other discomfort after impaction surgery.
3. Long-acting local anesthetics may delay the severity and onset of postoperative pain.
4. Long-acting, extended release opioids should not be used for acute pain.
5. Unless contraindicated, NSAIDs are first-choice analgesics for acute pain.
6. NSAIDs and acetaminophen can be used in combination to improve analgesic outcomes, although dosage must be carefully monitored.
7. Short-acting opioids may be appropriate for acute “breakthrough” pain.
8. Practitioners may be required to use prescription-drug monitoring programs, and these programs can be useful to detect opioid misuse or addiction.

9. All instructions for patient analgesia and analgesic prescriptions should be carefully documented.

Pharmacologic Characteristics of Oral Opioids for Acute Postoperative Dental Pain

There are three major classes of opioids—natural opium derivatives, semisynthetic opioids, and synthetic opioid agonists. In dentistry, the most important of these is the semisynthetic class, which includes codeine, hydrocodone, and oxycodone. Synthetic agents such as fentanyl and meperidine are used primarily by the intravenous route and are not covered in this chapter. The most commonly used pharmaceutical forms of opioids for the control of pain in dentistry are combinations of the opioid component with acetaminophen (e.g., Vicodin®). All of these agents act non-selectively at mu, kappa, and delta opioid receptors and possess analgesic, antitussive, and sedative therapeutic actions but lack the beneficial anti-inflammatory and antipyretic therapeutic actions seen with the NSAIDs. The addition of acetaminophen to an opioid pain-control regimen gains two specific advantages—first, the acetaminophen possesses antipyretic activity, and second, acetaminophen and opioids appear to act synergistically and therefore are capable of producing levels of pain

A. H. Jeske (✉)
UTHealth School of Dentistry, Houston, TX, USA
e-mail: Arthur.H.Jeske@uth.tmc.edu

relief not achievable with even higher doses of either agent taken alone.

Non-Analgesic Actions and Adverse Effects

Depending on the dose and the degree of opioid tolerance in an individual, the following non-analgesic effects are characteristics of most opioids: dizziness, somnolence, nausea, vomiting (caused by stimulation of the chemoreceptor trigger zone in the brainstem), respiratory depression (initially caused by reduction of the sensitivity of the respiratory center to carbon dioxide), miosis, constipation, cardiovascular depression, and, eventually, loss of consciousness, coma, and death (usually due to cardiorespiratory arrest). Virtually, all of these actions are the result of drug binding to the mu opioid receptor, and virtually, all can be reversed by the administration of sufficient doses of narcotic antagonists (e.g., naloxone).

Consideration must be given to the potential for respiratory depression in individuals who present with a compromised airway (e.g., Mallampati Class III) and those with obstructive sleep apnea (OSA). While there are many factors that determine the potential for postoperative opioid toxicity in patients with OSA, current evidence suggests a need for multimodal postoperative pain control in these individuals, including local anesthetics and non-opioid analgesics (e.g., NSAIDs) (Maund et al. 2011; American Society of Anesthesiologists Task Force 2014).

Comparative Efficacy of Opioids

High-level scientific evidence from systematic reviews of acute postoperative pain in adults suggests that based on number-needed-to-treat (NNT) that single opioids and opioid-acetaminophen combinations are generally equivalent to single NSAIDs but are inferior to pain relief outcomes associated with some NSAIDs and the combination of ibuprofen and

acetaminophen (Moore et al. 2015). These outcomes are particularly relevant to acute postoperative dental pain, as the majority of the randomized controlled trials that comprise these systematic reviews and meta-analyses were conducted using the third-molar impaction surgery pain model. A summary of the comparative efficacy of various analgesics based on number-needed-to-treat values for single doses of various opioids and opioid combination products suggests that oxycodone, in combination of higher doses of acetaminophen, appears to be worth of consideration when an NSAID cannot be used for acute postoperative dental pain (Table 1).

Some caution must be observed when interpreting the NNT values shown in Table 1.

NNT is a relative measure of the reduction of pain level following a single dose of an analgesic and does not quantify the subjective relief of pain that may be produced by drugs with significant CNS depressant effects, such as opioids. While a given opioid analgesic's NNT (as reported above) may be less favorable, that is, higher than a non-opioids, this value alone does not predict the entire pain-relieving effect, which varies with the opioid's alteration of the central processing of pain stimuli and the interpretation of the pain.

Table 1 Number-needed-to-treat (NNT) values for opioid and opioid-acetaminophen combination analgesics versus placebo, based on overview of systematic reviews and meta-analyses (Gaskell et al. 2009; Moore et al. 2015)

Drug	NNT	Confidence interval 95%
Oxycodone 5 mg + APAP 325 mg	5.5	na
Oxycodone 10 mg + APAP 650 mg	2.7	2.4–3.1
Oxycodone 10 mg + APAP 1000 mg	1.8	1.6–2.2
Oxycodone 15 mg	4.6	2.9–11
Codeine 30 mg + APAP 300 mg	6.9	4.8–12
Codeine 60 mg + APAP 600–650 mg	3.9	2.9–4.5
Codeine 60 mg + APAP 800–1000 mg	2.2	1.8–2.9
Codeine 60 mg	12	8.4–18

APAP acetaminophen/acetyl para amino phenol

This same phenomenon also makes the quantitative study of opioid analgesia, based on VAS instruments and other tests, more challenging.

Of interest for practitioners in the United States is the absence of systematic reviews and meta-analyses for pain relief outcomes using hydrocodone and hydrocodone-acetaminophen combinations. This is likely due to the fact that hydrocodone has been banned or is unavailable in many other countries for some time. It is possible to glean some moderate-quality evidence for the combination of hydrocodone and acetaminophen from a limited number of clinical trials. For example, in a comparison of celecoxib 400 mg with hydrocodone and hydrocodone-acetaminophen in outpatient orthopedic patients, hydrocodone compared favorably to placebo and performed equally well with celecoxib over the first 8 h postoperatively but was inferior to celecoxib at subsequent test intervals, and the hydrocodone products produced a significantly higher incidence of adverse effects. Similarly, international studies of hydromorphone (Dilaudid®) are not available in systematic reviews, and this agent, therefore, cannot be recommended as evidence-based pain control in dentistry.

Recently, the comparative efficacy of opioid analgesics in the management of dental pain has been further elucidated in two systematic reviews, both of which found that while 5 mg of oxycodone alone produced analgesia that was no better than placebo, it does produce effective analgesia when used in combination with acetaminophen (Miroshnychenko et al. 2023; Teoh et al. 2022).

Oxycodone, used in combination with effective doses of acetaminophen, currently appears to be the opioid of first choice. In contrast to hydrocodone, codeine, and tramadol, oxycodone does not depend on activation by CYP 2D6. A recent review of its pharmacokinetic and pharmacodynamic properties (Kinnunen et al. 2019) suggests that while approximately 19% of a dose of oxycodone is metabolized by CYP 2D6 to oxymorphone, which does not readily cross the blood barrier, and the contribution of active metabolites to analgesia produced by oxycodone appears to

be clinically insignificant (Kinnunen et al. 2019). Oxycodone is principally inactivated metabolically by CYP 3A4, and caution must be exercised if it is to be administered to patients taking inducers of CYP 3A4 (e.g., corticosteroids) or inhibitors (e.g., azole antifungal drugs, macrolide antibiotics).

Prescribing Considerations

1. For acute postoperative pain, only short-term administration of the analgesic should be needed.
2. A need for a sedative effect, especially in the first 24 h postoperatively, may warrant the use of an opioid analgesic in combination with acetaminophen or an NSAID, when the patient's activities would not be affected by possible CNS depression and when no other contraindications exist.
3. Warnings with opioid analgesic therapy should be issued verbally and in writing on the prescription. They should include the prohibition of alcohol intake during opioid treatment, as well as possible interactions with other CNS depressant drugs and development of allergic reactions, as well as possible nausea and vomiting. The adverse interactions between acetaminophen and alcohol underscore the need for warning the patient about alcohol ingestion.
4. Pregnancy constitutes a contraindication to the use of NSAIDs, particularly in the first and third trimesters. Among the commonly used NSAIDs, observed differences in GI irritation only become manifest after prolonged therapy (>30 days). Typically, another drug to reduce GI irritation (e.g., misoprostol) is only prescribed during longer-term administration.
5. Scientific evidence to support the selection of hydrocodone over another opioid, for example, oxycodone, is lacking. While there are a limited number of systematic reviews and meta-analyses for oxycodone, this type of evidence has not been available for hydrocodone, partly because the drug is not available in many countries other than the United States

(Moore et al. 2016). For this reason, and because oxycodone responses appear to be less susceptible to CYP-related pharmacogenetic variations, the author recommends the use of immediate-release forms of oxycodone as an evidence-based opioid analgesic to be used in combination with acetaminophen for the management of acute postoperative pain in adults.

6. Systemic diseases must be taken into consideration when prescribing opioids. Liver disease can compromise the metabolic disposition of opioids, as well as first-pass elimination during the initial absorption of orally administered opioids. In such cases, the dose should be reduced and the dosage interval increased to reduce the potential for toxicity. In patients with kidney disease, opioids with active metabolites (e.g., codeine, meperidine, and morphine) should be avoided, as these metabolites are ordinarily excreted in the urine (Wie 2017).
7. When prescribing oxycodone, the prescription should note that an immediate-release form, not an extended-release one (e.g. Oxycontin®) is to be issued to the patient.

An example of an evidence-based prescription for an opioid for short-term management of acute postoperative pain in an otherwise healthy adult without contraindications is illustrated in Fig. 1. It should be noted that the rationale for the use of this combination is based upon data from the overview analysis of systematic reviews by Moore et al. (Moore et al. 2015) and is designed to provide effective reduction in pain level and to take advantage of the synergistic actions of opioids combined with acetaminophen and the prolongation of effect at the relatively higher dose of each component.

The prescription illustrated here limits the total daily exposure of the patient to 3000 mg (3 doses of 1000 mg over 24 h), and it limits the total amount of opioid dispensed to less than ten tablets. If a longer duration of pain control is needed, the number of opioid dosage units can be increased proportionately.

Rx

Oxycodone 5 mg immediate-release tabs

Dispense 6 (six) tabs

Sig: Take 2 (two) tabs with 1,000 mg acetaminophen every 8 hours for acute dental pain.

Warnings: Do not drive or operate machinery while taking. Do not drink alcohol or use other CNS depressants while taking. May be habit forming.

Fig. 1 Sample prescription for acute postoperative dental pain using an opioid in combination with over-the-counter acetaminophen

Adverse Drug Interactions, Opioid Pharmacogenetics

Opioid analgesics are capable of adversely interacting with other dental and medical drugs, both through pharmacodynamic and pharmacokinetic mechanisms (Haas 1999). The most significant adverse interactions for commonly prescribed opioids are found in Table 2.

Less significant drug interactions are also possible—the clinician should consult the complete prescribing information for all drugs prescribed. For elderly and medically compromised patients with systemic disease that could impact drug metabolism and/or excretion, consultation with the patient's physician is recommended (Haas 1999).

Among the commonly prescribed opioids, three are prodrugs, that is, they are metabolically converted from an inactive/less active parent form into an active metabolite. These agents include codeine (metabolized to morphine), hydrocodone (metabolized to hydromorphone), and tramadol (metabolized to the active M-1 opioid structure). All of these reactions depend on

and none carry an FDA-approved label indication specifically for dental sedation. However, this “off-label” use has become accepted over time through case studies and retrospective safety analyses (Dionne et al. 2006; Ehrich et al. 1997).

The clinical pharmacologic properties of these representative sedative/hypnotics are found in Table 3.

Unlike diazepam and lorazepam, midazolam and triazolam have a rapid onset, even when used orally, a relatively short duration, and are subject to a high first-pass effect due to rapid inactivation of approximately 25% of the absorbed dose by hepatic cytochrome P450 enzymes (Becker 2011). It should be noted that administration of the oral dose form sublingually effectively bypasses initial hepatic degradation and results in higher peak plasma levels with intensified sedative actions (Becker 2011).

The safety and efficacy of benzodiazepines and related drugs (e.g., zolpidem) for sedation of anxious dental patients have been validated over the past two decades. Following the introduction of triazolam (Halcion®) on the US market in 1982 for the short-term management of insomnia, the dental profession began to use it for preoperative sedation of dental outpatients, recognizing that its short duration of action and absence of active metabolites were favorable when compared with longer acting benzodiazepines with active metabolites, such as diazepam. In 1997, its comparative safety and efficacy with diazepam in patients with documented anxiety undergoing endodontic treatment were studied (Ehrich et al. 1997). Oral doses of 0.25 mg triazolam were found to be more effective and equally safe with orally

administered 5 mg diazepam and without a prolonged recovery period. Subsequently, the Dental Organization for Conscious Sedation (DOCS) protocols were promulgated, with the recommendation that patient responses to the initial sedative doses be carefully monitored prior to administration of an additional dose of sedative (Gordon et al. 2007).

The most significant adverse drug interactions involving benzodiazepines involve excessive CNS depression with possible cardiorespiratory depression when they are co-administered with other CNS depressants, particularly opioids, alcohol, and barbiturates. Additionally, serious pharmacokinetic interactions may occur, particularly for benzodiazepines with high oral bioavailability and, therefore, high susceptibility to agents that reduce the “first-pass” effect (hepatic extraction of the drug as it passes from the gastrointestinal tract to the circulation) (e.g., midazolam and triazolam). Of particular note in dentistry are the effects of macrolide antibiotics and azole-type antifungal drugs. According to the manufacturer’s data for oral midazolam, erythromycin, administered at doses of 500 mg three times daily, increases the $C_{\max}$ of oral midazolam by approximately 170%, while once-daily doses of ketoconazole increase the $C_{\max}$ by over 300%. Increases in area under the curve (AUC) for these same interactions were 281–341% and 1490%, respectively. Obviously, the doses of midazolam administered to patients taking these antibiotics and antifungal agents would need to be significantly reduced to avoid excessive sedation. The emergency antidote for benzodiazepine overdose, flumazenil, is described in greater detail in chapter “Pediatric Considerations in Clinical Pharmacology”.

All benzodiazepines commonly used for dental outpatient sedation are Schedule IV controlled substances in the United States and may produce tolerance and dependence when used for prolonged periods. The short-term administration of these drugs for dental outpatient sedation is not associated with tolerance or dependence. One of these agents—alprazolam (Xanax®)—is currently among those drugs being marketed in large quantities through illicit channels, even in “coun-

Table 3 Pharmacologic properties of selected benzodiazepine sedative/hypnotic agents

Drug	Adult dose	Plasma half-life	Duration of action
Diazepam	5–15 mg	20–70 h	2–3 h
Lorazepam	2–4 mg	10–20 h	8–12 h
Midazolam ^a	0.25–0.5 mg/kg	1–5 h	30–45 min
Triazolam	0.125–0.5 mg	1–5 h	6–8 h

^a Approved in the United States for oral use only in patients under the age of 18

terfeit” forms. Benzodiazepines are now frequently implicated in cases of drug overdose deaths (Lembke et al. 2018).

Carisoprodol

Carisoprol (e.g., Soma®) is now classified by the US Drug Enforcement Administration as a Schedule IV controlled substance. It is included here because it may be diverted to illicit channels and co-abused with other addictive substances, including opioids and/or benzodiazepines. Carisoprodol is a centrally acting skeletal muscle relaxant whose precise mechanism of action is unknown. It is indicated as an adjunct to rest, physical therapy, analgesics, and other measures for the relief of discomfort of acute, painful musculoskeletal conditions. Adverse effects include a high incidence of somnolence and occasional tachycardia, facial flush, dizziness, headache, light-headedness, dermatitis, nausea, abdominal cramps, and dyspnea. Carisoprodol also possesses properties of physical and psychological dependence, and its frequent co-abuse with opioids and benzodiazepines as part of this “Holy Trinity” of co-abused drugs led to its being classified by US Drug Enforcement administration as a Schedule IV controlled substance in 2012 (Horsfall and Sprague 2017).

Carisoprodol interacts with benzodiazepines in the brain to produce a synergistic increased in dopamine in the nucleus accumbens with respiratory depression, increasing the likelihood of respiratory arrest when co-ingested with opioids (Horsfall and Sprague 2017). This reclassification reportedly has resulted in fewer cases of carisoprodol overdose in some states (Sun et al. 2017). These characteristics distinguish the drug from other types of commonly prescribed skeletal muscle relaxants, which are not currently classified as controlled substances.

Withdrawal from sustained abuse of carisoprodol results in a severe syndrome resembling neuroleptic malignant syndrome (Paul et al. 2016). Management of carisoprodol withdrawal and may require more extreme pharmacologic measures than simply administering an i.v. ben-

zodiazepine, including the infusions of high doses of propofol, fentanyl, ketamine, quetiapine, and haloperidol (Vo et al. 2017).

Prescription Monitoring Programs

Currently, there are many states in the United States that have now implemented prescription drug monitoring programs designed to continuously assess the prescribing patterns for controlled substances among health care providers, including veterinarians, in their jurisdictions. Frequently operated by state boards of pharmacy, these programs use electronic databases to track all prescriptions for controlled substances issued in their jurisdictions and can be used by state regulatory boards and law enforcement agencies to identify inappropriate patterns of prescribing. Examples of such inappropriate patterns could include the following:

- One patient receiving prescriptions for a controlled substance from multiple practitioners within the same time period
- Multiple patients with the same address receiving prescriptions for controlled substances
- Prescriptions for controlled substances that are not within the scope of practice of a particular type of practitioner (e.g., an amphetamine prescription issued by a dental provider)

Data on controlled substance prescriptions issued by dentists over a 2-year period in a US state (South Carolina) were analyzed and published in 2016 (McCauley et al. 2016). That study determined that 653,650 prescriptions were issued by dentists, and significantly, 113,818 patients had received multiple opioid prescriptions within a 30-day period. These data can be used to inform the dental profession about the extent to which prescribing controlled substances is problematic in a given geographic area, and prescription monitoring programs have the capability to issue alerts on certain patients who appear to be engaging in “doctor shopping,” drug

diversion, and drug abuse. A survey of members of the US National Dental Practice-Based Research Network found that higher levels of opioid prescribing were associated with less consistent implementation of the use of prescription drug monitoring programs (McCauley et al. 2018).

Conclusion

Dentists should continue to consider emerging evidence for the use of non-opioid analgesics, especially in view of the ever-increasing problem of opioid abuse and diversion.

While NSAIDs or NSAID-acetaminophen combinations provide excellent relief of acute dental pain, they cannot be used in a significant number of patients who have contraindications (e.g., gastrointestinal or renal disease), and opioids remain the mainstay of therapy in this population, preferably when used in combination with acetaminophen for acute postoperative dental pain in adults. In children, the traditional use of codeine in combination with acetaminophen is now questionable, based on the possibility that children who are ultrarapid metabolizers for CYP 2D6 may experience cardiorespiratory arrest when this drug is used.

Practitioners who carefully consider the CNS depressant properties of opioid analgesics can safely and effectively prescribe a single, immediate-release opioid with an effective dose of an over-the-counter NSAID or acetaminophen with confidence, based upon robust scientific evidence. Minimizing the amount of opioid dispensed to the patient and utilization of prescription monitoring programs where available can help reduce abuse and diversion of prescription opioids.

At present, single benzodiazepines remain drugs-of-choice for routine sedation of dental outpatients, based on known efficacy and safety, and relatively low risk of abuse when prescribed in very small quantities. However, the practitioner must now consider the increasing prevalence of drug overdose reactions involving benzodiazepines when prescribing them.

Carisoprodol has few indications in dentistry, and based on its recent classification as a Schedule IV controlled substance, the dentist should carefully consider the possibility of abuse and diversion associated with it.

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Antibiotics and Antibiotic Prophylaxis

Arthur H. Jeske

Introduction and Principles

Antibiotics play an important role in the management of odontogenic infections (primarily as adjuncts to surgical procedures) and for prevention of systemic infections arising from bacteremias associated with dental procedures in patients at risk for serious systemic infective complications (e.g., infective endocarditis). However, dental professionals must be vigilant in the use of these drugs to avoid their overuse and help address the serious issue of increasing antibiotic resistance (Centers for Disease Control and Prevention 2013). Since the approval of phenoxymethylpenicillin (penicillin V) by the Food and Drug Administration in 1956, dental infections in outpatients have been managed successfully with a penicillin or another oral narrow-spectrum antibiotic, which is effective against infections caused by susceptible gram-positive and anaerobic bacteria. Currently, only a few, selected classes of antibiotics are indicated for the dental management of oral infections, including penicillins, cephalosporins, lincosamides, and macrolides (American Association of Endodontists 2012; American Academy of Pediatric Dentistry 2014; Flynn 2011). Recently, three recent

publications have summarized the frequency of various antibiotic prescriptions written in the United States (Germak et al. 2017; Roberts et al. 2017) and internationally (Soares et al. 2021). The outcomes of these survey-based studies indicate that virtually all of the antibiotics considered to be appropriate for the management of odontogenic infections at the end of the twentieth century remain antibiotics-of-choice in contemporary dental practice. An exception to this generalization from the international survey outcomes is the continued presence of erythromycin as an antibiotic of choice (Segura-Egea et al. 2017).

In the United States, the unfavorable characteristics of erythromycin base and other erythromycin salts, including poor pharmacokinetics, irritancy in the gastrointestinal tract, and rapid development of bacterial resistance, are documented sufficiently so that its use in this country has largely been abandoned.

Tetracyclines, metronidazole, and macrolides may be used in the management of periodontitis, particularly refractory cases, where sub-antimicrobial doses of tetracyclines and macrolides appear to be beneficial (Haffajee et al. 2003; Hirsch et al. 2012). This chapter is focused on commonly used oral antibiotics, with an emphasis on high-level scientific evidence. The reader is referred to other sources for information on antiviral and antifungal agents.

A. H. Jeske (✉)
UTHealth School of Dentistry, Houston, TX, USA
e-mail: Arthur.H.Jeske@uth.tmc.edu

There are several important principles that must be understood prior to prescribing an antibiotic for a dental patient:

1. Antibiotics are adjuncts to debridement and/or surgical interventions (tooth extraction, pulpectomy, incision-and-drainage) and should not be used in place of these procedures in the management of infections.
2. Antibiotics are *not* effective against viruses or fungal organisms. Their successful use depends on an accurate diagnosis, which should confirm a bacterial etiology.
3. An antibiotic regimen should be based upon the signs and symptoms of the bacterial disease. The patient's responses to an antibiotic should be monitored using defined clinical endpoints (reduction of swelling, reduction of body temperature).
4. Prophylactic use of antibiotics in dentistry is limited (see Prescribing Considerations). The practitioner must be familiar with the current scientific evidence for this use, as well as current guidelines established by dental and medical professional organizations.
5. Some antibiotics may carry a high risk of allergic reactions and should be used only after careful consideration of the patient's medical and dental history.
6. There are relatively few randomized, controlled clinical trials of antibiotics in dentistry and, therefore, relatively few systematic reviews and meta-analyses on which to base therapeutic decisions.
7. The prescription of antibiotics is not appropriate as an interim measure when symptoms warrant immediate physical examination of the patient, particularly in children (Cherry et al. 2012).

Antibiotics Commonly Used in Dentistry

Penicillins There are many forms of penicillin, the most common being amoxicillin, a beta

lactam aminopenicillin. Aminopenicillins have an extended spectrum, which includes some gram-negative organisms that are not typically involved in odontogenic infections. Amoxicillin is now the most widely prescribed drug in this class, owing primarily to its pharmacokinetic traits. Because penicillins act by inhibition of bacterial cell wall synthesis, they have little systemic toxicity in mammals (mammalian cells lack a “cell wall”). However, penicillins may carry a relatively higher risk of allergy than other dental antibiotics. Penicillin VK was formerly recommended as the antibiotic of first choice for routine dental infections, but amoxicillin has largely replaced it in the United States.

Cephalosporins Since the development of cephalexin, considered a first-generation agent in the United States, there are now four generations of cephalosporins. Each later generation was synthesized to provide a broader spectrum of activity. While cephalosporins are indicated for antibiotic prophylaxis in selected patients, they are generally not preferred as routine agents in cases of odontogenic infections. They are included among alternative agents for antibiotic prophylaxis because of their effectiveness against gram-positive cocci.

In patients with allergy to penicillin, first-generation cephalosporins (e.g., cephalexin, Keflex®) carry a relatively low rate of cross-sensitivity of approximately 1%, but this can vary considerably for other generations of these agents (Campagna et al. 2012). In cases in which a cephalosporin is appropriate for the management of an odontogenic infection, an agent from the second-generation cephamycin subgroup should be selected, as this group possesses greater antibacterial efficacy against gram-negative anaerobes than other groups of cephalosporins (Molavi 1991). Agents in this subgroup include cefoxitin, cefotetan, and cefmetazole. Cephalosporins are bactericidal and possess a mechanism of action similar to that of the penicillins, that is, inhibition of bac-

terial cell wall synthesis by blocking the transpeptidation reaction.

Lincosamides Clindamycin is a lincosamide that is very effective against anaerobic and mixed aerobic-anaerobic infections. It possesses favorable pharmacokinetics and has become a widely used alternative for dental infections in cases of penicillin allergy. Lincosamides were initially implicated as a major causative agent for pseudomembranous colitis. It now appears that they have a risk for this complication similar to that of the cephalosporins and aminopenicillins. Lincosamides may be bactericidal in some bacterial species, particularly at higher dosages, and their mechanism of action involves inhibition of bacterial protein synthesis by binding to the same 50s ribosomal subunit as erythromycin.

Macrolides Erythromycin, clarithromycin, and azithromycin are members of this group and may be bacteriostatic or bactericidal, at higher concentrations. Macrolides inhibit protein synthesis by binding to the 50s ribosomal RNA. Macrolides possess less favorable characteristics for use in dentistry than the other antibiotic classes. Macrolides also bind to cytochrome P450 hepatic enzymes and can result in numerous seriously toxic drug interactions. Macrolides (azithromycin) are associated with cardiac arrhythmias, and erythromycin may also stimulate uncomfortable contractions of GI smooth muscle because of its similarity to a locally released GI pro-motility hormone, motilin. Except for use as alternatives for infective endocarditis prophylaxis, they are not preferred as first-choice or penicillin-alternative drugs for routine odontogenic infections. For the management of odontogenic infections with agents from this class, erythromycin base and its various salts are not recommended, and azithromycin, based

on its favorable pharmacokinetics, is the preferred representative for dental use of this class of antibiotic.

Nitroimidazoles Metronidazole was originally marketed for use in the treatment of protozoan infections that also has proven bactericidal activity against anaerobes. Because many oral infections (acute periodontal infections) are predominantly anaerobic in nature, metronidazole may be useful alone or in combination with amoxicillin. It has proven to be very effective when tested in vitro against periodontal pathogenic organisms. Because metronidazole is effective only against anaerobic organisms, its recommended role is as an adjunct to be used with a penicillin when a suboptimal clinical response to the penicillin occurs (Bali et al. 2015).

Tetracyclines Tetracyclines, including doxycycline, are broad-spectrum, bacteriostatic agents that inhibit bacterial protein synthesis by binding to the 30s ribosomal subunit. They chelate calcium ions and thus have a propensity to cause fluorescent tooth staining through incorporation into the enamel of developing teeth and even into remineralizing enamel of teeth that have already erupted (McKenna et al. 1999). While not typically indicated for routine odontogenic infections, tetracyclines at sub-antimicrobial doses modulate matrix metalloproteinases (collagenases) involved in the breakdown of extracellular structures and inflammation. They are used in selected cases of periodontitis refractory to conventional therapy procedures. Some patients may develop photosensitivity to these drugs, which can be severe (Jeske 2017).

The specific names and other characteristics of dentally useful antibiotics are shown in Table 1 (Jeske 2017).

Table 1 Classification and characteristics of common dental antibiotics for oral administration

Antibiotic	Class	Mechanism of action	Recommended adult oral dosage ^a	Special considerations
Penicillin VK	Beta lactam penicillin	Bactericidal	500 mg q 4 h	Absorption impaired by food
Amoxicillin	Aminopenicillin	Bactericidal	500 mg q 8 h	Absorption not impaired by food, available with beta lactamase inhibitor
Cephalexin	Cephalosporin	Bactericidal	2 g 30 min to 1 h before procedure	Risk of cross-allergy with penicillins is low; alternative agent for prophylaxis
Clindamycin	Lincosamide	Bactericidal	300 mg q 6 h	Excellent alternative in cases of penicillin allergy
Azithromycin	Macrolide	Bacteriostatic	500 mg day 1, then 250 mg 1 q day	Once daily dosing; Alternative agent for prophylaxis
Clarithromycin	Macrolide	Bacteriostatic	500 mg q 12 h	Alternative agent for prophylaxis
Metronidazole	Nitroimidazole	Bactericidal	500 mg q 8 h	Disulfiram-like reactions with alcohol; effective against anaerobes only
Doxycycline	Tetracycline	Bacteriostatic	20 mg q 12 h prior to meals	Adjunct for periodontal therapy; available in local delivery forms

^a See (American Academy of Pediatric Dentistry 2014) for information regarding pediatric dosages

Prescribing Considerations

1. Following the diagnosis of a bacterial infection and removal of the infected tissue (i.e., drainage or extraction or root canal therapy), the dental provider can begin antibiotic therapy with a standard dose of an orally administered first-choice agent (such as amoxicillin 500 mg) or an alternative agent. It is appropriate to initiate antibiotic therapy on an empirical basis (without obtaining culture and sensitivity testing). Culture and sensitivity testing requires additional time and may or may not identify specific etiologic pathogens. Culture and sensitivity testing should be considered if initial therapy is not effective. At this time, it appears that penicillin V is inferior when taken at 6-hour intervals. This is because of its pharmacokinetic profile with rapidly peaking serum levels followed by rapid decline. Penicillins are among a group of antibiotics that are termed “time dependent” for their pharmacodynamic action (bactericidal effect) (Craig 1998; Klutman 1996).
Put simply, the duration of time that penicillin V exceeds typical bacterial MICs (minimal inhibitory concentrations) relative to its dosing interval is insufficient at 6-h intervals,
2. The typical course of antibiotic therapy for dental infections runs for 5–7 days, unless symptoms persist. The patient should be monitored closely at the beginning of antibiotic therapy. Noticeable improvement should be expected within 24–48 h.
3. Because of reduced absorption in the presence of food, penicillin V should be prescribed 1 h before meals or 2 h after meals.
4. Penicillins do not appear to be effective for the management of symptomatic irreversible pulpitis (Keenan et al. 2005). Guidelines for the selection and use of antibiotics in adults for endodontic infections have been published by the American Association of Endodontists (American Association of Endodontists 2012).
5. The American Academy of Pediatric Dentistry has promulgated guidelines for antibiotic use in children (American Academy of Pediatric Dentistry 2014). Note that antibiotic dosages must be adjusted for the child’s body weight. Generally, the same antibiotics routinely used for odontogenic infections in adults are also the preferred agents in the pediatric patient population.

necessitating doses taken every 4 h. This dose interval is inconvenient when compared with the 6- to 8-h dosing intervals that can be used with most other dental antibiotics.

6. Warnings with antibiotic therapy should be issued verbally and in writing on the prescription. They should include the possible development of allergic reactions and diarrhea and other GI disturbances. It is recommended that refills not be authorized for antibiotic prescriptions so that unnecessary, prolonged exposure of the patient to the drug is avoided, minimizing the risk of adverse effects, as well as development of antibiotic-resistant bacterial strains.
7. Recent studies indicate that special consideration should be given to the prescription of antibiotics in obese patients. The data suggest that pharmacokinetic alterations in obese individuals, even following successful bariatric surgery, necessitate higher doses of amoxicillin in order to achieve expected therapeutic outcomes (Soares et al. 2021).

Antibiotic Prophylaxis

While there is little scientific evidence to support the use of prophylactic antibiotics to prevent postoperative complications, guidance from professional resources suggests its continued use in patients at highest risk of developing complications from infective endocarditis (Thornhill et al. 2018). Most regimens involve a single, pre-operative dose of a bactericidal agent with activity against *Streptococcus viridans*. There is limited evidence showing that a second dose will not enhance outcomes (Lopes et al. 2011). There is also limited evidence that antibiotic prophylaxis reduces complications following implant placement and no strong evidence supporting use to prevent complications of third-molar surgery (Esposito et al. 2013; Lodi et al. 2012). Antibiotic prophylaxis prior to dental treatment in patients with total joint arthroplasty (artificial joint) is controversial, and professional guidance now emphasizes good oral hygiene to prevent infective complications in these patients. At this time, there is insufficient scientific evidence on which to base the practice (American Academy of Orthopedic Surgeons and American Dental Association 2012).

However, when in doubt, the dentist is obligated to consult with the patient's physician(s) to determine the need for antibiotic prophylaxis and the appropriateness of the recommended regimen. Recently, a secular trend analysis of the incidence of infective endocarditis in a country in which antibiotic prophylaxis for patients at risk of infective endocarditis had been abandoned suggests that the abandonment of the practice may have contributed to a rise in cases of infective endocarditis. Further, it appears that this increase affected patients previously deemed not have conditions that put them at high risk for developing infective endocarditis (Dayer et al. 2015).

At this time, recent expert analyses of the practice of antibiotic prophylaxis to prevent infective endocarditis have been published with important major conclusions. Dayer et al. (Dayer et al. 2018) concluded that while antibiotic prophylaxis to prevent infective endocarditis produces a "marginal gain," the benefits of the prophylaxis outweigh the risks, especially in patients at high-risk for endocarditis, and even for those at moderate risk. This group further suggested that when all evidence is considered, the possibility that antibiotic prophylaxis has some small impact cannot be discounted. Finally, these authors indicated that antibiotic prophylaxis involves very little expense and the dosing regimens currently recommended minimize the risk of development of antibiotic resistance.

In a related publication, Thornhill et al. (2018) suggest that patients previously classified as "moderate risk" for infective endocarditis might also be considered as high-risk and that current guidelines for stratifying the risks of infective endocarditis related to dental procedures may require re-evaluation. This appears to have been confirmed indirectly in a report of an outbreak of bacterial endocarditis in a US oral surgery practice (Ross et al. 2018). In this report on 15 patients who developed infective endocarditis and who underwent oral surgical procedures, ten had predisposing cardiovascular conditions, and of these, five had mitral valve prolapse, a condition for which current guide-

lines no longer recommend prophylactic antibiotic (Wilson et al. 2007).

Currently, individuals who abuse drugs intravenously are considered to be at high risk of staphylococcal infective endocarditis, to the extent that these patients are being denied heart valve replacement surgery owing to the high risk of failure (Sanaiha et al. 2020). However, at this time, there are no professionally promulgated guidelines for the use of antibiotic prophylaxis in these types of patients, although dental practitioners should certainly be aware of this risk and the potential cardiac complications that may arise.

In 2021, the American Heart Association published a scientific statement on the current status of the prevention of viridans-group infective endocarditis and removed clindamycin from its list of recommended prophylactic antibiotics, replacing it with doxycycline 100 mg (Wilson et al. 2021).

In 2022, outcomes from a case cross-over analysis and cohort study of over 7 million US citizens confirmed that antibiotic prophylaxis reduces associations between oral surgical procedures and infective endocarditis in individuals at high risk for infective endocarditis ().

With regard to antibiotic prophylaxis to prevent infection of prosthetic joints, a recent cohort study performed on a database consisting of the entire population of England (55 million), there was no significant temporal association between late prosthetic joint infections and invasive dental procedures (Thornhill et al. 2022a). That group concluded that there is no rationale to administer antibiotics to prevent prosthetic joint infections in patients undergoing invasive dental procedures.

Adverse Effects

Antibiotics, as prescribed in dentistry, are generally well tolerated. With the exception of allergy, most adverse effects from antibiotics are related to their effects on the gastrointestinal tract. Virtually, all antibiotics may irritate the stomach or stimulate contractions of gastrointestinal smooth muscle, resulting in nausea, vomiting,

and cramping. They may also disrupt the normal flora, resulting in diarrhea, or lead to antibiotic-associated colitis and a potentially life-threatening overgrowth of *C. difficile*.

Symptoms with most cases of antibiotic-associated diarrhea dissipate when the antibiotic is discontinued. It is imperative that patients be cautioned against the use of antidiarrheal drugs and/or probiotics in place of medical diagnosis and management of this rare, but serious, complication. The development of any sign or symptom of an allergic reaction (rash, itching, and/or hives) requires that the antibiotic agent be discontinued immediately and the patient be evaluated medically (Beacher et al. 2015).

Penicillins are typically associated with a rate of allergy that is relatively higher than for other classes of antibiotics. These reactions may range from delayed-onset, mild forms (e.g., rash) to immediate-onset Type I anaphylaxis.

Robust guidance on the relative incidence of serious adverse reactions to oral antibiotics prescribed by dentists has been published (Thornhill et al. 2019). In this study, outcomes from the British National Healthcare System were surveyed over an 8-year period, which included nearly 29 million prescriptions for antibiotics written by dentists. Among the conclusions of that study, amoxicillin was deemed to be “remarkably safe,” while clindamycin had the worst adverse drug reaction profile, with a 30-fold higher incidence of fatal reactions. The ranking of the antibiotics in this study from least likely to most likely to cause an adverse drug reaction was (1) amoxicillin, (2) cephalosporins, (3) erythromycin, (4) tetracyclines, (5) azithromycin, (6) metronidazole, (7) amoxicillin with clavulanic acid, (8) clarithromycin, (9) penicillin V, and (10) clindamycin.

Adverse Drug Interactions

Antibiotics are capable of adversely interacting with other dental and medical drugs, both through pharmacodynamic and pharmacokinetic mechanisms. The most significant adverse pharmacodynamic interaction for commonly

Table 2 Clinically significant drug interactions involving antibiotics used in dentistry (modified from (Ciancio 2011))

Primary drug	Action	Interaction (and effect)
Alcohol	Metabolism decreased by	Metronidazole (severe nausea, vomiting)
Benzodiazepines	Enhanced by	<i>Erythromycin, clarithromycin</i> (increased CNS depression)
Carbamazepine	Enhanced by	<i>Erythromycin, clarithromycin</i> (increase carbamazepine toxicity)
Coumarins (including warfarin)	Enhanced by	<i>Erythromycin, clarithromycin, metronidazole</i> , penicillins, tetracyclines (increased risk of bleeding)
Digoxin	Enhanced by	<i>Erythromycin, clarithromycin</i> (increased toxicity of digoxin, including cardiac arrhythmias)
Lidocaine	Enhanced by	<i>Erythromycin, clarithromycin</i> (increased toxicity of lidocaine, CNS depression)
Penicillins	Antagonized by	Probenecid, salicylates, coumarin, diphenylhydantoin, griseofulvin (reduced efficacy against infection)
Statins	Enhanced by	<i>Erythromycin, clarithromycin</i> (increased statin toxicity, e.g., rhabdomyolysis)
Tetracyclines	Antagonized by	Antacids, iron (reduced absorption of tetracyclines)
Tetracyclines	Antagonizes	Penicillin (reduced efficacy against infection)
Theophylline	Potentiated by	<i>Erythromycin, clarithromycin</i> (increase toxicity of theophylline, possible cardiac arrhythmias)

prescribed antibiotics is the mutual antagonism that occurs when a bactericidal agent (penicillins, cephalosporins) is co-administered with a bacteriostatic agent (tetracycline). The recent scientific evidence does not support an adverse interaction between oral contraceptives and antibiotics used in dentistry (Taylor and Pemberton 2012).

Conversely, if drugs with similar mechanisms of action are administered together, a beneficial synergism may result. Combinations of antibiotics are not generally recommended in dentistry. However, the addition of *metronidazole* to a penicillin regimen may improve outcomes because of the selective action of metronidazole on strictly anaerobic organisms. Among the antibiotics discussed here, macrolides are the most likely to produce pharmacokinetic drug interactions. Serious adverse interactions of various classes of dental antibiotics are listed in Table 2 (Ciancio 2011).

Less significant drug interactions are also possible—the clinician is urged to consult the complete prescribing information for all drugs prescribed.

For medically compromised patients with systemic disease that could affect drug metabolism and/or excretion, consultation with the patient's physician is recommended.

Antibiotics for Periodontitis

Both systemic and locally applied antibiotics have been investigated for the management of periodontal diseases, focused primarily on refractory periodontitis (periodontitis that responds poorly to conventional therapy).

A recent systematic review of this subject indicates that the scientific evidence for systemic antibiotics in the treatment of refractory periodontitis does not support this indication, nor does it show that systemic antibiotics add an incremental benefit to conventional treatment along (e.g., mechanical debridement) (Santos et al. 2016). Until there are more studies with less heterogeneity and with parallel design, the use of systemic antibiotics for periodontal therapy remains controversial.

Conclusion

Antibiotics continue to play an important, albeit adjunctive, role in the management of routine odontogenic infections. They are safe and effective when prescribed at recommended doses and based on the patient's presenting signs, symptoms, and coexisting medical conditions. The

number of patients who are candidates for antibiotic prophylaxis is relatively small, and prophylactic use should be guided by the current recommendations of professional organizations, as based on scientific studies. Dentists should continue to consider emerging evidence for the use of low dosage antibiotics in cases of refractory periodontitis and other inflammatory diseases. At present, the topic of dental antibiotics can be summarized as follows:

1. Penicillins and aminopenicillins (e.g., amoxicillin) remains drugs-of-first-choice for the management of routine odontogenic infections.
2. Alternative antibiotics to be used in cases in which penicillins are contraindicated include clindamycin, azithromycin, metronidazole, and moxifloxacin.
3. Antibiotic resistance is problematic in medicine but is still of limited significance in the treatment of odontogenic infections in dental outpatients.
4. Based upon the consensus of experts, antibiotic prophylaxis remains indicated in limited groups of patients with specific, high-risk conditions for cardiac and prosthetic joint infections.
5. At the present time, there is no compelling scientific evidence that mandates changes in current guidelines for the use of antibiotics in dentistry.
6. New findings from retrospective case studies and secular trend analyses of populations may disrupt current practices of antibiotic prophylaxis to prevent infective endocarditis, and the practitioner is urged to regularly monitor the guidelines for antibiotic prophylaxis promulgated by professional medical and dental organizations.

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Pharmacologic Management of Patients with Drug-Related Coagulopathies

Issa A. Hanna, Alfredo R. Arribas,
Amir All-Atabakhsh, and John A. Valenza

Understanding Clinical Hemostasis: The Key to Patient Management

Understanding and applying clinical knowledge of hemostasis is essential in managing patients undergoing dental procedures who are on “blood-thinners.” This requires the clinician to make the distinction between the function of antiplatelet and anticoagulant medications. Patients utilize these medications for primary or secondary prevention of thromboembolic complications of vascular disease (Palareti and Cosmi 2009). The clinical implications and management will vary accordingly, and a portion of these patients will be on a combination of multiple antiplatelet or antiplatelet and anticoagulants.

The prevalence of antiplatelet and anticoagulant use is increasing due to an aging population and increased survival of cardiovascular or cerebrovascular insults. As a result, clinicians will regularly encounter this population of patients in their practices. This chapter will focus on patients who are using these medications for acquired medical problems and not on patients who utilize

these medicines for congenital coagulopathies, diseased-induced states (e.g., pregnancy, cancer, severe trauma or burns, etc.), or effects of medications that are pro-coagulant (e.g., birth control). The authors intend to provide the clinician with an understanding of the physiology of clotting, associated medication therapies, and, more importantly, evidence-based principles and recommendations on managing patients on “blood thinners.” We will provide an overview of the elements involved in clotting in response to vascular injury, especially due to dental procedures, which will further aid the clinician in making evidence-based clinical judgments when treating this patient population.

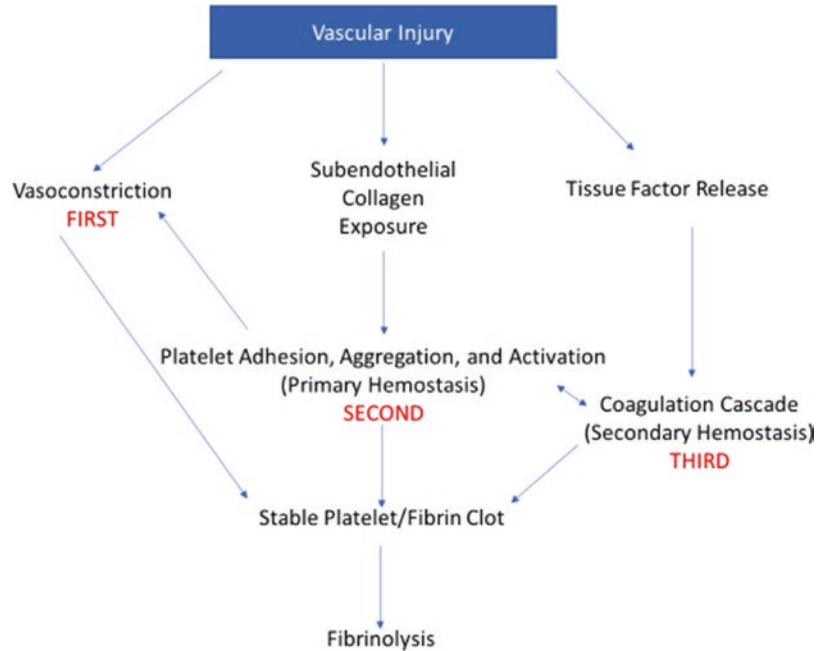
Clinical Hemostasis

Hemostasis is a multifactorial and dynamic process. After vessel injury, there is an immediate vasoconstriction phase to decrease the flow of blood, lasting approximately 20 min (Mingarrode-León et al. 2014). Injury to the vessel wall also leads to exposure of subendothelial collagen and the progression into the platelet phase of hemostasis.

Prior to understanding platelet function after vessel wall injury, the clinician should recognize that the uninjured, vascular endothelium inhibits platelet function, and this prevents a thrombotic event. When a vessel wall is disrupted, as with

I. A. Hanna (✉) · A. R. Arribas · A. All-Atabakhsh · J. A. Valenza
School of Dentistry, University of Texas,
Houston, TX, USA
e-mail: Issa.A.Hanna@uth.tmc.edu;
Alfredo.R.Arribas@uth.tmc.edu;
Amir.AllAtabakhsh@uth.tmc.edu;
John.A.Valenza@uth.tmc.edu

Fig. 1 Hemostatic response to vascular injury



surgical intervention, there is a series of events that lead to hemostasis. This series of events is described in Fig. 1.

Platelet Phase

The fundamental cell responsible for the production of platelets is the megakaryocyte, which is derived from hematopoietic stem cell precursors. The megakaryocyte produces and releases platelets and is primarily regulated by the enzyme thrombopoietin (TPO), which is principally produced in the liver, with some contribution from the kidney and bone marrow. The circulating platelet has a life span of 7–10 days with approximately 33% of platelets being stored in the spleen (Wilson et al. 2015). Normal platelet count is 150,000–450,000/uL, and a significant decrease in platelets may also represent risks associated with postoperative bleeding (Branehög et al. 1975).

Once circulating platelets are no longer inhibited from binding to the vessel wall, the promotion of adhesion of platelets to the exposed collagen occurs. This is initially mediated by exposed von Willebrand Factor (VWF), which is

naturally bound to the subendothelial collagen and released upon injury (note: additional VWF is also released by subsequent platelet activation) (Wilson et al. 2015). This helps form a platelet clot, which improves hemostasis (Mingarro-de-León et al. 2014). Aggregation of the platelet plug is then assisted by the activation of platelet glycoprotein (Gp) IIb/IIIa (α IIb β 3) receptor, which facilitates the binding of VWF and fibrinogen to these receptors (Yip et al. 2005). Figure 2 provides greater detail on the platelet cascade and the factors involved in obtaining a platelet plug. It is important to note that circulating endogenous thrombin, adenosine phosphate (ADP), epinephrine, and thromboxane A2 are important in platelet activation and clot formation (Howard et al. 2013). Inhibition of these select proteins has important clinical consequences on the formation of the platelet plug.

Coagulation Phase

Following the platelet phase is the coagulation phase (Fig. 3). The binding of coagulation factor X to receptors on the platelet surface occurs and accelerates conversion to factor Xa, which is crit-

Fig. 2 Platelet phase of hemostasis

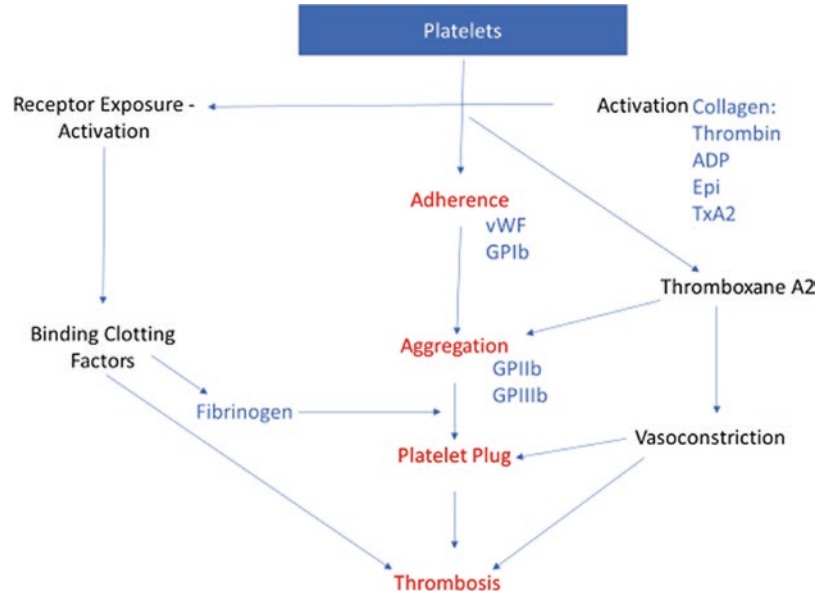
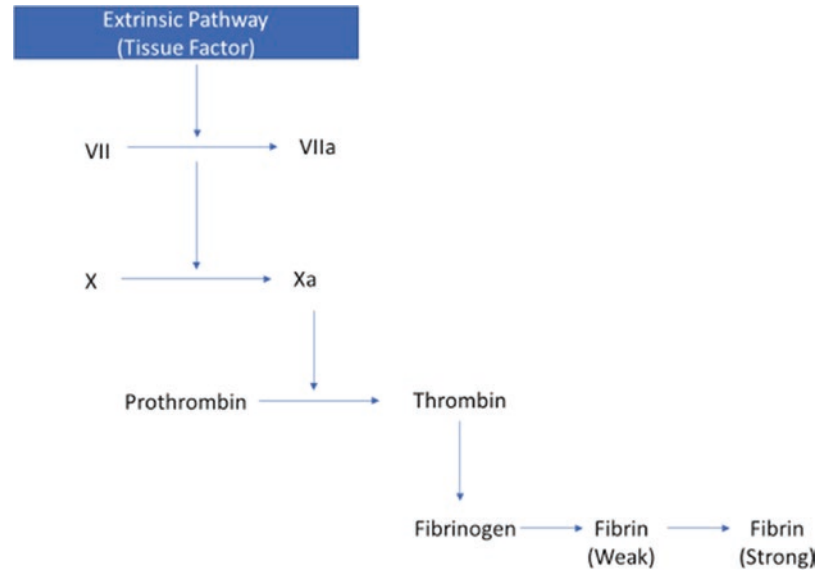


Fig. 3 Coagulation phase of hemostasis



ical in converting prothrombin to thrombin. Thrombin then acts to convert the soluble fibrinogen clot to a strong, insoluble fibrin plug (Mingarro-de-León et al. 2014). The coagulation phase terminates when the damaged vessel wall is repaired and/or the clot is stable. The clot then undergoes fibrinolysis, which is facilitated by the conversion of prekallikrein to kallikrein by Factor XII. Kallikrein then converts inactive plasminogen to active

plasmin, which then lyses the fibrin plug (Mingarro-de-León et al. 2014). Antiplatelet therapy targets the initial stages of clotting formation by disrupting platelet aggregation and the formation of a platelet plug, as described earlier in the chapter. Anticoagulation therapies target the latter stages of clot formation by inhibiting the function of clotting factors and disrupting the coagulation cascade (Bagot and Arya 2008).

Knowing the physiology of hemostasis, the dental clinician can better co-manage the use of blood thinners with the physician and plan how to approach providing dental care to patients on these medications.

When treating patients taking medications that disrupt platelet function, the clinician should anticipate the likelihood for prolonged intraoperative and/or immediate postoperative bleeding. With patients on anticoagulants, however, the clinician should expect relatively normal intraoperative or immediate postoperative hemostasis, but the potential for bleeding problems or complications hours later when the coagulation cascade occurs.

Antiplatelet Therapy

Indications for Antiplatelet Therapies

Understanding antiplatelet and anticoagulant therapies is directly tied to the underlying patho-

physiology of thrombosis. In the 1850s, Virchow described three phenomena that increased the likelihood of thrombosis. These concepts have been further refined over the past decades but are summarized as follows:

- Interrupted blood flow (stasis)
- Irritation of vessel lining (endothelial or vessel wall injury)
- Blood coagulation (hypercoagulopathy) (Bagot and Arya 2008)

Antiplatelet therapy is widely used to reduce the risk of myocardial infarction and in the treatment of patients with high-risk atherosclerotic vascular disease, acute coronary syndrome, and arterial/venous thrombosis. It is also utilized in patients with a history of prior percutaneous coronary interventions, coronary bypass surgery, atrial fibrillation, and stroke (Dinkova et al. 2013).

Table 1 illustrates these morbidities and how they contribute to an increased likelihood of thrombosis.

Table 1 Antiplatelet morbidities and how they contribute to risk of thrombosis (Dinkova et al. 2013)

Pathophysiology	Phenomena contributing to thrombosis	Adverse outcome
Atherosclerotic vascular disease/acute coronary syndrome/myocardial infarction	Vessel wall injury (due to atheromatous plaques deposited within the tunica intima) Interrupted blood flow (due to formation of atheroma in coronary arteries which disrupts laminar blood flow)	Myocardial infarction and its sequelae such as rupture of the myocardium
Venous thrombosis	Interrupted blood flow (due to stasis, immobilization, peripheral vascular disease, etc.) Vessel wall injury (due to trauma, surgery, etc.)	Superficial or deep vein thrombosis, which may lead to an embolism such as a pulmonary embolism
Arterial thrombosis	Interrupted blood flow (due to stasis, aneurysm, vessel wall dissection, etc.) Vessel wall injury (due to aneurysm, vessel wall dissection, etc.)	Arterial thrombosis which can lead to an arterial embolus
Percutaneous coronary Intervention (i.e., angioplasty and coronary stents)	Vessel wall injury (sclerosis of intravascular walls requiring widening of vessel diameter or placement of an indwelling intravascular stent) Interrupted blood flow (turbulent blood flow in the presence of stents)	Thrombosis of coronary vasculature, myocardial ischemia, myocardial infarction
Coronary bypass surgery	Interrupted blood flow (via surgical placement of a shunt to bypass a coronary artery blockage)	Thrombosis of bypass graft
Atrial fibrillation	Interrupted blood flow (inadequate atrial contraction leading to turbulent blood flow, which potentiates thrombosis)	Stroke (ischemic)
Transient ischemic activity (TIA)/stroke (ischemic)	Interrupted blood flow (due to thrombus formation in cerebral arteries or emboli from proximal arteries)	Cerebral ischemia and possible cerebral Infarction

Assessing Platelet Function

Platelet Function Tests

Historically, bleeding time (BT) has been used to determine platelet functionality. It was initially described *in vivo* by Duke in 1910. BT was regarded as the most useful screening tool for platelet function until the early 1990s. However, overwhelming literature suggests that prolonged BT does not translate into increased blood loss from surgery. Many studies have shown that a low dose of acetylsalicylic acid (aspirin) has an effect on bleeding time but has no clinical significance (Dinkova et al. 2013).

Other available platelet function tests can be subdivided into platelet adherence, aggregation, or activation studies. These studies are best utilized for special patient populations, such as in individuals with congenital platelet disorders (e.g. light transmission aggregometry [LTA]) or in trauma (e.g., thromboelastogram [TEG]).

Antiplatelet Medications

Multiple mediators precipitate platelet aggregation and adhesion. Antiplatelet medications have been developed to individually target these mediators to achieve their desired effects. Examples of these medications are outlined in Table 2.

COX Inhibitors: Aspirin

Acetylsalicylic acid was initially synthesized in the 1850s by acetylation of salicylic acid. Methods to synthesize acetylsalicylic were further refined over the next few decades. In 1899, scientists at Bayer named it “aspirin” and began large-scale distribution (Dinkova et al. 2013).

Aspirin irreversibly binds with platelets to inhibit aggregation by acetylation of serine resi-

dues of the COX enzymes, which renders the platelet nonfunctional for the duration of its life span. In turn, the generation of thromboxane A₂, which is a catalyst of platelet aggregation, is decreased. This inactivation of COX lasts up to 10 days (i.e., the entire life of the platelet) (Malmquist 2011).

There are no significant differences in COX enzyme inhibition between 81 or 325 mg doses. The differences are limited to anti-inflammatory and antipyretic activities, which are outside the scope of this chapter (Dinkova et al. 2013).

Clopidogrel/Ticagrelor/Prasugrel/Ticlopidine

Clopidogrel (Plavix®) is an antiplatelet agent belonging to the thienopyridine family. Clopidogrel blocks the P2Y₁₂ receptor on the platelet cell membranes, inhibiting ADP-induced platelet aggregation. It is a pro-drug and needs to be metabolized to its active metabolite by several hepatic cytochrome P450 isoenzymes.

Clopidogrel is indicated for monotherapy for the prevention of atherothrombotic events in patients suffering myocardial infarction, ischemic stroke, or peripheral arterial disease. It is also used in conjunction with aspirin in patients suffering from acute coronary syndrome (ACS) for non-ST segment elevation ACS (unstable angina or non-Q-wave myocardial infarction), including patients undergoing percutaneous coronary intervention (PCI) with stent placement and ST-segment elevation following acute myocardial infarction in medically treated patients eligible for thrombolytic therapy.

Platelet inhibition by clopidogrel is both dose- and time-dependent, and patients are usually given a loading dose of 300–600 mg and then maintained at 75 mg/day.

Table 2 Classes of antiplatelet medications (Dinkova et al. 2013)

COX inhibitors	Inhibitors of ADP-mediated activation of P2Y12 receptors	Prostaglandin analogues	Phosphodiesterase inhibitor	Fibrinogen receptor antagonist
Acetylsalicylic acid (Aspirin) ^a	Clopidogrel (Plavix) ^a	Alprostadil	Dipyridamole	Abciximab
	Prasugrel (Effient) ^a	Iloprost		Tirofiban
	Ticagrelor (Brilinta) ^a			
	Ticlopidine (Ticlid) ^a			

The duration of clopidogrel therapy varies according to the indication (Dinkova et al. 2013):

- Myocardial infarction: from a few days to less than 35 days
- Ischemic stroke: from 7 days to less than six months
- Following the insertion of a bare metal stent (BMS): 4–12 weeks
- Following the insertion of a drug-eluting stent (DES): 6–12 months

Ticagrelor (Brilinta®) is a new direct inhibitor of the platelet P2Y₁₂ receptor and therefore does not require metabolic activation. Unlike thienopyridines, ticagrelor binds reversibly to the P2Y₁₂ receptor and at a site that is independent of ADP. However, it still results in suppression of ADP-induced platelet activation by temporarily “locking” the receptor in an inactive state until it dissociates. Ticagrelor has a significantly faster onset and offset of antiplatelet activity compared with clopidogrel in subjects with stable coronary artery disease or acute coronary syndromes. Ticagrelor is indicated for the prevention of thrombotic events in patients with acute coronary syndrome or myocardial infarction with ST elevation. The drug is combined with acetylsalicylic acid unless the latter is contraindicated (Dinkova et al. 2013).

Prasugrel (Effient®) is the newest member of the thienopyridine family and also irreversibly binds to P2Y₁₂-receptor with a rapid onset and stronger inhibitory effect than clopidogrel. It is licensed for use with aspirin for the prevention of atherothrombotic events in patients with acute coronary syndrome (Dinkova et al. 2013).

Ticlopidine (Ticlid®) is an ADP receptor blocker, preventing the binding of fibrinogen to platelet glycoprotein. It is indicated for use in patients who cannot tolerate aspirin or with aspirin in DAPT (Dinkova et al. 2013). Ticlopidine has fallen out of favor because of bone marrow toxicity (Iqbal et al. 2023).

Management of Patients on Single and Dual Antiplatelet Therapy

Antiplatelet therapies are used to manage a multitude of medical conditions. Single antiplatelet therapy (SAPT) is commonly used to manage cerebrovascular accidents (CVA), cerebrovascular disease, coronary artery bypass grafts (CABG), percutaneous coronary intervention (PCI), or as a final treatment modality after a period of dual antiplatelet therapy (DAPT). Dual antiplatelet therapy is typically used for patients who have had a recent cardiac stent placed due to coronary artery disease. Though many patients often transition to SAPT after a period of time, a subset of high-risk patients will maintain long-term DAPT. It is also important for the dental clinician to confirm the type of stent and date of placement (patients are given a medical card after stent placement that specifies the type of stent inserted). Patients with bare metal stents are typically maintained on DAPT for a minimum of 1 month and usually maintained on a single antiplatelet therapy after that. Patients with drug-eluting stents are typically maintained on dual antiplatelet therapy for at least 12 months and then maintained on a single antiplatelet therapy afterward (Singh et al. 2013). It is important that DAPT remains uninterrupted in these time frames to prevent thromboses and associated morbidity (Grines et al. 2007).

Providing safe and predictable care to the dental patient who is on antiplatelet therapy can be confusing and, in many instances, without evidence-based direction. Many variables must be considered when making treatment decisions for these patients. These include the patient's overall state of health, the risk of bleeding from the planned surgical procedure, as well as the risks associated with cessation of any antiplatelet medications. The dental clinician should always consult the appropriate managing physician, whether it is a primary care provider (PCP) or a specialist (ADA 2018). When discussing the patient's care with the medical clinician, it is imperative to review the extent of the surgery,

anticipated bleeding and determine a risk assessment on whether to treat the patient from a medical standpoint. After reviewing the literature, the authors' goal is to provide the dental clinician with evidence-based data that will help them discuss the decision on managing the patient's antiplatelet medications, whether SAPT or DAPT, with the managing physician.

When determining the course as it relates to antiplatelet therapy and dental surgery, the clinician must consider that discontinuing therapy increases the risk of systemic morbidities. Prior studies have shown that discontinuing antiplatelet therapy for surgical intervention provides significantly increased systemic risks versus the risk of bleeding and complications of bleeding if antiplatelet therapy is maintained. Maulez et al., in a case-controlled study, looked at 309 patients who were diagnosed with an ischemic stroke or transient ischemic attack (TIA) undergoing long-term aspirin therapy before their event and 309 patients with an ischemic stroke (IS) and TIA who did not have aspirin therapy 6 months prior to their event (Maulaz et al. 2005). The results showed that aspirin interruption increased the risk of an IS/TIA by 3.4-fold with a 95% confidence interval (CI). Risks were even greater for patients with a prior diagnosis of coronary heart disease (Maulaz et al. 2005). When contemplating the cessation of aspirin or SAPT for dental procedures, the risk of an ischemic event and associated morbidities must be compared to the risk of hemorrhage from the surgical intervention. The risk of cessation of antiplatelet therapy is only one component in making treatment decisions for this patient population. The clinician must also have valid data to determine if patients who remain on antiplatelet therapy have a risk of significant bleeding after dental surgical procedures.

Traditionally, the dental clinician requests that the managing physician hold antiplatelet therapy prior to dental surgical procedures. We now know that cessation of antiplatelet therapy can have significant systemic consequences. In addition, many studies have shown that maintaining antiplatelet therapy has no significantly increased morbidity associated with postoperative bleed-

ing. Napenas et al. in 2013 published data from a literature review of 15 articles and found no clinically significant increase in the risk of postoperative bleeding in patients either on SAPT or DAPT that underwent invasive dental procedures (Napeñas et al. 2013).

In 2015, Dézsi et al. compared postoperative bleeding in patients undergoing dental extractions who were on SAPT or DAPT. Patients were observed in-office until stable clot formation was noted. Those on SAPT took up to 60 min to form a stable clot, whereas DAPT patients took as long as 130 min. However, clinical outcomes were no different in either group, and no morbidity was recorded. The authors concluded that dental extractions could be safely performed while maintaining single or dual antiplatelet therapy (Dézsi et al. 2015). These studies provide clear evidence that the risk of maintaining antiplatelet therapy for invasive dental procedures is low, and the systemic risk of discontinuing therapy is high. Treatment decisions must therefore consider the dental clinician's professional comfort level with hemostasis control, the complexity and invasiveness of the procedure, and the baseline risk of bleeding, especially with a patient on antiplatelet therapy. For complex dental surgery, it may be prudent for the dental clinician to stage procedures to decrease the associated risk of bleeding complications.

Cardona-Tortajada et al. monitored 155 patients on varied SAPT who underwent dental extractions. Twenty-six patients had minor bleeding complications, which were controlled by local hemostatic measures, and one patient had severe bleeding and required an emergency room visit. The antiplatelet drug had no influence on postoperative bleeding, yet the number of teeth extracted was statistically relevant to postoperative bleeding complications. Cardona-Tortajada et al. advised not to extract more than three teeth at a time and that these should either be adjacent or correlative and not in different parts of the dental arch (Cardona-Tortajada et al. 2009). These recommendations seem to make sense, as more extensive and invasive surgeries have an inherent increased risk of bleeding as a baseline,

which will only be amplified for the patient on antiplatelet therapy.

In summary, the data presented shows that cessation of antiplatelet therapy, whether SAPT or DAPT, has a greater risk of systemic morbidity than the risk of bleeding from dental surgery, while these therapies are maintained. Other important factors for the clinician to consider when making treatment decisions include patient medical comorbidities, patient medications, invasiveness and extent of the procedure, dentist's comfort level with hemostasis, and care of the medically complex patient, as well as patient's ability to recognize postoperative bleeding issues and having access to after-hours care. Patients on antiplatelet therapy may have an increased risk of postoperative bleeding, yet this bleeding is usually well controlled with local hemostatic measures and uncommonly associated with morbidity. Additionally, in more invasive and extensive surgeries, the clinician should stage the procedure to decrease the risk of significant post-op bleeding.

Anticoagulation Therapy

Indications for Anticoagulation Therapies

The dental clinician should be aware of the general indications for the use of anticoagulants,

which include atrial fibrillation, venous thromboembolisms, and the presence of a mechanical heart valve (Guyatt et al. 2012a,b). Unlike antiplatelet therapies, the efficacy of anticoagulant therapies has been traditionally monitored. We will discuss the changing paradigm in the direct oral anticoagulants section. The severity of the disease state usually dictates more aggressive anticoagulation (usually monitored by the patient's International Normalized Ratio (INR)). Table 3 summarizes the indications for anticoagulant therapies.

In these scenarios, the clinician is placed in a dilemma. Unplanned cessation of anticoagulant therapy can have devastating effects, such as fatal mechanical heart valve (MHV) thrombosis in 15% of patients, embolic stroke leading to death or major disability in 70% of patients, and with venous thromboembolism a case-fatality rate of approximately 5–9%. On the other hand, continuing anticoagulant therapy results in a major bleeding risk of 2–4%, of which there is a case-fatality rate of 8–10% (Spyropoulos and Douketis 2012).

Because of the more severe clinical consequences of thrombosis over major bleeding, a strategy that incurs three to ten more cases of major bleeds to prevent one case of thrombosis or stroke would, in theory, be clinically acceptable. This is based on the devastating consequences of a stroke as compared with a bleed, which in the majority of cases, can be surgically controlled.

Table 3 Suggested risk stratification for perioperative thromboembolism (Spyropoulos and Douketis 2012)

Risk category	MHV	Atrial fibrillation	Venous thromboembolism
High (>10%/year risk of ATE or 10%/month risk of VTE)	Any mechanical mitral valve	CHADS ₂ score 5 or 6	Recent (<3 months) VTE
	Caged-ball or tilting disc valve in mitral/aortic position	Recent (<3 months) stroke or TIA	Severe thrombophilia
Intermediate (4–10%/year risk of ATE or 4–10%/month risk of VTE)	Bilateral AVR with major risk factors for stroke	Rheumatic Disease	Deficiency of protein C, protein S or antithrombin
			Antiphospholipid antibodies
			Multiple thrombophilias
			VTE within past 3–12 months
Low (<4%/year risk of ATE or <2%/month risk of VTE)	Bilateral AVR with major risk factors for stroke	CHADS ₂ score 3 or 4	Recurrent VTE
			Non-severe thrombophilia
			Active cancer
			VTE>12 months ago

TIA transient ischemic attack, AVR aortic valve replacement, ATE atrial thromboembolism, VTE venous thromboembolism, MHV mechanical heart valve

Assessing Coagulation Function

We will briefly review concepts that aid the clinician in managing anticoagulated patients undergoing dental surgical procedures.

Prothrombin Time (PT)/International Normalized Ratio (INR)

The INR was developed to incorporate the international sensitivity index (ISI) values and attempt to make prothrombin time (PT) results uniformly usable. The working reference has been calibrated against internationally accepted standard reference preparations, which have an ISI value of 1.0. The ISI value is the exponent in the formula and is, therefore, critical for calculating the INR.

$$\text{INR} = (\text{patient PT} / \text{mean normal PT})^{\text{ISI}}$$

Consequently, small errors in the ISI assignment may affect the calculated INR substantially. An otherwise healthy patient who is not anticoagulated will have an INR value of 1 (Al-Mubarak et al. 2007).

Partial Thromboplastin Time (PTT)

Partial thromboplastin time (PTT) has been described as measuring the overall speed of a clot formation, correlating with the activity of the intrinsic clotting cascade. PTT must be interpreted with caution as it doesn't reflect the in vivo hemostatic response and the interaction between the vessel wall, platelets, fibrinogen, and circulating coagulation factors (WHO 1999).

Anticoagulation Medications: Warfarin

Current oral anticoagulant practice dates back almost 75 years to when vitamin K antagonists were discovered as a result of investigations into the cause of hemorrhagic disease in cattle. Characterized by a decrease in prothrombin levels, this disorder was caused by the ingestion of hay containing spoiled sweet clover. Hydroxycoumarin, a lactone, which was isolated from bacterial contaminants in the hay, interferes

with vitamin K metabolism, thereby causing a syndrome similar to vitamin K deficiency. This paved the way for discovering a more potent variant of coumarins, namely, warfarin, for use as a rodenticide. Warfarin is derived from WARF for Wisconsin Alumni Research Foundation and -arin for its connection to coumarin (Link 1959).

Warfarin interferes with the synthesis of the vitamin K-dependent clotting factors, which include prothrombin (factor II) and factors VII, IX, and X. The synthesis of the vitamin K-dependent anticoagulant proteins, proteins C and S, is also reduced by warfarin.

Mechanism of Action

All the vitamin K-dependent clotting factors possess glutamic acid residues at their N-termini. A posttranslational modification adds a carboxyl group to the γ -carbon of these residues to generate γ -carboxyglutamic acid.

Warfarin inhibits vitamin K epoxide reductase (VKOR), thereby blocking the γ -carboxylation process. This results in the synthesis of vitamin K-dependent clotting proteins that are only partially γ -carboxylated.

Warfarin acts as an anticoagulant because these partially γ -carboxylated proteins have reduced or absent biologic activity. The onset of action of warfarin is delayed until the newly synthesized clotting factors with reduced activity gradually replace their fully active counterparts.

The antithrombotic effect of warfarin depends on a reduction in the functional levels of factor X and prothrombin, clotting factors that have half-lives of 24 and 72 h, respectively (FDA 2010).

Pharmacology

Warfarin is a racemic mixture of R and S isomers and is rapidly and almost completely absorbed from the gastrointestinal tract. Racemic warfarin has a plasma half-life of 36–42 h, and more than 97% of circulating warfarin is bound to albumin. Only the small fraction of unbound warfarin is biologically active (Wilson et al. 2015).

Variability of anticoagulation with warfarin in between individuals is significantly dependent on the expression of a group of cytochrome P-450-2C9 (CYP2C9) enzymes, which is

responsible for oxidative metabolism (metabolic clearance) of the more active S-enantiomer, and the Vitamin K epoxide reductase enzyme (VKORC1), which reduces the Vitamin K, the cofactor necessary for gamma-carboxylation of Vitamin K-dependent clotting factors. This can predispose to overdosage conditions and a higher risk of bleeding.

In 2007, the FDA issued a black box warning, which updated the product label of warfarin by advising physicians to consider CYP2C9 and VKORC1 genetic tests to improve their initial estimate of warfarin dose (FDA 2010).

Interactions with Other Commonly Used Medications

The fundamental mechanisms of interaction between warfarin and antibiotics are twofold:

1. Antimicrobial agents disrupt gastrointestinal flora that synthesizes vitamin K (Onysko et al. 2016).
2. Antimicrobials inhibit cytochrome p450 (CYP450) enzymes (primarily CYP2C9 and 3A4), which are responsible for the metabolism of warfarin. The antibiotics most likely to interfere with warfarin are Trimethoprim and Sulfamethoxazole (TMP/SMX), ciprofloxacin, levofloxacin, metronidazole, fluconazole, azithromycin, and clarithromycin. Low-risk agents include clindamycin, cephalexin, and penicillin G. A 2008 study investigated the anticoagulation effects of a 10–20% preemptive warfarin dose reduction versus no dosing change in patients taking TMP/SMX or levofloxacin. The investigators found that the preemptive warfarin dose reduction (intervention) significantly decreased the number of supratherapeutic INR values above 4 when compared to controls (two of eight vs eight of nine) (Onysko et al. 2016). There is no consensus in the dental literature to support adjusting warfarin dosing in the presence of TMP/SMX or metronidazole in the pre or postoperative setting, but the clinician should be aware of the increased risk of bleeding if the patient has been recently prescribed antibiotics, especially TMP/SMX or metronidazole (Onysko et al. 2016).

Anticoagulation Medications: Heparin

Heparin is a proteoglycan that functions as a cofactor of the naturally occurring anticoagulant, antithrombin. Because the half-life of heparin is short (60 min), the therapeutic levels are maintained by intravenous bolus injections followed by monitored infusion. The therapeutic range is monitored by prolongation of the partial thromboplastin time (PTT). There are several different types of heparin; notable is low-molecular-weight heparin (LMWH) with a longer half-life and can be delivered subcutaneous once or twice a day in contrast to unfractionated heparin (FDA 2006). Enoxaparin (Lovenox®) and Dalteparin (Fragmin®) are commonly used LMWHs in clinical practice.

Patients who have been on long-term therapy with heparin do not require laboratory monitoring; however, when monitoring is required (such as intraoral bony surgery), an anti-Xa assay is used because the PTT is not predictably prolonged (Malmquist 2011).

Complications associated with heparin are limited, the most notable being heparin-induced thrombocytopenia (HIT). HIT is a rare, life-threatening complication of heparin due to an autoimmune response toward Platelet Factor 4 and heparin complex, which results in thrombocytopenia and arterial and venous thrombosis. The most common manifestations are bleeding and a drop in platelet count. It is typically treated with immediate cessation of heparin and the use of an alternative, non-heparin-based anticoagulant (Greinacher et al. 2017).

Direct Oral Anticoagulants (DOAC)

The direct oral anticoagulants (DOAC) are a newer class of drugs that act to attenuate coagulation by targeting a specific factor in the coagulation pathway. Due to the new nature of these drugs, the nomenclature in the literature is somewhat varied and without absolute consensus. In 2015, Barnes et al., for the International Society on Thrombosis and Hemostasis

Subcommittee on the Control of Anticoagulation, published a discussion of various terms and recommendations on nomenclature for these new drugs. The varied nomenclature discussed by Barnes et al. were novel oral anticoagulants (NOAC), target-specific oral anticoagulants (TSOAC), and direct acting oral anticoagulants (DOAC) (Barnes et al. 2015). The subcommittee recognized that the oldest and most commonly used term was novel oral anticoagulants (NOAC). Yet, they also recommend that the classification term for these drugs be transitioned to direct acting oral anticoagulants (DOAC) as this better describes their mechanism of action (Barnes et al. 2015). For the purpose of this chapter and to limit confusion in terminology, the term direct oral anticoagulants (DOAC) will be used. Yet the reader should be

aware and informed of the other terms used for this class of drugs.

These drugs either prevent the generation of thrombin (Factor IIa) or directly inhibit thrombin. The drugs that indirectly inhibit thrombin formation do so by targeting Factor Xa. Inhibition of Factor Xa interferes with the formation of the fibrin clot by blocking the conversion of fibrinogen to fibrin. Even more recent agents act by directly inhibiting thrombin in the clotting cascade, inactivating thrombin, and inhibiting thrombogenesis (Shi and Crowther 2019).

DOACs that are currently available include factor Xa inhibitors Rivaroxaban (Xarelto®), Apixaban (Eliquis®), Edoxaban (Savaysa®), and Betrixaban (Bevyxxa®) and direct thrombin inhibitor dabigatran (Pradaxa®) as noted in Table 4 (Thean and Alberghini 2016; Chen et al. 2020).

Table 4 Most commonly used novel oral anticoagulants (Barnes et al. 2015; Chen et al. 2020; Parasrampur and Truitt 2016)

Brand name	Generic name	Target factor	Indications (FDA approved)	Clearance/metabolism
Pradaxa	Dabigatran	IIa	Stroke prevention in NVAf (Nonvalvular Atrial fibrillation) Treatment of DVT and PE Prevention of recurrent DVT and PE prevention of thromboembolism after total hip replacement Atrial fibrillation	Renal (80%)
Xarelto	Rivaroxaban	Xa	Stroke prevention in NVAf (nonvalvular atrial fibrillation) Treatment of DVT and PE Prevention of recurrent DVT and PE Prevention of thromboembolism after total knee replacement and total hip replacement Prevention of thromboembolism in hospitalized acutely ill medical patients Prevention of major cardiovascular events in patients with chronic CAD/peripheral artery disease	Renal, hepatic
Eliquis	Apixaban	Xa	Stroke prevention in NVAf (Nonvalvular atrial fibrillation) Prevention of thromboembolism after total knee replacement and total hip replacement Treatment of DVT and PE Prevention of recurrent deep vein thrombosis and pulmonary embolism	Renal (27%) GI tract
Savaysa	Edoxaban	Xa	Stroke prevention in NVAf (nonvalvular atrial fibrillation) Treatment of DVT and PE	Renal (50%) Metabolism and biliary secretion (50%)
Bevyxxa	Betrixaban	Xa	Prevention of deep vein thrombosis and pulmonary embolism in adults hospitalized for an acute medical illness	GI (93–95%) (Renal 5–7%)

Dabigatran, rivaroxaban, apixaban, and edoxaban are FDA-approved for lowering the risk of stroke and embolism in nonvalvular atrial fibrillation (NVAf) as well as deep vein thrombosis and pulmonary embolism treatment/prophylaxis. Unique indications include betrixaban for prophylaxis of venous thromboembolism (VTE) in hospitalized patients for an acute medical illness and rivaroxaban in combination with aspirin to reduce major cardiovascular events in patients with chronic coronary artery disease (CAD) or peripheral artery disease (Chen et al. 2020). It should be noted that there are also multiple off-label indications for which these drugs are being used.

Reversal Agents for Anticoagulants

Fresh frozen plasma (FFP) is the most commonly used method in North America for rapidly reversing warfarin. The usual dose of FFP for anticoagulant reversal is 15 mL/kg of body weight. When urgent reversal of a therapeutic (rather than supratherapeutic) INR is required, a lower dose of 5–8 mL/kg may be appropriate. FFP rarely completely corrects the INR without the use of Vitamin K, irrespective of the FFP dose or the INR value. Risks of FFP include failure to completely reverse coagulopathy, immune-mediated thrombocytopenia, and anaphylactoid reactions, to name a few (Joseph et al. 2014; Frontera et al. 2014).

These limitations have led to recommendations that prothrombin complex concentrates (PCCs) be used in place of FFP for the treatment of anticoagulant-associated coagulation factor deficiencies. PCCs reverse coagulopathy laboratory markers from warfarin over coagulation more rapidly and effectively than FFP (Joseph et al. 2014; Frontera et al. 2014).

There are two different PCCs available, the 3-factor and the 4-factor concentrates. Both con-

tain factors II, IX, and X, yet the 4-factor concentrate also contains factor VII. The 4-factor concentrate, in theory, should be more effective at correcting INR due to factor VII inclusion. The recommended dose of prothrombin complex concentrate (PCC) for reversal of anticoagulation ranges from 25 to 100 U/kg depending on the product used (Joseph et al. 2014; Frontera et al. 2014).

Irrespective of the blood product administered to reverse warfarin anticoagulation, the patient should also receive intravenous vitamin K at a dose of 2.5–5 mg administered over 30 min; otherwise, the INR is unlikely to completely correct, and a “rebound coagulopathy” may develop after the transfused factors are cleared. Intravenous administration of vitamin K has a small risk of severe anaphylactoid reaction and should only be given by slow intravenous infusion (over 20–30 min), and oral vitamin K should be used whenever possible. Because more rapid reversal is achieved with intravenous administration, it is preferred in cases when urgent reversal is required, such as life-threatening bleeding or reversal of the warfarin effect in a patient diagnosed with acute HIT (Crowther and Warkentin 2008).

Preparation, delivery, and administration of FFP, PCC, and IV Vitamin K do not make them a practical method of reversal of a bleeding patient using anticoagulants in the typical office setting.

Specific Reversal Agents for DOACs

Ligand-specific and small molecule reversal agents have been more recently utilized, and long-term outcomes are still currently under investigation. These agents are likely to be primarily used in life-threatening bleeding and emergent surgery (Abed et al. 2017). In addition, these agents may allow the safer implementation of uninterrupted or minimally interrupted DOAC protocols for elective surgery.

There are currently two DOAC reversal agents approved for use in the United States. Andexanet alfa is approved for the reversal of apixaban and rivaroxaban, and idarucizumab is approved for the reversal of dabigatran (Cuker et al. 2019).

Idarucizumab (Praxbind®) is a monoclonal antibody that acts as a noncompetitive irreversible inhibitor of unbound and thrombin-bound dabigatran and its active metabolites. The compound has a high affinity, and it is a specific inhibitor of dabigatran action. The agent has a rapid onset mechanism of action and has been demonstrated to be safe and efficacious with a simple dosing regimen. Laboratory evidence of reversal is observed within minutes. Idarucizumab was approved by the FDA in 2015 and is widely incorporated into protocols for use in acute bleeding or emergent surgery (Dalal et al. 2016; Andresen et al. 2018).

Andexanet-alfa (Andexxa®) is a recombinant modified human factor Xa decoy protein. It binds with high affinity to Factor Xa inhibitors within 2 min of IV administration, thereby neutralizing the direct and indirect effects of Factor Xa. A bolus dose is followed by an infusion with the restoration of thrombotic activity being reflected by the change in thrombin generation and quantitative anti-FXa activity. Andexanet-alfa reverses the anticoagulant effects of small molecule anti-FXa agents (rivaroxaban and apixaban) as well as low molecular weight heparin and fondaparinux (the latter two being indirect FXa inhibitors). This was approved by the FDA in May of 2018 (Dalal et al. 2016; Andresen et al. 2018).

The utilization of DOAC reversal agents should be limited to indicated interventions, which usually include life-threatening bleeding and bleeding not amendable or responsive to other interventions (Cuker et al. 2019).

A North American panel of anticoagulation experts published a clinical guidance review in 2019 providing a comprehensive framework for DOAC reversal (Cuker et al. 2019).

The panel did recommend that the use of andexanet alfa be also extended to the off-label reversal of edoxaban and betrixaban (Shi and Crowther 2019). The panel's DOAC reversal guidelines are well summarized in Fig. 4.

In summary, the dental clinician must review the anticoagulated patient's overall health and the risk for developing a thrombosis, determine the potential for major postoperative bleeding, utilize the guidance of the patient's managing physician to direct anticoagulation management, and plan for intraoperative and postoperative contingencies in the consultation appointment. In instances where additional resources are required to safely carry out the surgical plan, the clinician should consider referral to a hospital-based provider or an oral and maxillofacial surgeon.

The management of nonmajor bleeding with DOACs typically involves holding the anticoagulant and implementing local control measures. DOACs half-lives are approximately 12 hours; therefore, holding a dose should result in a fast decline of the anticoagulant effect (Wigle et al. 2019).

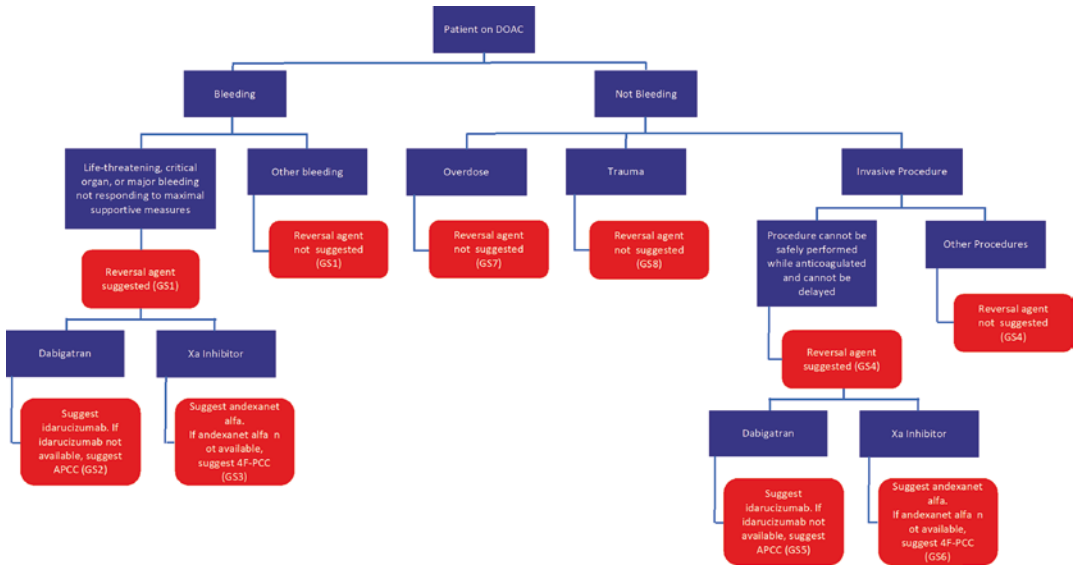


Fig. 4 Anticoagulation forum DOAC reversal guidelines summary (Cuker et al. 2019)

Management of Patients on Anticoagulation Therapy

Given the prevalence of patients on anticoagulation therapy presenting for oral surgical procedures, the clinician will be required to make clinical decisions to optimize the patient's outcomes (i.e., minimizing the risk of thrombosis and postoperative bleeding). These decisions include assessing the need to continue anticoagulation, assessing the likelihood of major bleeding with the planned surgical procedure, intraoperative hemostasis strategies, and appropriate postoperative surveillance.

Warfarin

The risk of thrombosis and determining the continuation of anticoagulation during surgical intervention always play a role in treatment decisions. The 9th edition of the ACCP (American College of Chest Physicians) Antithrombotic Therapy and Prevention of Thrombosis Guidelines recommends initially stratifying patients based on their risk of thrombosis (low, intermediate, or high) (Guyatt et al. 2012a, b). Subsequently, the clinician should assess the complexity of the

planned procedure. Inpatients who have a high risk of thrombosis, regardless of the complexity of the case, should undergo an anticoagulation bridging protocol to minimize the duration of not being anticoagulated. Data has shown heparin bridging therapy will reduce the postoperative VTE risk by approximately two-thirds. On the opposite end of the spectrum, in patients with a low risk of thrombosis, regardless of the complexity of the case, bridging is not typically indicated, and a strategy that considers stopping anticoagulation prior to surgery can be utilized. It is in the intermediate-risk patients where clinical judgment and experience of the clinician dictate what approach to bridging is warranted. It would be prudent not to bridge in cases where significant bleeding is predicted and therefore consider cessation of anticoagulation prior to surgery in concert with the supervising managing physician. In cases with a low risk of bleeding, such as most dental surgical procedures, it would be in the patient's overall best interest to continue anticoagulation by either continuing the patient's typical regimen or utilizing bridging protocols. The flowchart below (Fig. 5) is a useful guide to help the clinician with the decision process (Spyropoulos and Douketis 2012).

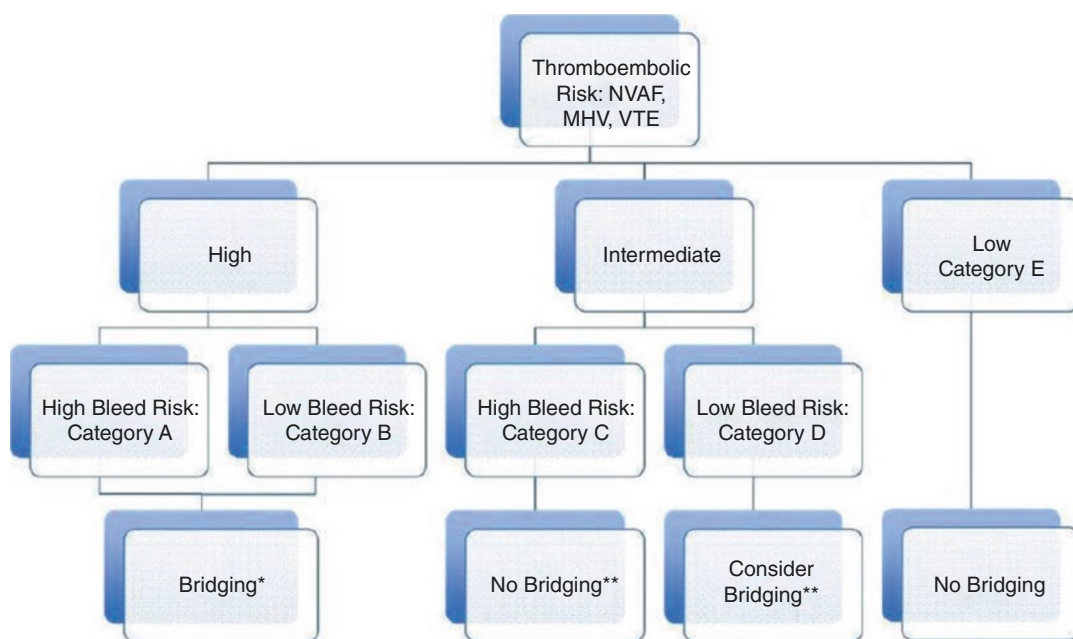


Fig. 5 Suggested periprocedural heparin bridging strategies for patients on chronic VKA therapy based on patient thromboembolic and procedural bleed risk (Spyropoulos and Douketis 2012) (reprinted with permission)

In scenarios where bridging is warranted, the managing physician will guide the dental clinician with a strategy that includes the timing of cessation of the patient's regimen and initiation of the bridging treatment (Spyropoulos and Douketis 2012). Many bridging protocols have been proposed; one such example that a physician may utilize is illustrated in Table 5.

Spyropoulos cited a recent systematic review of 7169 patients undergoing periprocedural bridging therapy and found a thromboembolism rate of 0.9% (95% CI, 0.0–3.4%) and a major bleeding rate of 4.3% (95% CI, 2.4–6.2%) primarily when LMWH bridging was used (Spyropoulos and Douketis 2012). The chapter's authors' professional approach to mitigating bleeding risks in these patients is based on the determined risk of postoperative bleeding, accounting for the extent of the surgery, and the assessment of systemic bleeding risks. The patient treated in an outpatient setting who is at lower risk of a bleeding event may require a

same-day follow up call and post-op day one follow-up appointment or call. Yet for a patient with a concern for higher risk of a bleeding event, the prudent clinician may choose to observe the patient for a period of time in the office after the procedure, followed by post-op calls, or an inpatient observation status when deemed needed.

In patients who are at intermediate to high risk of thrombosis, an alternative strategy to bridging may be to not interrupt their anticoagulation therapy. Wahl showed a 0.95% incidence of severe thrombosis when warfarin was discontinued for tooth extractions, most of which led to patient death (Wahl 1998). Akopov conducted a retrospective analysis of 197 patients presenting over a 12-month period with nonfatal cardioembolic cerebral infarction, 14 (7.1%) of these patients suffered infarcts related to the discontinuation of warfarin therapy for procedures (Akopov et al. 2005).

Maintaining warfarin anticoagulation in the perioperative setting requires the clinician to keep a few principles in mind:

Table 5 Anticoagulation bridging protocols (Spyropoulos and Douketis 2012) (reprinted with permission)

Day	Intervention
Preprocedural Intervention	
–7 to –10	Assess for perioperative bridging anticoagulation; classify patients as undergoing high-bleeding risk or low-bleeding risk procedure; check baseline labs (Hgb, platelet count, creatinine, INR)
–7	Stop aspirin (or other antiplatelet drugs)
–5 or –6	Stop warfarin
–3	Start LMWH at therapeutic or intermediate dose*
–1	Last preprocedural dose of LMWH administered no less than 24 h before start of surgery at half the total daily dose; assess INR before the procedure; proceed with surgery if INR < 1.5; if INR > 1.5 and < 1.8, consider low-dose oral vitamin K reversal (1–2.5 mg)
Day of procedural Intervention	
0 or +1	Resume maintenance dose of warfarin on evening of or morning after procedure†
Postprocedural Intervention	
+1	Low-bleeding risk: restart LMWH at previous dose; resume warfarin therapy High-bleeding risk: no LMWH administration; resume warfarin therapy
+2 or +3	Low-bleeding risk: LMWH administration continued High-bleeding risk: restart LMWH at previous dose
+4	Low-bleeding risk: INR testing (discontinue LMWH if INR > 1.9) High-bleeding risk: INR testing (discontinue LMWH if INR > 1.9)
+7 to +10	Low-bleeding risk: INR testing High bleeding risk: INR testing

*LMWH regimens include enoxaparin 1.5 mg/kg once daily or 1.0 mg/kg twice daily subcutaneously; dalteparin 200 IU/kg once daily or 100 IU/kg twice daily subcutaneously; and tinzaparin 175 IU/kg once daily subcutaneously. Intermediate-dose LMWH (ie, nadroparin 2850–5700 U twice daily subcutaneously; enoxaparin 40 mg twice daily subcutaneously) has been less studied in this setting

†Loading doses (ie, 2 times the daily maintenance dose) of warfarin have also been used.

- Patients who are anticoagulated are at an inherently increased risk of a bleeding event.
- Patients who are not in a therapeutic range are at increased risk of thrombosis.
- Patients who are at supratherapeutic levels have an even greater risk of significant post-surgical bleeding.

The question then lies in patients who are kept on warfarin therapy. Are there INR ranges that are safe from a dental surgical standpoint to treat with minimal risk of significant postoperative bleeding while maintaining the thrombosis prevention benefits of anticoagulation? Morimoto et al., in a 2011 prospective study, performed 433 extractions on 382 patients with a postoperative hemorrhage rate of 3.9%. Greater than 90% of the cases with postoperative bleeding occurred within six days of extraction, and 60% had an INR of 3.0 or greater. Bleeding events in patients with an INR of 2.0–3.0 were twofold lower compared with patients with an INR greater than 3.0. The study also noted that surgical factors that increased the risk of bleeding included acute inflammation and the need for surgical extraction (Morimoto et al. 2011). Weltman et al. performed a systematic review in 2015 of the management of patients undergoing dental extractions who were also receiving warfarin therapy. The authors' findings determined that patients whose INR was 3.0 or less can safely continue their warfarin regimen prior to dental extraction procedures without significant risk of bleeding. The study did also show that gelatin sponge packing did improve post-op bleeding in this patient population (Weltman et al. 2015).

The clinician must recognize that most medical diagnoses treated with warfarin require a therapeutic INR range between 2.0 and 3.0 (e.g., atrial fibrillation, DVT w/ or w/o PE), with mechanical heart valve patients requiring higher

INR ranges of 2.5 to 3.5. As the data reaffirms, patients with INR ranges of 3.0 or less can be safely treated with minimal risk of a significant postoperative bleed. Accounting for the fact that most indications for warfarin therapy call for a therapeutic INR range of 3 or lower, patients can be safely treated while minimizing the risk of thrombosis. Patients that present with supratherapeutic INR ranges (3 or greater) should be deferred by the dental clinician for optimization by their physician. It is critical to remember that PT/INR lab testing should be performed on the same day of treatment to ensure an accurate pre-operative INR.

If the patient is deemed at low risk of a thromboembolic event (as described earlier in this section), withdrawal of warfarin without bridging is typically accomplished two to seven days preoperatively. The variability of the withholding period depends on other systemic factors, including advanced age, systolic heart failure, and liver function.

In order to help the clinician to determine the safety of dental treatment for patients receiving warfarin, we have slightly modified a table originally published by Herman et al. in 1997, as noted in Table 6 (Hermann et al. 1997). We have adapted the original recommendations to align better with the current data that has been presented. These are recommendations to be considered by each clinician, who should use appropriate judgment when making final treatment decisions.

Regardless of the choice between stopping warfarin with or without bridging, anticoagulation can be restarted on the evening of the procedure if there is minimal concern for bleeding at the conclusion of the procedure. With bridging, the parenteral anticoagulant can be discontinued when the INR is within the therapeutic range.

Table 6 The warfarin patient-outpatient dental procedure safety recommendations (Hermann et al. 1997)

Dental Procedure	Suboptimal INR Range		Normal INR Range			Out of Range
	<1.5	1.5 to <2.0	2.0 to <2.5	2.5 to 3.0	>3.0 to 3.5	
				Mechanical Prosthetic Heart Valve		
			Atrial Fibrillation; Venous Thrombosis; Pulmonary Embolism			
Examination, Radiographs, Impressions, Orthodontics						
Simple Restorative Dentistry, Supragingival Prophylaxis						
Complex Restorative Dentistry, Scaling and Root Planning, Endodontics					Probably Safe with Local Measures *	
Simple Extraction, Biopsy			Local Measures *	Probably Safe with Local Measures *		
Multiple Extractions, Single Surgical Extraction (including Bony Impaction)			Local Measures *	Probably Safe with Local Measures *		
Single Dental Implant Placement, Gingivectomy, Minor Periodontal Flap Procedure		Local Measures *	Probably Safe with Local Measures *			
Full Arch or Full Mouth Extractions	Local Measures *	Probably Safe with Local Measures *				
Extraction of Multiple Bony Impactions, Placement of Multiple Dental Implants, Extensive Flap Surgery	Probably Safe with Local Measures *					

* Increased need for the use of local measures such as: sutures, Gelfoam[®], SURGICEL[®], topical thrombin, tranexamic acid

Management of Patients on Direct Oral Anticoagulants (DOACs)

Treating patients on direct oral anticoagulants (DOACs) presents additional challenges for the dental clinician. With increasing indications of use and market penetration, the number of patients taking these drugs is large and will continue to grow. Soon after the FDA approval of the first DOACs, Pradaxa in 2010 and Xarelto in 2011 and Firriolo and Hupp in 2012 published a clinical manuscript addressing the use of DOACs and implications for the management of the dental patient (Firriolo and Hupp 2012). At the time, the authors concluded that the clinical data present to make sound judgments on the management of these therapies and dental extractions was lacking and that clinical studies were needed. The authors reinforced discussing options, risks, and management with the patient's physician. They concluded as a precaution that in cases where it is deemed necessary to stop DOACs for surgery (i.e., risk of major bleeding), the medication should be stopped at least 24 h prior to the procedure. They also concluded that it should be restarted only after the risk of bleeding has become minimal within the first 24–48 h. Elad et al., in 2014, attempted to better clarify directive care for this population by performing a review of 18 randomized clinical trials (Elad et al. 2016). Elad et al. took into account factors such as the risk of procedural bleeding; issues of

reversing and monitoring these agents; risk of thromboembolism, as proposed by the ACCP (Guyatt et al. 2012a); and patient comorbidities. From this review, they concluded that there are four possible options when treating this patient population:

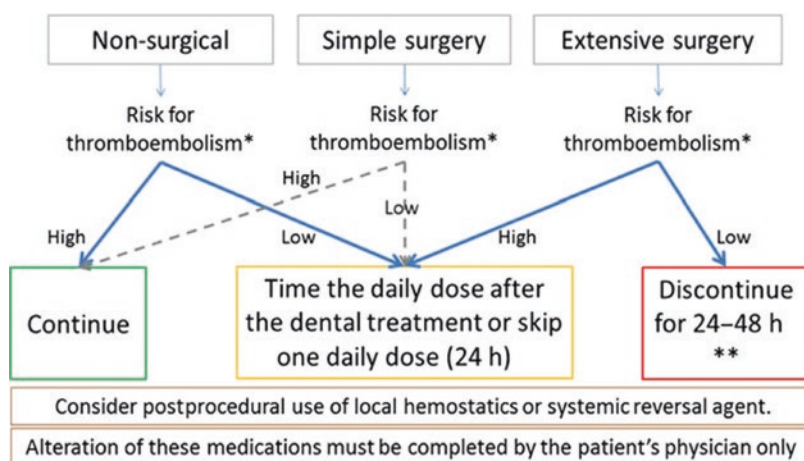
1. Continue DOAC
2. Perform the procedure as late in the day as possible
3. Discontinue use 24 h prior to treatment
4. Discontinue use 48 h prior to treatment

Their final recommendation was that patients on DOACs would most likely benefit from options 1 or 2 (Elad et al. 2016). Elad et al. developed a suggested treatment algorithm as noted in Fig. 6.

Elad et al. also cautioned the dental clinician in caring for these patients, stating that until there is greater evidence about the bleeding risk in patients on DOACs and, referral to a hospital-based dental clinic or an oral and maxillofacial surgeon should be considered.

A dental narrative published in the Australian Dental Journal in 2015 by Thean and Alberghini reaffirms those recommendations provided by Elad et al. (Thean and Alberghini 2016; Elad et al. 2016). Thean and Alberghini reinforced that the dental clinician must direct care under the guidance of the patient's physician and not unilaterally withhold anticoagulation therapy for dental treat-

Fig. 6 Suggested algorithm for management approach prior to dental intervention (Elad et al. 2016) (reprinted with permission)



ment. They recommended DOACs can be continued and bleeding controlled with local measures for procedures with low risk of bleeding, such as simple single extractions. These local measures included the use of Gelfoam® or SURGICEL® packing, mechanical pressure, suturing, and/or tranexamic acid mouthwash. For high-risk procedures, such as multiple extractions or surgical extractions, they also recommend the dental clinician refer to an oral and maxillofacial surgeon for care. For patients undergoing more extensive surgery, cessation of DOACs may be indicated (Thean and Alberghini 2016).

In 2012, Spyropoulos discussed the possible need for 2–3 days of a low-dose LMWH bridging regimen (e.g., enoxaparin 40 mg once daily) in postoperative patients who are unable to take oral medications (Spyropoulos and Douketis 2012). For patients undergoing cessation of DOAC therapy, Spyropoulos recommends cessation time periods based on renal clearance and surgical risk of bleeding. He defines simple dental extractions as having a low risk of bleeding (a 2-day risk of major bleed of 0–2%) and multiple tooth extractions as a high risk for bleeding (2-day risk of a major bleed of 2–4%). Based on the patient's renal function and the risk of procedural bleeding, he then provides preoperative drug cessation

recommendations as summarized in Table 7 (Spyropoulos and Douketis 2012).

In 2018, a review article by Lusk et al. evaluated the available evidence regarding how to safely manage DOAC therapy in patients requiring dental procedures with a low-to-moderate risk of bleeding. They concluded that among patients undergoing low-to-moderate risk dental procedures while receiving DOAC therapy, bleeding rates were low regardless of whether the DOAC was held or continued surrounding the procedure. Documented bleeding was mild and easily controlled by local hemostatic measures. Patients can safely continue DOAC therapy surrounding these low-to-moderate risk dental procedures (Lusk et al. 2018).

In their definition, low-risk procedures include local anesthetic administration, simple restorations, supragingival scaling, and single-tooth extraction. Moderate risk procedures include extractions of two to four teeth and local gingival surgery of five or fewer teeth. High-risk procedures include extractions of six or more teeth, osseous biopsy, placement of multiple implants, and generalized gingival surgery of six or more teeth.

In 2019, a systematic review and meta-analysis by Manfredi et al. evaluated the current

Table 7 Preoperative oral anticoagulant cessation recommendations (Spyropoulos and Douketis 2012) (reprinted with permission)

Drug (dose)*	Patient renal function	Low bleeding risk surgery† (2 or 3 drug half-lives between last dose and surgery)	High bleeding risk surgery‡ (4 or 5 drug half-lives between last dose and surgery)
Dabigatran (150 mg twice daily)			
$t_{1/2}$ = 14–17 h	Normal or mild impairment (CrCl > 50 mL/min)	Last dose: 2 days before surgery (skip 2 doses)	Last dose: 3 days before surgery (skip 4 doses)
$t_{1/2}$ = 16–18 h	Moderate impairment (CrCl 30–50 mL/min)	Last dose: 3 days before surgery (skip 4 doses)	Last dose: 4–5 days before surgery (skip 6–8 doses)
Rivaroxaban (20 mg once daily)			
$t_{1/2}$ = 8–9 h	Normal or mild impairment (CrCl > 50 mL/min)	Last dose: 2 days before surgery (skip 1 dose)	Last dose: 3 days before surgery (skip 2 doses)
$t_{1/2}$ = 9 h	Moderate impairment (CrCl 30–50 mL/min)	Last dose: 2 days before surgery (skip 1 dose)	Last dose: 3 days before surgery (skip 2 doses)
$t_{1/2}$ = 9–10 h	Severe impairment§ (CrCl 15–29.9 mL/min)	Last dose: 3 days before surgery (skip 2 doses)	Last dose: 4 days before surgery (skip 3 doses)
Apixaban (5 mg twice daily)			
$t_{1/2}$ = 7–8 h	Normal or mild impairment (CrCl > 50 mL/min)	Last dose: 2 days before surgery (skip 2 doses)	Last dose: 3 days before surgery (skip 4 doses)
$t_{1/2}$ = 17–18 h	Moderate impairment (CrCl 30–50 mL/min)	Last dose: 3 days before surgery (skip 4 doses)	Last dose: 4 days before surgery (skip 6 doses)

*Estimated $t_{1/2}$ based on renal clearance.

†Aiming for mild to moderate residual anticoagulant effect at surgery (< 12%–25%).

‡Aiming for no or minimal residual anticoagulant effect (< 3%–6%) at surgery.

§Patients receiving rivaroxaban, 15 mg once daily.

literature regarding the importance of discontinuing or not discontinuing (DOACs) before invasive oral procedures and to establish the frequency and type of postoperative bleeding events in patients. They concluded that the studies included in the review and analysis did not discern any important differences in postoperative bleeding events in patients who continued versus patients who discontinued DOACs, and no thromboembolic events were recorded (Manfredi et al. 2019). However, they cautioned that the results should be interpreted with caution because of the low quality of the studies evaluated.

Local Hemostatic Agents

What we have discussed so far are all preparations in anticipation for the planned surgical procedure. In any of the abovementioned scenarios (bridging, continuation of oral anticoagulant, or cessation of anticoagulation), the patient may continue to have more than expected intraoperative bleeding. Fortunately, there are agents available, as discussed below, that can help control the intraoperative oozing. It should be noted that none of these measures substitute the first response to any persistent bleeding, which is to apply firm pressure either manually or with the patient biting on folded gauze for at least 10 minutes.

Gelfoam®

Gelfoam® is a water-insoluble, porous, pliable product prepared from purified porcine skin capable of absorbing up to 45 times its weight of whole blood. It may be cut without fraying and is able to absorb and hold within its interstices many times its weight of blood and other fluids. It is theorized that Gelfoam® promotes coagulation both by surface area hemostasis and by providing a mechanical framework for coagulation. When used, it must be applied with moderate pressure to the site of bleeding. It should not be used in locations with a risk of infection (i.e., surgical extraction of an infected tooth), and it typically resorbs within 6 weeks (Achneck et al. 2010).

SURGICEL®

Another adjunct that can be used for capillary, venous, and small arterial bleeds is SURGICEL®, which is an oxidized regenerated cellulose agent. Due to the low pH level, it has acidic properties and achieves hemostasis via the denaturation of blood proteins, mechanical activation of the clotting cascade, and local vasoconstriction. Another distinguishing factor is the antibacterial properties of SURGICEL®. It should be used with caution in areas where there is potential of nerve exposure (i.e., inferior alveolar nerve, mental nerve, etc.) due to its acidic nature. Invariably, SURGICEL® should be retrieved after hemostasis is achieved unless its removal poses a significant risk of re-bleeding. This is to prevent complications such as a giant cell-type of reaction at the surgical site (Achneck et al. 2010).

Morimoto et al. reported 87 cases receiving antiplatelet drugs. Teeth were extracted without reducing antiplatelet therapy, oxidized cellulose was applied, and suturing was performed for local hemostasis. They concluded that sufficient hemostasis can be obtained in most cases of tooth extraction, and appropriate local hemostatic methods can be successful when postoperative hemorrhage occurs (Morimoto et al. 2008).

Fibrin Glue

Fibrin glue is effective as a hemostatic agent as it mimics the final pathway of the coagulation cascade, where fibrinogen is converted into fibrin under the action of thrombin, factor XIII, fibronectin, and ionized calcium (Brewer and Correa 2006). It combines two components: thrombin (principally human) and fibrinogen (normally plasma-derived) (Achneck et al. 2010). Fibrin glue is effective in managing bleeding in patients with high risk of prolonged and excessive bleeding after surgery. Traditionally, it has been used to manage bleeding in cardiovascular, hepatic, and splenic surgeries. It also has proven itself as a useful tool in the armamentarium for treating the dental surgery patient who is at high risk for bleeding during or after surgery (Mankad and

Codispoti 2001). Fibrin glue contains human or animal components, leading to hesitation of use in certain patients (Achneck et al. 2010).

Tranexamic Acid (TXA), ε-aminocaproic Acid (Amicar)

Tranexamic acid (TXA), and ε-aminocaproic acid (Amicar) are antifibrinolytic agents that block the proteolytic site of plasmin and inhibit plasminogen activator incorporation into the nascent fibrin clot. A 4.8% TXA solution has been proven to be very effective in reducing bleeding complications (Dinkova et al. 2013). This solution may be difficult to obtain from a pharmacy unless prior arrangements have been made. A recommended regimen is to hold 10 ml in the mouth for 2 min, 30 min pre-procedure, and then repeat every 2 h for six to ten doses as needed (Carter and Goss 2003).

Carter et al. demonstrated that a 2-day course of 4.8% TXA is as efficacious as a 5-day course in controlling hemostasis post-dental extractions in patients anticoagulated with warfarin (Carter and Goss 2003).

Topical Thrombin

Thrombin forms the foundation of a fibrin clot by promoting the conversion of fibrinogen to fibrin. Thrombin is commonly derived from bovine sources, yet bovine thrombin has been known to cause a significant immune response in some patients. As such, new formulations derived from human plasma or recombinant human sources are available and are being formulated (Achneck et al. 2010). The application of thrombin in controlling hemostasis during dental surgery can be performed by applying thrombin directly to the surgical site, yet it is best utilized by combining thrombin synergistically with a second hemostatic agent, such as Gelfoam®, as a carrier. This combination can then be packed into the surgical site for hemostatic control.

Conclusion

The data presented in this chapter demonstrates that in most clinical cases, the decision can be made to maintain antiplatelet or anticoagulation therapy for routine dental treatment, including “simple” extractions. This is especially true for patients who are at high medical risk of a thromboembolic event and/or patients requiring minor dental surgical procedures.

Intraoperative hemostatic measures would benefit any patient who is on one or more blood thinners, irrespective of the treatment algorithm chosen. However, it is most critical in the patient who maintains his/her blood thinner therapy during his/her operative course. Malmquist provided a list of key hemostatic measures that should be considered, that include the following (Malmquist 2011):

1. Avoid flap procedures
2. Limit surgical trauma, for example, limiting the number of teeth extracted or the sectioning of teeth
3. Curette-associated socket granulation tissue
4. Obtain primary closure when a flap has been elevated
5. Use non-resorbable sutures to control the tension on the flap and eliminate the possibility of premature breakdown of the suture material
6. Use topical hemostatic materials in the surgical site to reduce bleeding
7. Use electrocautery or laser to minimize bleeding
8. Apply an appropriately placed pressure dressing for a sufficient period of time
9. Use topical rinses, such as tranexamic acid, to inhibit fibrinolysis
10. Provide treatment early in the day, allowing for observation of bleeding throughout the day

Decisions should always be co-managed under the guidance of the treating physician and include the consideration of the following:

- The physiology and sequence of hemostasis
- The clinical indications for the use of these medications and the risks associated with the continued use or cessation of these agents when treating this patient population
- When can patients be maintained on these therapies
- If stopping antiplatelet and/or anticoagulation therapy is decided
- What are the increased systemic risks as they are weighed to the benefits of decreasing bleeding risks?

The management of the dental patient on antiplatelet and/or anticoagulation therapy can be complex. Yet with sound medical knowledge and good clinical decision-making, the practitioner should be able to navigate these challenging cases successfully.

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Pharmacology of Psychiatric and Neurologic Drugs

Miriam R. Robbins

Introduction

Psychiatric and neurologic diseases are common in the general population and therefore are commonly encountered in dental patients. Approximately 15% of the worldwide population are affected—a number that is expected to increase as life expectancy increase (Feigin et al. 2020). Such diseases may vary in severity and consequences, but treatment of these chronic conditions generally involves long-term use of multiple pharmacologic agents. Because of the similarity in the underlying pathophysiology of the diseases presented, there is overlap between the medications used. For example, anticonvulsant medications are also used to treat mood disorders. The majority of patients with Parkinson's disease (PD) and Alzheimer's disease (AD) will exhibit neuropsychiatric symptoms including depression, anxiety, agitation, apathy, and psychosis at some time during the course of disease. The focus of this chapter is to summarize potential drug interactions of clinical importance to the oral health care provider, based on a review of the most current evidence-based literature available with an emphasis on systematic reviews (Table 1). Although the information presented on various medications used to treat these disorders is presented here, it is intended to highlight the undesir-

able interactions of these drugs with drugs commonly prescribed by dentists. Comprehensive information on the pharmacodynamics and pharmacokinetic of these psychotropic and neurologic drugs can be accessed elsewhere.

Psychiatric Disorders

Psychiatric diseases are common in the United States. One in five US adults lives with a mental illness (57.8 million or 22.8% in 2021). Mental health surveys carried out in the United States suggest that during any 1-year period, approximately 26% of the population will have a mental disorder, and almost 50% of all people will have mental illness sometime during their lifetime (Nimh.nih.gov 2023a). A total of 1.21% of adults in the United States live with schizophrenia (Nimh.nih.gov 2023b). And 2.6% of adults in the United States live with bipolar disorder (Nimh.nih.gov 2023c). Also, 8.4% of adults in the United States—21 million—had at least one major depressive episode in the past year. (Nimh.nih.gov 2023d) Furthermore, 19.1% of adults in the United States experienced an anxiety disorder such as generalized anxiety, posttraumatic stress disorder, obsessive-compulsive disorder, and phobias (Nimh.nih.gov 2023e). Psychiatric disorders have high rates of co-occurrence with patients exhibiting more than one type of symptom (e.g., depression and anxiety frequently occur together). Additionally, patients may have medical comor-

M. R. Robbins (✉)
School of Dental Medicine, University of
Pennsylvania, Philadelphia, PA, USA
e-mail: mrrobb@upenn.edu

Table 1 Dental drug interactions

Dental drugs	Interacting drug(s)	Details of interaction	Recommendations	Key references
Aspirin and NSAIDs	Lithium	NSAIDs inhibit renal clearance of lithium, potentially resulting in lithium toxicity within 5–10 days	Best to use diflunisal Limit use of NSAIDs to 2–3 days Decrease dose of lithium	Bauer and Mitchner (2004), Hashimoto et al. (2002), Alda (2015)
	SSRIs (fluoxetine, sertraline, paroxetine) SRI (duloxetine) Donepezil Galantamine Rivastigmine	Increased risk of bleeding (especially GI) due to interference with platelet function	Limit use to 2–3 days. Avoid in patients with history of GI bleeding or bleeding diathesis	Serebruany (2006), Anglin et al. (2014), Mansour et al. (2006), Loke et al. (2008)
	Divalproex sodium Valproic acid	Synergistic effect on bleeding with aspirin	Avoid	Vasudev et al. (2010), Abdallah (2014), Manu et al. (2016)
Acetaminophen	Carbamazepine Phenytoin	Increased potential hepatotoxicity of acetaminophen and decrease its pharmacologic effects	Avoid concurrent use Max daily dose of acetaminophen <2 g	Patsalos and Perucca (2003), Perucca (2006)
	Lamotrigine	Reduction in therapeutic levels of lamotrigine	Limit use to 2–3 days	Patsalos and Perucca (2003)
Tramadol	Bupropion MAOIs SSRIs TCAs	Increased seizures and/or serotonin syndrome	Avoid	Sansone and Sansone (2009)
	Pramipexole	Increased CNS and/or respiratory depression	Monitor	
	MAOIs	Hypertension and autonomic reactions	Avoid	
Codeine Hydrocodone Oxycodone	SSRIs (fluoxetine, sertraline, paroxetine) Bupropion Haloperidol	Decreased efficacy of pain medication due to delayed metabolism (CYP2P6 inhibitors) of prodrug to active morphine derivative	Alternative pain medications	Becker (2008), Friedlander and Norman (2002))
	Antipsychotics Benzodiazepines TCAs	Increased central CNS depression	Reduce dose of narcotics, alternate pain medications	
Doxycycline Metronidazole Tetracycline	Lithium	Increased lithium levels; risk of toxicity	Avoid	Hashimoto et al. (2002), Alda (2015)
Metronidazole	Antipsychotics Carbamazepine Phenobarbital Phenytoin Primidone	Decreased seizure threshold	Use with caution, reduce dose	Kennedy et al. (2013), Ruiz Diaz et al. (2020), Patsalos and Perucca (2003), Perucca (2006)

Table 1 (continued)

Dental drugs	Interacting drug(s)	Details of interaction	Recommendations	Key references
Macrolides Erythromycin Clarithromycin	Benzodiazepines (except lorazepam) Carbamazepine Phenobarbital Phenytoin Valproate	Inhibition of AED/ psychotropic drug metabolism leading to increased CNS depression (CYP450 inhibition)	Avoid or monitor AED/psychotropic drug levels	Gubbins and Heldenbrand (2010), Patsalos and Perucca (2003), Perucca (2006)
	Valproate	Increased valproate concentration (CYP3A4 inhibition)	Avoid	Vasudev et al. (2010), Abdallah (2014)
	Fluoxetine Haloperidol Pramipexole Quetiapine Sertindole Venlafaxine Zotepine	QT prolongation	Avoid	Kennedy et al. (2013), Ruiz Diaz et al. (2020)
	Donepezil Galantamine	Decreased hepatic metabolism of cholinesterase inhibitors increasing cholinergic activity	Monitor, consult with physician if extended course is anticipated	Defilippi and Crismon (2003), Bentué-Ferrer et al. (2003)
	Carbamazepine Phenobarbital Phenytoin	Decreased doxycycline half-life by 50%	Use alternate antibiotic	Perucca (2006)
Fluconazole Ketoconazole Itraconazole	Carbamazepine Clonazepam Phenobarbital Phenytoin Quetiapine Benzodiazepines Donepezil Galantamine	Inhibition by antifungal of metabolism of drug (CYP450) inhibition) potentiating psychotropic effects and possibly toxic serum levels Decreased hepatic metabolism of cholinesterase inhibitors increasing cholinergic activity	Avoid or monitor AE/psychotropic drug levels Monitor, consult with physician if extended course is anticipated	Kennedy et al. (2013), Gubbins and Heldenbrand (2010), Defilippi and Crismon (2003), Perucca (2006)
Local anesthetics with epinephrine	Older generation antipsychotics (haloperidol and chlorpromazine). Not seen with newer atypical antipsychotics COMT inhibitors Entacapone Tolcapone TCAs	Blockage of α_1 - receptors can cause orthostatic hypotension that rarely can be worsened by the β_2 -receptor stimulation by epinephrine Delayed metabolism of both epinephrine and levonordefrin possibly potentiating cardiovascular effects Possible potentiation of the sympathetic effects of epinephrine or levonordefrin (more prevalent with levonordefrin)	Monitor vital signs during and after administration. Limit to two carpules of 1:10,000 epinephrine or less. Aspirate to avoid intravascular injection. Avoid use of epinephrine impregnated retraction cord. Avoid levonordefrin use with TCAs	Yagiela (1999), Brown and Rhodus (2005), Moodley (2017), Malamed (2014), Saraghi et al. (2017), Friedlander and Marder (2002), Balakrishnan and Ebenezer (2013)

bilities, either as direct effect of the underlying pathophysiology or as a side effect of chronic and/or debilitating disease. More than 50% of patients with substance abuse disorders have a co-occurring mental illness (Substance Abuse and Mental Health Services Administration 2015). Psychiatric disorders are frequently associated with rates of compliance with medical and dental treatment.

A mainstay of treatment for psychiatric disorders is psychotropic medications. Patient responses are variable, so the use of combination therapy is common. Patients may be on several different medications depending on their underlying symptoms, and there is crossover between the medications and disorders. Psychotropic medications may have short- and long-term adverse effects and frequently have drug interactions with clinical implications for medical and dental providers. Common types of medications include antidepressants, mood stabilizers, antipsychotics, and antianxiety drugs (Box 1).

Antidepressants

While the exact mechanism is not known, it is hypothesized that antidepressant drugs work by inhibiting the uptake of serotonin and/or norepinephrine in the synapse. Millions of patients take antidepressant medications in the United States not only to treat depression but also to treat anxiety, panic disorders, posttraumatic stress disorder, obsessive-compulsive disorder, and seasonal affect disorder. Some antidepressants are prescribed off-label to treat problems such as chronic pain, insomnia, ADHD, menstrual symptoms, migraines, and tobacco cessation (Fishbain 2000; Urits et al. 2019).

Antidepressant classes are named depending on their effect on reuptake or enzyme inhibition. Commonly prescribed antidepressants include selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), and bupropion. Tricyclics (TCAs), tetracyclics, and monoamine oxidase inhibitors (MAOIs) are older classes of antidepressants that are used less frequently because of higher inci-

dence of side effects (including xerostomia, orthostatic hypotension, and cardiac toxicities) and more adverse drug interactions. Today, TCAs are more commonly used in the management of chronic pain and insomnia due to their ability to elevate synaptic norepinephrine concentrations and sedative properties (Finnerup et al. 2015; Catalani et al. 2014).

MAOIs were the first group of drugs used to treat depression but are currently only used to treat patient resistant to other pharmacologic therapies. MAOIs act by inhibiting the activity of presynaptic monoamine oxidase (MAO; MAO-A and MAO-B), leading to increased epinephrine, norepinephrine, serotonin, and dopamine concentrations in the neuronal cytoplasm (Becker 2008), leading to downregulation of postsynaptic receptors (Bhat and Preethishree 2015). They fell out of favor because of serious dietary and drug restrictions that were needed to avoid life-threatening hypertensive episodes. These included dietary restrictions of any foods containing tyramine and any other drugs with an effect of serotonin uptake (Rabkin et al. 1984).

Tricyclic antidepressants gained popularity in the 1960s as the major antidepressant medication in the United States. In addition to limiting the reuptake of neurotransmitters, TCAs have alpha-1 antagonist, anticholinergic and antihistamine properties (Baldessarini 2006). Anticholinergic side effects include xerostomia, urinary retention, and constipation, while the alpha-1 blockade causes orthostatic hypotension. Metabolism occurs in the liver, via the cytochrome P450 pathway. Notable central nervous system effects of TCAs include sedation and a reduction in the seizure threshold. Cardiovascular effects of TCAs include tachycardia and changes in cardiac conduction possibly leading in some cases to cardiac arrhythmias and causing widening of the QRS complex and QT prolongation. TCAs display analgesic properties at doses significantly lower than those used to treat depression and are considered a first-line drug in the treatment of neuropathic pain, fibromyalgia, back pain, and chronic headaches (Fennema et al. 2017).

Box 1 Psychiatric Medications

Antidepressants Selective serotonin reuptake inhibitors (SSRIs) Tricyclics Monoamine oxidase inhibitors (MAOI) Serotonin norepinephrine reuptake inhibitor (SNRI) Serotonin-2 antagonists/reuptake inhibitor (SARI) Norepinephrine dopamine reuptake inhibitor (NDRI)	Citalopram (Celexa®) Escitalopram, (Lexapro®) Fluoxetine (Prozac®) Fluvoxamine (Luvox®) Paroxetine (Paxil®) Sertraline (Zoloft®)	Mood stabilizers	Carbamazepine (Tegretol®) Divalproex (Depakote®) Gabapentin (Neurontin®) Lamotrigine (Lamictal®) Lithium (Eskalith®, Lithonate®) Oxcarbazepine (Trileptal®) Topiramate (Topamax®) Valproic Acid (Depakene®, Epival®)
	Amitriptyline (Elavil®) Clomipramine (Anafranil®) Desipramine (Norpramin®) Doxepin (Sinequan®) Imipramine (Tofranil®) Nortriptyline (Pamelor®) Protriptyline (Vivactil®) Trimipramine (Surmontil®) Phenelzine (Nardil®) Tranylcypromine (Parnate®)	Antipsychotics	Typical (first generation) Chlorpromazine (Thorazine®) Flupenthixol (Fluanxol®) Fluphenazine (Modectate®) Haloperidol (Haldol®) Loxapine (Loxapac®) Mesoridazine (Serentil®) Pericyazine (Neuleptil®) Perphenazine (Trilafon®) Pimozide (Orap®) Pipotiazine (Piportil®) Prochlorperazine (Stemetil®) Thioridazine (Mellaril®) Thiothixene (Navane®) Trifluoperazine (Stelazine®) Zuclopenthixol (Clopixol®)
	Venlafaxine (Effexor®) Duloxetine (Cymbalta®)		Atypical (second generation) Aripiprazole (Abilify®) Clozapine (Clozaril®) Lurasidone (Latuda®) Olanzapine (Zyprexa®) Quetiapine (Seroquel®) Risperidone (Risperdal®)
	Trazodone (Desyrel®)	Anxiolytics	Benzodiazepines Alprazolam (Xanax®) Bromazepam (Lexotanil®) Chlordiazepoxide (Librium®) Clonazepam (Klonopin®) Clorazepate (Tranxene®) Diazepam (Valium®) Flurazepam (Dalmane®) Lorazepam (Ativan®) Nitrazepam (Magadon®) Oxazepam (Serax®) Temazepam (Restoril®) Triazolam (Halcion®)
	Bupropion (Wellbutrin®, Zyban®)		Non-benzodiazepine Buspirone (BuSpar®) Hydroxyzine (Vistaril®) Meprobamate (Miltown®) Pregabalin (Lyrica®) Propranolol (Inderal®)

Selective serotonin reuptake inhibitors (SSRI) inhibit the neuronal uptake of serotonin and generally have the least side effects of all classes of antidepressants. They are used to treat mild to moderate depression, obsessive-compulsive syndrome, panic disorder, social phobias, and post-traumatic stress disorder. With the exception of paroxetine, SSRIs have relatively minimal anticholinergic properties or sedative effects. Unlike TCAs, SSRIs do not affect the seizure threshold or cardiac conduction. Common side effects include nausea, diarrhea, headache, jitteriness and agitation, and insomnia. SSRIs (especially fluoxetine, paroxetine, and sertraline) are also potentially associated with an increase in sleep bruxism, although the exact etiology is currently unknown (Garrett and Hawley 2018).

Other atypical antidepressants include the SNRIs represented by venlafaxine and duloxetine. They resemble the tricyclic antidepressants in action and have similar side effects including sedations, nausea, insomnia, xerostomia, and constipation. They can cause elevations and heart rate and blood pressure because of norepinephrine uptake inhibition. They are also commonly used in the treatment of chronic neuropathic pain including diabetic neuropathy, fibromyalgia, and post herpetic neuralgia (Finnerup et al. 2015).

Dental Pharmacology Implications for Patients Medicated with Antidepressants

Potential drug interactions with antidepressants of dental concern include sedatives, narcotic and nonsteroidal anti-inflammatory drugs (NSAIDs) analgesics, and epinephrine/levonordefrin. Antidepressants have significant sedative properties, and caution should be taken when prescribing any other medication with sedative side effects. Antidepressants that elevate serotonin levels can precipitate seizures or serotonin syndrome if used in conjunction with other medications that increase serotonin levels. Serotonin syndrome is a potentially fatal consequence of excessive central nervous system serotonergic activity characterized by hyperre-

flexia, muscle rigidity and tremors, agitation, and confusion (Sansone and Sansone 2009). Additional symptoms can include diaphoresis, tachycardia, hyperthermia, hypertension, visual hallucinations, and coma. Dental drugs of concern include opioid analgesics (especially meperidine), anesthetic agents including fentanyl, tramadol, and erythromycin and should be avoided in patients taking serotonergic drugs.

SSRIs and SNRIs increase the risk of bleeding through the reduction of platelet levels of serotonin that causes decreased platelet aggregation and clot formation (Serebruany 2006). Patients taking SSRIs (especially fluoxetine, paroxetine, and sertraline) have an increased risk of upper gastrointestinal (GI) bleeding, and that this risk may be synergistically increased by concurrent use of NSAIDs (Anglin et al. 2014). Additionally, paroxetine, sertraline, and fluvoxamine inhibit the enzymes needed to metabolize NSAIDs including ibuprofen, naproxen, diclofenac, and celecoxib, further increasing the risk of significant bleeding (Mansour et al. 2006; Loke et al. 2008).

The SSRI antidepressants also can inhibit several of the cytochrome P-450 enzymes produced in the liver, impacting the biotransformation and clearance of other drugs. This occurs with varying intensity with fluoxetine and paroxetine exhibit the greatest degree of inhibition. Of clinical relevance is the inhibition of CYP2D6, which is responsible for metabolizing codeine, hydrocodone, and oxycodone from a prodrug to an active morphine metabolite. Patients medicated with these SSRIs may not experience the expected analgesic effect from codeine or its derivatives (Becker 2008; Friedlander and Norman 2002).

A long held and commonly believed drug interaction is between MAOIs and local anesthetics with vasoconstrictors. It was believed that inhibition of MAO could potentiate the activity of epinephrine by delaying its oxidation, leading to increased and potentially harmful cardiovascular reactions. Restriction on using local anesthetic with epinephrine or levonordefrin is frequently listed as a contraindication with MAOIs. However, review of the literature shows an absence of case reports to support this claim.

More recent reviews show no credible evidence of a significant interaction between MAOIs and epinephrine or levonordefrin (Yagiela 1999; Brown and Rhodus 2005), and this has been supported by human and animal studies and clinical experience.

A similar contraindication for the use of epinephrine and levonordefrin has been proposed for TCAs. Animal models show significant interaction between TCAs and adrenergic vasoconstrictors (Yagiela 1999) including increases in systolic blood pressure. However, there are no case reports in the English literature that translate into clinically adverse events at the doses of local anesthetic used during most dental procedures, especially with epinephrine (Moodley 2017; Malamed 2014; Saraghi et al. 2017). Despite this, most anesthesia texts and pharmacology sites continue to suggest limiting the use of local anesthetics with epinephrine to 0.05 mg and avoiding the use of anesthesia with levonordefrin all together. Additional human research is needed to determine if therapeutic doses of local anesthetic with vasoconstrictors are related to negative cardiovascular outcomes in patient taking TCAs.

Mood Stabilizers

Traditionally, mood stabilizers were primarily antimanic agents, but more recently, the definition has been expanded to include agents with efficacy in treating acute symptoms of manic and depressive symptoms as well as in preventing these symptoms in bipolar disorder (Bauer and Mitchner 2004). Lithium is the only psychotropic agent that is truly a mood stabilizer, and it is considered the first-line drug for the treatment of bipolar disorder. While the mechanism of action is not completely understood, but it is thought to somehow stabilize neuronal membranes, thereby reducing their rate of discharge and neurotransmitter turnover (Hashimoto et al. 2002; Alda 2015). Lithium is excreted solely by the kidneys and has a number of side effects including cardiac dysrhythmias, GI disturbances, and tremors. It has a very narrow therapeutic ratio and a high risk of toxicity, necessitating strict adherence to a

daily dosing schedule and careful monitoring of serum lithium levels. Early signs of lithium toxicity included nausea, vomiting, lethargy or drowsiness, slurred speech, and muscle weakness, which quickly progress to ataxia, dysrhythmias, seizures, renal failure, and coma if serum levels are not lowered. In addition to lithium, the atypical antipsychotic olanzapine is also used to treat bipolar depression in patients who have refractory disease or are unable to tolerate lithium's side effects.

Several anticonvulsant medications have also proven useful in managing bipolar affective disorders in addition to seizure disorders. It is thought that they work by stabilizing the neuronal membranes and helping to suppressing the spread of impulses to neighboring neuron pathways. Commonly used anticonvulsants include carbamazepine, topiramate, lamotrigine, gabapentin, and valproate (both as divalproex and valproic acid) (Melvin et al. 2008).

Dental Pharmacology Implications for Patients Medicated with Lithium and Anticonvulsants

NSAIDs and metronidazole are two dental drugs that can cause lithium toxicity by reducing renal clearance in a short period of time. Likewise, tetracycline and doxycycline can raise lithium levels through an unknown mechanism causing the potential of toxicity. Macrolides such as erythromycin and clarithromycin can lead to increased serum levels of carbamazepine and divalproex/valproic acid leading to toxicity. Alternative antibiotics and pain medications are recommended.

Valproate (divalproex and valproic acid) can cause thrombocytopenia, abnormal platelet function, and changes in coagulation factor levels, leading to an increased risk of bleeding (Vasudev et al. 2010; Abdallah 2014). NSAIDs and aspirin should be avoided in patients taking these drugs. Carbamazepine can also cause blood dyscrasias, including anemia, leukopenia, and agranulocytosis, and so patients taking this drug are usually carefully monitored via frequent CBCs (Manu et al. 2016).

Antipsychotics

Antipsychotics are psychotropic drugs used to treat psychotic symptoms such as delusions and hallucinations seen in schizophrenia, mania, delusional disorder, and psychotic depression. They are also frequently used in patients with Alzheimer's disease who have acute agitation. They are generally classified as typical (first generation) or atypical (second generation). They work as antagonists at dopamine receptors in various parts of the brain. Blockage of the D₂ receptors in the limbic system produces the antipsychotic result, but most of these drugs lack specificity for this receptor. Interaction with other dopamine receptors as well as other receptor systems, including histaminic, muscarinic, serotonergic, and α -adrenergic receptors produce many side effects (Miyamoto et al. 2012). These include extrapyramidal symptoms such as akathisia and tardive dyskinesia (more prevalent with typical antipsychotics), sedation, weight gain, xerostomia, orthostatic hypotension, cognitive impairment, prolactin effects, and cardiac changes (including QT interval prolongation) (Kennedy et al. 2013). Development of extrapyramidal symptoms with the typical antipsychotic medications is often treated by the addition of anticholinergic drugs, nonselective beta blockers, or benzodiazepines.

Dental Pharmacology Implications for Patients Medicated with Antipsychotics

Antipsychotic medications may potentiate other central nervous system depressants such as narcotic analgesics. Caution should be used when prescribing to prevent orthostatic hypotension and respiratory depression (Friedlander and Marder 2002). Blockage of α 1-receptors by the typical antipsychotics can cause orthostatic hypotension that theoretically could be worsened by the β 2-receptor stimulation by epinephrine. Local anesthetic with epinephrine should be limited to no more than 0.05 mg of

epinephrine in patients taking older antipsychotics like haloperidol and chlorpromazine. The macrolide antibiotics clarithromycin and erythromycin should be used cautiously in some of the newer atypical antipsychotics (like quetiapine) because of competing metabolism leading to increased serum levels of the antipsychotic. This can lead to QT interval prolongation and cardiac arrhythmias (Kennedy et al. 2013; Ruiz Diaz et al. 2020).

Benzodiazepines

Benzodiazepines are most commonly used to treat anxiety and are the drug of choice for generalized anxiety disorder. They are often used in combination with other antidepressants to treat panic disorders and other anxiety states. They work by indirectly enhancing γ -aminobutyric acid (GABA) neurotransmission and receptor sensitivity. Side effects include depressed respiration and cardiovascular tone, sedation, fatigue, and dependence (Kyrios 2014).

Dental Pharmacology Implications for Patients Medicated with Anxiolytics

Drug tolerance and dependence are of concern. Benzodiazepines have a wide therapeutic index, and therefore most dental drug interactions are unlikely to produce significant toxicity (Griffin III et al. 2013). Exceptions include macrolide antibiotics erythromycin and clarithromycin and the azole antifungals (fluconazole, ketoconazole, and itraconazole) that compete for the same hepatic cytochrome P450 enzyme needed to metabolize all benzodiazepines (except lorazepam). Co-administration can cause elevated serum levels and prolonged action, especially with triazolam and midazolam (Gubbins and Heldenbrand 2010). Additionally, caution should be used with other drugs that have the potential to enhance CNS and respiratory depression.

Parkinson's Disease

Parkinson's disease is a progressive multisystem and multi-symptomatic degenerative neurological condition believed to be the result of both genetic and environmental factors. It is due to a loss of dopaminergic neurons in the substantia nigra region of the brain. It is one of the fastest growing neurologic disorders worldwide due to the aging population. It is estimated that more than a million people in the United States have Parkinson disease, with the number expected to significantly increase in the coming years (Willis et al. 2022). This number doesn't account for the thousands of cases that go undetected. Men are one and a half times more likely to have Parkinson's than women (Marras et al. 2018). Parkinson's disease ranks among most common late-life neurodegenerative diseases (after Alzheimer's disease) and affects approximately 1.5–2.0% of people aged 60 years and older (Nussbaum and Ellis 2003; Wirdefeldt et al. 2011; Pringsheim et al. 2014).

Clinical manifestations of Parkinson's disease include rigidity and bradykinesia accompanied by a resting tremor. Additionally, neuropsychiatric symptoms including cognitive impairment, depression, anxiety, hallucinations, and psychosis are common (Aarsland et al. 1999; Aarsland et al. 2009; Miyasaki et al. 2006). Pharmacologic treatment of Parkinson disease can be divided into symptomatic and neuroprotective (disease-modifying) therapy aimed at controlling symptoms and maintaining functional independence. Treatment protocols aimed at controlling symptoms focus on increasing dopamine availability and preventing its breakdown by inhibiting acetylcholine. This is usually done by utilizing levodopa/carbidopa levodopa alone or with the addition of catechol-O-methyltransferase (COMT) inhibitors. Although levodopa is the most effective medication available for treating the motor symptoms of Parkinson disease, other neuroprotective medications such as monoamine oxidase type B inhibitors (MAOBIs), amantadine, anticholinergics, β -blockers, or dopamine agonists may be initiated first to avoid levodopa-related motor complications (Connolly and Lang 2014).

Neuropsychiatric symptoms are treated using SSRIs as the first drugs of choice, but tricyclic antidepressants, SNRIs, and antipsychotics may all be used depending on severity of the symptoms and patient's response to drugs (Menza et al. 2009; Bomasang-Layno et al. 2015; Suchowersky et al. 2006; Chen 2012) (Box 2). Common oral side effects of medications used to treat Parkinson's disease include orthostatic hypotension, xerostomia, and bruxism.

Box 2 Pharmacologic Treatment of Parkinson's Disease (Suchowersky et al. 2006; Chen 2012)

Tremor control Amantadine (Symmetrel®) Trihexyphenidyl (Artane®) Benztropine (Cogentin®) Propranolol (nonselective beta blocker)	Cholinesterase inhibitors Rivastigmine (Exelon®) Donepezil (Aricept®) Galantamine (Razadyne®) Tacrine (Cognex®)
Dopamine precursor Levodopa/Carbidopa Sinemet® Parcopa® Stalevo® Rytary®	N-methyl D aspartate antagonist Memantine (Namenda®) Memantine and donepezil (Namzaric®)
Dopamine agonists Pramipexole (Mirapex®) Ropinirole (Requip®) Apomorphine (Apokyn®) Rotigotine (Neupro®) Bromocriptine (Parlodel®)	Antidepressants Citalopram (Celexa®) Fluoxetine (Prozac®) Paroxetine (Paxil®) Sertraline (Zoloft®) Trazodone (Desyre1®)
MAO-B inhibitor Selegiline (Eldepryl®) Rasagiline (Azilect®) Safinamide (Xadago®) Zydis selegiline HCL (Zelapar®)	Anxiolytics Lorazepam (Ativan®) Oxazepam (Serax®) Alprazolam (Xanax®) Buspirone (BuSpar®)
COMT inhibitors Entacapone (Comtan®) Tolcapone (Tasmar®) Opicapone (Ongentys®)	Antipsychotics Aripiprazole (Abilify®) Clozapine (Clozaril®) Olanzapine (Zyprexa®) Quetiapine (Seroquel®)
Anticholinergics Trihexyphenidyl (Artane®) Benztropine (Cogentin®) Biperiden (Akineton®)	Risperidone (Risperdal®) Ziprasidone (Geodon®)

Dental Pharmacology Implications for Patients Medicated for Parkinson's Disease

Potential drug interactions of concern include COMT inhibitors and local anesthetics with vasopressors. These drugs reversibly block catechol-O-methyltransferase inhibiting the inactivation of exogenously administered epinephrine and levonordefrin contained in a local anesthetic solution. Tachycardia, hypertension, and arrhythmias have been reported. Vasoconstrictors should be used with caution. Vital signs should be monitored during and after administration of the first carpule, and total dose should be limited to 2 or less carpules of 1:100,000 (0.034 mg) of epinephrine. Care should be taken to aspirate to prevent intravascular injection (Malamed 2014; Balakrishnan and Ebenezer 2013). Retraction cord impregnated with epinephrine should not be used. Most anti-parkinsonian drugs aimed at increasing the amount of dopamine present cause CNS depression, and therefore any additional sedative could have additive effect (Friedlander et al. 2009). Other drug reactions with the psychotropic drugs used in Parkinson are covered previously.

Alzheimer's Disease

Alzheimer's disease (AD) is a progressive neurodegenerative condition that accounts for 80% of all cases of dementia in the United States (Alz. org 2018). AD is characterized by a progressive and irreversible deterioration in cognitive abilities including memory and abstract thought and functionality that impairs activities of daily living ultimately leading to death. The exact etiology of AD isn't known, but it is believed to be multifactorial, involving several genetic, environmental, and biologic risk factors. The formation of cerebral plaques composed of intracellular neurofibrillary tangles of tau proteins and extracellular amyloid protein deposits is a key pathologic finding. These plaques lead to neuroinflammation that results in the progressive damage and

destruction of cortical neurons (Selkoe and Schenk 2003; Walsh and Selkoe 2004; Kumar and Singh 2015). An estimated 5.7 million Americans are living with Alzheimer's dementia in 2018 including an estimated 5.5 million people age 65 and older and approximately 200,000 individuals under age 65 who have younger-onset Alzheimer's. Alzheimer's disease is the sixth-leading cause of death in the United States and the fifth-leading cause of death among those age 65 and older (Alzheimer's Association 2018).

The rate of progression of the cognitive changes of AD is variable from patient to patient, but on average, a person with AD lives four to 8 years after diagnosis. Symptoms often predate a formal diagnosis by several years. Diagnosis is made on the basis of a history of pattern of symptoms over time including a clinical history of progressive dementia and the exclusion of other causes. Along with cognitive changes seen in AD, psychiatric and behavioral symptoms are common in patients with AD and contribute substantially to the morbidity of the illness. During the early stages, patients may experience personality changes such as irritability, depression and anxiety. In the later stages, symptoms such as apathy, psychomotor impairment, and psychosis become more prevalent. Delusions or hallucinations appear in 30–50% of AD patients, and as high as 80% of patients exhibit agitated or aggressive behavior (Ropacki and Jeste 2005; Lopez et al. 2001).

Pharmacologic treatment of AD is aimed at modulating the decline of cognitive, functional, and behavioral symptoms by increasing the concentration of neurotransmitters in the brain (Box 3) (Kumar and Singh 2015; Casey 2015; Orgeta et al. 2017). Most of the current pharmacologic treatments available reduce or control some of the cognitive and behavioral symptoms and do not alter the progressive pathophysiology of the disease or stop the damage and destructions of neurons. There are five FDA-approved drugs for the treatment of AD—three cholinesterase inhibitors (donepezil, galantamine, and rivastigmine), one N-methyl D aspartate (NMDA) antagonist (memantine) and one combination drug (meman-

tine combined with donepezil). Early to moderate stage AD is treated using the cholinesterase inhibitors, which prevent the breakdown of acetylcholine and can delay worsening of symptoms and improved cognitive function for one to five years for about 50% of the patients that take them (Han et al. 2017). Common side effects include GI distress (nausea, vomiting, and diarrhea) as well as dizziness and fatigue (Noetzli and Eap 2013). Memantine either alone or with donepezil is used in more advanced disease and may delay worsening of symptoms by modulating the glutamate-induced overactivation of the NMDA receptor that contributes to neuronal damage and death (Kishi et al. 2017).

Box 3 Pharmacologic Management of Alzheimer’s Disease (Casey 2015; Orgeta et al. 2017; Van Dyck et al. 2023)

First-line medications	Donepezil (Aricept®)
cholinesterase inhibitors	Galantamine (Razadyne®)
	Rivastigmine (Exelon®)
N-methyl-D-aspartate receptor antagonists	Memantine (Namenda®)
Combination	Memantine and donepezil (Namzaric®)
Amyloid beta-directed antibody	Aducanumab (Aduhelm®)
Adjunctive medications	Lecanemab (Leqembi®)
Antidepressants	Citalopram (Celexa®)
Anxiolytics/hypnotics	Fluoxetine (Prozac®)
Antipsychotics	Paroxetine (Paxil®)
Mood stabilizers	Sertraline (Zoloft®)
	Trazodone (Desyrel®)
	Venlafaxine (Effexor®)
	Lorazepam (Ativan®)
	Alprazolam (Xanax®)
	Zolpidem (Ambien®)
	Triazolam (Halcion®)
	Aripiprazole (Abilify®)
	Clozapine (Clozaril®)
	Haloperidol (Haldol®)
	Quetiapine (Seroquel®)
	Carbamazepine (Tegretol®)
	Divalproex sodium (Depakote®)
	Lamotrigine (Lamictal®)
	Levetiracetam (Keppra®)

Aducanumab (Aduhelm™) and lecanemab (Leqembi®) are two new immunotherapies

recently approved to treat early Alzheimer’s that are the first to target the fundamental pathophysiology of the disease. A third drug Donanemab is currently in Phase III clinical trials. These monoclonal antibodies bind to soluble amyloid-beta protofibrils, reducing amyloid plaque accumulation resulting in a slowing in cognitive and functional decline when compared to placebo (Van Dyck et al. 2023). They are given via intravenous infusions. Side effects include infusion-related reactions including chills, fever, rash, and body aches and amyloid-related imaging abnormalities with edema (ARIA) and brain bleeds. It is too early to determine if there are other significant long-term side effects associated with this class of medications.

As AD progresses, behavioral symptoms and mood disorders often necessitate the initiation of psychotropic medications to modify these effects. These include all of the major groups of psychiatric medications including antidepressants (tricyclic, SSRI, and SNRI) for depression and apathy; anxiolytics for anxiety, restlessness, verbally disruptive behavior, and resistance; antipsychotic medications for hallucinations, delusions, aggression, agitation, hostility, and uncooperativeness; and mood stabilizers for agitation and sleeping aids for AD-associated sleep disturbances (Sink et al. 2005; Gauthier et al. 2010). While atypical antipsychotics are generally the first drugs of choice, they have been associated with increased mortality in patients with dementia and behavioral symptoms, and their effectiveness is low with no significant difference from placebo. Risperidone and olanzapine in particular have been linked to an elevated risk of cerebrovascular adverse events (Mulsant et al. 2005; Giron et al. 2001; Olin et al. 2002; Schneider et al. 2006; Wang et al. 2015). These drugs all cause some degree of xerostomia, which can increase the rate of caries and periodontal disease. The use of antipsychotic medications (especially the first-generation neuroleptics) increases the risk of development of extrapyramidal symptoms that can lead to involuntary jaw movements making it difficult for the patient to eat, speak, and tolerate a removable prosthesis if needed.

Dental Pharmacology Implications for Patients with Alzheimer's Disease

Increased potential for adverse drug interactions may occur with the AD patient on multiple medications.

Care should be taken before prescribing antimicrobials, analgesics, and anxiolytics. Co-administration of galantamine with macrolide antibiotics or azole antifungals may decrease the metabolism of the cholinesterase inhibitors leading to both peripheral and central hypercholinergic effects, such as agitation, excitation, bradycardia, and loss of consciousness (Defilippi and Crismon 2003; Bentué-Ferrer et al. 2003). Many of the potential drug interactions of concern relate to the psychiatric drugs used to treat the behavioral aspect of AD and are covered earlier in the chapter (Pasqualetti et al. 2015).

Seizure Disorders

A seizure is the result of spontaneous excessive electric discharge of cerebral neurons that results in alteration in level of consciousness, motor activity, sensory changes, and behavioral abnormalities. Approximately 10% of Americans will experience a seizure in their lifetime. An isolated seizure can be due to many things including a high fever, drug or alcohol withdrawal, or a metabolic imbalance and is not classified as epilepsy.

Epilepsy refers to a group of common neurologic disorders characterized by chronic and recurrent paroxysmal seizure activity (at least two unprovoked episodes of unknown etiology) (Epilepsy Foundation 2018). It is the fourth most common neurological disorder in the United States. In 2015, the Center for Disease Control estimated that 1.2% of the total US population (~3.4 million people) had active epilepsy (Zack and Kobau 2017). Incidence is highest among young children (<2 years of age and the elderly (>70 years of age). There are over 30 different types of seizures and over 60 different types of epilepsy. Each is defined by such factors as cause,

seizure type, age of onset, and clinical manifestations. There is frequent overlap of signs and symptoms, sometimes making the final diagnosis of the type of epilepsy difficult. Many patients have more than one type of seizure and the features of each type of seizures may change over time (Robbins 2009).

The International League Against Epilepsy developed an international classification of epileptic seizures based on clinical and electroencephalographic features in 1981, which was revised in 2017 (Fisher et al. 2017). Seizures are divided into focal (previously partial) or generalized depending if one or both hemispheres of the brain are involved. Subcategories include motor or non-motor onset (for both) and retained or impaired awareness for focal seizures. Antiepileptic drugs (AEDs) are the mainstay of treatment (Box 4) (Glauser et al. 2016; Schmidt and Schachter 2014). They work by suppressing seizure occurrence but do not correct the underlying pathologic process that produces the seizures in the first place. Therefore, patients with epilepsy frequently need lifelong treatment. There are many available AEDs, some of which are used as monotherapy and others that are used in combination with other medications. They are categorized as first, second, and third generation based on when they were introduced for use, with the majority of the older or first-generation AEDs metabolized by the liver. These first-generation medications (carbamazepine, ethosuximide, phenobarbital, phenytoin, primidone, and valproic acid) are still commonly used but have a higher rate of undesirable dose-dependent side effects, including cognitive and sedating effects. Carbamazepine, ethosuximide, phenobarbital, phenytoin, and primidone are potent inducers (valproic acid is an inhibitor) of drug metabolizing hepatic enzymes leading to clinically important adverse drug interactions. Compared to first-generation AEDs, the newer or second-generation agents generally have wider therapeutic ranges and fewer serious adverse effects (Löscher et al. 2013; Zhuo et al. 2017). Since fewer of these AEDs are metabolized by the liver, protein binding and drug interactions are not as prob-

Box 4 Antiepileptic (AED) Drugs (Epilepsy Foundation 2018; Zack and Kobau 2017)

Narrow-spectrum AEDs (focal seizures)

- Carbamazepine (Tegretol®)
- Felbamate (Felbatol®)
- Gabapentin (Neurontin®)
- Lacosamide (Vimpat®)
- Oxcarbazepine (Trileptal®)
- Phenobarbital (Luminal®)
- Phenytoin (Dilantin®)
- Pregabalin (Lyrica®)
- Primidone (Mysoline®)
- Tiagabine (Gabitril®)
- Vigabatrin (Sabril®)

Broad-spectrum AEDs (focal and generalized seizures)

- Clonazepam (Klonopin®)
- Divalproex (Depakote®)
- Ethosuximide (Zarontin®)
- Lamotrigine (Lamictal®)
- Levetiracetam (Keppra®)
- Rufinamide (Bonze®)
- Topiramate (Topamax®)
- Valproic acid (Depakene®)
- Zonisamide (Zonegran®)

lematic as some of the older medications (Chen et al. 2018).

The second-generation agents are also prescribed “off-label” for non-seizure disorders, such as chronic neuropathic pain (including migraine), fibromyalgia, and trigeminal neuralgia. When used to treat pain, these agents may be referred to “analgesics.” Their actions and use are significantly different from opiates or nonsteroidal anti-inflammatory drugs. Carbamazepine, gabapentin, lamotrigine, topiramate, and valproate are used as mood stabilizers in the treatment of bipolar disorder (Melvin et al. 2008). Anticonvulsants have been evaluated in some disorders of impulse control, such as impulsive aggressiveness (Brodie et al. 2016).

Dental Pharmacology Implications for Patients Taking AEDs

Anticonvulsants may increase hepatic microsomal enzyme activity, which can reduce the blood concentration of other drugs metabolized by the same enzyme system. Carbamazepine, phenytoin, primidone, and phenobarbital are particularly strong hepatic enzyme inducers leading to reduced serum concentrations of the co-administered drug and reduced pharmacological effect. Affected drugs of clinical importance in dentistry include antibiotics (doxycycline, metronidazole), benzodiazepines (alprazolam, clonazepam, diazepam, and midazolam), and steroids (dexamethasone and prednisone) (Patsalos and Perucca 2003). Clarithromycin, erythromycin, fluconazole, ketoconazole, and metronidazole all inhibit the metabolism of carbamazepine, leading to potentially toxic level (Perucca 2006). Miconazole and fluconazole can substantially increase plasma concentrations of phenytoin. Valproate (divalproex and valproic acid) can cause thrombocytopenia, abnormal platelet function, and changes in coagulation factor levels, leading to an increased risk of bleeding (Vasudev et al. 2010; Abdallah 2014). NSAIDs and aspirin should be avoided in patients taking these drugs.

Conclusion

Neurologic and psychiatric illnesses are estimated to affect as many as 1.5 billion people worldwide. This number is expected to increase as the population ages—a number that is expected to grow as life expectancy increases (Gooch et al. 2017). Treatment of these chronic conditions involves long-term use of multiple pharmacologic agents, raising the concern of side effects and possible drug interactions with medications used during the delivery of dental care. With the continued introduction of new therapeutic classes of drugs for these conditions, the potential for adverse drug interactions will continue to grow. Appropriate preoperative assessment of dental patients should always include obtaining a

through medical history that includes a current complete list of medications. It is incumbent on the oral health-care practitioner that an analysis of possible drug interactions be done prior to the administration or prescription of any drugs used during the provision of dental care.

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Endocrine Drugs of Significance in Dentistry

Arthur H. Jeske

Adrenocorticosteroids

In dentistry, corticosteroids with anti-inflammatory activity are used in a variety of inflammatory conditions (e.g., pulpitis) and to control postoperative morbidity (e.g., oral surgery). Among the most frequently used and studied agents are dexamethasone (e.g., Decadron®), methylprednisolone, and prednisolone.

In endodontics, evidence from several recent systematic reviews suggests that steroids appear to be effective in cases of irreversible pulpitis (Aminoshariae et al. 2016) and, depending on the type and dose of agent, can be effective for management of postoperative pain of endodontic origin (Shamszadeh et al. 2018). Among beta-methasone, dexamethasone, prednisolone, and methylprednisolone, prednisolone appeared to produce significantly greater pain reduction at 6 h, possibly associated with its more rapid diffusion into cells, where nuclear DNA transcription is affected by these agents (Shamszadeh et al. 2018). Anti-inflammatory corticosteroids available in the United States and their dosages are summarized in Table 1.

The perioperative use of corticosteroids in oral surgery is well documented, with demon-

strated effectiveness for the reduction of swelling and trismus, although outcomes for pain reduction are equivocal (Arora et al. 2018; Taylor et al. 2018). Administration of the corticosteroid achieves greater effectiveness when the parenteral, rather than other, route is utilized and by administration of the corticosteroid prior to the surgical procedure (Al-Dajani 2017; Herrera-Briones et al. 2013), and preoperative administration is likely more successful because of the relatively slow onset of action of these agents, associated with their principal action in altering gene transcription. With regard to dose of synthetic corticosteroids, a recent prospective randomized clinical study demonstrated that there was no difference between 4-mg and 8-mg doses of dexamethasone in suppression of postoperative edema following third-molar surgery,

Table 1 Summary of anti-inflammatory corticosteroids

Agent	Anti-inflammatory activity ^a	Equivalent dose (mg) ^b
Hydrocortisone	1	20
Cortisone	0.8	25
Prednisone	4	5
Prednisolone	5	5
Methylprednisolone	5	4
Betamethasone	25–40	0.6
Dexamethasone	30	0.75

^aRelative to hydrocortisone

^bDose equivalent to hydrocortisone

A. H. Jeske (✉)

UTHealth School of Dentistry, Houston, TX, USA

e-mail: Arthur.H.Jeske@uth.tmc.edu

although the same study was unable to detect a significant analgesic action of the drug (Arora et al. 2018).

Currently, the American Association of Oral and Maxillofacial Surgeons recommends consideration of the use of a perioperative corticosteroid (dexamethasone) to limit swelling and decrease postoperative discomfort after third-molar extractions as part of its white paper recommendations on opioid prescribing and acute and postoperative pain management (Arora et al. 2018), without specification of dosage.

Management of Patient At-Risk of Acute Adrenal Insufficiency

Prolonged medical therapy with corticosteroids is a common intervention for a wide variety of diseases and conditions, including autoimmune diseases (e.g., lupus erythematosus, allergic reactions), eye and gastrointestinal diseases, musculoskeletal conditions, neurologic disorders, organ transplantation, and, of course, replacement of corticosteroids associated with adrenal cortical disorders, such as autoimmune Addison's disease. Natural adrenocorticosteroids are classified as glucocorticoids (with effects on intermediary metabolism and immunity, e.g., cortisol), mineralocorticoids (having salt-retentive functions, e.g., aldosterone), and androgenic/estrogenic hormones (Katzung and Trevor 2015). Glucocorticoids have significant suppressive effects on inflammatory processes, by affecting the numbers, distribution, and functions of leukocytes and by reducing cytokines and chemokines. They also inhibit macrophage activity and antigen-presenting cells and block phospholipase A₂ with concomitant reduction of prostaglandin synthesis (Katzung and Trevor 2015).

It is generally held that corticosteroid therapy lasting longer than 2 weeks can result in adrenal cortical suppression due to negative feedback of the supplemental hormone on the secretory cells of the adrenal cortex. It is important to note that this suppression is also associated inhalational use of corticosteroids for chronic asthma.

Symptoms associated with acute adrenal insufficiency may occur during stressful dental procedures, and while rare, they can be life-threatening. This condition is classified as primary adrenal insufficiency (e.g., destruction or atrophy of the adrenal gland), secondary adrenal insufficiency (e.g., pituitary disorders), Cushing syndrome (e.g., long-term corticosteroid use as anti-rejection therapy in organ transplantation), and adrenal crisis. Adrenal crisis is of great concern when managing patients undergoing stressful or prolonged procedures, such as quadrant surgeries. Clinically, adrenal crisis is due to inadequate secretion of endogenous corticosteroids during stress and involves several organ systems, including the GI tract (cramping, nausea, vomiting), with hypotension, weakness, fatigue, headache, cyanosis, dehydration, and muscle and joint pain. The underlying biochemical changes include hyponatremia, eosinophilia, hypoglycemia, and azotemia. Therapy must be immediate and aggressive and usually involves administration of fluids and electrolytes and infusion of high doses of hydrocortisone (Miller et al. 2001). Risk factors for the development of adrenal crisis include adrenal insufficiency, pain, infection, invasive procedure, general anesthesia, and poor health status at the time of the procedure. Fortunately, this condition occurs only rarely, with an estimated risk of less than 1 case per 650,000 with adrenal insufficiency. This is based on a recent systematic review that identified six documented case reports over a 66-year period (Khalaf et al. 2013). More recently, a triple-blind, randomized, controlled trial has determined that in dental patients undergoing minor surgical procedures with local anesthesia and taking long-term glucocorticoid for conditions other than primary adrenal insufficiency, "routine blanket" glucocorticoid supplementation is unnecessary (Ramasamy and Madhan 2023). With regard to the adverse effects that may be associated with the perioperative administration of dexamethasone, Polderman et al. recently determined that this corticosteroid probably does not increase the risk of infection following surgical procedures, although the evidence for an effect on wound healing remains inconclusive, and mild increases in blood sugar

Table 2 Corticosteroid supplementation for various types of dental procedure risk category in patients with adrenal insufficiency (Miller et al. 2001)

Corticosteroid supplementation for types of dental procedures by risk category in patients with adrenal insufficiency
Negligible risk Nonsurgical dental procedures Supplementation regimen: none required
Mild risk Minor oral surgery (limited number of simple extractions, biopsies) Supplementation regimen: 25 mg Hydrocortisone equivalent Day of procedure
Moderate-to-major risk Major oral surgery (multiple extractions, quadrant periodontal surgery, impaction surgery, general anesthesia, procedures >1 h) Supplementation regimen: 50–100 mg Hydrocortisone equivalent day of procedure and for at least 1 day postoperatively

can occur, which can be greater in patients with diabetes mellitus (Polderman et al. 2018).

In spite of the low incidence of adrenal crisis in patients with adrenal insufficiency who are undergoing dental procedures, recommendations for perioperative corticosteroid supplementation persist, despite findings that routine dental procedures done under local anesthesia are very unlikely to precipitate adrenal crisis (Miller et al. 2001). Dental outcomes notwithstanding the incidence of clinically significant adrenal insufficiency appear to be significantly higher in surgical intensive care unit populations. In one prospective study of patients seen for trauma, general surgery, urology, and gynecologic oncology over a 9-month period, the overall rate of adrenal insufficiency was 0.66% (Barquist and Kirton 1997). On this basis, recommendations for supplementation have been based upon stratified risks, as presented in Table 2.

Oral Contraceptives, Estrogen/Progestogens

Oral contraceptive drugs are among the most widely prescribed pharmaceutical products throughout the world. Recent estimates based on

a large-scale cohort in the United Kingdom indicate that 16.2% of women aged 12–49 years used a combination oral contraceptive and an additional 5.6% used a progestogen-only birth control pill (Cea-Soriano et al. 2014). In the United States, based on estimates from 2018, prescriptions for estrogens, ethinyl estradiol, and various contraceptive combination products typically represent several preparations among the most widely used drugs.

For many years, the relationship between the use of estrogens and periodontal disease and postsurgical complications (e.g., acute alveolar osteitis) has been known and studied in dentistry. Additionally, the increased risk of pregnancy in female dental patients taking oral contraceptives caused by antibiotic therapy has raised medicolegal concerns in the dental profession.

Antibiotic/Oral Contraceptive Interaction.

There are three potential mechanisms by which the effectiveness of oral contraceptives may be reduced by antibiotics (Taylor and Pemberton 2012):

1. Diarrhea and vomiting. Both vomiting and persistent diarrhea result in reduce absorption of combined oral contraceptives (e.g., progestogen with ethinyl estradiol) and progestogen-only formulations. In such cases, it is recommended that if vomiting occurs within 2 h of taking the medication, another dose should be taken immediately, and in the case of diarrhea, additional contraceptive measures should be used until recover is complete.
2. Enzymatic induction by antibiotics. Rifamycin-type antibiotics (e.g., rifampicin, rifabutin) can induce hepatic microsomal oxidase enzymes, which breakdown ethinyl estradiol, and the resultant decrease in circulating ethinyl estradiol levels reduce its contraceptive effect. While this interaction has been scientifically documented to result in alterations to estrogen pharmacokinetics, no studies to date have evaluated actual pregnancy risks associated with the alteration of blood levels of oral contraceptive (Simmons et al. 2018a).

3. Disruption of gastrointestinal flora associated with antibiotics. Ethinyl estradiol also undergoes enterohepatic cycling necessary for the activation by gut bacteria of the hormone prior to its being reabsorbed, although progestogens do not undergo this recycling. Therefore, concerns have been raised about non-rifamycin-type antibiotics, including common dentally prescribed agents such as penicillins, and their potential to reduce the effectiveness of oral contraceptives containing ethinyl estradiol. However, a recent systematic review with 29 included studies indicated that evidence derived from clinical and pharmacokinetic studies does not support the existence of drug interactions between hormonal contraception and non-rifamycin antibiotics (Simmons et al. 2018b). Concurrently, another individual study of the impact of dicloxacillin on pregnancy risk among 364 Danish females taking oral contraceptives and using dicloxacillin prior to conception determined an odds ratio for unintended pregnancy of 1.18 (95% CI 0.84–1.65), although the risk was slightly higher for users of progestogen-only contraceptives (OR 1.83, 95% CI 0.63–5.34) (Pottgard et al. 2018). Interestingly, virtually all authors of publications related to this subject advise caution and the recommendation for supplementary contraceptive measures for some or all conditions in which females are exposed to antibiotics while taking oral contraceptive agents (Taylor and Pemberton 2012).

Oral Contraceptives and Periodontal Disease. Oral contraceptive use has for quite some time been associated with gingival inflammation (Heasman and Hughes 2014), apparently beginning in the early era of oral contraception in which estrogen dosages were relatively high. However, the most recent studies demonstrate that in the era of “modern” low-dose oral contraceptive formulations, these drugs do not place patients at an increased risk for gingivitis or periodontitis (Preshaw 2000).

Oral Contraceptives and Risk of Postoperative Localized Alveolitis. The increased risk for localized alveolitis in users of oral contraceptive drugs has been well documented and known to be positively associated with the dose of estrogen in the contraceptive. It is also documented that the risk for this post-extraction complication can be reduced in users of oral contraceptives if tooth extraction is performed during the 23rd through 28th day of the pill cycle (Catellani et al. 1980). Recently, a meta-analysis has provided more robust statistical estimates of the likelihood of alveolar osteitis in oral contraceptive users (Bienek and Filliben 2016). In this review that included 29 studies, it was found that among females, oral contraceptive use nearly doubled the rate of alveolar osteitis (13.9% among users, 7.5% among nonusers). Additionally, this study demonstrated a greater risk generally for females than males, and the vast majority of the studies reviewed determined that smokers had a significantly higher risk of the complication. Therefore, the current scientific evidence suggests a need for additional vigilance in monitoring patients for the development of localized alveolitis following tooth extraction if they take oral contraceptives.

Thyroid and Anti-Thyroid Agents

The appropriate dental management of patients with thyroid disease rests upon understanding the physiological effects of the disease, recognition of the signs and symptoms that represent inadequate control of the disease, and safely using dental therapeutic agents, with an emphasis on avoiding adverse drug interactions and side effects. Worldwide, the prevalence of overt hyperthyroidism in countries with sufficient iodine supplementation ranges from 0.2% to 1.3% (Preshaw 2000). In the United States, based on the 2002 National Health and Nutrition Examination Survey (NHANES III), over hyperthyroidism occurred in 0.5% of the population, with 0.7% having subclinical hyperthyroidism, and there is a significantly higher incidence in females than males (2.7% vs. 0.23%) (Taylor et al. 2018).

There are three major classes of agents used to manage thyroid disease in outpatients (Table 3) (Katzung and Trevor 2015).

The major concerns for the dental management of patients with thyroid disease center on assessment of the stability of the patient's disease, compliance of the patient with thyroid supplementation (in cases of hypothyroidism), physical assessment of the patient at the time of the dental procedure (e.g., vital signs), recognition of the signs and symptoms of hypo- and hyperthyroidism, and planning for potential emergencies that might be expected in patients with thyroid disease. One of the major concerns has been related to the potential interaction between thyroid supplementation and vasoconstrictors used in local anesthetics.

In dental patients, in addition to the patient's medical history, hypothyroidism is associated with hypotension, tachycardia, lethargy, cold intolerance, myxedema, reduced respiratory rate, and weight gain. Significantly, hypothyroidism may also put the patient at an increased risk of seizures. Hyperthyroidism is associated with fatigue, heat intolerance, tachycardia, palpitations, nervousness, proptosis, tremor, warm skin, and weight loss.

The most recent guidelines for the management of patients with thyroid disease suggest that it be done in three phases—pretreatment, intraoperative, and posttreatment (Pinto and Glick 2002). Prior to dental treatment, the type of thyroid disease should be determined, as well as the possible presence of comorbid conditions, such as cardiovascular disease. In any case in which thyroid disease is suspected but untreated, elective dental therapy should be postponed until medical evaluation and stabilization have been completed. It is recommended that medical control of the disease be determined by obtaining

baseline thyroid-stimulating hormone levels and complete blood counts. The type of medication(s) used to treat the disease should also be determined and potential adverse drug interactions evaluated. Baseline vital signs must be assessed prior to undertaking dental treatment.

In a small study of subjects with short-term hypothyroidism assessing cardiovascular and other responses to epinephrine infusion, hypothyroid individuals had a normal response to epinephrine, in spite of their initially reduced baseline pulse rates and metabolic parameter (e.g., blood glucose levels) (Johnson et al. 1995). The investigators concluded that under physiological conditions, there did not appear to be any synergistic relationship between thyroid status and catecholamines.

During treatment, vital signs should be regularly monitored and appointment duration limited to the patient's tolerance. Local anesthetics with epinephrine are not contraindicated when medical therapy has resulted in a normal (euthyroid) state. However, one review article suggests that summation of the cardiovascular effects of epinephrine with those of excessive blood levels of thyroid hormone can occur, and the interaction was rated as category 4 (major or minor, with "possible," not established, documentation) (Yagiela 1999). Current drug literature databases (drugs.com, see Chap. "Biologic Agents") rate the interactions as "moderate," with a need for monitoring for signs and symptoms of adverse effects. A recent case report suggests that rapid intervention in suspected cases of thyroid storm can result in uneventful recovery, although medical assistance should be summoned (Lee et al. 2016).

Following treatment, caution should be exercised in regard to the increased sensitivity of the hypothyroid patient to central nervous system

Table 3 Major classes of drugs used in the treatment of thyroid disease

Drug class	Indication	Toxicity(ies)
Thyroid preparations (Levothyroxine, Liothyronine)	Hypothyroidism	Symptoms of thyroid excess (text)
Antithyroid preparations (Methimazole, Propylthiouracil)	Hyperthyroidism	GI distress, rash, agranulocytosis, hepatitis, hypothyroidism
Beta blockers (Propranolol)	Hyperthyroidism	Asthma, AV block, hypotension, bradycardia

depression, for example, with opioid analgesics. In hyperthyroid patients, precautions should be taken in regard to the use of NSAID analgesics or aspirin, due to the possible presence of cardiovascular disease and the reduced binding of thyroxine (T_4) to T_4 -binding globulin, resulting in increased levels of free thyroxine and possible thyrotoxicosis (Yagiela 1999).

Insulin

Drugs used in the management of blood glucose for control of diabetes mellitus are in wide use, and patient compliance with insulin regimens is essential in preventing loss of glycemic control and (significantly for dentistry) acute excesses (hyperglycemia) and deficiencies (hypoglycemia) in blood glucose that create emergency situations. While the management of these emergencies is beyond the scope of this chapter, it is important to understand the current scientific basis for use of the hormone insulin and its implications.

Currently, two types of insulin delivery are in general use by patients with diabetes mellitus.

Multiple daily injections have been in use for many years, while continuous subcutaneous insulin infusion device was introduced more recently. Typically, these insulin delivery techniques are based on self-monitoring of blood glucose (e.g., glucometer testing of blood samples) or real-time continuous glucose monitoring. When scrutinized in systematic reviews, clinical trials using hemoglobin A1c levels to assess glycemic control have determined that both multiple daily insulin injections and continuous subcutaneous infusion have similar effects on glycemic control, while real-time continuous glucose monitoring is superior to self-monitoring of blood glucose, again based on HbA1C level (Yeh et al. 2012).

With regard to insulin overdoses, a recent systematic review determined that while the majority of cases can be managed in the outpatient setting when recognized and treated promptly, cases involving large doses of insulin (median

dose of 900 international units) require hospitalization (Johansen and Christensen 2018). In such cases, the median hospital stay was 94 h (range 12–721 h), and treatment could have involved IV or IM glucagon, IV or IM octreotide, surgical excision, IV corticosteroids, and oral administration of complex carbohydrates, in addition to IV glucose administration. Complications of these overdoses included intermittent cerebral impairment, hypokalemia and other electrolyte anomalies, cardiac arrhythmias, and liver function. Accordingly, the management of patients who experience insulin overdose reactions in the dental office must be carefully diagnosed and managed aggressively, which may include transport and admission to the hospital.

Conclusion

The pharmacological management of dental patients with endocrine disorders varies with the severity of the disease and the type(s) of medications and/or hormonal supplements prescribed by physicians managing the endocrine disease. In patients with endocrine disorders, a careful medical history and consultation with patients' physicians are necessary not only to safely administer and prescribe dental drugs but also to assess the patient's physical status, ability to tolerate dental procedures, and anticipate emergencies that might arise in conjunction with dental treatment.

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Pharmacologic Treatment of Common Oral Mucosal Inflammatory and Ulcerative Diseases

Nadarajah Vigneswaran and Susan Muller

Oral Mucosal Inflammatory and Ulcerative Diseases of Infectious Origin

Herpes Simplex Virus Type 1 (HSV1) Infections

Description

Primary herpetic gingivostomatitis usually arises in children and young adults. Primary infection may be mild or asymptomatic; however, clinical features can present in up to 30% of patients (Arduino and Porter 2006). Infection is almost always associated with HSV-1 although primary orofacial HSV-2 infection has been reported in some older patients due to sexual practices.

Clinical Features

Primary HSV Infection

- There is a wide range of signs and symptoms including myalgia or malaise followed in 1–3 days by mucocutaneous vesicular eruptions of the lips, tongue, hard and soft palates,

buccal mucosa, and gingiva. The blisters quickly breakdown to form painful shallow ulcers surfaced by a yellow to gray pseudo-membrane with a red halo. The ulcers generally heal within 2 weeks without scarring. Other clinical findings may include lymphadenopathy, headache, coated tongue and mal-odor, loss of appetite, and hypersalivation.

- Differential diagnoses of primary HSV include herpetiform recurrent aphthous ulcerations, necrotizing ulcerative gingivitis, infectious mononucleosis, erythema multiforme minor, Cocksackie virus, and Varicella Zoster Virus infection (Table 1).

Recurrent HSV Infection

- Reactivation of the latent HSV can be triggered by many causes or may be spontaneous. Reported trigger factors include fever, invasive dental procedures, UV light, emotional stress, fatigue, immunosuppression, and trauma.
- Recurrent HSV in immunocompetent individuals typically occurs on the lips and keratinized oral mucosa such as hard palate and attached gingiva. The lesions begin as red macules that form vesicles, which are highly infectious. The vesicles break forming scabs or ulcers. In the oral cavity, due to masticatory forces, vesicles may not be seen. In addition, rather than a single ulcer, small crops of shallow ulcers can occur, chiefly on the palate. The ulcers generally heal within 2 weeks.

N. Vigneswaran (✉)

Department of Diagnostic and Biomedical Sciences
University of Texas School of Dentistry, Houston,
TX, USA

e-mail: Nadarajah.Vigneswaran@uth.tmc.edu

S. Muller

Atlanta Oral Pathology, DeKalb Medical,
Decatur, GA, USA

Table 1 Differential diagnoses for oral inflammatory and ulcerative diseases based on clinical presentation and etiology

Type	Etiology		
	Infectious	Immunologic	Other
<i>Primary acute</i>	Primary herpetic gingivostomatitis	Erythema multiforme	Traumatic ulcers
<i>Primary chronic</i>	Mucositis: Candidiasis Ulcers: Cytomegalovirus-induced oral ulcers Deep fungal infections: Oral histoplasmosis and blastomycosis	None	Oral squamous cell carcinoma Necrotizing sialometaplasia Traumatic ulcerative granuloma with stromal eosinophilia (TUGSE) Ulcerations with sequestration
<i>Recurrent acute</i>	Recurrent oral herpes simplex and herpes zoster virus infections	Recurrent aphthous stomatitis Herpes-associated erythema multiforme	None
<i>Recurrent chronic</i>	Chronic herpes simplex virus infection	Lichen planus Mucous membrane pemphigoid Pemphigus vulgaris Paraneoplastic pemphigus/ paraneoplastic autoimmune multi-organ syndrome (PNP/PAMS) Drug-mediated oral reactions	None

Chronic (Recrudescent) Herpetic Ulcers

- It is associated with immunosuppression, including solid organ and bone marrow transplants, AIDS, and long-term immunosuppressive drug therapy. Unlike recurrent HSV1 intraoral lesions in immunocompetent individuals, chronic HSV lesions can present on both keratinized and nonkeratinized mucosa and hence may be difficult to distinguish from other oral mucosal disorders including recurrent aphthous ulcers. The lesions can last for prolonged periods of time and may appear as crateriform ulcers or can exhibit a distinct linear ulceration. The ulcers may be red or have a pseudomembranous cover.

Diagnosis

- Usually, diagnosis of primary and recurrent HSV infection is based on clinical history and patient presentation. When the presentation is atypical (i.e., chronic herpetic ulcers), confirmatory laboratory testing is required.
- A Papanicolaou (Pap) test or Tzanck test can show virally altered epithelial cells obtained from the lesion. However, false negatives can occur as the test reportedly detects <60% of infections.

- Detection of viral DNA of HSV by polymerase chain reaction (PCR) is more sensitive than tissue culture and is not contingent upon the presence of viable virus.
- Serological testing of HSV is not always accurate and may give both false negatives and false positives, particularly in type-specific antibody tests for HSV-1 and HSV-2.

Therapeutic Guidelines

- Primary HSV1 infection is a self-limiting condition in an immunocompetent individual and needs only symptomatic treatment. It is important to ensure adequate hydration and nutrition (Arduino and Porter 2006). Other topical therapies include ice, lip balms, and over-the-counter topical anesthetic preparations.
- In immunocompetent individuals, therapy may be of limited benefit for primary HSV1 infection. Systemic antiviral therapy has proven beneficial if the drug is started within 48 h of onset (Magel et al. 2013, Radulescu 2016) (Table 2).
- Prophylactic antiviral therapy in immunosuppressive patients at high risk for chronic HSV is usually administered (Table 2). Sunscreen, especially to the lips, may help in preventing recurrent HSV infection.

Table 2 Commonly prescribed pharmacologic therapy for primary and recurrent oral HSV infection

Disease status	Acyclovir	Valacyclovir	Famciclovir
Immunocompetent patient			
Primary HSV infection	200 mg 5× day for 10 days	1000 mg BID for 10 days	250 mg TID for 10 days
Recurrent HSV infection	200 mg 5× day for 5 days	500 mg BID for 3 days	125 mg BID for 5 days
HSV-prophylaxis/suppression ^a	400 mg BID	500 mg QD	250 mg BID
Immunocompromised patient			
Primary HSV infection	400 mg 5× day for 10 days	N/A	N/A
Recurrent HSV infection	400 mg TID for 7–10 days	500 mg BID for 7 days	500 mg BID for 7 days
HSV-prophylaxis/suppression ^a	400 mg BID	500 mg BID	500 mg BID

^aOff-label use for suppressive anti-viral therapy for recurrent erythema multiforme. Note: use all systemic HSV medications with caution when prescribing to patients with impaired renal function and hepatic disease. For use in pre-pubescent children use acyclovir

- Both topical (acyclovir 5% ointment and penciclovir 1% cream) and systemic antiviral therapies (acyclovir, famciclovir, and valacyclovir) are approved for treating primary and recurrent oral HSV infections (Table 2).

Oral Candidiasis and Associated Oral Lesions

Description

Oral candidiasis, a fungal infection, is the most common opportunistic infection affecting the oral cavity. *Candida albicans* (*C. albicans*) is a commensal organism and is part of the normal oral flora, found in at least 35% of healthy, asymptomatic mouths. The organism becomes a pathogen when the immune system is compromised, or mucosal barriers are disrupted causing disease. Local predisposing factors include removable dentures, xerostomia, and local immunosuppression such as topical steroid or antibiotic use. Systemic predisposing factors include immunosuppression due to disease including diabetes, HIV infection, B12, or iron deficiency anemia and medications such as corticosteroids, broad spectrum antibiotics, and chemotherapy. Most oral infections are caused by *C. albicans*, but other species have been isolated including *C. tropicalis*, *C. glabrata*, and *C. krusei*. *C. albicans* is a dimorphic organism and can grow as yeast or in hyphal form, which can colonize the superficial layers of the oral epithelium.

Clinical Features

Oral candidiasis has numerous clinical presentations, and the patient may have more than one presentation at the same time.

Pseudomembranous Candidiasis

- It is the most readily recognized form of oral candidiasis that presents with white/yellow curd-like adherent plaques that can be removed with scraping. The underlying mucosa is erythematous. This form of candidiasis can occur anywhere in the oral cavity although tongue, buccal vestibule, and oropharynx are typically infected. Patients may complain of an altered taste and may have mild pain.
- Pseudomembranous candidiasis may occur as localized infection in the soft palate in asthmatic patients using inhaled corticosteroids. Oral candidiasis associated with inhaled corticosteroid use is seen more commonly in patients using high potency corticosteroid fluticasone than those treated with low potent beclomethasone.

Atrophic (Erythematous) Candidiasis

- It presents as diffuse erythema lacking any of the white curd-like pseudomembranes. Acute erythematous candidiasis results from persistent pseudomembranous candidiasis.
- The palate and dorsal tongue are the most common sites for chronic atrophic candidiasis. It commonly presents as generalized diffuse erythema with depapillation (atrophic glossitis) of the dorsal surface of the tongue. Typically, the dorsal tongue appears red and atrophic or bald with loss of the filiform papillae. Affected patients frequently complain of chronic tongue soreness and burning sensation, particularly when eating spicy foods.
- Chronic erythematous candidiasis is the most common type of candidal infection seen in the

elderly and in individuals wearing removable complete or partial dentures with acrylic denture base, commonly known as denture stomatitis.

Denture Stomatitis

- It presents as diffuse erythema under a full or partial acrylic denture and is caused by poor denture hygiene and is most commonly seen in 24-h denture wearers. The maxillary alveolus and hard palate are the most commonly affected sites. Denture stomatitis can be mistaken for a contact allergy as the erythema corresponds to the denture margins. Chronic irritation from poor fitting dentures and failure to remove the dentures while sleeping are the main causes for the development of chronic candidal infection.
- Candidal adherence to the acrylic denture bases occurs directly to the poorly fitting dentures with roughened surfaces, and preexisting bacterial plaque leads to the formation of candidal biofilm. Formation of the candidal biofilms in denture bases promotes transition from blastospores to hyphae, increasing their virulence and infectivity. Hence, poorly fitting dentures acting as reservoirs for candidal organisms contribute to recurrence of this form of candidiasis in long-term denture wearers.

Median Rhomboid Glossitis

- It presents as a central area of erythema along the midline dorsal tongue with atrophic papillae. Steroid inhalers, smoking, and poor oral hygiene are predisposing factors. Occasionally, a similar lesion can be observed on the palate where the affected tongue comes in contact.
- Median rhomboid glossitis may be asymptomatic, or patients may complain of altered taste or discomfort when eating spicy or acidic foods.

Angular Cheilitis

- It presents as erythematous fissured areas at the corners of the mouth and is a mixed fungal and bacterial infection. Predisposing factors include loss of vertical dimension, whereby

the mucosa of the upper and lower lips overlap, creating a constant warm and moist environment promoting infection.

- Although angular cheilitis can occur alone, it is often seen with intraoral candidiasis, usually the atrophic form including denture stomatitis. Patients complain of discomfort particularly when opening the mouth. Circumoral type of erythematous candidiasis, involving entire upper and lower lips and perioral skin, is seen in patients who habitually lick their lips or frequently use petroleum-based lip balms for dry lips.

Chronic Hyperplastic Candidiasis

- Presents as isolated thick white plaques that do not wipe off. Clinically, it can be confused with oral leukoplakia. It frequently occurs in individuals without any local or systemic predisposing factors, involving the retrocommisural area or lateral surface of the tongue.
- Hyperplastic candidiasis mimicking leukoplakia would resolve completely after a course of antifungal treatment. In contrast, conventional leukoplakia with dysplasia (pre-malignant lesion), which frequently have secondary candidal infection/colonization, will not resolve with a course of antifungal treatment.

Diagnosis

- Most forms of oral candidiasis can be diagnosed based on clinical findings, along with detailed medical and dental history. If the patients have had a recent history of antibiotic or steroid use, or if the presentation is an area of bright erythema underneath a denture, the diagnosis is often straightforward.
- Chronic hyperplastic candidiasis can mimic oral leukoplakia and may require biopsy to confirm the infectious nature of the clinical findings. Oral exfoliative cytology can confirm the presence of candidal hyphae and yeast forms.
- Fungal cultures can also be used, but this is rarely done except in individuals who are resistant to standard therapy. A culture can identify the exact species of *Candida* and their drug sensitivity.

Therapeutic Guidelines

- Determining the predisposing factor(s) causing oral candidiasis is the important first step, as even with appropriate antifungal medication, recurrent candidiasis can be an ongoing problem.
- Patients with complete and removable partial dentures should remove the dentures while sleeping. Frequent denture disinfection is recommended to prevent the recurrence of denture stomatitis. This can be accomplished by soaking the denture overnight in chlorhexidine, which has fungal static properties (Silva et al. 2012).
- Alternative methods for disinfecting acrylic denture bases include soaking the denture in sodium hypochlorite (1%) for 10 min or subjecting the dentures for microwave irradiation (800 W) immersed in water for 6 min (Silva et al. 2012).
- Candidiasis in patients with xerostomia may benefit from a multipronged approach as hyposalivation can have many causes. The most common cause is drug-induced xerostomia, but radiation to the head and neck and Sjögren Syndrome are also associated with xerostomia. Patients should be encouraged to maintain good oral hygiene, proper hydration, and frequent rinsing to keep the mouth moist. Over-the-counter saliva substitutes may be helpful, and in some cases, pilocarpine or cevimeline, which stimulate salivary flow, can be prescribed.
- Patients with oral candidiasis secondary to steroid inhalers should be educated on the importance of rinsing the mouth thoroughly after inhaler use or using inhalers with spacer devices.
- Patients need to be reminded to remove their dentures when using the topical treatment as the medication will not reach the affected area. In addition, denture wearers can line the inside of the denture with either nystatin or clotrimazole cream daily as an appropriate antifungal treatment for the denture.
- Topical nystatin, ketoconazole, and clotrimazole creams are also useful for angular cheilitis (Table 3). If there is a significant inflammatory component, topical antifungal-corticosteroid combination therapy with either clotrimazole-betamethasone cream or nystatin-triamcinolone cream can be used as the steroid will quickly reduce the inflammation.
- Topical or systemic antifungal therapies are usually the first line of treatment in oropharyngeal candidiasis, which are listed in Table 3 (Gupta 2013). Nystatin oral suspension or pastilles and clotrimazole troches used as topical therapy is usually effective in resolving oral candidiasis. The patient needs to be

Table 3 Commonly prescribed pharmacologic therapy for oropharyngeal candidiasis and angular cheilitis

Disease	Nystatin	Clotrimazole	Fluconazole	Posaconazole ^a
Oropharyngeal candidiasis	Oral suspension; 100,000 units/mL. 1tsp, 4× daily for 2-weeks, hold for 3 min, expectorate, no food/liquid/rinsing for 30 min ^b	Troches, 10 mg 5× day for 2-weeks, NPO 30 min ^b	Tablet, 100 mg, BID on day 1, then 100 mg/day for 13 days ^c	Oral suspension; 40 mg/mL. 100 mg (2.5 mL) PO BID on day 1, then 100 mg PO QD for 13 days ^b
Angular cheilitis	Cream; 100,000 units/g; apply topically 4× day for 2 weeks	Clotrimazole and betamethasone dipropionate; 1%/0.05% Apply topically 4× day for 2 weeks	N/A	N/A

^aProphylaxis for invasive candidiasis in severely immunocompromised and allogeneic hematopoietic stem cells transplant (aHSCT) patients and treatment for oropharyngeal candidiasis refractory to fluconazole

^bNote: If the patients wear dentures, these must be removed before rinsing

^cNote: Be aware of possible drug interactions with: warfarin, statins, oral hypoglycemic agents; may need a 4 week course in selected cases

reminded not to rinse, drink, or eat for at least 30 min after using the topical antifungal medications to prevent the drug from being diluted or washed away. Topical miconazole, a muco-adhesive tablet sold under the brand name of Oravig, is applied to the maxillary vestibule near the canine fossa once a day.

- Systemic medications for oral candidiasis include fluconazole, ketoconazole, itraconazole, posaconazole, voriconazole, and echinocandins; however, fluconazole is the first-line systemic treatment (Gupta 2013) (Table 3).
- Fluconazole is effective in oropharyngeal candidiasis and in patients who have failed topical antifungal treatment. Fluconazole is used to treat candidiasis in patients with immunosuppression from HIV infection, cancer therapy, solid organ and marrow transplants, and autoimmune disease. Fluconazole can also be used as a preventive agent in this population who are susceptible to recurrent infections. It is important to be aware of drug interactions before prescribing fluconazole, including warfarin, statins, phenytoin, proton pump inhibitors, and sulfonylureas. Fluconazole should be used with caution in patients who have impaired liver function.
- Posaconazole is recommended as prophylaxis for invasive candidiasis in severely immunocompromised and allogeneic hematopoietic stem cells transplant (aHSCT) patients. Posaconazole is the treatment of choice for oropharyngeal candidiasis refractory to fluconazole (Gupta 2013) (Table 3).

Immune-Mediated Oral Mucosal Inflammatory and Ulcerative Diseases

Recurrent Aphthous Stomatitis

Description

Recurrent aphthous stomatitis (RAS) commonly known as aphthous ulcers or canker sores is a self-limiting, chronic oral mucosal disease that presents as solitary single or multiple painful ulcers

(Vigneswaran and Rodu 2008; Stoopler and Sollecito 2014; Vigneswaran and Muller 2020). RAS is the most frequent form of recurrent solitary oral ulcers affecting approximately 20% of the general population with a slight female predilection (Vigneswaran and Rodu 2008; Vigneswaran and Muller 2020). Most RAS patients are healthy young (age range 10–19 years) individuals without any underlying systemic immune deregulation (Vigneswaran and Rodu 2008).

Clinical Findings

Recurrent Aphthous Stomatitis (RAS)

- RAS is usually preceded by prodromal symptoms of burning or tingling for up to 48 h before lesion onset. The ulcers generally occur on nonkeratinized mucosa and appear as an ulcer with a white-yellow pseudomembranous covering surrounded by an erythematous halo.
- RAS is classified into three forms based on size, site, duration, and the tendency to heal with scarring: minor (MiRAS), major (MaRAS), and herpetiform (HeRAS) types.
- Minor RAS (MiRAS) is the most common type seen in 80% of RAS patients. The MiRAS ulcers are usually less than 1 cm in diameter and heal within 2 weeks.
- Major RAS (MaRAS) is a rare and more severe form of RAS, presenting as continuously recurring ulcers measuring more than 1 cm in diameter with a healing time exceeding 2–3 weeks. These ulcers tend to heal with scarring and have predilection to lips and posterior portion of the oral cavity (soft palate and oropharyngeal fauces).
- Herpetiform RAS (HeRAS) is the least common form of RAS that appear as numerous small ulcers (2–3 mm in diameter) that can affect multiple intraoral sites. HeRAS tends to occur at a later age (> 20 years) than other clinical types with is more common in females than males.

Behçet's Syndrome

- Behçet's syndrome is a rare systemic vasculitis characterized by oral and genital ulcers and ocular lesions, including uveitis. Vascular

manifestations include venous thrombosis in up to 30% of cases (Escudier et al. 2006).

- Arthritis or arthralgia is frequently the presenting feature most commonly of the knees and ankles along with other systemic manifestations. The cause is unknown but is believed to be due to an autoimmune process triggered by an environmental or infectious agent in a genetically predisposed patient. HLA-B51 on chromosome 6p is the most strongly associated risk factor for Behçet's syndrome.
- Patients presenting with constant recurrence of >3 major aphthous ulcer are diagnosed as "complex aphthosis," which is considered to be a "forme fruste" (attenuated manifestation) of Behçet's syndrome (Jorizzo et al. 1985; Letsinger et al. 2005).

Periodic Fever Adenitis Pharyngitis Aphthous Ulcer (PFAPA) Syndrome

- PFAPA is a recurrent fever syndrome characterized by episodes of fever lasting between 3 and 6 days, associated with at least one additional clinical finding: aphthous stomatitis, cervical adenitis, and pharyngitis (Wang et al. 2021). PFAPA regularly recurs every 3–8 weeks. The age of onset varies but usually presents in children and adolescents. Treatment options for PFAPA include nonsteroidal anti-inflammatory drugs, corticosteroids, and tonsillectomy (Wang et al. 2021).

Drug-Induced Aphthous-Like Oral Ulcers

- In rare instances, older patients without a history of RAS develop aphthous-like ulcers after starting to take a new drug. Anti-angina drug nicorandil is the most frequently cited drug to cause aphthous-like ulcerations (Vigneswaran and Muller 2020).
- Nonsteroidal anti-inflammatory drugs such as diclofenac and flurbiprofen have been reported to cause oral and genital ulcers. The angiotensin-converting enzyme (ACE) inhibitor captopril and the angiotensin II receptor blocker losartan may also cause aphthous-like ulcers (Vigneswaran and Muller 2020).

- Other systemic drugs implicated in oral ulcers include a number of nonsteroidal anti-inflammatory drugs (i.e., naproxen, flurbiprofen, piroxicam, and rofecoxib), proton pump inhibitor esomeprazole (Nexium), and immunosuppressive drugs (i.e., sirolimus, everolimus, methotrexate) (Madinier et al. 2000).

Diagnosis

- In most cases, RAS can be diagnosed based on history and clinical presentation (Table 1). There is no specific laboratory test for diagnosing RAS. However, it is important to differentiate oral ulcers due to local factors, infections, autoimmunity/hypersensitivity, and malignancy (Table 1). Microscopic features of RAS ulcers are nonspecific and may appear similar to a traumatic ulcer. If the ulcers appear atypical, then a biopsy is recommended to exclude ulcers of infectious origin or malignancy.
- HeRAS can clinically be confused with intraoral recurrent HSV-1 infection. Unlike HeRAS, which arises in nonkeratinized mucosa, recurrent HSV-1 ulcers in an immunocompetent patient occur on keratinized mucosa.
- The diagnosis of Behçet's syndrome is by clinical presentation and exclusion of other diagnoses. No specific laboratory tests exist for Behçet's syndrome. The International Study Group guidelines for diagnosing BD include frequent recurrence of oral aphthosis (mandatory criteria), associated with at least two other criteria among a list of genital, cutaneous, and ocular manifestations (i.e., genital ulcers, erythema nodosum, uveitis).
- No specific test exists for the diagnosis of PFAPA which results in delayed diagnosis. Response to corticosteroid therapy is often used to confirm diagnosis. Laboratory testing can rule out cyclic neutropenia or one of the hereditary periodic fever syndromes.
- Identifying and discontinuing the offending medication with resolution of ulcers without recurrence is the diagnostic criteria for drug-induced aphthous-like ulcers.

Therapeutic Guidelines

- Although there is no permanent cure for RAS, recurrence of oral ulcers becomes less frequent and may even resolve completely with advancing age.
- Patients should be advised to avoid trauma that can arise with braces, dental appliances, and malposed dentition. Patients should be encouraged to use a soft-bristled toothbrush and toothpaste without the detergent sodium lauryl sulfate (SLS). Some patients will report an association with certain foods such as nuts, chocolate, fruits, and vegetables, as well as ingredients in some toothpastes, gums, and hard candies. The culprit is different for different patients, and keeping a food diary may help to isolate a specific item with an outbreak of aphthous ulcers.
- For drug-induced aphthous-like ulcer, the offending medication should be discontinued, but symptomatic treatment of the ulcer is similar to aphthous ulcers.
- Drug treatment for RAS is generally based on the severity of disease, the impact on the patient's quality of life, accompanying systemic medical conditions, and potential adverse side effects of the therapeutic agent. In evaluating the treatment efficacy of drug therapy for RAS, it is important to realize that aphthous ulcers heal spontaneously and that the frequency of recurrence, lesion duration, and degree of pain perception vary among patients. This makes it difficult to critically evaluate treatment regimens that claim a reduction in frequency of recurrence, pain, or time to healing. Furthermore, many treatment recommendations are based on individual case studies without adequate controls and thus are of questionable validity.
- Categories of aphthous ulcer medications include over-the-counter topical pain preparations, antimicrobial agents, topical and intralesional glucocorticoids, systemic glucocorticoids, and systemic immunosuppressive agents (Table 4). In addition, some reports advocate physical manipulation by surgery or cauterizing agents.
- Use of antimicrobial oral rinses (i.e., chlorhexidine), topical analgesics, topical occlusive pastes, and topical corticosteroid remains the first line of treatment for minor RAS.
- Over-the-counter drugs such as Zilactin-B® adhesive gel (contains 10% benzocaine) and Orabase-B® paste (contains 20% benzocaine) provide protective coatings with topical anesthetics. For symptomatic relief, apply either of these drugs topically to the ulcers four times day until resolution of the ulcers. Topical application of 5% amlexanox (Aphthasol), which is an anti-allergic and anti-inflammatory drug reduces pain and accelerates ulcer healing and is recommended for patients who develop MiRAS infrequently but have a low tolerance to pain.
- In North America, topical, intralesional, and systemic glucocorticoids are the most commonly prescribed therapy for RAS depend-

Table 4 Commonly prescribed pharmacologic therapy for recurrent aphthous stomatitis (RAS)

Disease	Topical corticosteroid	Intralesional corticosteroid injection	Systemic corticosteroid	Pentoxifylline
Minor RAS	Clobetasol propionate or fluocinonide 0.05% gel/cream BID-TID	N/A	N/A	N/A
Major RAS < 3 ulcers Less frequent	^a Clobetasol propionate 0.05% gel 4–5× day for 2-weeks	N/A	N/A	N/A
Complex aphthosis and Major RAS of Behcet's syndrome	^a Dexamethasone oral solution, 0.05 mg/5 mL 4× daily for 10 days; hold for 3 min, expectorate, no food/liquid or rinsing for 30 min	Triamcinolone acetone 10 mg/mL (Kenalog-10); 0.4 mg/cm ²	^a Prednisolone 1.0 mg/kg or 0.16 mg/kg dexamethasone oral solution with a rapid taper for 2–3 weeks	400 mg, tid for 3–6 months for maintenance to prevent or reduce recurrences

^aPrescribe chlorhexidine oral rinse or fluconazole for antifungal prophylaxis

ing on the frequency of the recurrence, type, and number of ulcers (Wolverton 2013) (Table 4).

- Corticosteroid-sparing therapies for severe major RAS also known as complex aphthosis and major RAS associated with Behcet's syndrome include vasoactive agent pentoxifylline, immunosuppressive drugs such as thalidomide, azathioprine, cyclophosphamide, methotrexate, and cyclosporine A (Table 4). Severe cases of corticosteroid-refractory cases of major RAS can be treated also with the anti-TNF- α drugs such as Infliximab, Etanercept, and Adalimumab and anti-interleukin 1 drug Anakinra (Liu et al. 2022). These drugs should be reserved for use by dermatologists, internists, and rheumatologists because of the serious adverse effects associated with these drugs.

Erythema Multiforme and Related Disorders

Description

Erythema multiforme (EM) is an acute, self-limiting immune-mediated mucocutaneous disease presenting with distinct cutaneous targetoid lesions with or without oral, ocular and genital mucosal lesions (Bagan et al. 2005; Lamoreux et al. 2006). EM is considered a cell-mediated type IV hypersensitivity reaction triggered most commonly by infection. Recurrent herpes simplex virus (HSV) infection is the trigger for 90% EM cases (Sokumbi and Wetter 2012; Yuan and Woo 2015).

Clinical Findings

- Erythema multiforme is more common among younger males, and it is rare in young children (<3 years) and older adults (>50 years) (Farthing et al. 2005).
- EM typically presents with an abrupt onset of cutaneous eruptions and mucosal ulcerations without any prodromal symptoms. Based on the increasing severity, EM is subdivided into three clinical types: (1) EM minor: less severe form, with localized cutaneous lesions

with minimal or no mucosal involvement; (2) EM major: severe form with generalized cutaneous lesions with mucosal involvement; and (3) Stevens-Johnson Syndrome (SJS) and toxic epidermal necrolysis (TEN): most severe form; mucosal lesions are similar to EM major with different types of cutaneous lesions.

- Infectious agents are the most common cause for precipitating EM in children and young adults, whereas EM in adults is triggered by medications and environmental chemicals.
- Drugs are the most common cause for triggering EM major, Stevens-Johnson syndrome, and toxic epidermal necrolysis. The common drugs implicated in precipitating EM are anti-convulsants, antibiotics, and nonsteroidal anti-inflammatory drugs.
- Almost 50% of EM cases are idiopathic, occurring without any known triggers. Subclinical HSV infection is considered the most likely trigger for idiopathic EM. Frequent occurrence of EM (> 6 episodes per year) is linked in most cases to HSV infections, which is known as recurrent herpes-associated EM (HAEM).
- The "target" or "iris" lesions are the hallmark of EM and appear as fixed, symmetrical, round lesions with concentric rings and central duskiness, especially on the palms. Skin lesions present as itchy dull-red, purpuric macules, or urticarial plaque that may progress to vesicle or bulla. The Nikolsky sign is negative in the affected skin or mucosa. Oral mucosal lesions present as painful diffuse erythema, with or without hemorrhagic ulcers.
- The oral mucosal involvement in Stevens-Johnson syndrome is more severe and more extensive than that of EM major. Oral ulcers of EM major commonly present as solitary ulcers closely resembling aphthous ulcers.

EM-Related Conditions

Reactive infectious mucocutaneous eruptions (RIME): It presents as severe erythematous erosive and ulcerative lesions involving oral, genital and ocular mucosal surfaces with limited or no cutaneous involvement (Song et al. 2021; Pan

and Hussain 2023). Encountered commonly among male children and adolescents with a mean age of 12 years, RIME is triggered by non-HSV respiratory viral infection (Pan and Hussain 2023).

Mycoplasma pneumoniae-induced rash and mucositis (MIRM): *Mycoplasma pneumoniae* (MP) is a respiratory pathogen implicated in the community acquired pneumonia. Approximately 25% of the patients afflicted with *Mycoplasma pneumoniae* present with extrapulmonary manifestations, which includes oral erosive and ulcerative mucositis with or without cutaneous rashes. MIRM is frequently diagnosed during winter months among male children and adolescents. Patients initially experience prodromal symptoms including fever, malaise, and cough followed by oral erosive and ulcerative mucositis (Lofgren and Lenkeit 2021).

Diagnosis

- EM is diagnosed on the clinical findings, and there is no specific laboratory test for diagnosing erythema multiforme. Laboratory testing and cultures may be indicated in recurrent cases of EM to identify the microbial pathogen precipitating the EM. HSV-PCR can be performed in the swab obtained from oral lesions to confirm the diagnosis.
- Oral lesions of RIME and MIRM closely mimic oral lesions of immunologically mediated vesiculobullous diseases. Biopsies of non-ulcerated mucosa for routine histopathologic and direct immunofluorescent studies are necessary to confirm the diagnosis of erythema multiforme and to rule out any other immunologically mediated mucocutaneous disease (Pan and Hussain 2023).

Therapeutic Guidelines

- Erythema multiforme has an uncomplicated clinical course in most patients, except in patients with an immunocompromised status who are at risk for developing secondary bacterial infections of the skin or the mucosa. The most important step in the treatment of patients with EM is identifying and eliminating the causative drug(s) or agent.

- Antimicrobial (Chlorhexidine 0.12%) and local anesthetic oral rinses are recommended to reduce the pain and prevent secondary infection of the oral ulcerations.
- Severe cases of erythema multiforme major, SJS, and TEN require hospitalization in a burn unit for managing complications such as impaired oral intake, dehydration, and secondary infection.
- There is no consensus on the use of systemic corticosteroid for treating SJS and TEN patients. Use of systemic corticosteroid in SJS and TEN patients with widespread skin sloughing (>10% of total body surface) is associated with higher fatality rates due to increased risk for infection and sepsis.
- If systemic corticosteroid therapy is indicated for SJS and TEN patients, it should be started early in the disease course with 2–2.5 mg/kg/daily IV methylprednisolone in divided doses followed by rapid tapering until new blister formations cease. Corticosteroid-sparing treatments for SJS and TEN patients include cyclosporine (5 mg/kg/daily) and IV-immunoglobulins.
- Severity of oral mucosal involvement in EM may vary significantly. Patients with severe oral mucosal ulcerations may need a short course of prednisolone (40–60/day) tapered over 2–3 weeks. Patients with minor oral ulcerations and erosions can be managed with a high-potency topical corticosteroid gel (i.e., Clobetasol 0.05%) or topical corticosteroid oral rinse (Dexamethasone oral solution, 0.5 mg/5 mL). Therapeutic dose and duration are similar to the ones used for treating major RAS (see Table 4).
- The standard of care for patients with HSV-associated recurrent EM and idiopathic recurrent EM is antiviral prophylaxis with acyclovir (400 mg PO, bid) or Valacyclovir (500 mg PO, bid) or famciclovir (250 mg PO, BID) for more than 6 months (Table 2).
- Recurrent EM unresponsive to antiviral prophylaxis may need treatment with systemic immune modulating drugs such as azathioprine, dapsone, mycophenolate

mofetil, and cyclosporine. However, the effectiveness of these drugs in controlling the recurrence of EM has not been validated in well-controlled clinical trials. Moreover, these drugs are associated with significant side effects and hence should be used cautiously.

- Treatments for **MIRM** include systemic macrolide antibiotics, intravenous immunoglobulin, and cyclosporin (Chen and Li 2022).
- Currently, there is no viral prophylactic therapy for **RIME** because it can be triggered by several types of respiratory viral pathogens, which include adenovirus, enterovirus, seasonal, and COVID coronavirus.

Oral Lichen Planus and Lichenoid Oral Mucositis

Description

Lichen planus (LP) is a relatively common immunologically mediated mucocutaneous inflammatory and ulcerative disease affecting approximately 1–5% of the adult population (Gorouhi et al. 2014). Although LP is suspected to be an autoimmune disease, the target antigen is unknown (Didona et al. 2022). Autoantibodies against Dsg-1 and -3, a pathognomonic feature of pemphigus vulgaris, have been reported to be positive in patients with oral lichen planus (Didona and Hertl 2022). However, the pathogenic role of these autoantibodies in oral lichen planus is not known. LP affects skin, nails, hair, and mucosal surface lined by stratified squamous epithelium but does not involve the internal organs (Gorouhi et al. 2014). The prevalence of oral LP is higher in patients with chronic hepatitis C virus (HCV) infections, and hence, oral LP is considered one of the extrahepatic manifestations of chronic HCV (Nagao et al. 2016).

Clinical Findings

- Mucosal LP most commonly involves the oral cavity (oral LP) and in rare instances may affect the oropharynx, esophagus, and vulvo-vaginal mucosa.

- Oral LP usually develops in adults over the age of 40 with a strong female predilection. The following intraoral sites are, in descending order, the most common sites of involvement for LP: gingiva, buccal mucosa, tongue, lips, and palate.
- Oral LP has several clinical types that include reticular, erosive, and ulcerative variants. The reticular variant is the most common clinical type of oral LP and presents as lacy white keratotic striae and patches. These lesions exhibit symmetrical distributions involving bilateral buccal mucosa, lateral and ventral surface of the tongue, and lips. The reticular variant of LP is asymptomatic and usually detected as an incidental finding during a routine oral examination.
- Erosive LP presents as painful, diffuse erosive and ulcerative areas of erythema with white striae and patches at the periphery. Erosive LP frequently affects the gingiva presenting as desquamative gingivitis. Chronic irritation or trauma exacerbates oral lesions of lichen planus, which is known as an isomorphic response, or Koebner's phenomenon. Lichen planus involving the gingiva (desquamative gingivitis) tends to be more active in areas with fixed prosthesis (i.e., crown, bridges, and dental implants) compared to natural teeth.
- Only 15–20% of patients with oral lichen planus present with lichen planus affecting the skin (Didona et al. 2022). On the other hand, 60% of patients with cutaneous lichen planus will have oral mucosal involvement. Genital mucosal involvement is seen in 25–30% of patients with oral lichen planus (Didona et al. 2022).

Diagnosis

- The clinical differential diagnoses for oral LP include a lichenoid reaction to medications, contact hypersensitivity reaction to local allergens (i.e., cinnamon flavoring) or amalgam, discoid lupus erythematosus, chronic graft-versus-host disease (cGVHD), and oral potentially malignant disorders (Gillenwater et al. 2013).

- The most common group of drugs implicated in triggering oral lichenoid drug reactions are as follows:
 - Antihypertensive and antiarrhythmic agents (i.e., hydrochlorothiazide, beta blockers, angiotensin-converting enzyme inhibitors, Quinidine).
 - Anticonvulsants (i.e., Carbamazepine).
 - Antimalarial agents.
 - Nonsteroidal anti-inflammatory drugs.
 - Drugs used in the treatment of arthritis and gout (i.e., Sulfadiazine, Allopurinol, Penicillamine, Gold).
 - Anti-programmed cell death 1 (PD-1) and anti-programmed cell death-ligand 1 (PD-L1) immunotherapy.
 - Therapeutic monoclonal antibodies used for treating chronic immune mediated inflammatory diseases.
- A diagnosis of lichen planus is made on the basis of the typical clinical appearance and an incisional biopsy of the representative and non-ulcerated lesion. An incisional biopsy for routine histopathology and direct immunofluorescence testing is necessary to distinguish oral LP from oral epithelial dysplasia, lupus erythematosus, and mucous membrane pemphigoid.
- Clinicopathologic correlation is necessary to distinguish idiopathic oral LP from lichenoid mucositis triggered by medications, amalgam, local allergens (i.e., cinnamon), chronic GVHD, and chronic hepatitis C infections.
- Oral potentially malignant disorders with epithelial dysplasia and proliferative verrucous leukoplakias may present with a dense band of lymphocytic infiltrate at the epithelial-connective tissue interface (interface mucositis) closely mimicking lichen planus resulting in misdiagnosis as lichen planus.
- Lichenoid drug reactions are a group of potentially malignant disorders such as proliferative verrucous leukoplakia can be misdiagnosed clinically and microscopically as oral LP (Muller 2017). Hence, patients with both symptomatic and asymptomatic oral LP should undergo regular periodic follow-up and biopsy of any suspicious lesions.
- Although no curative treatments are currently available for treating oral LP, topical or systemic immunomodulating drugs are used to control symptomatic oral lesions with inflammation, atrophy, and erosion and to aid healing of the ulcerated lesions (Gonzalez-Moles et al. 2018).
- If a lichenoid drug reaction is suspected, changing the offending medication may result in the resolution of the lesions. Lichenoid contact hypersensitivity reaction to amalgam would resolve when the amalgam restoration is replaced with composite. If cinnamon-induced contact lichenoid hypersensitivity reaction is suspected, patients should be advised to discontinue the use of cinnamon flavored oral hygiene products, foods, and chewing gums. Elimination of chronic mucosal trauma and dry mouth would also help in preventing exacerbation of oral LP.
- Patients diagnosed with oral LP may need to be screened for chronic hepatitis C infection. Treatment of chronic hepatitis C infection using “direct-acting” antiviral medications is reported to cure the associated oral LP lesions (Scelza et al. 2022).
- Corticosteroid remains the first line of therapy for oral LP. Localized and less severe oral LP lesions are best managed by the use of a high-potency topical corticosteroid gel or oral solutions (Table 5).
- Lichen planus involving the gingiva (desquamative gingivitis) is best managed using a topical high-potency corticosteroid applied using medication carrier trays. Use of a medication carrier tray to apply corticosteroid topically to the affected gingiva not only improves the local absorption of the medication but also prevents the drug dissolving in the saliva and being swallowed.

Therapeutic Guidelines

- The treatment of patients with oral LP must be individualized based on the severity of symptoms, extent of the oral lesions, and response to treatment (Table 5). Patients presenting with asymptomatic oral LP do not need any drug therapy. Evolving oral potentially malig-

Table 5 Commonly prescribed pharmacologic therapy for oral lichen planus (OLP)

Disease	Topical corticosteroid	Intralesional corticosteroid injection	Systemic corticosteroid	Tacrolimus	Others
Erosive OLP-mild and localized	^a Fluocinonide or Clobetasol propionate gel 0.05%; 3–4 times/day for 3–4 weeks	N/A	N/A	^b Tacrolimus ointment; 0.1%, 3–4 times/day for 2–3 weeks	N/A
^c Gingivitis-associated OLP	^a Clobetasol propionate gel 0.05%; 2–3 times/day using medication carrier tray for 2–3 weeks	N/A	N/A	^b Tacrolimus ointment; 0.1%, 2–3 times/day for 2 weeks using medication carrier tray	^b Doxycycline 100 mg, QD for 2–6 months
Erosive OLP-moderate and widespread	^a Dexamethasone oral solution, 0.05 mg/5 mL 4× daily for 2–3 weeks swish and spit	N/A	N/A	^b Tacrolimus aqueous oral rinse; 1 mg/1 L of water; rinse for 2 min and spit; 4× daily for 3 weeks	N/A
Severe erosive ulcerative OLP	N/A	Localized: Triamcinolone acetonide 40 mg/mL (Kenalog-40); 0.1–0.2 mL/cm ²	^a Widespread: Prednisolone 1.0 mg/kg or 0.16 mg/kg dexamethasone oral solution with slow tapering for 3–6 weeks	N/A	^b Hydroxychloroquine (Plaquenil) 200–400 mg/day for 1–6 months

^aPrescribe chlorhexidine oral rinse or fluconazole for antifungal prophylaxis

^bOff-label use; increased risk of visual field defects, morbilliform rash

^cFor managing gingivitis associated with lichen planus doxycycline (anti-collagenolytic effect) is used for the initial 2–6 months of treatment and then topical steroids are used for maintenance. Important: remember that doxycycline may decrease the effectiveness of birth control pills, so those patients will need supplemental birth control. Also, remember to warn about possible photosensitivity

- Severe generalized erosive and ulcerative oral LP may need treatment with systemic corticosteroid therapy, either prednisolone (0.5 mg/Kg) or equivalent dose of dexamethasone oral solution for 4–6 weeks (Table 5). The dosage and duration of the treatment vary depending on the severity and the extent of the lesions. Intralesional injection of triamcinolone acetonide suspension is the preferred choice of treatment for severe localized ulcerative lesions of oral LP, minimizing the adverse effects associated with systemic corticosteroid treatment (Table 5).
- Prophylactic antifungal treatment should be co-administered with topical or systemic corticosteroid treatment for oral LP to prevent the development of candidiasis secondary to corticosteroid treatment (Table 5).
- Patients with oral LP who are unresponsive to corticosteroid therapy or develop severe adverse effects can be treated with other topical and systemic immunomodulating drugs, such as cyclosporine, tacrolimus, mycophenolate, methotrexate, and hydroxychloroquine (Plaquenil) (Table 5).
- Various types of biological therapies targeting the underlying immune-dysregulation have been used to treat oral LP refractory to conventional treatments. Apremilast is an oral phosphodiesterase type 4 inhibitor that diminishes the production of pro-inflammatory cytokines TNF-alpha, IFN-gamma, IL-2, IL-5, IL-8, and IL-12 implicated in the pathogenesis of LP. Potential therapeutic benefits of apremilast for oral LP were initially reported in case series and recently a multi-

center retrospective study (Didona et al. 2022). Other biologicals used for treating refractory oral LP include anti-TNF-alpha (i.e., infliximab, etanercept, adalimumab), anti-IL17 (i.e., Secukinumab), anti-IL12/13 (i.e., ustekinumab), and anti-IL-23 (i.e., guselkumab) drugs (Didona et al. 2022). Recent studies have reported the use of small-molecule inhibitors targeting Janus kinase (JK) pathway (i.e., tofacitinib, baricitinib, ruxolitinib, upadacitinib) in treating LP (Didona et al. 2022). However, only 10% patients with oral LP participated in the JK-inhibitor studies.

- Unfortunately, larger randomized trials to support the effectiveness of these biologicals are lacking, and many treatments are recommended based on anecdotal evidence. Most of these medications are associated with several adverse effects including immunosuppression and should be used under close medical supervision. Moreover, biological targeting TNF-alpha and small molecular inhibitors of Janus kinase used for treating other autoimmune diseases are reported to trigger oral LP (lichenoid drug reaction).
- In a number of cases, oral LP may undergo spontaneous remission or may remain inactive for years with occasional flare-up resulting in symptomatic lesions. Flare up of oral LP is associated with a long list of triggers, the most important being stress and anxiety (Chen et al. 2017).

Mucous Membrane Pemphigoid

Description

Mucous membrane pemphigoid (MMP), also known as cicatricial pemphigoid, is a chronic autoimmune disease characterized by development of painful vesiculobullous lesions predominantly involving the mucosa. MMP is a humoral autoimmune disorder in which affected patients have autoantibodies directed against specific adhesion molecules localized within the hemidesmosomes of the basal keratinocytes and in the basal lamina of the basement membrane

(Bickle et al. 2002; Bagan et al. 2005; Xu et al. 2013; Sultan et al. 2017; Du et al. 2022). The true incidence of MMP is unknown, but it is a rare disease compared to oral lichen planus. Other muco-cutaneous immunobullous diseases that may reveal similar clinical and microscopic findings to MMP include bullous pemphigoid (BP), linear IgA disease, epidermolysis bullosa acquisita and lichen planus pemphigoides. Approximately 20% of the patients with bullous pemphigoid have oral lesions. Lichen planus pemphigoides share microscopic features of lichen planus (i.e., lichenoid infiltrate), bound and circulating autoantibodies against BPAg1 (Hubner et al. 2019).

Clinical Findings

- MMP is more common among women with a female to male ratio of 2:1 and frequently affects the elderly population (age range: 60–80 years).
- MMP may involve all mucosal sites surfaced by stratified squamous epithelium but rarely affects the skin. Oral (90% of cases) and ocular mucosa (65% of the cases) are the most commonly affected sites. MMP involving the genital, esophageal, nasopharyngeal, and laryngeal mucosal surfaces are less frequent.
- MMP patients with lesions limited to oral mucosa have a better prognosis compared to patients with MMP involving other mucosal surfaces. The disease severity and extent of the involvement is highly variable from mild cases with gingival lesions only (low risk) to severe cases with ocular, genital, and esophageal mucosa (high risk) with increased morbidity and poor prognosis (Du et al. 2022).
- Oral mucosal lesions of MMP initially present as blisters and bullae, which are short-lived and rapidly break into painful ulcers.
- The most common oral manifestation of MMP is desquamative gingivitis, which presents as swollen bright red attached gingiva with focally denuded and ulcerated areas extending into the unattached gingival mucosa. More

than 85% of patients with MMP have gingival involvement. In contrast, only 25% of patients with oral lichen planus and 18% of patients with pemphigus vulgaris have gingival involvement.

- The palate represents the second most common site affected by MMP and the affected area appears as localized erythema with superficial chronic ulcers covered by necrotic pseudomembrane, which may heal with scars.
- Scarring of the ocular lesions of MMP may lead to blindness.

Diagnosis

- Clinical, histopathologic, and immunopathologic studies are necessary for diagnosing MMP and distinguishing from other immune-mediated mucocutaneous diseases such as lichen planus, pemphigus vulgaris, and erythema multiforme.
- An incisional biopsy of the affected and non-ulcerated mucosa should be taken for routine histopathologic and direct immunofluorescence testing. Microscopically, MMP exhibits a subepithelial split with a chronic inflammatory cell infiltrate with a few neutrophils and eosinophils.
- Direct immunofluorescence studies are positive for a linear deposition of IgG, C3, and occasionally IgA along the basement membrane zone. Indirect immunofluorescence testing for the detection of circulating autoantibodies is not indicated for MMP because the titer of these autoantibodies is low in MMP patients and does not correlate with the disease activity.

Therapeutic Guidelines

- Patients diagnosed with MMP involving the oral mucosa should consult an ophthalmologist to rule out conjunctival involvement. MMP patients experiencing soreness, pain, and ulcerations of genital mucosa or other anatomical sites should be referred to the appropriate specialist for further evaluation, assessment, and/or treatment (Taylor et al.

2015).

- Similar to other immunologically mediated mucocutaneous diseases affecting the oral mucosa, topical and systemic corticosteroids remain the mainstay of MMP therapy. For choosing the appropriate therapy, MMP patients are subdivided into “low risk” and “high risk” based on the severity and the extent of the disease involvement.
- Low-risk MMP patients with limited oral mucosal and gingival involvement are typically managed by moderate- to high-potency topical corticosteroids (Table 6).
- Desquamative gingival lesions associated with MMP are managed more effectively with topical corticosteroid gels applied using the medication carrier trays. Low-risk MMP patients with diffuse oral ulcerations who are not responding adequately for topical corticosteroid treatment are treated with a low dose of systemic corticosteroid therapy with antifungal prophylaxis (Table 6). Alternatively, localized oral ulcerative MMP can be treated with an intralesional injection of triamcinolone acetonide (Kalinska-Bienias et al. 2016).
- High-risk MMP patients with severe disease are treated with a combination systemic corticosteroid and cyclophosphamide.
- Corticosteroid-sparing treatments are necessary in low-risk MMP patients to reduce corticosteroid-related adverse effects, especially in patients with diabetes mellitus.
- Treatment with a combination of doxycycline and nicotinamide is effective, safe, and well-tolerated in low-risk MMP patients with mild oral disease.
- Corticosteroid-sparing agents used in the treatment of high-risk MMP patients include dapsone, azathioprine, mycophenolate mofetil, rituximab, and intravenous immunoglobulin (IVIg).
- There is insufficient evidence to determine optimal therapies for MMP and the advantages of using corticosteroid-sparing therapies over systemic corticosteroid monotherapy.

Table 6 Commonly prescribed pharmacologic therapy for mucous membrane pemphigoid (MMP)

^a Gingivitis associated MMP	Low-risk MMP with mild to moderate oral ulcerations	High-risk MMP with severe oral ulcerations, ocular and other mucosal involvement
^b Clobetasol propionate gel 0.05%; 2–3 times/day using medication carrier tray for 2–3 weeks	^b Mild disease with widespread: Dexamethasone oral solution, 0.05 mg/5 mL 4× daily for 2–3 weeks swish and spit	^b Prednisone (0.5–1.5 mg/kg/day with slow tapering for 3–6 months) + cyclophosphamide (1–2 mg/kg/day for 1 year)
^c Doxycycline 100 mg, QD for 2–6 months	Localized chronic and painful ulcers: Triamcinolone acetonide 40 mg/mL (Kenalog-40); 0.1–0.2 mL/cm ² ; can be repeated every 4–8 weeks	^b Prednisone (0.5–1.5 mg/kg/day with slow tapering for 3–6 months) + azathioprine (1–2 mg/kg/day for 1 year)
	^b Moderate disease with widespread: Prednisolone 1.0 mg/kg or dexamethasone oral solution (0.16 mg/kg) with slow tapering for 3–6 weeks	
	Mild disease with widespread: Steroid sparing Rx: ^c tetracycline (1–2 g/day) + nicotinamide (2–2.5 g/day) 1–2 months	
	Moderate disease with widespread: Steroid sparing Rx Dapsone, start with 25 mg/day and increase daily dose by 25 mg every 2 weeks to 75–100 mg/day	

^aFor managing gingivitis associated with MMP doxycycline (anti-collagenolytic effect) is used for the initial 2–6 months of treatment and then topical steroids are used for maintenance. Important: remember that doxycycline may decrease the effectiveness of birth control pills, so those patients will need supplemental birth control. Also, remember to warn about possible photosensitivity

^bPrescribe chlorhexidine oral rinse or fluconazole for antifungal prophylaxis

^cOff-label use

Pemphigus Vulgaris

Description

Pemphigus vulgaris (PV) is an extremely debilitating, potentially life-threatening, chronic vesiculobullous disease of the skin and mucous membrane (Black et al. 2005; Didona et al. 2019). It is a humorally mediated autoimmune disorder caused by circulating antibodies against desmosomal proteins, desmogleins-1 and -3, resulting in acantholysis and intraepithelial blistering. In rare instances, PV can be induced by certain drugs such as penicillamine and angiotensin-converting enzyme inhibitors.

Paraneoplastic pemphigus (PNP), also known as paraneoplastic autoimmune multi-organ syndrome (PNP/PAMS), is a rare life-threatening autoimmune disease with mucocutaneous and multi-organ involvement seen in patients with lymphoproliferative malignancies (i.e., leukemia or lymphoma) (Amber et al. 2018; Flood et al.

2018). PNP is a rare disorder that clinically and microscopically resembles pemphigus, and hence, it is regarded as a “mucocutaneous phenotype” of PAMS (Amber et al. 2018). Both humoral and cell-mediated autoimmune responses are implicated in the pathogenesis of PNP/PAMS contrasting the PV, which exclusively mediated by humoral immunity (Amber et al. 2018). Patients with PNP have serum autoantibodies against members of the plakin family of proteins, namely, envoplakin, periplakin, desmoplakin I and II, and, less frequently, with BP230, plectin, and epiplakin (Amber et al. 2018).

Clinical Findings

- PV is a rare vesiculobullous and ulcerative disorder affecting the oral mucosa with an estimated incidence of one to ten people per million worldwide.
- Although PV mainly afflicts adults between the ages of 40–60 years, it may be seen in chil-

dren and young adults. PV does not have a gender predilection. PV is more common among persons of Jewish, Mediterranean, and Asian descent.

- The oral cavity is the primary and initial site of disease manifestation in the majority of PV patients; skin lesions do not appear until months or even years later. The bullous lesions of the skin remain localized at early stages and become widespread after 6–12 months if left untreated. PV may also affect anogenital mucosa, nasopharynx, pharynx, and esophagus.
- Oral lesions of PV develop as vesicles or small bullae that burst rapidly leaving painful erosions and ulcers. Established PV lesions may resemble major aphthous ulcers or large superficial ulcers surfaced by a necrotic pseudomembrane with erythematous and irregular outlines.
- The most common intraoral sites affected by PV in descending order are soft palate, hard palate, buccal mucosa, ventral tongue, and floor of the mouth. Gingival involvement by PV (desquamative gingivitis associated with PV) is less frequent (15–20%) compared to mucous membrane pemphigoid and lichen planus.
- Mucocutaneous manifestations of paraneoplastic pemphigus/paraneoplastic autoimmune multi-organ syndrome (PNP/PAMS) are highly variable and polymorphic. Five distinct clinical manifestations of PNP/PAMS are recognized in the skin, which are (1) pemphigus-like lesions, (2) pemphigoid-like lesions, (3) lichen planus-like lesions, (5) erythema multiforme-like lesions, and (6) graft versus host disease-like lesions. All patients with PNP/PAMS have oral involvement and present with severe oral vesiculoulcerative mucositis affecting the entire oral cavity (pan-stomatitis), in contrast to pemphigus vulgaris, which preferentially affects the palate, buccal mucosa, and ventral surface of the tongue. PNP/PAMS presents as hemorrhagic erosive/ulcerative mucositis resembling a severe case of erythema multiforme. Oral ulcerative mucositis is the earliest manifestation of PNP/PAMS that tends to be treatment resistant. Systemic manifestations of PAMS frequently

involve both upper and lower respiratory tracts, and PAMS patients present with progressive dyspnea due to obstructive lung disease and bronchiolitis obliterans, which is one of the leading causes of death in PNP/PAMS patients, next to sepsis secondary to immunosuppressive treatment. The mortality rate of PNP/PAMS is significantly high (50–90%) compared to pemphigus vulgaris with a mortality rate of approximately 5–15% (Amber et al. 2018; Lim et al. 2021).

Diagnosis

- Clinical, histopathological, direct, and indirect immunofluorescence examinations are necessary to diagnose PV and distinguish it from other immune-mediated mucocutaneous diseases. Gentle rubbing of perilesional mucosa leads to the stripping of the epithelial surface known as a positive Nikolsky sign.
- Microscopically, PV is characterized by suprabasilar clefting with intraepithelial acantholysis resulting in free-floating Tzanck cells. The underlying lamina propria typically exhibits mild chronic inflammation with occasional eosinophils. Direct immunofluorescence (DIF) examination reveals intercellular IgG and C3 positivity, which is considered to be the gold standard for diagnosing pemphigus vulgaris. Indirect immunofluorescence testing is positive for circulating autoantibodies that bind to the intracellular junction of stratified squamous epithelium of normal esophagus. Enzyme-linked immunosorbent assay (ELISA) can be used to quantify the serum autoantibodies against Dsg-1 and Dsg-3 for confirming the diagnosis and therapeutic monitoring (Schmidt et al. 2010).
- Immunoblotting studies and indirect immunofluorescence testing using rat bladder transitional epithelium as substrate are necessary to distinguish PV from PNP/PAMS.

Therapeutic Guidelines

- Treatment with high-dose corticosteroid alone or in combination with other systemic immunosuppressive drugs is the standard of care for PV (Harman, Albert et al. 2003;

- Ingen-Housz-Oro et al. 2015). Patients with PV should be under the care of a dermatologist with expertise in managing patients with PV (Didona et al. 2019; Lim et al. 2021). Patients diagnosed with PV should be treated immediately with prednisolone (1 mg per kg per day) irrespective of extent and severity of the disease (Table 7). The initial dose of corticosteroid should be continued for a minimum period of 2-weeks before attempting to taper the dosage (Didona et al. 2019; Lim et al. 2021). Rapid tapering of corticosteroid treatment should be avoided to prevent severe flare-ups. Coordination with the patient's primary care physician is important to manage corticosteroid-induced adverse effects, especially in patients with a history of diabetes mellitus.
- Other immunosuppressive drugs that can be used in combination with corticosteroids to treat more severe and refractory cases of PV include azathioprine, cyclophosphamide, methotrexate, mycophenolate, and rituximab (Table 7). This treatment should be continued until the cessation of new bulla formation, and oral mucosa and skin surfaces are negative for Nikolsky sign. Subsequently, the dosage can be reduced to half and continued until all of the lesions are resolved.
 - PV patients may need to be on a minimum effective maintenance low-dose corticosteroid as a long-term management strategy. PV patients undergoing treatment with systemic corticosteroids and other immunosuppressive drugs are at high risk for developing oral candidiasis and concurrent treatment with anti-fungal medication is critical (Table 7).
 - Other immunosuppressive and immunomodulating drugs currently used in the treatment of PV with refractory systemic corticosteroid include cyclophosphamide, dapsone, methotrexate, intravenous immunoglobulin, rituximab, and infliximab (Ingen-Housz-Oro et al. 2015; Didona et al. 2019; Lim et al. 2021).
 - Therapeutic plasma exchange (plasmapheresis), immunoabsorption for rapid removal of circulating autoantibodies against Dsg-1 and Dsg-3, and extracorporeal photochemotherapy are used as adjuvant therapies for selected cases of PV that are refractory conventional drug treatments (Lim et al. 2021).
 - Treatment for PNP/PAMS is challenging, especially for oral lesions, which are corticosteroid treatment resistant. Early detection and treatment of underlying malignancy is critical for the resolution of the PNP/PAMS and long-term survival of the patients. Systemic pred-

Table 7 Commonly prescribed pharmacologic therapy for pemphigus vulgaris (PV)

Gingivitis-associated Pemphigus vulgaris	PV-limited to oral mucosa	Severe PV with cutaneous and oral mucosal involvement
Clobetasol propionate gel 0.05%; 2–3 times/day using medication carrier tray for 2–3 weeks + ^c chlorhexidine oral rinse for antifungal prophylaxis	^a Prednisolone 1.0 mg/kg or 0.16 mg/kg dexamethasone oral solution with slow tapering for 3–6 weeks	^a Prednisone (2.0–2.5 mg/kg/day with slow tapering for 3–6 months) + mycophenolate mofetil (500 mg to 2 g/day)
^b Doxycycline 100 mg, QD for 2–6 months	Localized chronic and painful ulcers: Triamcinolone acetonide 40 mg/mL (Kenalog-40); 0.1–0.2 mL/cm ² ; can be repeated every 4–8 weeks	Prednisone (2.0–2.5 mg/kg/day with slow tapering for 3–6 months) + azathioprine (2–3 mg/kg/day for 1 year)
	Steroid sparing Rx Mycophenolate mofetil, 500–1500 mg; 4–8 weeks	

^aFor managing gingivitis associated with pemphigus vulgaris, doxycycline (anti-collagenolytic effect) is used for the initial 2–6 months of treatment and then topical steroids are used for maintenance. Important: Remember that doxycycline may decrease the effectiveness of birth control pills, so those patients will need supplemental birth control. Also, remember to warn about possible photosensitivity

^bOff-label use

^cPrescribe chlorhexidine oral rinse or fluconazole for antifungal prophylaxis

nisolone (1.5 mg/kg/day) in combination with steroid-sparing agents (i.e., azathioprine, mycophenolate, cyclosporine, cyclophosphamide, and methotrexate) are used for treating patients with PNP/PAMS with inconsistent therapeutic response and increased risk for life-threatening complications such as severe infections/sepsis (Amber et al. 2018). Systemic steroids usually have a beneficial effect on cutaneous lesions; however, oral ulcerative mucositis and bronchiolitis obliterans are nonresponsive to systemic prednisolone (Amber et al. 2018). Early detection and treatment of underlying malignancies may aid in PNP/PAMS resolution (Antiga et al. 2023).

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Pediatric Considerations in Clinical Pharmacology

Cristiane S. Fonteles 

Introduction

During childhood, age-related differences between children and adults may increase toxicity and reduce pharmacological responses to various drugs of clinical use in dentistry. Specific age groups, including neonates (≤ 40 weeks post-conception), infants (0–6 months old), infants (6–12 months old), toddlers (1–3 years old), children (4–6 years old), children (7–12 years old), and adolescents (13–18 years old) (Abdel-Rahman et al. 2012), carry inherent stages of anatomical, physiological, and biochemical maturation that may account for important variations in drug kinetics, frequently generating drug-exposure differences of clinical significance and, subsequently, enhanced toxicity in comparison to adult populations, compromising therapeutics. To minimize the influence of pediatric developmental changes over time on treatment outcomes, therapeutic trials must be designed to factor in each age category, from neonates to adolescence. This premise has been advocated and incentivized by the Food and Drug Administration (FDA) for decades.

Prior to the late 1990s, children were commonly excluded from therapeutic trials, and labels failed to include information pertaining specifically to pediatric populations. Until the present day, the pharmacokinetics of many commonly prescribed drugs in pediatrics are based on extrapolations from data obtained from clinical trials in adult populations, and though off-label prescriptions have declined considerably compared to earlier years, up until the present, it remains a reality in the United States, Canada, and Europe (Juárez-Hernández and Carleton 2022). Inasmuch as there is a paucity of labels containing information that address pediatric age groups, for years, the FDA has placed effort into bridging this gap by incentivizing and requesting pharmaceutical companies to carry out dose-finding studies, as well as to assess safety and efficacy of drugs in children, in order to update current labels and improve labeling information for novel medications. The Best Pharmaceuticals for Children Act (BPCA) and the Pediatric Research Equity Act (PREA) attempt to mitigate these issues and change this scenario (Malkawi et al. 2022). In spite of efforts and advances obtained thus far, pediatric drug development remains a challenge to be overcome, and the term “therapeutic orphans” first mentioned in 1960s (Shirkey 1968) still loiters as a reality for the pediatric population in many parts of the world.

C. S. Fonteles (✉)
Postgraduate Program in Dentistry, Faculty of
Pharmacy, Dentistry and Nursing, Federal University
of Ceara, Fortaleza, Brazil
e-mail: cfontele@ufc.br

Drug Administration and Pediatric Compliance

Historically, medications have been preferably administered by mouth to children and adults. Safety, convenience and reduced cost are some of the reasons why the oral route is favored by parents and healthcare professionals. However, compliance is an undeniable requirement in the administration of medications by the oral route, so in addition to providing safety and stable bioavailability, formulations must be designed to please the pediatric patient in order for pharmacotherapy to work. Specific formulation and prescription-linked factors may concomitantly impact compliance, increasing the odds of treatment refusal by different pediatric age groups. For instance, previous work demonstrates that size and quantity of solid formulations, dispensed volume of liquid medications, texture, and the ability of the formulation to mask bitter taste of active ingredients and deliver palatability may strongly influence the child's ability to accept and cope with drug therapy (Venables et al. 2015). Out of all these factors, taste remains by far the greatest reason for noncompliance to prescribed medications. These factors directly correlate with age, and the younger the child, the more challenging oral pharmacotherapy can become (Venables et al. 2015).

Other previously reported barriers to pediatric medication compliance include prescribed treatment regimen (dose, frequency, and duration), side effects, and parental perception of treatment results (Wilson et al. 2021). Informing parents of the most frequently expected side effects and requesting feedback of treatment results from both parents and children should also help guarantee treatment adherence. In light of these findings, the success of pediatric prescribing warrants professionals to choose a formulation tailored for each patient, considering the child's age, developmental phase, medical history, and illness. Furthermore, to increase pediatric compliance and prevent administration errors by parents, the dosing schedule should be simplified, by reducing frequency and duration of treatment, by limiting the number of prescribed medications, and by

avoiding unnecessary multidrug combinations (Yin et al. 2021).

Liquid medications and suspensions are the most frequently prescribed formulations for young children and also the type of dose form routinely associated with incorrect or imprecise administration by parents compared with solid formulations, such as tablets and capsules that are traditionally dispensed in milligrams (mg) (Smith et al. 2014). Though solid formulations present the advantages of encompassing greater technological advancements, stability, lower risk of medication errors, and cost-effectiveness, the inability of young children to swallow solid formulations validates the preference for liquids (Lajoinie et al. 2014). To minimize medication errors and increase safety and compliance, prescriptions of liquid formulations require the adoption of metric measurements for weight (kilograms, kg) and volume (milliliters, mL). Oral or written instructions to parents should advise on the choice of adequate dosing tools to be used for the prescribed volume, such as oral syringes when small volumes must be dispensed, avoiding nonmetric units and unmarked dosing utensils, such as spoons that may not precisely dispense the correct volume (Yin et al. 2021). In these instances, it has been suggested that volumes should not surpass a threshold of 5 mL and 10 mL per administered dose for children under and above 5 years of age, respectively (Committee For Medicinal Products For Human Use (CHMP) 2023). Also, meeting children's taste preferences by previous assessment improves his or her tolerance of larger volumes (Venables et al. 2015). Failure to acknowledge these premises not only lessens treatment adherence but also ultimately compromises treatment outcome and safety.

Pediatric Formulations

The caveat of reduced bioavailability of drugs in specific age groups generates the need for age-appropriate formulations and routes of administration capable of delivering adequate and stable levels of a given medication at the site of action, overcoming differences imposed by physical and

physiologic factors such as anthropometric measurements, ontogeny of digestive and metabolic pathways, protein expression and bonds, among others. Furthermore, these formulations must be suited with the right ingredients for specific routes of administration, also taking into account the choice of dosing regimen, safety, stability, and regulatory aspects (Malkawi et al. 2022).

The incorporation of excipients with a low likelihood of inducing toxicity and allergic reactions is another difficult challenge for the industry, since not all excipients compounded into adult formulations are safe for pediatric use. Excipients or inactive ingredients are components of a drug formulation that are not meant to and therefore should not exert active pharmaceutical effect, nor generate unneeded toxicity (Reker et al. 2019). Excipients play an important role in the manufacture of pharmaceutical formulations, to assure palatability to children, drug stability, and solubility while disguising the poor taste of the active ingredient(s). However, many of the commonly used inactive ingredients are toxic or potentially allergenic to children. Allergic reactions to these inactive ingredients are less common than adverse reactions or intolerance. Nonetheless, when present, hypersensitivity reactions are frequently immunoglobulin-E mediated, causing urticaria, angioedema, bronchospasm, and anaphylaxis in children (Reker et al. 2019). Many inactive ingredients added to oral formulations are deemed allergenic, and these include food substances like lactose and starch, polymers like polyethylene glycol and povidone, disintegrant-like carboxymethylcellulose, and dyes such as sunset yellow FCF and Allura red. The following excipients were identified as having a greater chance to cause pediatric toxicity: propylene glycol, ethanol, polysorbate 80, benzyl alcohol, parabens, benzalkonium chloride, aspartame, sorbitol, benzoic acid, and sodium benzoate (Rouaz et al. 2021). Lactose, gluten, and peanut oil are also examples of substances that are frequently allergenic or toxic to children but can be present in drug formulations. Intolerance to excipients is more widely observed in the population than hypersensitivity reactions

and may affect drug absorption, causing gastrointestinal symptoms in children, such as diarrhea, nausea, and vomiting (Reker et al. 2019). These symptoms may be easily misconstrued as being a toxic reaction to a medication's active ingredient. Thus, it is important to be aware of these variables and possible reactions when prescribing to children.

Pediatric formulations encompass oral dosage forms (e.g., solutions, suspensions and emulsions, tablets) and oral transmucosal buccal dosage forms (e.g., chewable tablets, gummies, lozenges, lollipops) (Malkawi et al. 2022). Oral forms are most commonly administered as solutions and suspensions because they are easy to swallow and often have a more pleasant taste. In these formulations, the active components are solubilized or suspended in aqueous solutions containing excipients to mask taste and result in appropriate drug delivery to the gastrointestinal tract (Malkawi et al. 2022). These formulations face greater physiological challenges to be absorbed, distributed, metabolized, and excreted.

Transmucosal dosage forms are administered through intranasal, buccal, sublingual, and rectal routes. These formulations require absorption across the mucosa and carry the advantage of minimizing first pass metabolism in addition to avoiding degradation in the gastrointestinal tract. One of difficulties faced with the oral transmucosal formulations is the contact time required to reach absorption, considering that children are often unable to keep these formulations in place intraorally for the required time, limiting absorption. Since the oral mucosa is highly vascularized, drugs administered by means of these formulations rapidly reach detectable levels in blood, and peak concentrations can be achieved within 10–15 min (Madhav et al. 2009). The rectal route has the added advantages of allowing administration of higher doses, without risk of vomiting episodes. This can be particularly helpful in the operating room and in-patient settings. Transmucosal formulations constitute excellent noninvasive, alternative routes of administration for office-based sedation, allowing faster onset and recovery time (Wolfe and Braude 2010).

Drug Absorption, Disposition, and Bioavailability in Pediatric Patients

During childhood, the right drug provided in a less-than-ideal form or dose to the wrong age-group can increase toxicity, ultimately determining a successful or unsuccessful therapeutic outcome. Drug absorption and disposition (distribution, metabolism, and excretion) fluctuate according to various physiological parameters that are influenced by sex, ethnicity, fed versus fasting state, presence of disease, and age (Johnson et al. 2006). This is especially true during the first month of life, but as a result of growth and development, constant physiological changes extend throughout infancy and childhood, frequently modifying how various drugs are absorbed, distributed, metabolized, and eliminated from the body. Thus, gastric and intestinal pH, gastric emptying time, gastrointestinal transit time, drug distribution, metabolism, and clearance are some of the parameters that strongly influence the speed and efficiency of drug bioavailability in pediatric patients that will be discussed in this section.

Gastric acid secretion results from the interplay of neuronal, hormonal, and endocrine pathways (Prozialeck and Wershil 2017). Gastric pH measurements made soon after birth may provide neutral values (pH 6–7) due to the presence of swallowed amniotic fluid inside the stomach (Avery et al. 1966), but fasting pH declines and remains acidic (< 3) within the following weeks, comparable to those seen in adults, though a wide pH fluctuation is described in the literature, likely due to the buffering effect caused by frequent feedings during this time (Ali et al. 2014; Chapron et al. 2022). Preterm infants (24–29 gestational age) have been reported to present acidic gastric pH, and an inverse correlation between intragastric pH and gestational age was previously observed (Kelly et al. 1993). Most drugs are classified as weak acids or bases (Charifson and Walters 2014). Higher gastric pH in newborns will favor greater concentrations of weak

acids like penicillin G ($pK_a = 2.73$) and ampicillin ($pK_a = 2.5, 7.3$ at 23°C) and reduce concentrations of itraconazole ($pK_a = 3.7$), a highly lipophilic weak base (Batchelor and Marriott 2015; Lu and Rosenbaum 2014).

Fasting versus feeding conditions exert great influence on the digestive function of neonates and infants (Kwatra et al. 2020). It was previously postulated that neonates have a slower digestive system than adults, with an extended gastric emptying time of calorie-containing liquids that matures to adult values by 6–8 months of life (Strolin Benedetti and Baltes 2003). However, a comparable gastric emptying time was observed in children and adults under fed conditions, and food type appears to be an important modulator of gastric emptying time (Bonner et al. 2015; Kwatra et al. 2020; Yu et al. 2014). Once drugs reach the intestines, pH resembles adult levels (6.4–7.4) and exhibits less variability than gastric pH (Fallingborg et al. 1990). A meta-regression analysis found no significant age-related difference in intestinal transit time (Maharaj and Edginton 2016).

Food composition and eating habits are additionally important in infants and young children, impacting drug absorption. Breastfed infants have fat as the main source of caloric intake, which as they grow is slowly replaced by carbohydrates (Batchelor 2015). Meals with a high fat content may enhance solubilization and ultimately absorption of lipophilic or poorly water-soluble drugs, augmenting bioavailability and treatment efficacy (Cheng and Wong 2020). Absorption of penicillin V has been shown to be reduced when administered in an oily liquid preparation (Finkel et al. 1981). Peak plasma concentrations of penicillin and cephalexin were reduced by 60% in the presence of milk, as opposed to drug administration under fasting conditions in 2- to 46-month-old infants and children (McCracken et al. 1978). High-fat intake has been demonstrated to increase griseofulvin bioavailability (Crounse 1961). On the other hand, high-carbohydrate meals were shown to affect acetaminophen absorption, causing a delay

in mean peak plasma concentrations (Forrest et al. 1982).

In newborns, human serum albumin closely matches adult levels (75–80%). Conversely, alpha-1-acid glycoprotein is approximately half that of the adult concentration at birth (McNamara and Alcorn 2002). The reduced plasma concentrations of these proteins, coupled with their limited drug-binding capacity, can increase the unbound fraction of specific drugs, affecting distribution (Fernandez et al. 2011). Furthermore, increased bilirubin concentrations in the plasma of neonates and infants will compete for albumin-binding sites, either increasing plasma concentration of unbound drug or generating bilirubin toxicity (Andersen et al. 1956).

Body composition changes with time and drug distribution in the young relate to total body water content. In newborns, 75–85% (higher values in premature infants) of the body weight is composed of water, which is significantly greater than observed values in children and adults. The water content slowly reduces to approximately 60% by 12 months of age, becoming similar to adults (Yaffe 1986). This impacts the extracellular fluid compartment (plasma and interstitial space) that is mainly composed of water and represents 40% of body weight at birth, but declines to 27% by age one. The intracellular fluid compartment also suffers fluctuations until 3 months of age. Overall, water content will rise and fall to adult values from ages 1 to 3 (Yaffe 1986). Clinically, a water-soluble drug will have greater volume of distribution per body weight in infants than adults. In addition, lower plasma concentrations of hydrophilic substances can be caused by dose reductions based on weight (Chapron et al. 2022). Body fat content also differs between neonates/infants and adults. Neonates have less body fat content, whereas infants have more body fat than older children and adults, which favors distribution of lipophilic drugs, such as diazepam. The blood-brain barrier is not fully formed, being more permeable than in older children and adults, facilitating permeation of lipophilic drugs.

Bioavailability of different drugs can also be influenced by metabolism and efflux transporters present in the intestinal lumen. Efflux transporters appear to mature early in life. However, intestinal and hepatic clearance of drugs do not mature early and may modify drug clearance. The cytochrome P450 (CYP) family of enzymes are key players in hepatic and extrahepatic (mainly within the gut wall) drug metabolism. The rate and efficiency of hepatic metabolism of drugs is a product of blood flow, protein binding, and enzyme activity. In young children, the liver (5% at birth) constitutes a greater percentage of body weight compared with adults, contributing toward an increased hepatic blood flow during this time (Yaffe 1986).

During infancy and childhood, certain metabolic pathways (CYP1A2, CYP2C9, and CYP3A4) may surpass, be similar to or lower (CYP2C19, CYP2D6, UGTs, or NAT2) than their activity during adulthood (Anderson and Lynn 2009). Expression of CYP1A2 is markedly low in neonates, approximating 50% of adult expression by 9-month age (Johnson et al. 2006). CYP2B6 is minimally expressed in neonates and rises to 50% of adult expression by approximately 15 months of age (Anderson and Lynn 2009). The CYP2C subfamily represents approximately 18% of adult hepatic CYP content and metabolizes many drugs of importance in the pediatric dental clinic, such as ibuprofen, diclofenac, and diazepam. The main expressed isoforms include CYP2C9, CYP2C19, and CYP2C8. Great variation was previously observed in the expression of CYP2C9, from the neonatal period up to 5 months of age, and adult levels were reached prior to 12 months, while CYP2C19 showed progressive increase from birth to 5 months and demonstrated great variation from 5 months to 10 years of age (Koukouritaki et al. 2004). Age and genotype are strong determinants of CYP2D6 expression. This isoform accounts for approximately 2% of adult hepatic CYP content and plays a very important role in the metabolism of close to 20% of frequently used and clinically relevant drugs. In regard to ethnic differences in enzyme activity,

there are different percentages of poor metabolizers among African Americans (up to 19%), Caucasians (5–10%), and Asians (approximately 1%) (Stevens et al. 2008). Activity of CYP2D6 increases for the first 2 weeks postnatally and remains stable up until age 12 months, following highly polymorphic patterns of expression of poor, intermediate, extensive, or ultrarapid metabolizers (Stevens et al. 2008). Another very important CYP isoform is CYP3A4, which represents the most abundant CYP isoform in the liver and intestines. Expression of CYP3A4 occurs in the first weeks of life, and based on in vitro data, CYP3A4 activity reaches 72% of adult enzymatic activity by age 1 year (Johnson et al. 2008). Midazolam is an important example of a drug frequently used in dentistry for children

with potential to significantly reduce clearance secondary to intestinal metabolism by the as-yet immature CYP3A4 in infants. In vivo derived ontogeny data from a recent meta-analysis of CYPs demonstrated enzymatic activity above adult values for CYP1A2, CYP2C9, and CYP3A4, whereas CYP2B6 and CYP2D6 did not show remarkable changes (Anderson and Lynn 2009). All other CYPs expressed a 50% activity or less by 4-month age (Upreti and Wahlstrom 2016). Anderson and Lynn (2009) (Anderson and Lynn 2009) predicted that due to physiologic variability, drugs metabolized by CYP2C19 and CYP2D6 require weight-corrected doses that resemble adult dosing. Age-related pharmacokinetics are summarized in Table 1.

Table 1 Age-related pharmacokinetic variations. (Table adapted from Anderson and Lynn (2009), Batchelor and Marriott (2015), Fernandez et al. (2011), Lu and Rosenbaum (2014))

Physiological parameters		Age-related differences		Example of affected drugs		Observation		References	
Absorption	Gastric pH	↑ at birth, ↓ thereafter		Anticonvulsant	Phenytoin, ↓F	Higher concentration of weak acids, and lower concentrations of weak bases in infants than in older children and adults		Batchelor and Marriott (2015), Lu and Rosenbaum (2014)	
				Barbiturate	Phenobarbital, ↓F				
				Antiviral	Ganciclovir, ↓F				
				Antifungal	Itraconazole, ↓Cmax				
				Antibiotics	Penicillin G, ↑F Ampicillin, ↑F				
Distribution	Gastric emptying time	↑ ≤ 6 months-age		Antibiotics	Sulfonamides, delayed absorption	In all age groups, gastric emptying becomes slower as volume of food intake increases More affected by meal type than by age		Bonner et al. (2015), Kwatra et al. (2020), Lu and Rosenbaum (2014)	
				Barbiturates	Phenobarbital, delayed absorption				
	Intestinal bile concentration	↓ < 1 month		Corticosteroids	Hydrocortisone, ↓F	Important for drug solubility		Batchelor and Marriott (2015)	
	Plasma protein binding	↓ ≤ 12 months-age		NSAIDs	Salicylates, ↑ unbound fraction	Plasma protein concentrations, reduced binding capabilities, other molecules that compete for same protein binding sites		Lu and Rosenbaum (2014), McNamara and Alcorn (2002)	
				Antibacterial	Sulfonamides, ↑ unbound fraction Ampicillin, ↑ unbound fraction				
	Body composition	↓ Body fat, ↑ water content <1 month-age; ↑ body fat >3 months of age		Anticonvulsants	Phenytoin, ↑ unbound fraction	The Vd will depend on drug's hydrophilicity or lipophilicity, and body composition at time of administration		Batchelor and Marriott (2015)	
				Benzodiazepines	Diazepam, ↑Vd in infants				

(continued)

Table 1 (continued)

Physiological parameters		Age-related differences		Example of affected drugs		Observation	References
Metabolism	CYP2C9	~ Expression ≤ 5 months-age, adult levels ≥ 12 months-age	Anticoagulant	Warfarin, $\uparrow$ CL < 12 -years-age			Anderson and Lynn (2009), Fernandez et al. (2011), Koukouritaki et al. (2004)
			NSAIDs	Celecoxib, $\uparrow$ CL 7–16 years-age			
	CYP2C19	$\uparrow$ Levels ≤ 5 months-age, ~ expression 5 months–10 years-age	Proton-pump inhibitors	Omeprazole, pantoprazole, $\downarrow$ CL < 1 month-age			Anderson and Lynn (2009), Fernandez et al. (2011), Koukouritaki et al. (2004)
	CYP2D6	$\uparrow$ Activity ≤ 2 weeks-age, stable ≤ 12 -months	Analgesic Antihistamine Anticonvulsants Benzodiazepines	Codeine, tramadol Diphenhydramine Carbamazepine Midazolam		< 1 -month-age: $\downarrow$ weight-corrected doses > 1 -month-age: Weight-corrected doses approximately same as adults	Fernandez et al. (2011), Stevens et al. (2008)
Excretion	Renal CL	$\downarrow$ Activity < 12 months-age	Benzodiazepines	Midazolam, $\downarrow$ CL preterm infants, $\uparrow$ CL from 2–12 years-age compared with 12–16 years-age			Anderson and Lynn (2009), Chapron et al. (2022), Fernandez et al. (2011)
			Local anesthetic	Lidocaine, $\uparrow$ $1/2$ -life in children compared with adults			
			Antihistaminic	Cetirizine, excreted mostly unchanged, $\uparrow$ CL			
						Slower maturation of tubular secretion than GRF. Post-conceptual age must be considered to assess renal function. Greater reduction of renal CL in premature infants	Anderson and Lynn (2009), Chapron et al. (2022), Lu and Rosenbaum (2014)

$\uparrow$ increased values, $\downarrow$ decreased values, ~ variable values, CYP cytochrome P450, UGT uridine diphosphate glucuronosyl transferase, (+) increased pediatric dose compared with adults, (–) decreased pediatric dose compared with adults, *same* pediatric dose is same as adults, *n.a.* not applicable or not available, *F* bioavailability, *Vd* volume of distribution, *CL* clearance

Saliva

Saliva plays a role in absorption of drugs administered by oral and transmucosal (buccal and sublingual) routes. The rate of saliva secretion decreases with age, so the younger the patient, the greater the quantity of secreted unstimulated saliva (Lopez-Jornet and Bermejo-Fenoll 1994). Salivary secretion increases from birth to 6-month age and slowly decreases, reaching by 12-month age values that are similar to flow rates described for older children. Most importantly, when secretory values were adjusted for weight, higher secretory levels are observed (Collares and Fernandes 2015). The approximate range of unstimulated whole saliva secretion in children 6–15 years of age was reported by different authors to be within 0.32–1.05 mL/min (Forcella et al. 2018; Gutman and Ben-Aryeh 1974; Rotteveel et al. 2004), whereas stimulated flow rate is 0.2–3.0 mL/min (Leonor et al. 2009). Adults aged 20–35 years demonstrated mean flow rates of 0.58 ± 0.24 (unstimulated) and 1.51 ± 0.72 mL/min (stimulated) (Gittings et al. 2015), while unstimulated flow rates in adults aged 60–76 years of age were within 0.09–0.65 mL/min and presented high variability. Longitudinal data show a certain stability in unstimulated and stimulated salivary flow rates in children aged 7–12 years and tend to follow constant patterns of low, medium, or high stimulated secretion throughout childhood (Salvo et al. 2007; Sánchez-Pérez et al. 2016).

Stimulated saliva has a greater secretory volume, is less viscous, retains greater water content and higher pH levels, and has superior buffering ability than resting saliva in young adults (Gittings et al. 2015). Once drugs are orally administered, salivary glands are stimulated to secrete saliva. Salivary pH, viscosity, and buffering capacity are also variables that may influence drug dissolution, absorption, and bioavailability along with salivary secretory patterns. More viscous saliva reduces drug dissolution. High or low pH also affects drug dissolution depending on its dissociation constant, among other properties, and buffering capacity of saliva determines a drug's ability to resist pH variations in the oral

environment (Gittings et al. 2015). The influence of salivary parameters in drug dissolution in pediatric populations of different age groups warrants investigation. Moreover, unbound portions of certain drugs diffuse from plasma into saliva, making saliva a relevant tool to monitor concentration of various substances in neonates and infants (Hutchinson et al. 2018).

Frequently Prescribed Medications for Children in Dentistry

Antibiotics

Antibiotics are the most frequently prescribed drugs in the pediatric dental setting, and odontogenic infections remain the most common reason for prescribing these medications (Goel et al. 2020). In dentistry, pediatric prescriptions are usually aimed at treatment of acute manifestations of chronic conditions, such as dental caries and periodontal disease, which are also the most frequent cause of pediatric emergency dental visits (Zou et al. 2018). Other conditions that commonly require antibiotic prescription for children include salivary gland infections, intraoral lacerations, aggressive periodontitis, prevention of infection following dental avulsions, and prophylaxis against bacterial endocarditis (American Academy of Pediatric Dentistry 2022). Particularly, during childhood, pulpal necrosis secondary to caries or dental trauma favors bacterial access into the pulp, furcation, periodontal, and periradicular tissues. Pediatric anatomic and immunologic factors may favor the spread of infection through a path of least resistance with infiltration of tissues and spaces, eventually causing facial cellulitis or abscesses (Lin and Lu 2006; Okoje et al. 2018; Schnabl et al. 2019). Absence of an early and suitable therapeutic approach allows microbial pathogens to further migrate and become the source of fascial space infections. These infections have the potential of developing into rare, dangerous, rapidly progressing complications, among which are Ludwig's angina, cavernous sinus thrombosis, brain abscesses, and sepsis (Lajolo et al. 2019;

Parkins 2018), and the number of children that develop sepsis from odontogenic etiology increases with caries experience. Previous work has identified a high prevalence of gram-positive facultative and anaerobic bacteria (e.g., *Staphylococcus* spp., *Streptococcus* spp., *Actinomyces naeslundii*), gram-negative facultative anaerobes (e.g., *Haemophilus parainfluenzae*, *Klebsiella pneumoniae*), and aerobes (e.g., *Neisseria subflava*) associated with odontogenic cellulitis in children (Ślotwińska-Pawlaczyk et al. 2023; Unkel et al. 1997). Gram-positive aerobes were also identified (e.g., diphtheroids) (Unkel et al. 1997).

When prescribing, special attention must be given to the child's immunity, preexisting comorbidities, source of infection, affected anatomical spaces, and signs and symptoms that are consistent with systemic response to infection. Presence of fever in association with facial swelling is important initial indicators of systemic involvement and warrant careful attention when choosing antibiotics, treatment regimen, route of drug administration (intravenous versus oral), and need for hospitalization, especially in young children that present with these findings and an added history of dental pain (Ślotwińska-Pawlaczyk et al. 2023). Other symptoms that commonly follow and should be taken into account include malaise, dysphagia, and other complications, such as dehydration and breathing difficulties.

There is no scientific evidence supporting antibiotic prescription to treat uncomplicated localized dental abscesses in children without clear evidence of a spreading infection or signs and symptoms consistent with systemic involvement (Leroy et al. 2021). Prescription of antibiotics should be avoided when odontogenic infections remain limited to the pulp and periradicular tissues. Blood flow within these confined anatomic sites is reduced or absent due to pulp necrosis or accumulation of fluids (pus), limiting drug bioavailability in these areas, which favor the development of bacterial resistance. In these instances, removal of the cause of infection (e.g., pulpotomies, pulpectomy, dental extraction, incision, and drainage) should provide the means to improve the clinical condition. Nonetheless, to

reduce morbidity, guidelines support antibiotic prescription when clinical signs and symptoms indicate the spread of infection into other anatomical sites (American Academy of Pediatric Dentistry 2022; IAPD 2021). In most cases, antimicrobial drug therapy is not recommended as a substitute for dental management but should be considered an adjunct to the indicated treatment approach.

Since infections that originate within the oral cavity are intrinsically polymicrobial, an extended-spectrum penicillin, that is, amoxicillin with or without clavulanic acid, remains the first line of treatment in outpatient therapy of odontogenic infections, intraoral lesions, and salivary gland infections, as well as in the prevention of metastatic infections and bacterial endocarditis. Amoxicillin has a low incidence of associated adverse reactions. A recent systematic review of common amoxicillin-related side effects in pediatric patients with meta-analysis of placebo-controlled clinical trials (Gillies et al. 2015) reported diarrhea as a frequent finding in children taking amoxicillin-clavulanic acid, whereas superinfections with candidiasis were common in children taking amoxicillin with or without clavulanic acid. The use of amoxicillin and amoxicillin/clavulanic acid did not increase rashes, nausea, vomiting, or abnormal liver function tests in this study. Nonetheless, comparison between amoxicillin monotherapy and amoxicillin/clavulanic acid showed an association of the latter with higher rates of severe cutaneous hypersensitivity reactions (Stevens-Johnson syndrome), hematological disorders, and liver toxicity (Salvo et al. 2007). In another recent work, amoxicillin (with or without clavulanic acid) accounted for a greater number of penicillin-associated hypersensitivity reactions (PHR) compared with other beta-lactam antibiotics (Azevedo et al. 2019). Amoxicillin has also been implicated in tooth discoloration in children under 8 years of age (Wang et al. 2023). These findings suggest that differences in toxicity between amoxicillin and amoxicillin/clavulanic acid must be considered when prescribing.

In the event of PHR, cephalosporins and macrolides can be prescribed. The prescription of

cephalosporins should be avoided in the presence of true immunoglobulin E (IgE)-mediated PHR (e.g., positive skin test or drug provocation test) and in children with a medical history significant for anaphylaxis, severe hypersensitivity, and drug reactions, especially in the presence of systemic symptoms (Ponvert et al. 2011). However, in case of non-IgE-mediated PHR, cephalosporins can be prescribed with a good margin of safety. A child is at the highest risk of cross-reactivity when structural similarities between R1 side chains are the greatest. Based on a recent meta-analysis performed with calculated similarity scores between R1 side chains of penicillins, and cephalosporins (Picard et al. 2019), the risk of a child with PHR to react to at least one cephalosporin varies from 16.45% (high score), 5.60% (intermediate score), and 2.11% (low score). First-generation (cefadroxil, cephalixin, and cefatrizine) and second-generation (cefprozil, cefaclor) amino-cephalosporins carry identical R1 side chains with amoxicillin and ampicillin and should be avoided in patients with PHR. Risk of cross-reactivity was the lowest with cefazolin (first generation); cefuroxime (second generation); cefixime, cefotaxime, ceftriaxone, ceftazidime, cefpodoxime, and ceftibuten third (third generation); and cefepime (fourth generation).

Recent data linking a single dose of clindamycin to an increased rate of life-threatening adverse drug reactions, most of which associated with *Clostridioides difficile* infection (Thornhill et al. 2015), led the American Heart Association (AHA) to no longer recommend clindamycin as an alternative in the prophylaxis of infective endocarditis and to include doxycycline as a safer option in the advent of hypersensitivity to penicillin, cephalosporin or macrolides (Wilson et al. 2021). First marketed in the late 1960s, doxycycline was for many years labelled as having the same adverse effects as first-generation tetracyclines in the developing dentition and was therefore contraindicated for use in children under the age of 8 years. This paradigm changed with recent data (Todd et al. 2015; Volovitz et al. 2007). Doxycycline has low calcium affinity, demonstrating reduced risk of dental staining in comparison with other tetracyclines, and is now

considered safe for use in pediatric populations for up to 21 days (American Academy of Pediatrics (AAP) 2021; Stultz and Eiland 2019).

Azithromycin and clarithromycin are second-generation macrolides with good safety margins, uncommon reports of allergenicity, and broad antibacterial spectra that cover most bacteria implicated in odontogenic and other forms of oral infection. In addition, immunomodulation may be an added benefit of this class of antibiotics (Zimmermann et al. 2018). Previous work showed a higher frequency of hypersensitivity reactions to azithromycin compared with clarithromycin among children suspected of macrolide allergy (Barni et al. 2015). Cross-reactivity between azithromycin and clarithromycin has also been described (Mori et al. 2014). The most common adverse events related to azithromycin in children are gastrointestinal. Data from prospective studies show that the risk of diarrhea caused by azithromycin is reduced in comparison with amoxicillin/clavulanate and greater than penicillin V (Zeng et al. 2020). Azithromycin is a proarrhythmic drug that has been shown to exert a slight absolute increase in the risk of death by cardiovascular reasons, including sudden cardiac deaths and death by any causes, during 5-day treatment in adults, predominantly among patients at a higher risk of cardiovascular disease (Ray et al. 2012). In pediatric populations, the risk of cardiotoxicity by azithromycin relates to an increased risk of polymorphic ventricular tachycardia caused by QTc prolongation (torsades des pointes) and the highest the prescribed dose (10–30 mg/kg/day), the greater the cardiotoxicity-related risk. Doses up to 10 mg/kg/day were associated with a lower incidence of cardiotoxicity (Zeng et al. 2020) in children.

Pediatric Pain Management

Pain assessment in the pediatric population is usually a difficult task, considering that most children either provide unreliable information due to young age and have limited communication skills or an unwillingness to offer information as a result of stress, fear, and anxiety

generated by pain and discomfort. The child's inability to communicate imposes great suffering and emotional burden on the patient and their families. These limitations make pain management in pediatric patients challenging, often leading dentists to prescribe based on parents' interpretation and their own professional understanding of the child's experience/perception of pain. Furthermore, the failure to recognize pain in children leads pain to be frequently mismanaged in hospitalized and outpatient settings (Orzalesi 2018). Behavioral assessment by instruments like the Frankl scale during a dental visit can be a helpful tool. Age-appropriate pain assessment scales, such as the Wong-Baker FACES, Faces Pain Scale, and visual analogue scale (VAS), can enable the child to communicate their level of pain in emergency and postoperative settings (Garra et al. 2010; Jain et al. 2012). Assessment of physiological parameters such as the child's heart rate, blood pressure, respiratory and oxygen saturation rates, and sweating of the hands can also be useful (Jain et al. 2012). Timely pain assessment is a critical step in managing pain in a dental setting and must be carefully considered before prescribing to pediatric patients.

Management of acute pain has a more favorable outcome when a multimodal approach is adopted, encompassing pharmacologic and non-pharmacologic concepts (Brown et al. 2019; Hyland et al. 2021). Therefore, planning antinociception for children and adolescents in dentistry should follow a similar rationale by anticipating presence/level of pain likely to be experienced during and after surgical procedures and planning pre-, trans-, and postoperative approaches to pain control. This concept gains further significance when the possibility of moderate-to-severe pain is anticipated. The combination of adequate local anesthesia and preemptive analgesia reduces the need for postoperative pain analgesics and has an opioid-sparing effect, shortening treatment regimen, which diminishes risk of adverse drug events, producing greater analgesia with less analgesics. The arsenal of drugs from which we can choose to manage mild-to-moderate pain in pediatric

patients (opioids and non-opioids) should spare opioids when possible and favors non-opioids to reduce unwanted adverse events (American Academy of Pediatric Dentistry 2022).

Four basic premises previously stated by the World Health Organization lay the foundation for successful pediatric analgesia (Vargas-Schaffer 2010): (1) preferably prescribe oral forms of medications, when possible; (2) use regular intervals when prescribing based on pain intensity and duration of analgesia provided by the drug of choice; (3) the analgesic regimen should be based on the level of pain identified by the patient, using pain assessment scales; (4) dosing should be customized for each patient and aimed to provide adequate pain relief by balancing out the dose that provides best analgesia at minimal adverse drug effects; and (5) the patient's progress should be monitored in detail and analgesic regimen readjusted as required after the initial 24 h.

The choice of dosing regimen for pain control in pediatric populations remains to be established (Pillai Riddell and Craig 2003). One might argue that the "as needed" (PRN) regimen may lead to administration of significantly less medication than scheduled administration (ATC) in the management of postoperative pain, though at the expense of greater and sometimes unnecessary exposure to analgesics, ATC in postoperative pain may assure continuous relief and better pain management (Chiaretti et al. 2013). The use of ATC alternating regimen of ibuprofen (10 mg/kg) and acetaminophen (10 mg/kg) every 2 h, following surgery in children with cerebral palsy, was compared with a retrospective group of 20 children subjected to the same surgical procedure but receiving PRN epidural morphine every 3–4 h (Tubbs et al. 2007). Their results showed the need for a single morphine dose in only 1/22 patients in the study group, demonstrating an opioid-sparing effect, shorter hospitalization time, and less requirement for antiemetics. Another study showed the lack of efficacy of ATC regimens to adequately control pain following tonsillectomies (Sutters et al. 2004), and more recent work showed superior pain control post-tonsillectomies using ATC compared with a PRN regimen of acetaminophen/hydrocodone

(Sutters et al. 2010). Interestingly, there is not enough evidence in the literature to affirm that pediatric postoperative pain management would be more effective if prescribed analgesics were administered under ATC versus PRN regimens (Hobson et al. 2015). Traditionally, the practice of postoperative pain management in hospital settings has given preference to ATC regimens, under the understanding that scheduled, round-the-clock prescriptions prevent pain, and by doing so provide comfort and lessen the need for other medications. However, it is the authors opinion that PRN schedules in the management of mild-to-moderate pain may suffice in most clinical outpatient cases, particularly when preemptive analgesia is adopted, potentially reducing the amount of consumed analgesics and side effects. Further studies are warranted to create specific protocols for children in dentistry.

Non-opioid Analgesics

Clinical trials evaluating adverse events caused by commonly prescribed analgesics in pediatric populations remain scarce, and most of the available data relate to adult populations. A systematic review of studies conducted in eight different countries, published between 1991 and 2014, reported on adverse drug events caused by acetaminophen and nonsteroidal anti-inflammatory drugs, NSAIDs (ibuprofen, naproxen, ketoprofen, nimesulide, ketorolac), opioids (morphine, codeine, oxycodone, tramadol), and NSAID/opioid combinations (ibuprofen/oxycodone, ibuprofen/codeine, acetaminophen/codeine) in children and adolescents (Hartling et al. 2016). Acetaminophen and ibuprofen exhibited comparably low rates (<10%) of adverse events affecting the gastrointestinal tract (e.g., nausea, vomiting, diarrhea), whereas children taking naproxen were at a higher risk (>10%) than acetaminophen and ibuprofen of manifesting nausea and vomiting. Ibuprofen demonstrated a greater risk of drowsiness/sleepiness than acetaminophen. Ketoprofen showed a greater risk of adverse events affecting the skin than ibuprofen and acetaminophen. Opioid monotherapy (oxycodone,

morphine, codeine) showed the lowest safety, with a higher risk of adverse events affecting the gastrointestinal tract, skin (e.g., itchiness, rash, and pruritus), and central nervous system (e.g., headache, drowsiness, dizziness). Codeine monotherapy doubled the risk of dermatologic symptoms compared with NSAIDs, possibly secondary to drug-induced histamine release. Ibuprofen/oxycodone, ibuprofen/codeine, and acetaminophen/codeine combinations demonstrated a greater margin of safety than opioid monotherapy. The overall risk of adverse events caused by acetaminophen and NSAIDs was low compared with opioid monotherapy. Severe side effects are uncommon in the pediatric population, but use should be carefully considered, especially in cases of hepatic and renal dysfunction or coagulation disorders.

Acetaminophen and ibuprofen are the most commonly used over-the-counter pain and fever medications by children under 12 years of age in the United States (Lu and Rosenbaum 2014; Vernacchio et al. 2009). Acetaminophen (N-acetyl-p-aminophenol, APAP, or paracetamol, PARA) is a pro-drug that acts on pain and fever by various mechanisms not yet fully understood. Upon entrance into the body, acetaminophen is subjected to a complex metabolic pathway that is essential for its pharmacodynamics but also carries a potential for toxicity. Roughly 80% of a therapeutic dose is converted to glucuronide and sulfate conjugates. Glucuronide conjugation is deficient in young children and infants (Prescott 1980). Therefore, children up to 10 years of age form less glucuronide conjugates but produce larger amounts of sulphate metabolites (Forrest et al. 1982). Deacetylation produces a *p*-aminophenol derivative able to cross the blood-brain barrier to be subsequently converted to AM404 (*N*-arachidonoyl phenolamine) by conjugation with arachidonic acid via the fatty acid amide hydrolase (FAAH) (Högestätt et al. 2005). For pharmacodynamic purposes, AM404 is a key metabolite that (1) transiently activates transient receptor potential vanilloid subtype 1 (TRPV1) and/or cannabinoid CB1 receptors in the brain and spinal cord; (2) weakly inhibits cyclooxygenase (preferably COX-2 or physiologically-

dependent isoform preference), explaining its lack of anti-inflammatory action; (3) and opioid and serotonergic pathways (Esh et al. 2021; Ohashi and Kohno 2020).

A small percentage of acetaminophen undergoes oxidation catalyzed by P450 enzymes, mainly CYP2E1, forming the toxic breakdown product N-acetyl-p-benzo-quinone imine (NAPQI). This metabolite is inactivated by conjugation with the sulfhydryl groups of glutathione to form nontoxic metabolites, cysteine and mercapturate, that are excreted in the urine. Depletion of glutathione stores by either toxic amounts of acetaminophen or systemic conditions that may diminish glutathione stores cause toxic NAPQI to accumulate and bind cellular proteins, thereby incurring cell injury (Mitchell et al. 1973). Obesity, liver steatosis, malnutrition, and glutathione synthetase deficiency are among systemic conditions that may deplete glutathione stores, increasing the risk of hepatotoxicity induced by accumulation of NAPQI (Hodgman and Garrard 2012; Tokatli et al. 2007). N-Acetylcysteine (NAC) is the antidote for acetaminophen-induced hepatotoxicity and acts by increasing glutathione supply (Hodgman and Garrard 2012). Chances of avoiding severe hepatic damage are greater if NAC is administered within the first 8 h of toxicity (Reid and Hazell 2003). It is important to note that acetaminophen administration of a single dose above 7 g in adults and 150 mg/kg in children, or cumulative repeated doses put the patient at a high risk of liver toxicity. For safety reasons, the maximum therapeutic dosages of 4 g/24 h for adults and 50–75 mg/kg/24 h children should be respected (Table 2). Overall, there are no significant differences in the excretion of acetaminophen between children and adults.

Acetaminophen is rapidly and widely absorbed by the oral route in a dose-dependent fashion, does not significantly bind plasma proteins, and has a plasma half-life of 1.9–2.5 h (Forrest et al. 1982). Choosing the right mode of administration for children is important to assure the best combination of good pharmacokinetics and effective pharmacodynamics. The oral and rectal routes are both safe modes of drug admin-

istration. Acetaminophen is highly bioavailable once administered by the oral route (Eandi et al. 1984), but rectal administration is frequently used to attain faster onset of action and has advantages that include ease of administration and low cost compared with parenteral administration and is a well-tolerated option for children unable to take medications by mouth (Thibault et al. 2023). The rectal route produces high inter-individual variations in pharmacokinetics (Montgomery et al. 1995). Since higher and less variable blood concentrations are provided by intravenous administration, this can be a safe and effective alternative in hospital settings when stable and higher drug levels are required (Dela Cruz Ubaldo et al. 2014). A risk of transient hypotension during intravenous administration can be a concern (Thibault et al. 2023). Absolute bioavailability (using the intravenous route as a reference) of acetaminophen following the oral route of administration is high (60–70%) regardless of the chosen formulation and presence of gastric contents, whereas the rectal route produces lower (30–40%) and unpredictable bioavailability after single-dose administration to healthy adults. Studies evaluating the relative bioavailability of rectal compared with oral acetaminophen formulations for pain control in children are few and inconsistent in their results (Anderson et al. 1996; van der Marel et al. 2001) and do not provide enough evidence to support equivalence of analgesic efficacy between the rectal and oral routes in children. Nonetheless, a recent systematic review and metaanalysis demonstrated comparable efficacy of rectal versus oral acetaminophen in fever control, with no significant differences in temperature decline 3 h post administration, maximum temperature reduction, or average time taken to cause temperature reduction of 1 °C (Goldstein et al. 2008).

Due to the risk involved in opioid use in pediatric postoperative pain, many pediatric clinical trials have focused on comparing the analgesic efficacy of NSAIDs and opioids. The central effects of acetaminophen coupled with the opioid-sparing properties of both acetaminophen and NSAIDs have justified clinical trials testing the analgesic efficacy of alternating ibuprofen

Table 2 Commonly prescribed dosage forms, dosages and potentially toxic excipients of nonopioid analgesics used to treat or prevent mild-to-moderate, moderate and moderate-to-severe pediatric pain. in dentistry. (Table adapted from (Lee 2018; Jeske 2021; Reker et al. 2019; Rouz et al. 2021)

Medications	Dosage forms	Use	Dosages	Excipients
Acetaminophen (Tylenol, Tempra, Panadol, Ofirmev, among others)	Oral capsule, oral tablet: 325, 500 mg	Mild-to-moderate pain	Oral/rectal <i>Neonates:</i> 10–15 mg/kg/dose Q6–8h as needed <i>Children:</i> 10–15 mg/kg/dose Q4–6h as needed. PR loading dose: 40–45 mg/kg dose. Max: 75 mg/kg/24 h not exceeding 4 g/24 h <i>Adults:</i> 325–650 mg Q4–6h or 1 g i.i.d. or q.i.d. max: 4 g/24 h, 5 doses/24 h	Starch Povidone Butylparaben High fructose corn syrup Carboxymethylcellulose sodium propylene glycol Sodium benzoate Sucralose FD&C red no. 40D&C red no. 33 FD&C blue no. 1
	Oral granule (effervescent): 650 mg			
	Oral liquid 160 mg/5 mL, 325 mg/10.15 mL, 500 mg/15 mL, 650 mg/20.3 mL			
	Oral suspension 160 mg/5 mL, 650 mg/20.3 mL			
	Oral tablet (chewable): 160 mg; 80 mg			
	Oral tablet (disintegrating): 160, 325, 80 mg			
	Oral tablet (extended release): 650 mg			
Ketorolac tromethamine (generic equivalents of Toradol, Sprix nasal spray)	Suppositories: 80, 120, 325, 650 mg	Acute, moderate-to-severe (short-term use, 5 days max)	Intravenous <i>Infant/children < 2 year:</i> 7.5–15 mg/kg/dose Q6h. Max:60 mg/kg/24 h <i>Child (2–12 year) and adolescents/adults ≤ 50 kg:</i> 15 mg/kg Q6h OR 12.5 mg/kg Q4hmax single dose: 15 mg/kgmin dosing interval: Q4hmax dose: 75 mg/kg/24 h, not to exceed 3750 mg/24 h <i>Adolescents (>12 year) and adults ≥ 50 kg:</i> 1 g Q6h OR 650 mg Q4h max single dose: 1 g min dosing interval: Q4 hmax dose: 4 g/24 h Oral <i>Adults and adolescent ≥ 16 year (>50 kg):</i> 10 mg Q6 h. Max dose: 40 mg/24 h	Alcohol (injectable solution)
	Film-coated tablet: 10 mg			
	Injectable solution ampoules: 15 mg/mL (1 mL); 30 mg/mL (1, 2, 10 mL); 60 mg/2 mL			
Naproxen/naproxen sodium (Naprosyn, Anaprox, Naprelan, Aleve among others)	Nasal spray 15.75 mg/100 microL spray; 8 sprays/1.7 g bottle	Acute, chronic, mild-to-moderate pain	Intravenous <i>Children 2–16 year:</i> 0.5 mg/kg/dose Q6–8h Max dose: 30 mg Q6h, not to exceed 120 mg/24 h <i>Single dose for moderate-severe pain:</i> 0.5 mg/kg. Max dose: 15 mg <i>Children ≥ 17 and adults (>50 kg): Multiple dose:</i> 30 mg Q6h. Max dose: 120 mg/24 h <i>single dose:</i> 30 mg <i>Children ≥ 17 and adults (<50 kg or renal impairment): Multiple dose:</i> 15 mg Q6h. Max dose: 60 mg/24 h <i>single dose:</i> 15 mg Intranasal <i>Adults ≥ 50 kg:</i> 31.5 mg Q6–8h (max dose: 126 mg) <i><50 kg:</i> 15.75 mg Q6–8 h (max dose: 63 mg) Oral <i>Child ≥ 2 year:</i> 5–7 mg/kg/dose Q8–12h. Max dose: 1 g/24 h <i>Adolescents and adults:</i> 250 mg naproxen Q8–12h PRN, or 500 mg (275–550 mg naproxen sodium) Q12h PRN. Max dose: 1.25 g/24 h Controlled release tablets: 750–1000 mg q.d.	Sucrose Magnesium aluminum silicate Methylparaben FD&C yellow no. 6 (sunset yellow)
	Oral tablets: 250, 375, 500, 750 mg			
	Delayed-release tablets: 375, 500 mg			
	Naproxen sodium oral tablet: 220, 275, 375, 500, 550, 750 mg Oral suspension: 125 mg/5 mL			

Table 2 (continued)

Medications	Dosage forms	Use	Dosages	Excipients
Ibuprofen (Advil, Motrin, Caldolor)	Solid formulations: Oral capsule: 200 mg Oral tablet: 100, 200, 400, 600, 800 mg Oral tablet (chewable) 50,100 mg	Acute, chronic, mild-to-moderate pain	Oral <i>Infant and children</i> ≥ 6-Mos: 5–10 mg/kg/dose Q6–8h. Max single dose: 400 mg, max dose: 40 mg/kg/24 h <i>Children</i> ≥12-year and <i>adults</i> : 200–400 mg/dose Q4–6h P.R.N. (max dose: 1.2 g/24 h)	Butylparaben D&C red #33 FD&C yellow #6 High fructose corn syrup Polysorbate 80 Propylene glycol Sodium benzoate Injection solution: Sodium acetate trihydrate and water for injections
	Liquid formulations Oral suspension: 100 mg/5 mL, 50 mg/1.25 mL Oral solution: 5 mg/mL			
	Intravenous solution 100 mg/mL (4, 8 mL)			
			Intravenous <i>Infants and children</i> 6–12 months: 10 mg/kg/dose, max of 400 mg/dose Q4–6 h P.R.N. Max dose: 40 mg/kg/24 h <i>Children and adolescents</i> 12–17 year: 400 mg/dose Q4–6h P.R.N. Max dose: 2.4 g/24 h <i>Adults</i> : 400–800 mg/dose Q6h P.R.N. Max dose: 3.2 g/24 h	

and acetaminophen regimens in children. The results showed superior efficacy of this alternating protocol in comparison with ibuprofen or acetaminophen monotherapies (Liu and Ulualp 2015).

Ketorolac is a very effective NSAID for moderate and moderate-to-severe acute onset pain, with available FDA-approved IV and intranasal formulations for pediatric use. Mucosal atomization devices are available for intranasal formulation for children. This NSAID has a significant opioid-sparing effect, with the benefit of acting against inflammation, and has been advocated for use in postoperative and preemptive pain management in children and may be used in oral and dental pain for children (American Academy of Pediatric Dentistry 2022). In spite of its advantages, dentists should be cautioned about its high cyclooxygenase-1 predilection (Warner et al. 1999) and risk of increased trans and postoperative bleeding when used preemptively. Risk of gastrointestinal side effects must also be carefully considered. In the United States, the oral form is approved only for use following a parenteral regimen.

Asthma and Other Allergic Disorders

An emergent body of scientific evidence has pointed toward a possible increase in the risk of asthma for children and adolescents exposed to acetaminophen prenatally and during childhood (Beasley et al. 2008, 2011; Perzanowski et al. 2010; Shaheen et al. 2005). Phase 3 of the International Study of Asthma and Allergies in Childhood (ISAAC) gave rise to two retrospective, cross-sectional studies with 6–7 (parents/guardians completed written questionnaires) and 13–14 (children completed written/video questionnaires) year old children. The first published work (Beasley et al. 2008) demonstrated that exposure to acetaminophen as an anti-fever medication in the first year of life showed an association of moderate magnitude (odds ratio 1.5, 95% CI 1.4–1.6) with an increased risk of asthma symptoms, including a dose-dependent risk of

severe asthma symptoms, rhinoconjunctivitis, and eczema at 6–7 years of age. Similar results were observed among 13–14 years old (Beasley et al. 2011), with a greater risk of asthma symptoms, rhinoconjunctivitis, and eczema dependent upon exposure to acetaminophen. A Taiwanese study used the National Health Insurance Research Database in 2005 to evaluate severe asthma exacerbation in children taking fever medications (acetaminophen versus NSAID) and demonstrated a higher risk of asthma exacerbation in children taking acetaminophen for fever control, compared with NSAIDs and absence of fever medication (Chung et al. 2020).

A randomized, double-blind, clinical trial was conducted to evaluate whether the use of ibuprofen by asthmatic children increased the risk of asthma exacerbation (Lesko et al. 2002). Two groups of children (aged 6 months to 12 years) were assigned to receive either acetaminophen (12 mg/kg) or ibuprofen (5 or 10 mg/kg). Children who took ibuprofen had lower rates of both hospitalization and outpatient visits for asthma relative to those who took acetaminophen either because ibuprofen reduced or acetaminophen increased risk of asthma-related morbidity. The proportion of children at risk of developing asthma-related symptoms (based on hospitalization or outpatient visit) was highest for children who received acetaminophen at average doses ≥ 11 mg/kg.

An earlier clinical trial failed to demonstrate a difference in adverse effects (including asthma/bronchiolitis) between 2-year-old children taking acetaminophen or ibuprofen (Lesko and Mitchell 1999). In addition, a cross-sectional study evaluated questionnaires and medical files of children aged 6 months to 6 years who were admitted to the emergency department with fever due to various causes and found no association between either acetaminophen or ibuprofen and an increased risk for wheezing (Matok et al. 2017). It is important to mention that in this study significantly less children with wheezing used ibuprofen. Most studies reporting an association between prenatal or early childhood administration of acetaminophen and risk of asthma or asthma symptoms are observational and retro-

spective. Further evidence provided from randomized clinical trials, controlling for important confounders, is needed.

Conclusion

There is still a paucity of data on pediatric-related pharmacokinetics of various medications, but much has improved due to efforts from drug-related agencies, especially in the United States and Europe. Pediatric prescription warrants cautious choice of age-appropriate drug formulations. Special care must be taken when prescribing to infants and children below age 2. Transmucosal dose forms provide a safe and effective alternative as drug delivery systems in children unable to take oral medications. Hypersensitivity reactions to drugs in pediatric patients must consider the risk of reactions to active as well as inactive ingredients. When prescribing to children, the dentist should (1) use metric measurements (mg/kg and mg/mL); (2) advise parents/caretakers on use of oral syringes when small volumes must be dispensed, avoiding nonmetric units and unmarked dosing utensils, such as kitchen spoons; (3) prescribe liquid medications of a maximum volume of 5 mL for children up to 5 years of age; and (4) provide oral and written instructions to parents/caretakers and children, when age-appropriate. Current scientific evidence dictates that antibiotic prescription in dentistry for healthy children should be restricted to clinical situations that indicate spread of infection. Non-opioids are preferable and safer for the management of mild and moderate pain. Acetaminophen/ibuprofen associations have demonstrated opioid-sparing effects and are preferable to opioid monotherapy in children. Scientific evidence suggests an association exists between acetaminophen administration prenatally and up to 12 months postnatally with asthma symptoms in children, but there is still a lack of randomized clinical trials controlling for genetic confounders to support changes in treatment recommendations.

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Biologic Agents

Katherine France

Introduction

Biologic agents are novel immunomodulatory medications targeted to control disease processes through modulation of inflammatory and immunologic pathways. They are a relatively recently developed class of drugs that has expanded in number and scope over the course of the twenty-first century, growing to be one of the most employed and fastest growing types of medications today. Biologic agents are employed across areas of medicine, including frequently in the treatment of autoimmune diseases and cancers. Their impact on dental treatment and their use in dental contexts remain poorly characterized, with only limited evidence focused directly on this setting to date.

Biologic agents are defined broadly as therapeutics isolated from natural sources, which imbues both their function and their risk of adverse effects. While the scope and nature of this specification varies across classes, one commonly used definition comes from the National Cancer Institute (NCI) of the United States (US) National Institutes of Health (NIH), which defines a biologic agent as: “a substance that is made from a living organism or its products and is used in the prevention, diagnosis, or treatment

of cancer and other diseases.” While broad, this characterization encompasses all isolated natural products used in medical treatment, including, as per the NCI, “antibodies, interleukins, and vaccines” (NCI 2021). For the ongoing treatment of chronic conditions, the medications tend to be delivered repeatedly over time, and it is these agents rather than vaccines or single-dose treatments that will be the focus of this chapter.

Each biologic agent is formulated to target specific immunologic pathways, either through activating or suppressing elements of the immune system through the provision of isolated, biologically active products. An expanding understanding of the physiologic processes underlying a variety of diseases, as well as a growing understanding of how they may be targeted, has allowed for maturation of this field into new contexts and more selective use of particular agents in different diseases (Strand et al. 2021). On the other hand, growth of this field has also allowed for increasing recognition of immunogenicity and its ability to limit the use of these agents, especially over prolonged time periods. There is also a concern, both in children and adults, that prolonged use of these agents may lead to immunosuppression (Axelrod and Adams 2021). This concern is shared with many of the traditional medications used to treat autoimmune and malignant diseases. Despite the targeted impacts of biologic agents, acting in a focused manner on well-defined processes, this potential for long-

K. France (✉)
Penn Dental Medicine, Philadelphia, PA, USA
e-mail: kfrance@dental.upenn.edu

term immune activation or suppression is increasingly recognized.

As the understanding of the potential of these drugs has grown over the last few decades, so have their market distribution and share. Innovative products that target constitutively active pathways, these agents are broadly thought of, and generally found to be, relatively safer alternatives to traditional treatments like disease-modifying antirheumatic drugs (DMARDs) and conventional chemotherapy for cancer treatment (Maini and Taylor 2000). Their formulations and use are also innovative, contributing both to the spread of their use and to their cost, which tends to be significantly higher than their traditional counterparts. In fact, as of 2022, it was estimated that biologic agents accounted for 35.9% of the market for treatment of autoimmune diseases and 42.2% of the market for treatment of cancer, with use in cancer contexts expected to grow the fastest over the coming years. Overall, the market for biologics accounted for \$324.78 billion US dollars in 2020 and is expected to grow to \$749.62 billion by 2028 as the number of products and range of applications continues to expand. This equates to a compounded annual growth rate of 10.8% as demand for novel therapeutic options and governmental support increases (BioSpace 2022).

In addition, the growth of these agents is explained in part by the increase in chronic disease that has occurred in the general population, including the increased diagnosis of especially rare diseases and the continued increased in life expectancy, which results in a larger population impacted by chronic diseases and receiving long-term targeted treatment. Further, the population receiving biologic agents, to be further characterized below, may include both pediatric and adult patients and, particularly soon after diagnosis, may require repeated adjustments to the medication used, or may require treatment with multiple agents in this class and across classes over time (Burgess et al. 2022).

However, across all areas of medicine and particularly in dentistry, the full range of adverse effects experienced by and risks to patients taking these medications remains to be elucidated.

As biologic agents continue their expansion in the market and define their uses in various disease contexts, long-term use of these agents in larger populations may clarify as-yet unknown or incompletely investigated concerns. This may hinder their overall growth, limit their use in various settings, and narrow their scope of applicability across conditions. However, given that this category is defined by their organic, or “biologic,” nature and not by a single mechanism of action, any limit may well be expected to pertain only to certain members of this class. The varying mechanisms as well as variable modes of delivery and impact on different physiologic processes will continue to mean that any concerns identified are expected to be agent- or mechanism-specific and not generalizable to all biologic agents.

Classification of Biologic Agents

Biologic agents are distilled from an organic source and named in part based on their source of origin, with the suffix of the agent distinguishing the origin. Monoclonal antibodies, a cellular expansion of an isolated or fabricated antibody that comprises the medication, are the most common type of biologic agent, accounting for 39.0% of the market (BioSpace 2022). These are isolated and clonally expanded populations derived from various origins. The suffix “-mab,” most commonly found, denotes a human source, while “-zumab” denotes a humanized source and “-ximab” a chimeric source, frequently a fusion of mouse and human antibodies. Biologic agents also include variant fusion proteins, which tend to carry the suffix “-cept” and are fabricated through the combination of distinct antibody components (Yeoh et al. 2019).

The specific agents are further characterized by the molecules they target. The most common classes of biologic agent by prevalence and the most expansive in number are the tumor necrosis factor (TNF)- α inhibitors, lymphocyte modulators, and interleukin inhibitors (Maini and Taylor 2000). As evident in their categorization, these drugs target central components of the immune

system, decreasing overactive signaling of these biologic compounds as found in diverse disease contexts (Murdaca et al. 2009). These are far from the only classes of biologic agents available, however, with additional classes targeting various receptors such as human epidermal growth factor receptor 2 (HER2), activated in breast cancer (i.e., trastuzumab), and vascular endothelial growth factor (VEGF), which is overactive in both multiple malignancies and in various ocular diseases such as age related macular degeneration (i.e., ranibizumab) (Wynn and Tang 2022; Jain 2014). Biologics may also target immunoglobulins, such as the binding of IgE by omalizumab in the treatment of persistent asthma (Jardieu and Fick Jr. 1999). These are only a few examples of the mechanisms and uses of biologic agents. The scope of these medications is prohibitively broad to review in full in this chapter.

Principles of Prescribing

Before starting treatment with biologic agents, patients are required to undergo careful pretreatment screenings inspired by the immunomodulatory actions of these medicines. When treatment with these agents is considered, patient selection proceeds carefully to minimize the risks of immune activation or suppression as well as to pre-emptively evaluate any risks of adverse effects. This may include consideration of past treatments used and their successes, the nature and severity of the disease being treated, the availability of alternative strategies (medications, surgical treatments, or other interventions), and logistical considerations such as cost and availability of the agent in question (Chebli et al. 2014).

During pretreatment evaluation, several general principles apply across biologic agents. Underlying diseases that may complicate treatment, including untreated infection, uncontrolled heart failure, and undiagnosed or untreated comorbid autoimmune processes, particularly multiple sclerosis, must all be ruled out as they can be activated or worsened by treatment with biologics (O'Neill and Scully 2012a). Patients

are required to have all routine vaccinations up to date to minimize the risk of preventable disease during immunomodulation (Kane 2011). Patients are also screened for infectious conditions that may be activated by this treatment, generally including tuberculosis, human immunodeficiency virus (HIV), hepatitis B and C, and varicella zoster virus (VZV). Baseline laboratory testing is another central component of patient selection and preparation through its evaluation of the patient's general health status and risks (Koo et al. 2011). This includes complete blood count (CBC), basic or comprehensive metabolic panel (BMP/CMP) including liver function testing, C-reactive protein and possibly other tests to evaluate systemic inflammation, and tests such as VZV antibody to determine long-term risks during treatment (Findeisen et al. 2021).

Delivery of biologic agents and frequency of dosing is widely variable according to agent and mechanism. In essence, however, all members of this class are protein-based drugs, and the size of the active molecules included means that they require delivery via infusion or injection and, with few exceptions, tend not to be orally administered (O'Neill and Scully 2012a). Dosing of commonly used biologic agents ranges from every 1–2 weeks to every 6 months depending on the disease being targeted and the response to the agent, with flexibility according to disease activity (Edwards et al. 2017). The delivery mechanisms of these agents contribute to their costs as well as to the logistical hurdles required to obtain them, particularly for those given via infusion, which must be delivered in a specialized center by trained healthcare providers, who also monitor for any immediate adverse events (Skalko-Basnet 2014).

Adverse reactions to biologic agents include both reactions that develop immediately on administration and those that develop over time with continued use (Patel and Khan 2017). Infusion reactions occur commonly during administration of biologic agents and can include fevers, chills, rigors, muscle pain, nausea, and vomiting as well as other symptoms such as gastrointestinal upset, flushing and pruritis, tachycardia, increased blood pressure, and palpitations.

Each of these tend to be temporary and result in minimal to no long-term impact on the patient, and infusion reactions may be decreased over repeated exposure to the agent and with supplemental medications provided during the infusion such as corticosteroids and antihistamines. In addition, infusion rates can be decreased to improve patient tolerance, and dosage adjustments can be considered as needed. Infusion reactions are a relatively common finding in patients receiving biologic agents formulated in this manner (Khan 2016). Other adverse reactions to biologic agents vary according to agent, class, and patient, including their background and other medical conditions.

The most concerning adverse reactions associated with biologic agents tend to take the forms of infection (including reactivation of latent infections), increased risk of malignancy, and worsening of congestive heart failure. Other concerns can include development or worsening of hematologic disorders and dermatologic disorders (Shivaji et al. 2019). A Cochrane Review of adverse effects associated with biologics found that there was a statistically higher overall rate of adverse effects in patients taking biologic agents compared to patients receiving alternative treatment regimens, as well as statistically elevated rates of medication withdrawal when using these agents (Singh et al. 2011). Infection is a central concern for patients receiving biologics and can include life-threatening infections due to immune compromise, reactivation of tuberculosis and other latent infectious organisms such as acute reactivation of herpesviruses, and development of new infections (Bharmal and Chia 2022). Older adult patients taking biologic agents have been found to be at statistically significantly increased risk of infection, although the same patients were at no associated increased risk of death (Borren and Ananthakrishnan 2019).

Increased rate of development and progression of malignancy has long been theorized with these agents given their impact on the immune system (Jain and Singh 2013). While there are cases suggesting increased rates of lymphoma or development of malignancies (including oral squamous cell carcinoma) associated with treat-

ment with biologic agents, Cochrane reports low confidence in this finding, and the rates have no clear association with aging (Singh et al. 2011; Borren and Ananthakrishnan 2019). Worsening of congestive heart failure does seem to be associated with treatment using biologics, which is why screening for this condition is recommended before commencement of the drugs. Some have also suggested that these patients may be at increased risk of developing other autoimmune conditions due to the impact of the biologic agent on overall immune signaling, but little evidence to date bears out this concern (Ito et al. 2020). As described below, however, some oral autoimmune reactions have been theorized to result from treatment with biologic agents.

Impact on Dental Treatment

Currently, very little is known about the impact of biologic agents on dental treatment and dental outcomes. Minimal research has been conducted, and the research that does exist is limited in scope and generalizability through small sample sizes, flawed study designs, and lack of reproduction of results to date. For example, no research exists to date on the impact of biologic agents on restorative need, endodontic treatment indication or tolerance, or the impact, if any, of these agents on oral hygiene and general oral status including salivary flow rates and composition of the oral microbiome. The evidence that does exist is reviewed here and is expected to increase in the coming years.

The largest amount of available evidence relating to biologic agents and dental medicine exists in the periodontal literature. It has long been recognized that interleukin (IL)-1 and TNF- α , among other signaling molecules, play a damaging role in the development and progression of periodontal disease through increasing pro-inflammatory signaling and decreasing bone remodeling (Graves and Cochran 2003). In patients treated with biologic agents, the majority of evidence comes from those receiving anti-TNF- α treatment. In patients treated with adalimumab for rheumatoid arthritis (RA), for

example, one study found a decrease in periodontal disease parameters including gingival index (GI), bleeding on probing (BOP), and probing depths (PD), as well as decreased RA disease activity despite similar plaque levels in patients with and without biologic therapy (Kobayashi et al. 2014). Similarly suggesting positive periodontal disease outcomes from biologic agents, another study found improvements in both psoriatic arthritis disease activity and periodontal parameters after TNF- α treatment in a cohort that at baseline exhibited moderate to severe periodontal disease (Ancuta et al. 2017). Patients treated with TNF- α antagonists were also found to have decreased systemic inflammation as measured by erythrocyte sedimentation rate (ESR) and TNF- α levels and improved periodontal parameters when receiving both nonsurgical periodontal therapy and TNF- α inhibitors compared to those receiving alternate medication or no periodontal treatment (Ortiz et al. 2009). Of interest, the patients receiving routine periodontal therapy in this study also displayed improved RA disease control after treatment. Other studies have failed to demonstrate a positive benefit of TNF- α antagonism, but a recent systematic review combined the evidence available and found that patients with RA treated with TNF- α antagonists had improved periodontal health during treatment including early decreases in BOP and GI with later improvements of PD and clinical attachment loss (CAL) (Zamri and de Vries 2020). Other medications, including the anti-CD20 agent rituximab and the IL-6 inhibitor tocilizumab, have also demonstrated positive periodontal disease outcomes including decreased inflammation (Papathanasiou et al. 2021). In general, the mechanisms of action of biologic agents and their use to control systemic inflammation and inflammatory conditions would suggest the potential for a positive impact on periodontal parameters during treatment.

During orthodontic treatment, a theoretical chance of delayed tooth movement has been proposed. This suggestion was based solely on the mechanisms of action of biologic agents and their interplay with the anatomic processes active during orthodontia, however, and has yet to be

confirmed using clinical data (Krishnan and Davidovitch 2006).

Relative to the ability of patients taking biologics to tolerate dental treatment, general recommendations include monitoring of disease control to understand patient status and ability to withstand the stress of dental procedures, as well as to heal appropriately after procedures (Georgakopoulou and Scully 2015a). One small study has evaluated adverse effects after dental extractions in patients taking biologic agents at a single academic center and found that major and minor complications occurred during 10.9% of encounters across classes of biologic agent (Davila et al. 2022). Of these, pain was the most common complication and was found to be over-represented in patients taking anti-VEGF agents. Another study completed using an animal model suggested a possible decrease in bone remodeling in subjects treated with the TNF- α antagonist infliximab (Ferreira-Junior et al. 2020). This study reported more scar tissue formation in mice receiving this agent, as well as lower circulating neutrophil, monocyte, and osteoclast levels and decreased reactivity of TNF- α , receptor activator of nuclear factor kappa beta ligand (RANKL), and osteoprotegerin. These findings in sum would suggest a possibility of delayed or compromised healing in patients on biologics but need to be confirmed and replicated.

In other surgical settings, research on adverse outcomes has been mixed. In inflammatory bowel diseases, a common indication for biologic agent therapy, studies have reached mixed conclusions of variable significance about the impact of the medications on patient outcomes (Zaghiyan et al. 2015). In Crohn's disease, both adults and children treated with TNF- α antagonists exhibit a trend toward higher rates of complication, although the overall rate of complication has been found to not be significantly different in patients treated with biologic agents compared to patients treated with other medications (Ahmed Ali et al. 2014; Yung et al. 2018; Law et al. 2018; Kotze et al. 2018; Lightner et al. 2019a). Within biologic agents, there is also no clear difference in risk according to class of agent used (Lightner et al. 2019b; Shah et al. 2021; Yang et al. 2012).

Given the trend toward increased risk in patients treated with TNF- α antagonists, however, evaluation of a patient's circulating levels of TNF- α has been recommended before surgical intervention, with the consideration of a drug holiday or pause in treatment for patients with Crohn's disease that have high medication levels (Zaghiyan et al. 2015). In ulcerative colitis, unlike in Crohn's, the low bioavailability of the medication nullifies this recommendation.

Studies have also evaluated surgical risks for patients taking biologic agents in other disease contexts. In rheumatoid arthritis, one of the most common indications for biologic therapy, research to date has found no single agent that carries a higher rate of complication after surgical intervention (George et al. 2019; Diaper et al. 2017). In psoriasis, similarly, a consensus of treating dermatologists has recommended continuation of biologic therapy during minor surgeries (Menter et al. 2019). The scope and nature of this recommendation may translate to dentistry given the nature of the interventions in dental medicine.

Oral Complications of Biologic Agents

During treatment with biologic agents, several oral adverse events, in addition to the systemic risks reviewed above, have been identified. A few reviews have compiled and discussed this evidence, which comes primarily from case reports and small case series as well as occasionally from the combined adverse event reporting that accompanies approval and prescribing information for individual agents (France et al. 2023; Georgakopoulou and Scully 2015b). The overall results are reviewed below and expect to be further clarified over time.

Osteonecrosis is a condition commonly associated with antiresorptive agents (termed medication related osteonecrosis of the jaw, MRONJ), as well as with radiation therapy for head and neck cancers (termed osteoradionecrosis, ORN) (Ruggiero et al. 2022). Although bisphosphonates are the medications that were first and

remain most frequently associated with the development of MRONJ, this serious complication has also been identified in association with treatment using antiresorptive biologic agents, antiangiogenic agents including the VEGF inhibitors, and other biologics (Kanwar et al. 2020). Denosumab, a humanized monoclonal antibody that targets IgG2, which binds to the RANK ligand (RANKL), is the most common antiresorptive agent. Denosumab exerts its therapeutic effect by reversibly inhibiting signaling associated with bone remodeling through the RANKL/osteoprotegerin pathway. This medication is frequently prescribed for treatment of osteoporosis, in which case it is often dosed in 6-month intervals, or for prevention of fractures in patients with solid tumors that have metastasized to bone, when it is commonly given in higher doses and at shorter intervals. Given its reversible inhibition, the effects of denosumab wear off after 12–24 months without treatment. In these patients, MRONJ has been observed with a frequency of 0.7–1.9%, and further data is needed to determine the relative wisdom of discontinuing treatment in patients who need dental treatment given the unclear long-term impacts of this therapy and incompletely characterized changes to MRONJ risk over time (Eguia et al. 2020).

Antiangiogenic agents including VEGF inhibitors have also been associated with development of MRONJ (Zarringhalam et al. 2017). The causative mechanism in these cases is theorized to relate to the therapeutic decrease in new vasculature caused by these agents as well as to a possible decrease in macrophage honing and osteoclast maturation (Georgakopoulou et al. 2018). The first case of a patient developing MRONJ during treatment with VEGF antagonists was reported in 2008 in a patient receiving bevacizumab, with additional cases following during treatment with aflibercept and others (Van Poznak 2010). In addition, patients taking these agents may be at risk of impaired wound healing, although this concern is mainly theoretical at this time. Biologic agents of other classes, including the TNF- α inhibitors infliximab and adalimumab and the lymphocyte modulator rituximab, have also been associated with MRONJ (Eguia et al. 2020).

These latter cases lack a clear mechanistic explanation for how the medications would contribute to MRONJ risk, and all biologic agents associated with MRONJ require further long-term data and controlled evidence to fully understand the overall risks and their modifiability over time.

Other oral complications reported during treatment with biologic agents include development of angioedema, oral ulcers, oral infections including candidiasis and viral infection reactivation, development of oral lichenoid reactions, and manifestation of oral sarcoidosis among others (France et al. 2023). An increased risk of oral infection such as oral candidiasis is in keeping with an overall increased risk of infection in these patients and therefore may have a mechanistic connection to the biologic agent. Oral lichenoid reactions in these patients have been associated temporally with the onset of the medications and are theorized to relate to the immune modulation inherent to the use of these agents (Georgakopoulou and Scully 2015b). These reactions have been associated particularly with agents targeting programmed cell death protein 1 (PD-1) and its ligand (PDL-1), commonly used in treatment of oral and oropharyngeal squamous cell carcinoma and other malignancies (i.e., pembrolizumab, nivolumab, atezolizumab) (Ai et al. 2020). Lichen planus is a T cell-mediated reaction and thus shares a mechanistic pathway with certain biologic agents, while lichenoid mucositis can be reactively triggered by a variety of medications and external substances (Yuan and Woo 2020). Although biologic agents have also been proposed for use in the treatment of oral lichen planus (see below), there does exist a physiologic rationale for how modulation of the T-cell pathway could lead to development of lichenoid reactions. Erythema multiforme has also been found to develop during treatment with TNF- α antagonists, likely also through modulation of inflammatory pathways (Goncalves et al. 2019).

Ulcerated mucositis and oral ulcers have similarly been associated with checkpoint inhibitors including the PD-1 and PDL-1 inhibitors above as well as inhibitors of cytotoxic T lymphocyte associated protein 4 (CTLA-4, i.e., ipilimumab). Patients taking these medications have also been

found to develop palatal pigmentation during treatment (Yuan and Woo 2020). This complication is shared with small molecule inhibitors, which are similar to but distinct from biologic agents (Georgakopoulou et al. 2018). Patients taking these and other targeted biologic agents for cancer treatment have also been found to exhibit xerostomia and hyposalivation as well as mucositis. Mucositis is a common complication resulting from various treatments for malignancy and in its assorted stages ranges from erythema to widespread ulceration that can prevent a patient's ability to tolerate oral alimentation (Georgakopoulou and Scully 2015b).

As with the general risks of malignancy, data on the development of oral malignancies is sparse and mixed. However, there have been several cases of newly diagnosed oral squamous cell carcinoma and malignant melanoma in patients taking TNF- α antagonists as well as rituximab (France et al. 2023). Lymphoma of various sites has also been associated with the use of biologic agents and has been found to present in the oral cavity (Baecklund et al. 2006).

Biologics in the Treatment of Oral Conditions

Biologic agents have been used off-label in the treatment of oral diseases, primarily in immune-mediated and autoimmune diseases with oral manifestations. Many of these examples involve treatment of patients with evidence of disease in the oral cavity in addition to other body systems, although other evidence does exist on the use of biologic agents in oral-only disease. While many agents may have potential to provide benefit to these patients, use of a biologic agent should be considered only by providers who are appropriately knowledgeable of the risks of these medications and who are experienced in their use, such as certain oral medicine specialists. In addition, the logistical considerations of delivery of these drugs, including the cost and in many cases the need for delivery through an infusion center, often limit their use in outpatient oral medicine practice.

Pemphigus vulgaris (PV) is the oral mucosal disease with the most evidence for effective treatment with biologic agents (Mays et al. 2019). Two randomized controlled trials have proven the efficacy of rituximab in the treatment of PV, one of which proved that treatment with rituximab was superior to steroids alone and led to US Food and Drug Agency (FDA) approval of this medication for this use (Joly et al. 2017; Kanwar et al. 2014; Press Release 2018). There is also promising evidence from a randomized controlled trial on the treatment of PV with intravenous immunoglobulin (IVIg) and some evidence of the use of local adjunctive treatment with platelet-rich plasma (PRP) (Mays et al. 2019).

In bullous and mucous membrane pemphigoid, there is a lower level of still convincing evidence of the efficacy of rituximab for control of the oral and systemic manifestations of these diseases (Mays et al. 2019). Bullous pemphigoid also has been proven to respond to omalizumab, which binds free IgE in serum, but given this disease's infrequent oral presentation and a lack of evidence on the use of omalizumab to treat oral mucosal lesions of pemphigoid, this treatment should be considered carefully before use (Kremer et al. 2019). A recent systematic review did find benefit to IVIg and TNF- α inhibitors as well as rituximab for the treatment of pemphigoid, although with the chance of incomplete resolution or relapse (Lytvyn et al. 2022). In fact, many patients with pemphigoid are found to relapse after treatment and may need to receive a combination of biologic agents, corticosteroids, and other immunosuppressants, albeit often at lower doses than when used as monotherapy, to maintain appropriate disease control (Veilleux and Shear 2017). Based on all of this evidence, in the United Kingdom, the National Health Service (NHS) has approved rituximab for use in both pemphigus vulgaris and pemphigoid (NHS England 2016).

Multiple studies have evaluated the efficacy of biologic agents for the treatment of Sjögren's syndrome, both for overall disease control and specifically for improvement of the oral component of the disease. These studies have resulted in mixed evidence with the need for confirmatory

studies to further evaluate appropriate use (Gueiros et al. 2019). Specifically, belimumab has been found to improve both salivary function and the symptoms of xerostomia, the subjective experience of dry mouth (De Vita et al. 2015). Abatacept also shows promise for the improvement of both subjective and objective symptoms. Rituximab use in Sjögren's syndrome improves objective measures including ultrasound scores and salivary function but shows less benefit to subjective symptoms (Fisher et al. 2018; Meijer et al. 2010). A newer agent, iscalimab, an anti-CD40 biologic, showed significant promise in early studies for improvement of oral symptoms and signs of Sjögren's syndrome and is in continued development for possible use in this and other settings (GlobalData 2023).

Other disease contexts with oral components and isolated oral conditions so far have significantly less evidence on the use of biologic agents. In the treatment of advanced oral and oropharyngeal squamous cell carcinoma, cetuximab targets epidermal growth factor receptor (EGFR) and is found to potentiate treatment of conventional chemotherapeutics in hopes to stop tumor growth and improve patient responses to radiation therapy (Naruse et al. 2016). Other biologic agents have also been evaluated in the treatment of oral cancers for their ability to reduce inflammation and pro-inflammatory signaling and modulate the tumor microenvironment. These drugs remain in testing (Brierly et al. 2023).

In Behçet's disease and the treatment of recurrent aphthous stomatitis, the TNF- α inhibitors infliximab, adalimumab, and etanercept have all been found to be helpful as adjuvant therapies or as monotherapy for control of oral ulcers, including major aphthous ulcers (Connolly et al. 2005; Vujević and Zirwas 2005; Robinson 2003; Melikoglu et al. 2005). In oral lichen planus, T-cell mediation provides theoretical promise for treatment with biologic agents, which have shown mixed results in clinical testing (Zhang et al. 2011). This includes reports of patients developing lichenoid lesions during treatment with biologics as well as some evidence of benefit with various biologics (France et al. 2023). To date, however, there is

no consensus on recommended treatment options in this class for patients with oral lichen planus. In orofacial granulomatosis, infliximab (as used in intestinal disease) may control oral symptoms, particularly on a short-term basis. In Steven's Johnson Syndrome, chronic graft versus host disease, and others, some evidence exist and require confirmation and reproduction of therapeutic effect (France et al. 2023; Elliott et al. 2011; O'Neill and Scully 2012b). While none of these diseases have to date shown definitive response under biologic agent therapy, studies are ongoing and employ both established and new biologic agents.

Conclusion

Biologic agents are a continually expanding and increasingly prevalent class of medications used to treat a wide variety of diseases affecting a full range of organ systems, primarily those with malignant or immunologic etiology. New agents are continually developed and introduced, and expanded applications are investigated for available biologic agents, continually increasing the reach and functionality of these agents. The first commercially available biologic agent, etanercept, was approved by the US FDA in 1998, providing at most 25 years of data on any of these medications (Curtis and Singh 2011). With longer periods of use, long-term effects of this class are beginning to become apparent, including development of local and systemic adverse events, development of resistance, and immune activation with prolonged use of these drugs. In addition, practical considerations including the cost, coordination of infusion or injection of these large molecules, and pretreatment preparation continue to limit the availability of biologic agents, particularly in populations with more limited health-care resources.

Mechanisms of action of biologic agents are also quite diverse as individual agents are grouped in this category solely based on their selective targeting of a naturally occurring physiologic pathway. This also limits the ability to compare and understand these agents and increases the

research required to fully characterize this class. These limitations aside, biologic agents, are heralded for the role they will continue to play in shaping the future of medical care and are increasingly recognized for an associated impact on dental medicine.

While data is currently limited, the impact of biologic agents on oral disease processes, treatment of oral lesions, and ability to tolerate dental care are beginning to be recognized. With increased attention to this area in future years, additional data is expected to further clarify any positive or negative impacts of this class of medication on dental health and the ability of patients to safely undergo dental treatment. Currently, best practice recommendations include for any oral health-care provider to monitor the patient's overall health status and disease activity and to prioritize treatment during periods of relative control. In addition, patients should as always be monitored for development of oral infections and should be treated in a timely manner as needed. Ideal oral health in these patients may contribute to overall disease control and quality of life, metrics that tend to be positively impacted by the efficacy and tolerability of biologic agents.

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Basic Emergency Drugs and Non-intravenous Routes of Administration

Arthur H. Jeske

Emergency drugs in the dental outpatient setting are considered to be adjunctive to physical measures, for example, airway opening and oxygenation. As such, the array of drugs listed here is a basic list, and many of these agents can be obtained through purchase of emergency kits. Before purchasing such kits, the dentist should consider the following:

1. No single kit meets the requirements of all dental offices, especially in regard to reversal (“antidotal”) agents.
2. Emergency drugs go out of date at different intervals, and expiration dates must be monitored accordingly.
3. Dose forms must be compatible with the abilities of the dental teams to use various routes of administration. This can be problematic when drugs indicated for intravenous use only are administered by nonvascular routes, which reduces efficacy and prolongs onset.
4. The training of the dentist(s) and staff members, requirements of the practice’s licensing jurisdiction, and the types of patients in the practice and the sedation modalities employed in the practice (Rosenberg 2010).

A. H. Jeske (✉)
UTHealth School of Dentistry, Houston, TX, USA
e-mail: Arthur.H.Jeske@uth.tmc.edu

Antihistamines

Antihistamines are appropriate for mild, delayed-onset allergic reactions, as are corticosteroids. They lack important actions necessary to alleviate the rapid pathophysiologic effects of Type 1 (anaphylactic) allergic reactions. In anaphylaxis, antihistamines, corticosteroids, and beta agonists are secondary treatments (LoVerde et al. 2018). Outcomes from the administration of oral or intramuscular antihistamines include sedation and anticholinergic actions, which must be considered prior to discharge and in the follow-up care of the patient.

Antiplatelet Drugs (Aspirin, Clopidogrel)

According to the American Heart Association’s Advanced Cardiac Life Support protocols, an antiplatelet drug is an important component of the first intervention in cases of acute coronary syndrome (American Heart Association 2016). Current, high-quality scientific evidence supports the beneficial actions of orally administered aspirin in patients with acute coronary syndrome, while other agents historically indicated in this emergency condition (morphine, oxygen, nitrates) are supported only by limited evidence and may actually result in less favorable post-event outcomes (McCarthy et al.

2017). A recent systematic review also highlighted the potential application of a clinical decision tool to optimize the use of antiplatelet agents in cases of acute coronary syndrome (Reynard and Body 2017). Nonsteroidal anti-inflammatory drugs should not be used in place of aspirin or clopidogrel for an antiplatelet action in cardiac-related emergency situations (American Heart Association 2016).

Antiseizure Agents

In the dental office setting, benzodiazepines are recommended as being both safe and effective for the management of status epilepticus prior to hospital admission. Typically, midazolam or lorazepam is recommended for terminating the potentially life-threatening seizures of status epilepticus. Outcomes from the RAMPART Study of interventions for this serious emergency, based on a total sample of 800 patients, indicated that intramuscular administration of 10 mg midazolam in adults (5 mg in children <40 kg) was equivalent to the administration of 4 mg of lorazepam administered intravenously in adults (2 mg in children <40 kg) and that the overall interval until seizure termination (including both the time actual drug administration and the onset of action of the drug) was essentially the same in both the i.m. and i.v. drug groups (Silbergeit et al. 2013). A recent systematic review of various routes of administration of midazolam and diazepam for status epilepticus has confirmed that non-intravenous midazolam is as effective and safe as intravenous or rectal diazepam in terminating early status epilepticus in children and probably also in adults (Brigo et al. 2015). Other drugs evaluated for this indication include lorazepam, phenobarbital, phenytoin, and paraldehyde (Appleton et al. 2008).

Atropine

Like many of the drugs described in this chapter, atropine is available in preloaded syringes. While atropine is not typically included in com-

mercially available drug kits, it is an important agent for the management of acute bradycardia, as may occur in local anesthetic overdose reactions. Clinically, significant bradycardia is defined as a heart rate lower than 60 bpm in awake individuals and nonathletes (American Heart Association 2016; Barstow and McDivitt 2017). When bradycardia occurs acutely, atropine is the drug of choice for its management (Deal 2013). Available in prefilled syringes, the typical dose for management of bradycardia in an adult is 0.4 mg intramuscularly, and an aggregate dose of 2 mg will produce total loss of vagal control of the heart, which must be considered when repeat doses are administered. Atropine autoinjectors (e.g., AtroPen®) are available with several concentrations of atropine, from 0.25 mg to 2 mg per administration.

Beta Adrenergic Agonists

The principal indication for beta agonists (e.g., isoproterenol, albuterol) is the management of acute bronchial constriction. These agents are typically available in dental emergency drug kits in inhalational dose forms, which can accurately meter the dosage. Importantly, beta agonists are powerful cardiac stimulants and may cause excessive cardiac workload when improperly administered. In the management of acute bronchial constriction, albuterol is typically administered by inhalation, using a propellant-driven aerosol liquid. The dose is commonly 90 µg per actuation of the inhaler device. Adverse effects of albuterol can be serious and include cardiac stimulation, due to activation of cardiac beta receptors. It is important to note that corticosteroid-based inhalers used by asthmatic outpatients are not rescue inhalers, due to the long onset of action of the corticosteroid. A recent systematic review has confirmed that inhaled, short-acting beta-2 adrenergic agonists such as albuterol remain the mainstay of treatment for acute asthma and that the metered-dose inhaler delivers improved lung function as the nebulizer or intravenous administration (Green 2011).

Corticosteroids

Glucocorticoids are typically stocked in dental office emergency kits for the management of anaphylaxis. While they are regularly used in the medical management of asthma and as adjuncts to epinephrine and antihistamines in cases of anaphylaxis, their therapeutic benefit and adverse effects in anaphylaxis have not been scientifically validated (Choo et al. 2012). Their therapeutic effect appears to be optimal when administered intravenously, likely due to the relatively slow onset of action when administered by other routes. Typically, a corticosteroid with a high ratio of anti-inflammatory to salt-retaining activity is preferred, such as dexamethasone, administered intravenously or intraosseously.

Epinephrine

Despite lack of validation from randomized and quasi-randomized controlled clinical trials (Sheikh et al. 2008), epinephrine remains the drug of first choice for the management of Type 1 allergic reactions, that is, anaphylaxis (American Heart Association 2016). Its benefits derive from direct agonist effects at beta-2 adrenoceptors that mediate bronchial dilation, as well as cardiac stimulation mediated by beta-1 adrenoceptors and alpha-1-mediated vasoconstriction (American Heart Association 2016). The optimal route of administration in nonhospital settings intramuscular, usually at a dose 0.3 mg for adults. Epinephrine is widely available in pre-filled autoinjectors but can be obtained in ampuls and vials. Recent reviews suggest that autoinjectors are associated with accidental injection of thumbs and insufficient needle lengths in obese patients, while syringe-administered epinephrine is associated with inadvertent subcutaneous or i.v. administration, as well as dosing errors (Chime et al. 2017). The adult dose is 0.3 mg administered intramuscularly, and the pediatric dose is 0.15 mg, also i.m. These doses are accurately metered by autoinjector devices, if the devices are used properly.

Glucagon

Like glucose, glucagon is used to manage acute hypoglycemia, but unlike glucose, it requires parenteral administration and may be indicated when the patient lapses into an unconsciousness as a result of hypoglycemia and can no longer safely be given oral glucose. According to a recent systematic review (Villani et al. 2017), 1 mg glucagon administered intramuscularly can be effective for the emergency management of hypoglycemia in the unresponsive adult (Vaccine Administration n.d.). Glucagon is available as a lyophilized solid and must be solubilized using the manufacturer-provide diluting solution. It is important to note that the efficacy of glucagon is dependent on the presence of glycogen in the patient's liver, where glucagon stimulates its breakdown to glucose. There are few adverse effects of glucagon when administered in a single dose for the management of acute hypoglycemia. However, glucagon is best reserved for use when intravenous glucose is not available or cannot be given (see next section, 11.9 Glucose).

A new form of glucagon, dasiglucagon (Zegalogue®), is now available in prefilled syringes for the treatment of severe hypoglycemia in adults and children over the age of 6 and has been shown to have a higher rate of absorption and longer half-life than traditional forms of glucagon (Xu et al. 2021).

Glucose

In the past, table sugar and sweet beverages (e.g., cola, orange juice) were considered “front-line” sources of glucose for diabetic patients experiencing acute hypoglycemia. However, the sugars found in substances like cake icing (sucrose) are more complex than the simple sugar glucose molecule. Nevertheless, mild hypoglycemia appears to be manageable with carbohydrate-containing foods and beverages (Evert 2014). A study of hypoglycemic children with Type 1 diabetes found that sucrose derived from candy is equally effective with glucose tablets for elevat-

ing blood glucose, while fruit-derived fructose was less effective than either glucose or sucrose (Husband et al. 2010). A recent systematic review found that high-quality scientific evidence for the management of hypoglycemia is lacking, and therefore, recommendations for various treatments are, therefore, limited (Villani et al. 2017). However, there is consensus that 15–20 g of oral sucrose or glucose (i.e., fast-acting carbohydrates) should be used in conscious patients with glucose levels less than 70 mg/dL, and those with blood glucose levels less than 50 mg/dL should eat 30 g of fast-acting carbohydrates, and this dose can be repeated in 10–15 min, as indicated (Cornelius 2017). The recommended treatment for unresponsive patients is 10% glucose administered intravenously or 1 mg of glucagon administered intramuscularly. Both 50 mL of dextrose 50% or 500 mL of dextrose 5% will elevate blood glucose levels approximately 75–125 mg/dL (Cornelius 2017).

Nitrates

Nitrates, such as nitroglycerin, were considered front-line agents for combination drug therapy (MONA) in cases of acute coronary syndrome by virtue of reducing peripheral vascular resistance with corresponding decreases in cardiac workload and myocardial oxygen demand (American Heart Association 2016). It should not be administered if systolic blood pressure is less than 90 mmHg, or if there is marked bradycardia or tachycardia (McCarthy et al. 2017). Because nitroglycerin appears to possess an analgesic action, it is important to note that relief of chest pain after the administration of nitroglycerin is not diagnostic of acute coronary syndrome. The victim is given one sublingual tablet or one spray dose every 3–5 min as appropriate and indicated, up to a total of three administrations. Recently published scientific evidence suggests that the benefits of nitrate administration in acute coronary syndrome may be only marginal, although its use in acute coronary syndrome continues (LoVerde et al. 2018).

Oxygen

In many emergencies in which airway compromise or other respiratory embarrassment may occur (e.g., overdose of CNS depressant drugs with depression of the respiratory center), oxygen is considered an important first-choice agent when combined with ventilation in cases involving significant hypoxemia ($\text{PaO}_2 < 94\%$). However, recent evidence has called into question the standard practice of administering high-flow oxygen in acute coronary syndrome. In such cases, oxygen would be indicated only in hypoxic patients with an O_2 Saturation less than 90% (McCarthy et al. 2017). Concerns about the use of high-flow oxygen therapy in normoxic patients with acute coronary syndrome include reduction of cardiac output and left ventricular perfusion, increased coronary vascular resistance, and the potential development of reactive oxygen radicals, which can cause cardiac dysrhythmias (McCarthy et al. 2017).

Reversal (Antidotal) Drugs

Naloxone

Naloxone (e.g., Narcan®) is a nonselective opioid receptor antagonist that is now being marketed in multiple preparations as a result of the current opioid overdose crisis in the United States. Naloxone is characterized clinically by a relatively short duration and the complete reversal of all actions of opioids in patients, including pain control, and may precipitate severe withdrawal in individuals who are addicted to opioids. Naloxone is typically administered parenterally, either i.m. or i.v., with the i.v. route producing the most reliable actions with the fastest onset of action. In nonhospital settings, naloxone is typically used by first responders by the intramuscular route, with a prefilled autoinjector (e.g., Evzio®) at a dose of 0.4 mg. However, recent case reports suggest that individuals who have acutely overdosed with fentanyl or fentanyl analogs may require much higher, repeated doses as high as 4 mg (Tomassoni et al. 2017). With the recent

appearance of nitazene opioids in the illicit drug market, antagonists with higher affinity than naloxone for the mu opioid receptor are needed for intoxication with higher-potency opioid agonists. Nalmefene (e.g., Revex[®]) has been introduced in the United States as a longer-acting, high-affinity opioid antagonist, and a nasal spray form is also being developed. At this time, naloxone can be administered by the nasal route with good efficacy (Krieter et al. 2019). A recent systematic review of the effect of the route of administration of naloxone in cases of out-of-hospital opioid overdose found that a higher concentration intranasal dose of naloxone (2 mg/mL) appears to have efficacy equivalent to that of intramuscular naloxone with a comparable rate of adverse events (e.g., agitation) (Chou et al. 2017). The intranasal route of administration can be accomplished with a commercially available preparation, illustrated in Fig. 1.

Flumazenil

Flumazenil (e.g., Romazicon[®]) is a nonselective benzodiazepine receptor antagonist, which, like naloxone, lacks inherent efficacy at these receptors (pure antagonist). Also, like naloxone, it has a relatively short duration and the potential to induce severe withdrawal in individuals who have taken benzodiazepines long term, including seizures, and may precipitate seizure activity in

patients with epilepsy. Recently, the need for adherence to the FDA's label indication, "for intravenous use only" has been emphasized (Weaver 2011). Typically, a 1- to 3-mg i.v. dose should be safe and effective when administered in cases of benzodiazepine-induced unconsciousness and apnea. Importantly, the short duration (20–45 min) of a single dose of flumazenil can result in recurrence of benzodiazepine-induced CNS depression, which can also be significant if the flumazenil has been administered to simply reawaken the patient to reduce recovery time. The actual dose of flumazenil required for emergency management will vary directly with the total dose of benzodiazepine that has been administered to the patient for sedation. Small doses (e.g., 0.2 mg) have been incorrectly recommended by the sublingual or intramuscular route, as both this dose and these routes of administration are unlikely to achieve therapeutic concentrations in time to prevent serious, permanent injury to the patient (Weaver 2011).

Vascular Access/Intraosseous Administration

Developed initially by the US military forces for vascular administration of drugs and fluids under battlefield conditions, intraosseous drug administration is now a mainstream route that is routinely employed by emergency medical technicians and is included in the American Heart Association's for Advanced Cardiac Life Support guidance for several management protocols, which cite the following advantages of this route of administration (American Heart Association 2016):

- Access can be established in all age groups.
- Access can often be achieved in 30–60 s.
- Preferred over endotracheal route.
- May be easier to establish than the endotracheal route in cardiac arrest.
- Any ACLS drug that is administered intravenously can be given IO.

Fig. 1 Commercially available naloxone nasal spray preparation, with atomization device. (Image courtesy of Adapt Pharma, Inc.)



Intraosseous infusion can be accomplished at two sites in both adult and pediatric patients

(tibial and humeral sites only). The driver is illustrated in Fig. 2, and a schematic drawing of the indwelling needle, tubing connection to the needle, and the syringe is seen in Fig. 3. The non-collapsible marrow venous plexus in the proximal head of the tibia, as well as the humerus, allows rapid administration and vascular uptake of medications delivered, including crystalloids, colloids, and even blood (Leidel et al. 2009). However, because the humerus provides more rapid delivery to the heart, it may be preferred for specific cardiovascular emergencies.

Caution must be exercised in performing intraosseous emergency medications in conscious patients, as the pressure from the drug infusion can elicit temporary, sharp pain. While a small amount of a plain lidocaine solution can be

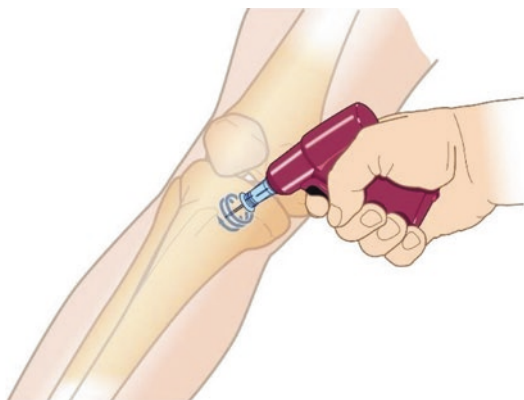


Fig. 2 Intraosseous driver insertion in proximal head of tibia. (Image courtesy of Teleflex, Inc.)

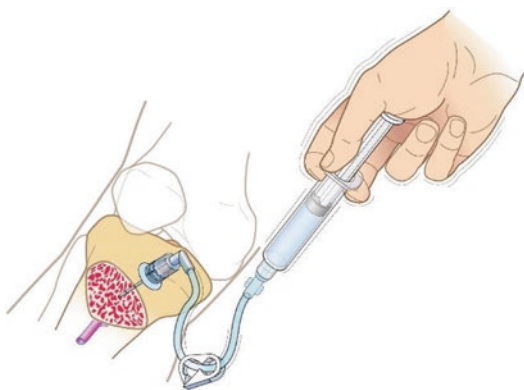


Fig. 3 Intraosseous injection technique at proximal head of tibia, syringe-to-cannula-indwelling needle in place (image courtesy of Teleflex, Inc.)

administered prior to administration of the actual interventional drug, the single administration of an urgently required emergency drug in the emergent prehospital setting (e.g., a reversal agent) could be significantly delayed if this step is taken.

Intranasal (Inhalational) Administration

The intranasal route of drug administration has been used in medicine for decades, including uses in over-the-counter preparations and prescription drugs for respiratory diseases (e.g., asthma). Recently, there has been a resurgence in interest in this route of administration for emergency drugs, including naloxone for opioid overdoses (Corrigan et al. 2015). Based on currently available studies, the advantages of this route include the following:

- Needles not required, less painful.
- Rapid drug delivery.
- Pharmacokinetics more favorable than i.m. in obese and elderly patients.
- Minimizes risk of needlestick injuries and spread of blood-borne diseases.

There are, however, several limitations to this technique, including the following:

- Limited volume of medication (<1 mL).
- Limited data/evidence for safety and efficacy.
- Costs greater than those of i.v.
- Contraindicated in cases of nasal trauma or recent use of nasal vasoconstrictors.
- Less reliable than i.v. or i.o. routes.
- Patient acceptance (palatability) variable.
- May cause nasal mucosal irritation.

At this time, two emergency drugs commonly stocked in dental offices—midazolam and naloxone—may be effectively used by the intranasal route, for the management of seizures and opioid overdose, respectively, when the preferred intravenous route is unavailable or impractical (Rech et al. 2017). The armamentarium for intranasal drug administration includes a conventional



Fig. 4 Medication atomization device (MAD), showing nasal cone, attached to conventional Luer-lock disposable syringe. (Image courtesy of Teleflex, Inc.)

syringe and a special medication atomization device that is attached to the syringe after the medication is drawn up and that provides a seal when inserted into the nasal orifice (Fig. 4).

Other drugs that have been documented to be effective when administered intranasally include fentanyl, sufentanil, ketamine, hydromorphone, flumazenil, and glucagon (Corrigan et al. 2015).

Intramuscular Administration

Perhaps, the simplest parenteral route of drug administration, i.m. injection is reliable and relatively safe and is used with autoinjectors for bystander treatments of anaphylaxis and opioid overdoses. Recent scientific evidence from studies in children and adolescent patients has confirmed that the intramuscular administration of midazolam with a commercial autoinjector device is as effective as rectally or intravenously administered diazepam for the acute management of seizures (Mula 2017). However, the proper armamentarium may include filter needles (for drugs drawn from ampuls) and may require various gauges and needle lengths, depending on the age/size of the target muscle (Vaccine Administration n.d.). Evidence-base recommendations for adult dosages and routes of administration of emergency drugs are summarized in Table 1.

Table 1 Current evidence-based doses and routes of administration recommended for basic emergency drugs in adults without contraindications^a

Drug	Dose	Route(s)
Albuterol (bronchodilator)	90 µg	Inhalation
Antihistamine (diphenhydramine)	50–100 mg	i.m.
Antiplatelet drug (clopidogrel)	300 mg	Oral
Antiplatelet drug (non-enteric-coated aspirin)	160–325 mg	Oral, chewed
Anti-seizure drug (lorazepam)	4 mg	i.v.
Antiseizure drug (midazolam)	10 mg	i.m., intranasal
Atropine	0.4 mg	i.m.
Corticosteroid (dexamethasone)		i.m. or i.v.
Epinephrine	0.3 mg	i.m.
Flumazenil		i.v., i.o., intranasal
Glucagon	1 mg	i.m.
Glucose	15–20 g	Oral
Naloxone	0.4 mg	i.m.
Nitroglycerin	0.4 mg	Sublingual
Oxygen 100% (high-flow, non-rebreathing bag-valve-mask)	15 lpm	Inhalation

^aInitial dose only; re-administration should be based on recurrence of emergent symptoms, physiologic status of the patient, etc

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Internet Resources for Dental Pharmacology

Arthur H. Jeske

The list of internet-based resources for information on dental pharmacology is not intended to be comprehensive, and the site URLs and their content may change without notice. They are included here because they meet three criteria:

1. They are publicly accessible at no cost.
2. They are based strictly on scientific or US government-vetted information.
3. In most cases, they contain summaries that can easily be understood and communicated to members of the healthcare team and patients.

The reader should note that some for-profit (.com) sites are included, but they offer useful, no-cost features of importance to dental pharmacology. Neither the author nor Springer Publications has any financial interest in these sites.

American Academy of Oral Medicine

aaom.com

Perhaps, the best tool available at this website is access to the authoritative journal, *Oral Surgery*

A. H. Jeske (✉)
UTHealth School of Dentistry, Houston, TX, USA
e-mail: Arthur.H.Jeske@uth.tmc.edu

Oral Medicine Oral Pathology Oral Radiology. This can be accessed using the top tool button labeled “Publications and News,” then accessing “OOOO Journal.” Past issues of this journal are archived and can be easily searched within the site using the search feature.

American Academy of Pediatric Dentistry

aapd.org

The AAPD publishes an extensive list of clinical guidelines at its internet site, and most of these documents are regularly revised, ideally on their every-3-year cycle. Those AAPD Guidelines with particular relevance to dental pharmacology can be found under the category “Publications,” “Oral Health Policies and Recommendations,” “Clinical Practice Guidelines,” and “Oral Health Policies” and include the following:

1. Use of Local Anesthesia for Pediatric Dental Patients.
2. Use of Nitrous Oxide for Pediatric Dental Patients.
3. Use of Antibiotic Therapy for Pediatric Dental Patients.
4. Antibiotic Prophylaxis for Dental Patients at Risk for Infection.
5. Useful Medications for Oral Conditions.

American Heart Association

heart.org

Of greatest importance at this website is a variety of patient and practitioner resources for the management of patients with cardiovascular diseases that increase the risk of serious systemic and cardiovascular device infections. While there is a great deal of information at this site, many of its features are written in plain English for the lay person, and it can easily be searched using the simple alphabetical search tool at the top of the web page. The most relevant documents available from the AHA site can be found under the search key “Guidelines and Statements” and include the following:

1. Infective endocarditis (include guidelines for antibiotic prophylaxis for dental patients and a downloadable “wallet card” for patients, in both English and Spanish).
2. Cardiac Medications At-A-Glance (includes an overview of all categories of cardiovascular drugs, with explanations of their mechanisms of action, indications, etc.)

American Academy of Periodontology

perio.org

This site contains several features with relevance to dental pharmacology. At the website, select “Research and Science” from the top toolbar, then “AAP Clinical and Scientific Papers.” At this page, the dentist can select “Position Papers” for information on the following topics:

1. 2017 Classification of Periodontal and Peri-Implant Diseases and Conditions.
2. Systemic Antibiotics in Periodontics.
3. Tobacco Use and the Periodontal Patient.

Also at this page, by selecting “Academy Statements,” the following relevant papers may be accessed:

1. Periostat® as an Adjunct to Scaling and Root Planing.
2. Use of Moderate Sedation by Periodontists.

American Association of Endodontists

aae.org

No dental specialty utilizes antibiotics, anti-inflammatory drugs, analgesics, and local anesthetics to a greater extent in the outpatient setting than endodontics. The authoritative US resource for guidelines in this area is the American Association of Endodontists (AAE), which regularly publishes its *Colleagues for Excellence* information newsletter, and all issues of this publication are archived at the AAE website.

Searches at this site related to dental pharmacology would begin at the “For Professionals” tool. The dentist can then select “Publications & Research” to access the “*Journal of Endodontics*” archives and the “*ENDODONTICS: Colleagues for Excellence*” publications. This is the current location for the document, “Endodontic and Antibiotic Update.”

Selection of the tab “Clinical Resources” followed by “Guidelines & Position Statements” allows access to two other useful documents related to antibiotic use:

1. *AAE Guidance on the Use of Systemic Antibiotics in Endodontics.*
2. *AAE Guidance on Antibiotic Prophylaxis for Patients at Risk of Systemic Disease.*

American Association of Oral and Maxillofacial Surgeons

aaoms.org

Two very significant publications reside at this website. The first is the AAOMS’ *Position Paper on Medication-Related Osteonecrosis of the Jaw* (recently updated) and the *White Paper on Opioid Prescribing* are perhaps the most significant resources for the general dentist. The paper on

MRONJ is addressed in this book in Chap. “Introduction to Contemporary Dental Pharmacology and Special Topics”, and the latter publication on opioid prescribing is described in greater detail in Chap. “Opioid Analgesics, Benzodiazepines, and Other Controlled Substances”.

American Dental Association (ADA) Center for Evidence-Based Dentistry

ada.org/en/resources/research/science-and-research-institute/evidence-based-dental-research

This site contains Clinical Practice Guidelines, Dental Evidence, and Systematic Reviews developed under the auspices of the ADA’s Center for Evidence-Based Dentistry. The Clinical Guidelines feature of this website represent the strongest evidence for various dental interventions, materials, and procedures,

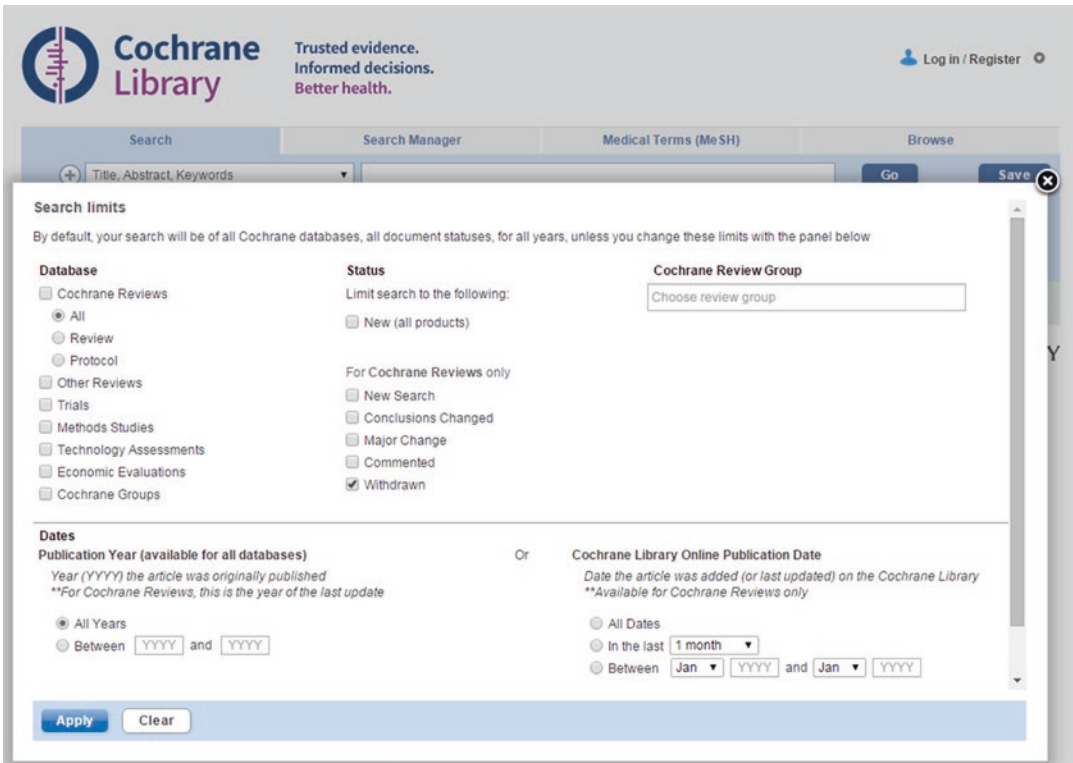
as they are based on outcomes from multiple systematic reviews and are vetted by ADA expertise, including the ADA Council on Scientific Affairs, ADA Division of Science.

Cochrane Library

cochrane.org

Widely regarded as the world’s most authoritative library of systematic reviews, the Cochrane Library, established in 1941, only accepts reviews based on randomized controlled trials (RCTs) (Fig. 1). In spite of this high standard, the health-care professional can access thousands of systematic reviews of medical and dental interventions. The site features a “Browse by Topic” search tool, under which two topics particularly suited to this textbook can be found:

1. Pain and anesthesia.
2. Dentistry and oral health.



The screenshot displays the Cochrane Library homepage with a search bar at the top. Below the search bar, there is a section titled "Search limits" which allows users to filter their search results. The filters are organized into three main categories: Database, Status, and Dates.

Database: Users can select from various Cochrane databases. The "All" option is selected by default. Other options include Review, Protocol, Other Reviews, Trials, Methods Studies, Technology Assessments, Economic Evaluations, and Cochrane Groups.

Status: Users can limit their search to specific document statuses. The "New (all products)" option is selected by default. Other options include For Cochrane Reviews only, New Search, Conclusions Changed, Major Change, Commented, and Withdrawn.

Dates: Users can filter by publication year or Cochrane Library Online Publication Date. The "All Years" option is selected by default for publication year. For Cochrane Library Online Publication Date, users can select "All Dates", "In the last 1 month", or "Between" two dates.

At the bottom of the search limits section, there are "Apply" and "Clear" buttons.

Fig. 1 Cochrane library homepage

Under this item (Pieper et al. 2015), the reader can access a number of dentally relevant reviews. For example, “Pharmacological interventions for preventing dry mouth and salivary gland dysfunction following radiotherapy.” After selecting a review of interest, the dentist can select the abstract, which briefly describes the background for the systematic review, the objectives, the search methods, the selection criteria (used for identifying those studies which were included and excluded), the methods used to collect and analyze data, the main results of the systematic review, the authors’ conclusions, and an easy-to-interpret “Plain language summary.” Finally, a search tool allows the reader to access the entire systematic review (“Get access to the full text of the systematic review”). Throughout this book, the reader can see the use of Cochrane systematic reviews (e.g., in Chap. “Non-opioid Analgesics in Dental Practice”).

Drug Enforcement Administration (United States)

<https://www.dea.gov>

This site is primarily concerned with legal, regulatory, and administrative information related to controlled substances for the United States. For practitioners who prescribe controlled substances, there are several very helpful tools and documents that reside here. Examples include the following major headings, with selected resources following:

Drug Info

1. Drug Fact Sheets.
2. Drug Scheduling.
3. Controlled Substances Act.

Resource Center

1. Drug disposal.
2. How do I... (this feature addresses commonly asked questions, including how to change registration address, etc.)

The reader is cautioned that this agency, and its web-based information is only applicable to juris-

dictions within the United States and US-related locations within DEA jurisdiction. Also, the ordering, administration, and prescription of controlled substances in the United States is also regulated by applicable state law and rules and regulations of dental and medical licensing boards.

Drugs.com

drugs.com

This commercial website is very useful and efficient when information on adverse drug interactions is needed. At top of the website, the toolbar contains an “Interactions Checker.” The practitioner enters a series of drug names and then clicks “Check for Interactions.” The software does not limit the number of drugs that can be entered in this feature. The information is provided both in plain-English and professional format and includes a severity rating in regard to specific interactions, as well as advice of management of interactions. Like other websites of this nature, it also contains package insert information on prescription drugs, as well as links to a variety of apps and important patient-oriented descriptions of common diseases through links to the *Harvard Health Guides* and *Mayo Clinic Disease References*. The site also has an “advanced search feature,” a phonetic search feature and drug information downloadable in Spanish.

In a “Q&A” section, questions about various drugs can be posted and responded to, and there is a list of support groups for patients with various serious systemic diseases.

Additional important features of this website include the following:

- Master list of drugs A–Z.
- Drugs by condition.
- Drugs by class.
- Drug comparisons.
- Generic drugs.
- OTC drugs.
- International drugs.
- Natural products.
- Veterinary products.
- Drug side effects.

- Dosage guides.
- Drugs in pregnancy.
- Breastfeeding warnings.
- Pricing and coupon availability.
- Inactive ingredients.

Finally, there is a “Pro Edition” tool with an A–Z professional drug list, FDA prescribing data, AHFS Drug Information Monographs, A–Z drugs facts, natural products, Stedman’s Medical Dictionary, a list of current medical conferences, and a comprehensive list of international pharmaceutical companies.

This website is one of several internet-based drug information services, which also includes commercial healthcare and pharmaceutical advertising and is described here as an example of such sites, rather than as an endorsement by the author.

Food and Drug Administration (United States)

www.fda.gov/

As the official informational website of the US Food and Drug Administration, this site is very likely the most information-intensive one of all those included in this chapter. For the busy dental practitioner, it can be somewhat challenging to navigate. The recommended starting point is the tool button “Drugs” at the top of the first page. Second, the tab “Find Information about a Drug” opens a wide variety of resources, and particularly useful are links to “Consumer-friendly Information” and the link “Drugs@FDA,” which leads to the full label information for all FDA-approved drugs, including approval history. Another potentially useful tab is “Orange Book Search,” which provides access to therapeutic equivalencies of different brands of a particular medication.

National Library of Medicine/ PubMed

www.ncbi.nlm.nih.gov/pubmed

Among the most powerful scientific search engines ever developed, the US National Library

of Medicine’s PubMed database is relatively easy to use and allows customization of search criteria to make literature searches very efficient. A complete description of the operation of this website is beyond the scope of this textbook. However, the reader is advised to take advantage of several features of this website:

1. On the first page of the site, under the “Using PubMed,” “PubMed Tutorials” can be accessed to become more familiar with use of the site.
2. On the first page of the site, under the “PubMed Tools” category, a “Clinical Queries” feature can be activated to help guide a very specific search of relevant articles.
3. On the first page of the site, under “More Resources,” the “MeSH” tool can be activated (Medical Subject Headings), to help make searches by subject more efficient.

After a specific abstract has been located by entering a search term in the top search field, a list of “Similar Articles” appears to the right of the main search outcome screen, and the availability of free, complete texts of articles is noted in the upper right of the screen, as well as just below the abstract, and these may allow downloads of the article in pdf format at no cost or direct the user to a fee-based subscription site.

University of Washington

dental.washington.edu/dept-oral-med/special-needs/patients-with-special-needs/

While not comprehensive for all disease entities, this site contains descriptions of appropriate dental and medical considerations and treatment modifications required for the management of many patients with special needs. Once at the homepage, the tool “Additional Guidance” can be selected, which then leads to “Special Needs Fact Sheets.” At this page, there are four categories of information designed for “Parent/Caregiver,” “Children,” “Adults,” and “Medical.” While this site does not comprehen-

sively cover all serious systemic conditions, it addresses many of the more common ones, which carry serious implications for possible modifications in dental management and dental drug therapy.

Evaluation of Randomized Controlled Trials and Systematic Reviews

There is a wealth of literature related to various methodologies used for the evaluation of the validity of systematic reviews and randomized controlled trials. Perhaps, the simplest and most reliable for systematic reviews is the AMSTAR tool (Assessment of Multiple Systematic Reviews), which has been published in an updated, refined format as AMSTAR 2 (Shea et al. 2017). The AMSTAR instrument is based on a simple list of 11 items, each of which is categorized into a standardized set of four possible responses: “yes,” “no,” “can’t answer,” and “not applicable,” and a summary score is determined as the sum of all of the “yes” responses. Recently, the AMSTAR instrument and a modified form of it were critically evaluated, and it was determined that AMSTAR is simple, reliable, and valid, although additional research is needed for systematic reviews of mixed study designs, such as diagnostic accuracy test studies, studies of disease etiologies, and studies of prognoses (Pieper et al. 2015).

With regard to the reporting of randomized trials, an instrument termed “CONSORT” is now widely accepted as a standard reporting format and is acknowledged routinely in many journals in which the CONSORT-compliant studies are published (Begg et al. 1996; Moher et al. 2001). Practitioners should familiarize themselves with the fundamental evaluations provided in these publications prior to making changes in patient care based on published literature.

Conclusion

There are many reliable, scientifically vetted internet resources to obtain a variety of information related to dental pharmacology. Most are free of cost and commercial interests and, if used efficiently, can significantly augment the dental practitioner’s access to reliable and, perhaps most importantly, up-to-date information. At this transitional time in the dental profession, that is, moving from practice based largely on historical uses of drugs and the application of medical drug information to off-label dental prescribing scenarios to evidence-based dental practice, the practitioner must carefully assess both the qualitative and quantitative attributes of scientific literature and filter it based on the various levels of scientific evidence. While it is often tempting to adopt a procedure, material or drug to dental practice based solely on the recommendation of an “expert” speaker (level 6 evidence), such decisions must be weighed carefully against the availability—or lack thereof—of higher levels of scientific evidence from randomized controlled trials (RCTs) and systematic reviews (level 1 scientific evidence). *Caveat emptor.*

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